AKTU

B.Tech : I-Year

Electronics Engg.

UNIT - 1 : (i) Semiconductor Diode (ii) Diode Application
(iii) Special Purpose two terminal Devices

ONE SHOT LECTURE

- PDF NOTES (Theory + Derivation + Solved Numerical + PYQs)
- Most Important Questions
- DPP

By Gulshan sir

GW GATEWAY CLASSES

Gateway Classes

AKTU B.Tech I-Year, II-Year, III-Year

Gateway Classes

GW GATEWAY CLASSES

Courses in Application for the Session:2025-26

II-Year Courses will be available in App → Last Exam of I-Year

III-Year Courses will be available in App → Last Exam of II-Year

I-Year Courses available in App → Now Onwards

Gateway Classes

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Subjects By Gulshan sir in App

S.NO	SUBJECT	SEMESTER
1	Engineering Mathematics – I	I
2	Engineering Mathematics – II	II
3	Engineering Physics	I/II
4	Fundamental of electronics engineering	I/II
5	Engineering Mathematics – III	III / IV
6	Engineering Mathematics – IV	III / IV
7	Discrete Mathematics OR Discrete Structures and Theory of Logic (DSTL)	III

Gateway Classes

BEC201 : FUNDAMENTALS OF ELECTRONICS ENGINEERING

UNIT-1

Semiconductor Diode

Depletion layer, V-I characteristics, ideal and practical Diodes, Diode Equivalent Circuits, Zener Diodes breakdown mechanism (Zener and avalanche)

Diode Application

Diode Configuration, Half and Full Wave rectification, Clippers, Clampers, Zener diode as shunt regulator, Voltage-Multiplier Circuits

Special Purpose two terminal Devices

Light-Emitting Diodes, Photo Diodes, Varactor Diodes, Tunnel Diodes.

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 1

Today's Target

- Basics of Semiconductors
- DPP
- PYQs

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GATEWAY CLASSES

Gateway Classes

Electronic Device

- Any device whose action is based on the controlled flow of electrons through it is called an Electronic device

Electronics

- The branch of engineering that deals with the study of these electronic devices is called electronics
- Electronic devices are the basic building blocks of all the electronic circuits

Gateway Classes

Classification of solid Materials on the basis of their electrical properties

(1) Conductors

- Large number of mobile charge carriers or free electrons
- High conductivity or low resistivity

Example: Silver, Copper, Aluminium

(2) Insulators

- No charge carriers or free electrons
- Low conductivity or high resistivity

Example: Glass, Quartz, Rubber etc.

(3) Semiconductors

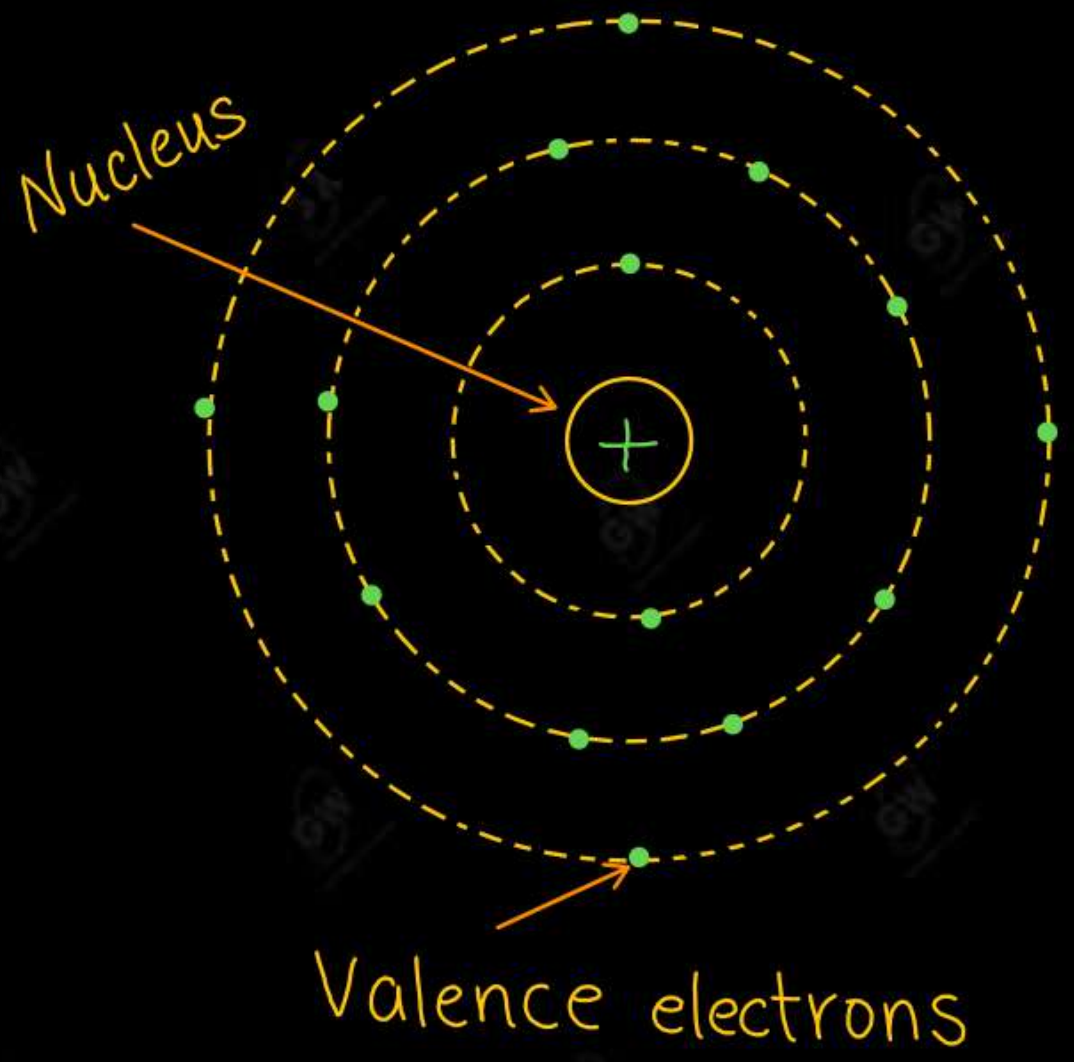
- Conductivity or resistivity lies between conductor and insulator

Example: Silicon (Si) and Germanium (Ge)

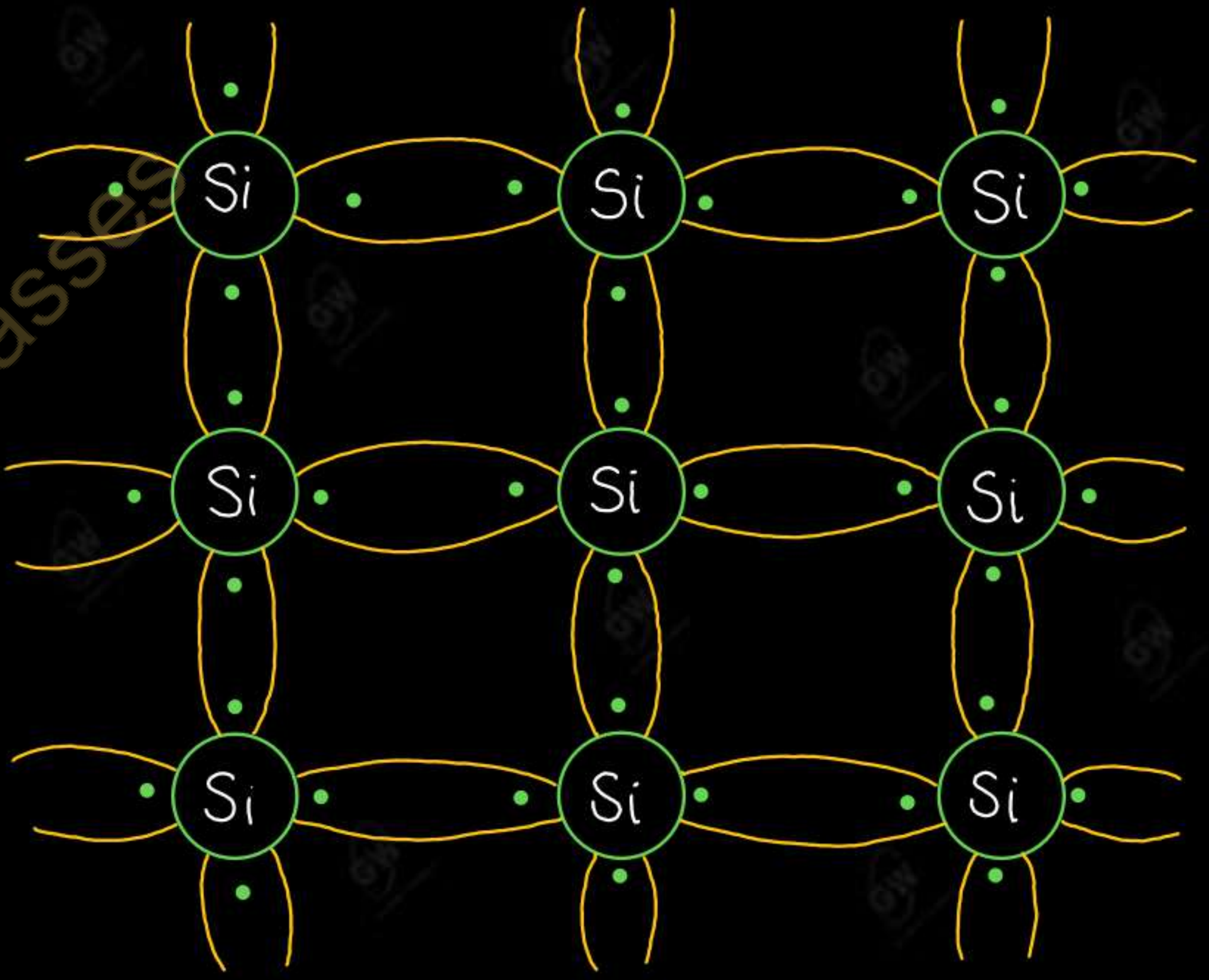
Gateway Classes

Silicon (Si)

- Atomic Number = 14, Number of proton = 14, Number of electron = 14
- Electron are distributes as 2, 8, 4 in the I, II, and III orbit respectively
- Valence electron = 4 (Tetravalent)



Atomic Structure of Si atom

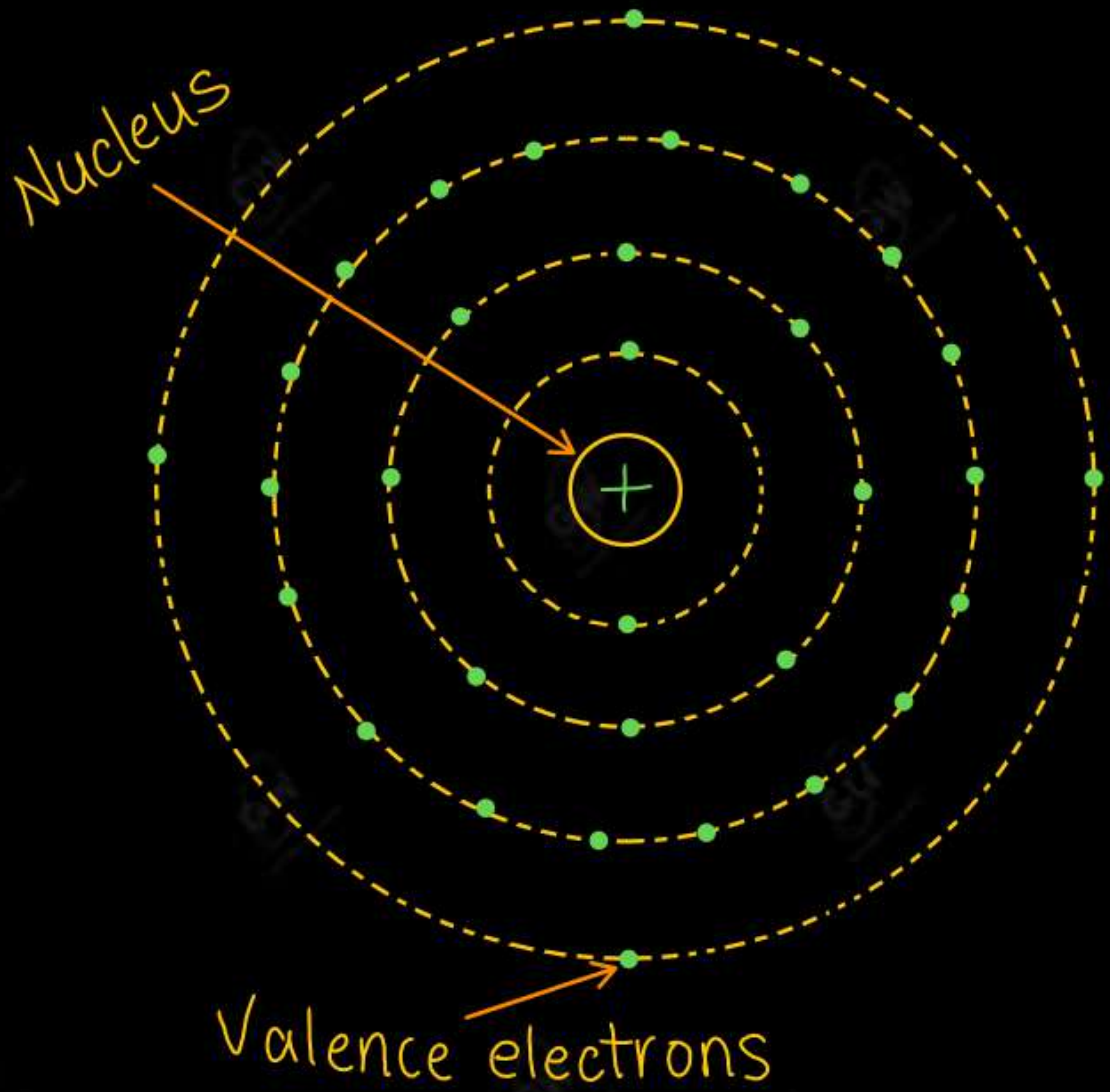


Crystal structure of Si atom

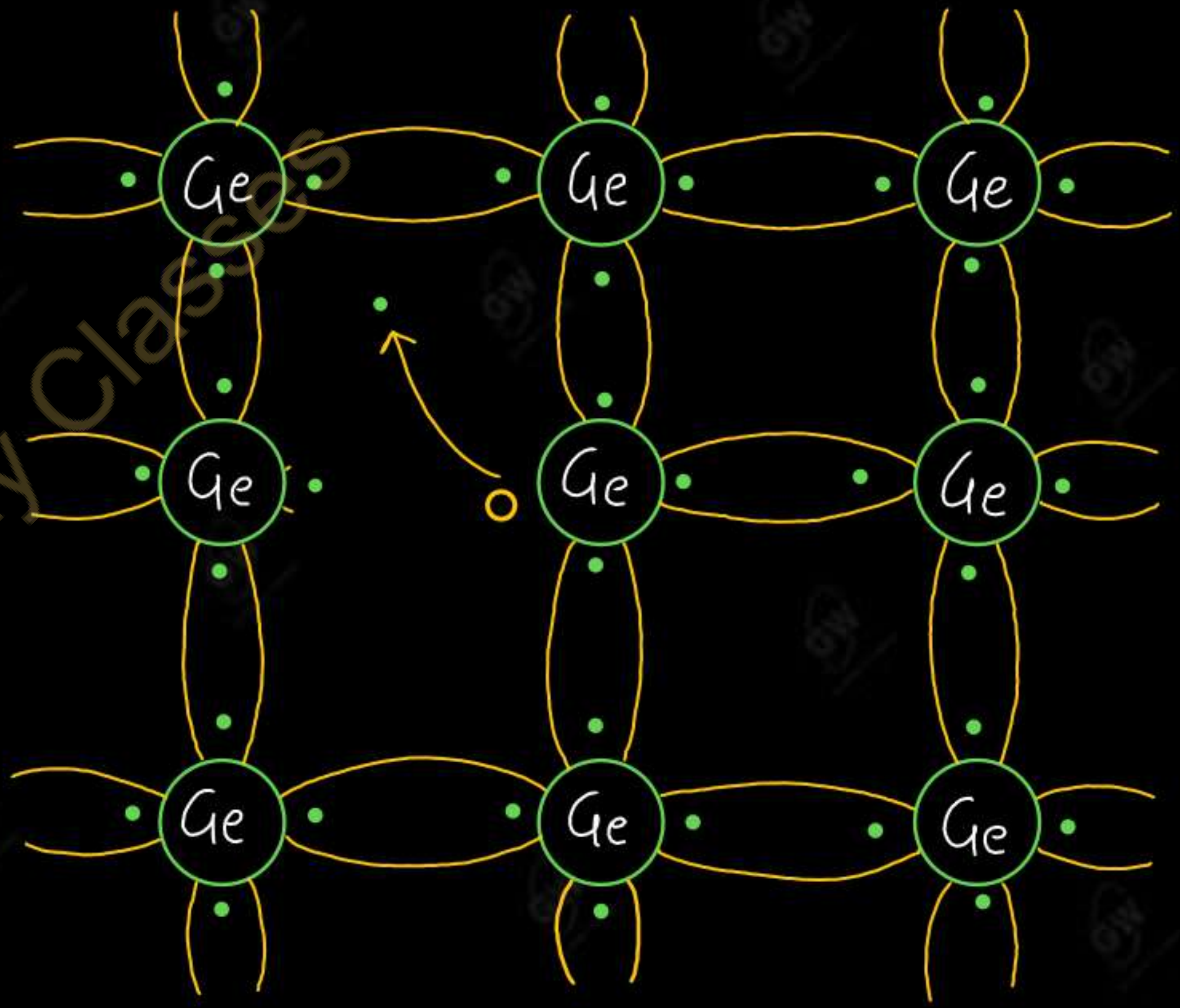
Gateway Classes

Germanium (Ge)

- Atomic Number = 32, Number of proton = 32, Number of electron = 32
- Electron are distributes as 2, 8, 18, 4 in the I, II, III and IV orbit respectively
- Valence electron = 4 (Tetravalent)



Atomic Structure of Ge atom

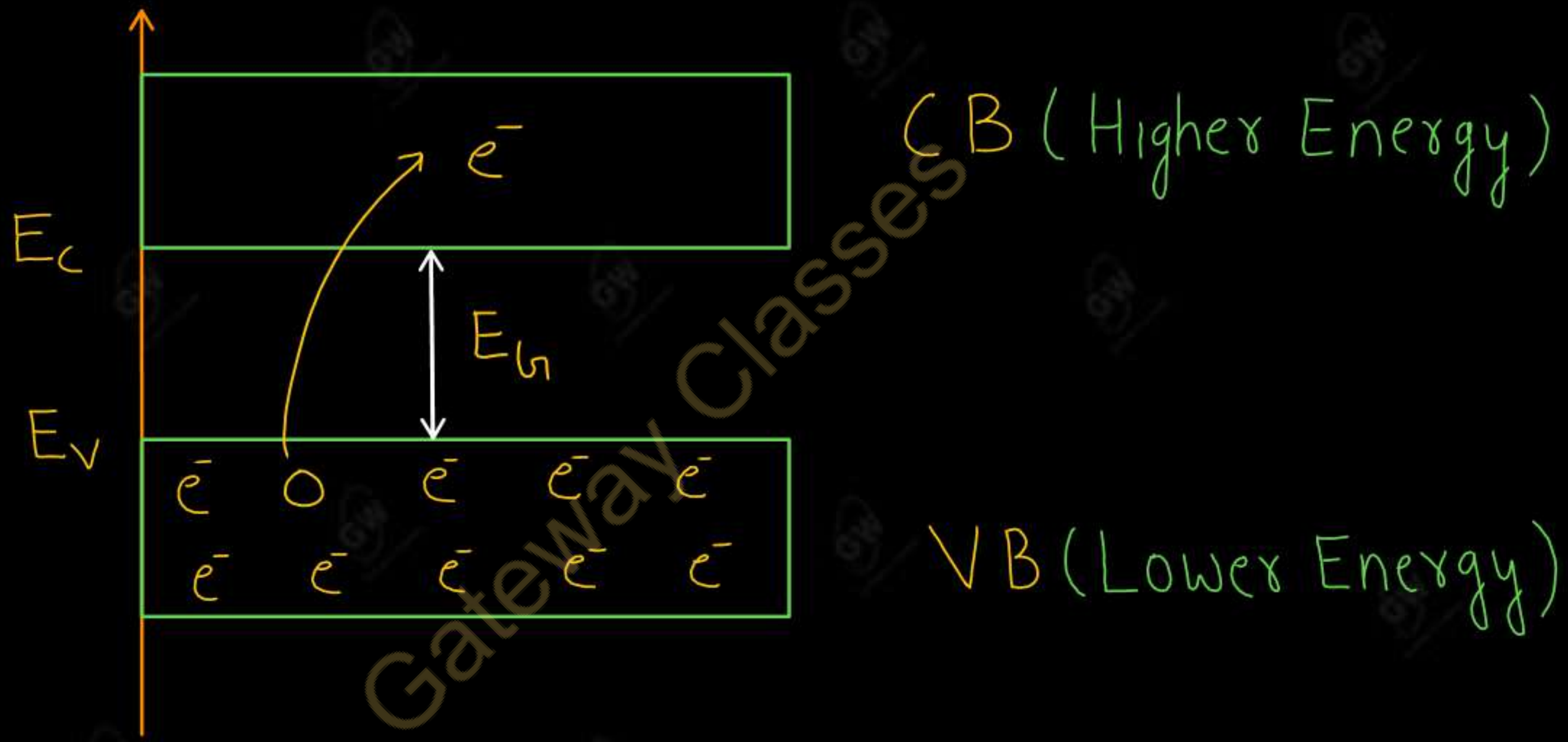


Crystal structure of Ge atom

Gateway Classes

Energy band theory

- Valence shell energies fluctuates in a solids (Range of energies)
- The range of energies possessed by electrons of the same orbit in a solid is called energy band



Valence band

- The energy band which possess all bo nded Valence electrons is called Valence band

Conduction band

- The energy band which possess free electrons is called Conduction band

Note : (1) Electrons in conduction band are responsible for the flow of current

(2) Holes in valence band are also responsible for the flow of current

(3) Electrons are negative charge carriers

(4) Holes act as positive charge carriers

Forbidden energy gap OR Band gap energy (E_g)

- Minimum energy required to move electrons from valence band to conduction band

OR

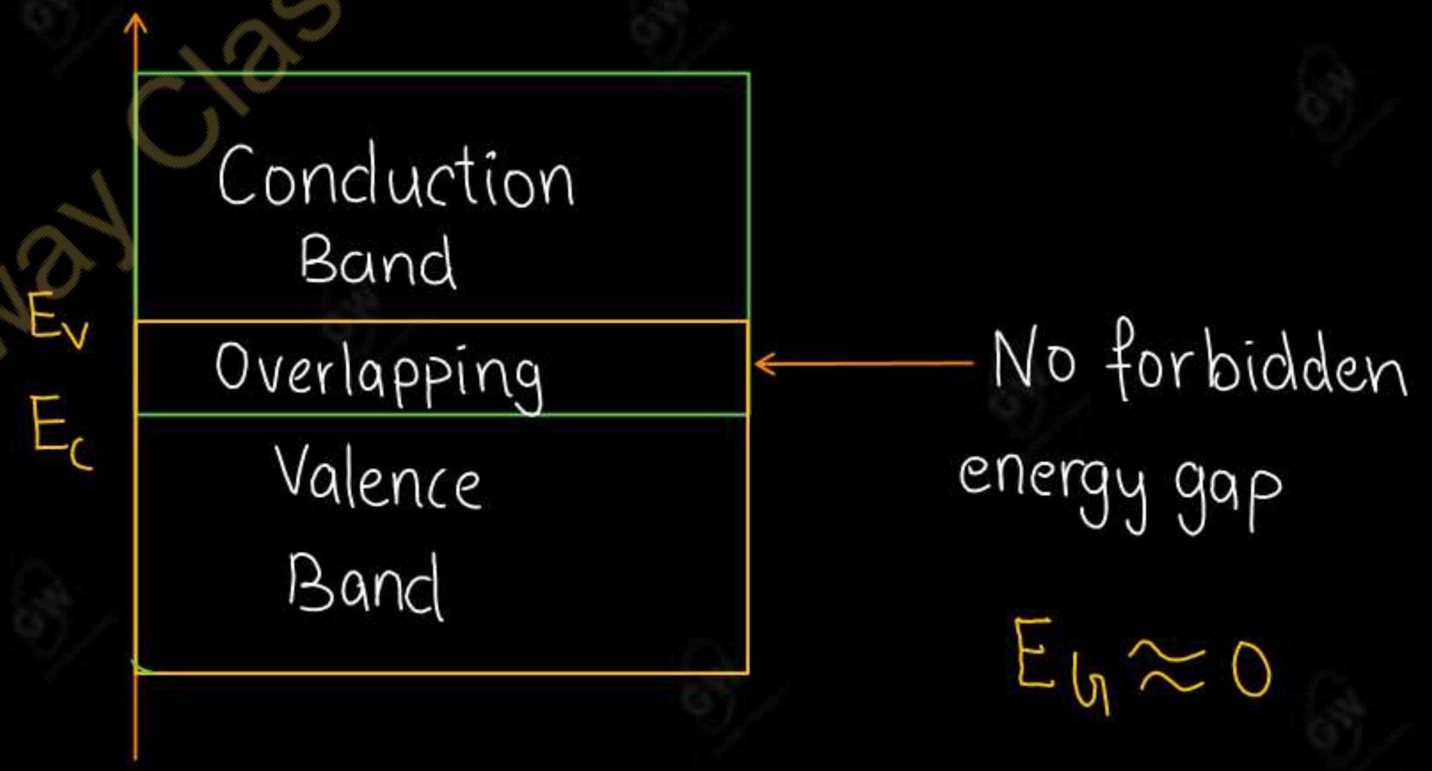
There is an energy gap between Valence band and conduction band known as Forbidden energy gap

Gateway Classes

(1) Conductors

- Forbidden energy gap between Valence band and Conduction band is Zero (No Forbidden energy gap)
- In fact, Valence band and conduction band overlap each other
- Hence, it is very easy for a Valence band electron to become a Conduction band electron
- For this reason, the electrical conductivity of conductors is very high

Examples: Copper, Silver, Aluminum



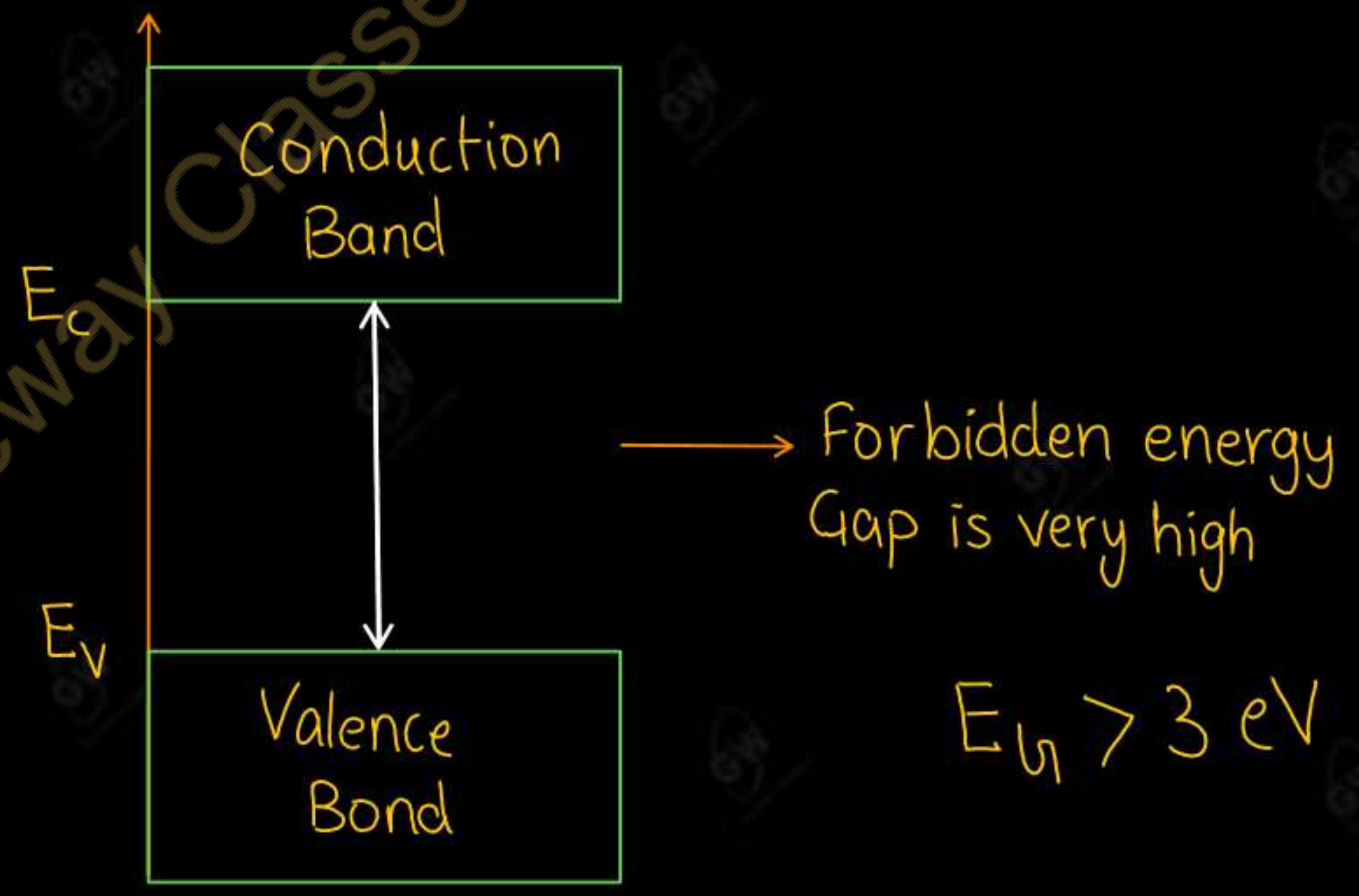
Energy band diagram of metals

Gateway Classes

(2) Insulators

- The forbidden energy gap between Valence band and Conduction band is very large
- Due to this wide gap, it is almost impossible for an electron to cross the gap to go from valence band to conduction band
- For this reason, the electrical conductivity of insulators is very low

Example: Glass, Rubber etc.



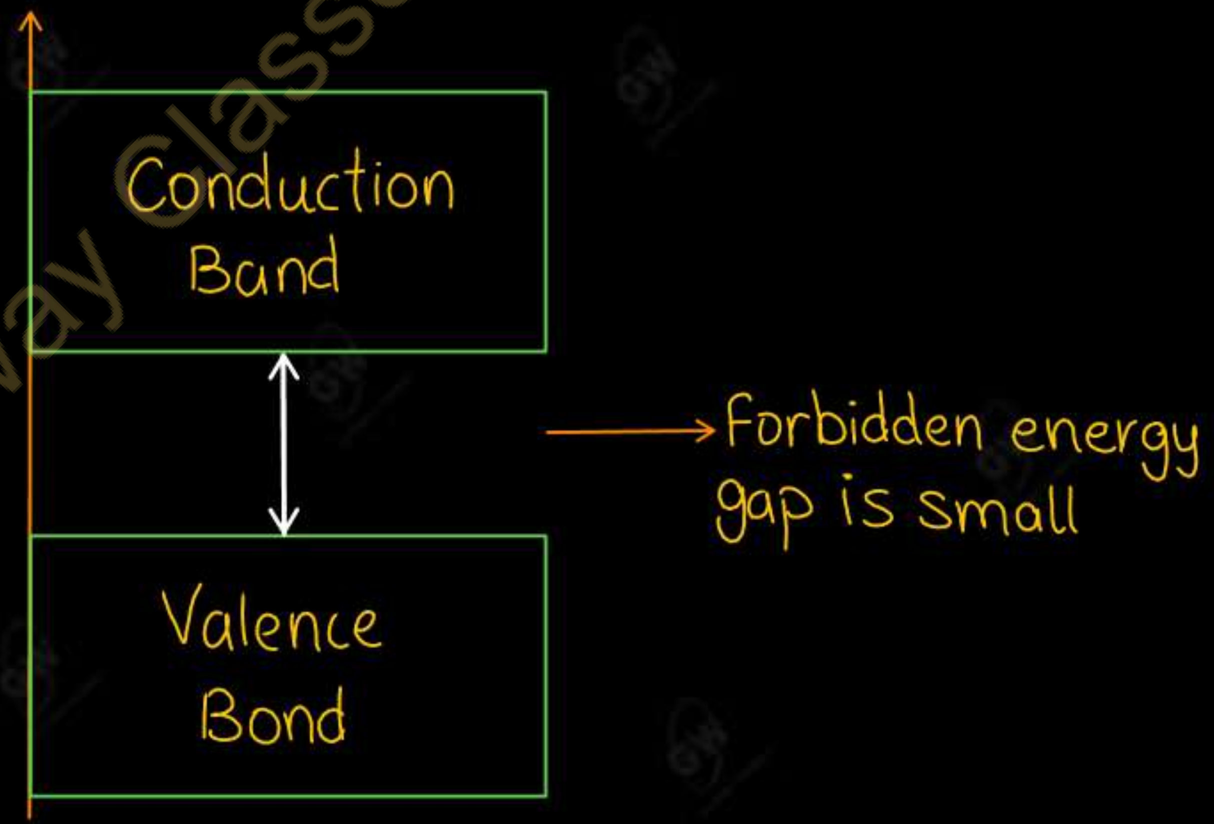
Energy band diagram of Insulator

Gateway Classes

(3) Semiconductor

- The forbidden energy gap between the valence band and conduction band is not very wide
- For Si, Forbidden energy gap $E_G = 1.1 \text{ eV}$ and For Ge, Forbidden gap $E_G = 0.7 \text{ eV}$
- For this reason, the electrical conductivity of semiconductor lies between conductor and insulator

Example : Silicon (Si) and Germanium (Ge)



Energy band diagram of semiconductor

Gateway Classes

Effect of temperature on conductivity of semiconductor

(i) At low temperature (0 K)

- Valence band is completely filled
- Conduction band is completely empty
- Thus, a semiconductor behaves as an insulator at low temperature

(ii) At room temperature (300 K)

- At room temperature some electron jump to conduction band due to thermal energy.
- Valence band is partially filled
- Conduction band is also partially filled
- Hence at room temperature, semiconductors are able to conduct some electric current

Note : (1) As temperature increases, conductivity of semiconductor increases.

(2) As temperature increases, conductivity of conductor decreases.

Gateway Classes

GW GATEWAY CLASSES

Classification of Semiconductors

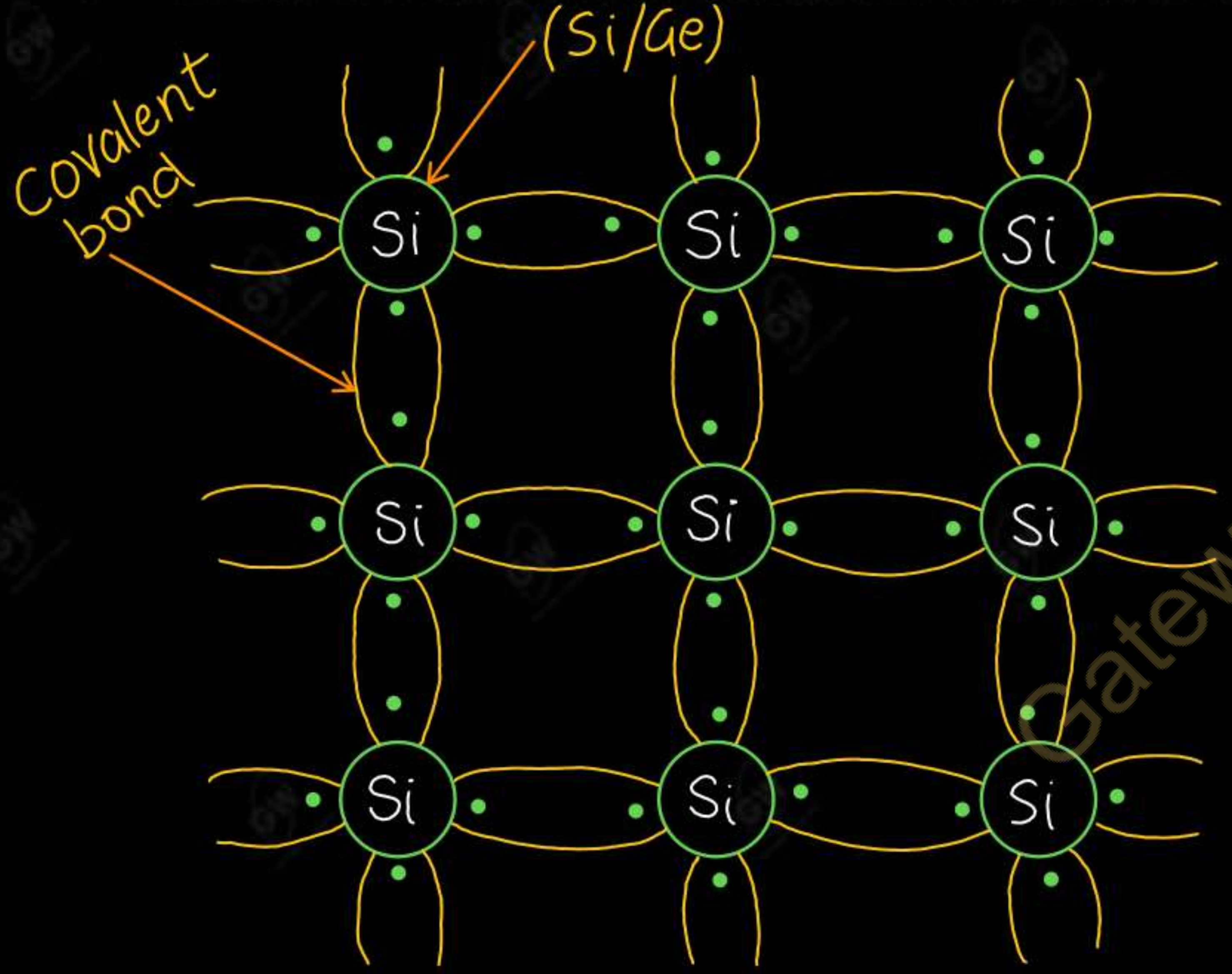
- **Semiconductors**
  - Intrinsic Semiconductor
  - Extrinsic Semiconductor
    - N-type Semiconductor
    - P-type Semiconductor

Gateway Classes

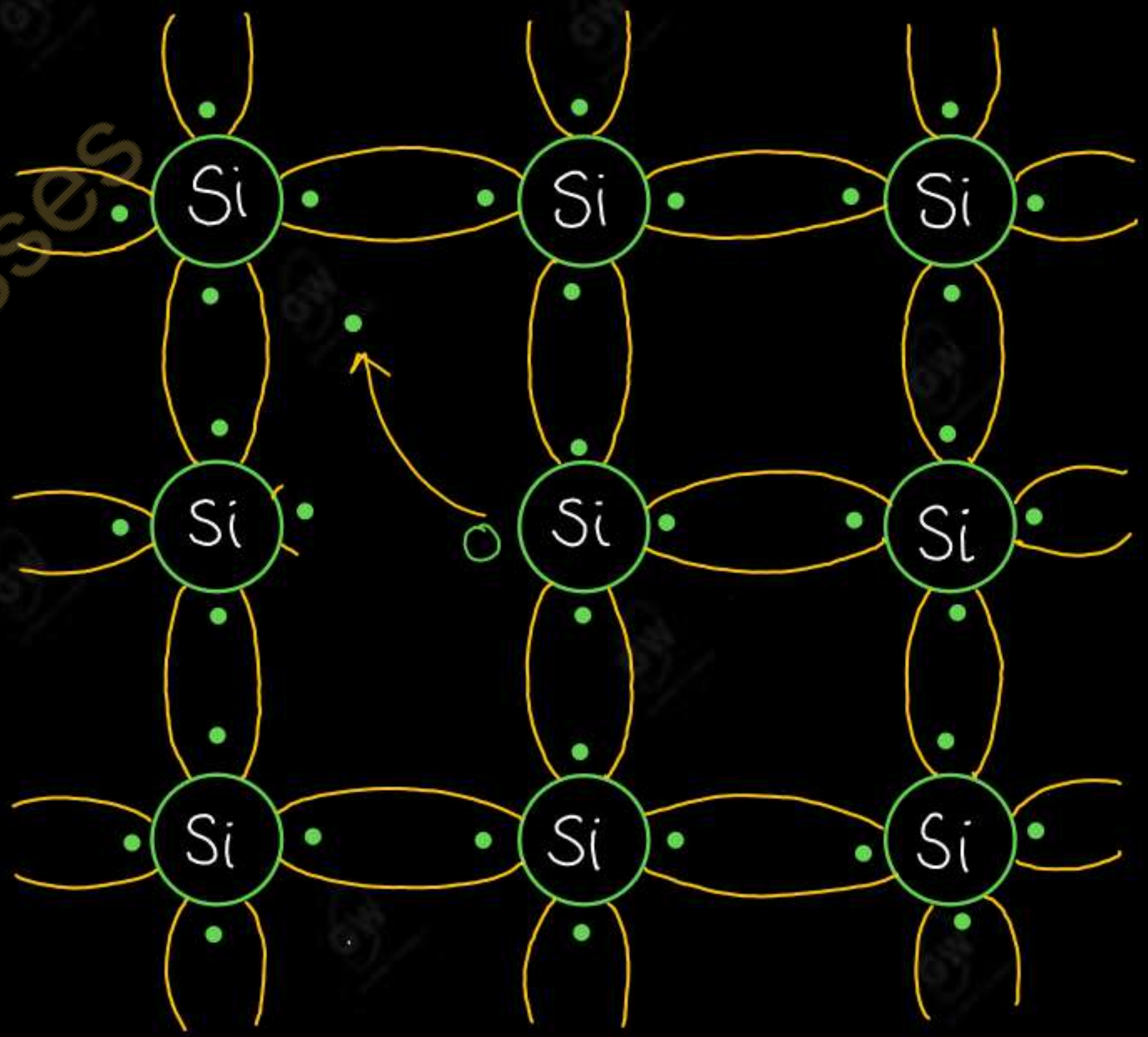
Intrinsic Semiconductor

- A semiconductor in an extremely pure form is known as Intrinsic Semiconductor.
- The Silicon (Si) and Germanium (Ge) are the two most widely used intrinsic semiconductor
- At low temperature (0 K), all the valence electrons are tightly held by covalent bonds with another atoms.

- Hence electrons cannot move through the crystal structure or we can say no free electrons are available to conduct electricity
- Thus, the intrinsic semiconductors behaves as insulator at absolute zero (0 K)temperature.



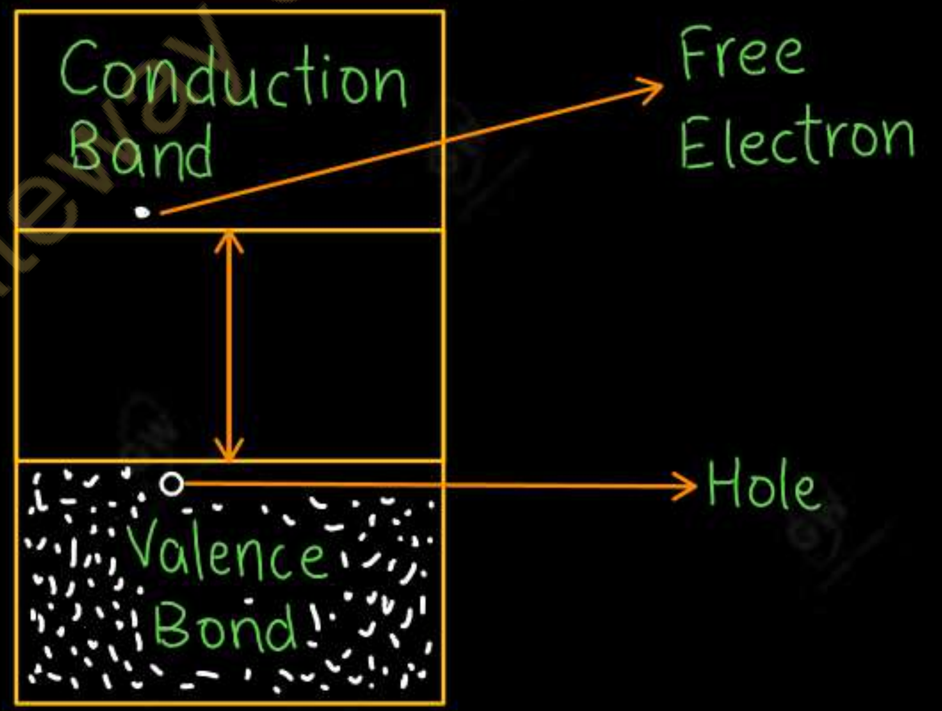
Crystalline structure at 0K



Crystalline structure at 300K

Gateway Classes

- But at room temperature (300K), some of covalent bond break and the electron becomes free to move through the crystal.
- Due to this electron movement a vacancy is also produced known as hole.
- Thus, electron and hole are produced in pair (Concentration of free electrons = Concentration of holes)
- This type of simultaneous generation of electrons and holes due to temperature is called Thermal Generation
- A pure semiconductor have small conductivity at room temperature and not suitable for practical operation



Energy band diagram

Gateway Classes

2. Extrinsic Semi-Conductor

- An intrinsic Semiconductor is converted in to extrinsic Semiconductor by adding impurities.

  The process of adding Impurities to intrinsic semiconductor is called doping and impure semiconductor is called extrinsic Semiconductor.

- Types of impurities added to intrinsic semiconductor

  (i) Trivalent Impurities or Acceptor impurities : They have 3 valence electrons (Example : B, Al, Ga)

  (ii) Pentavalent Impurities or Donor impurities : They have 5 valence electrons (Example : P, As, Sb)

- Need of Doping

  Doping is done to increases the conductivity of semiconductor devices. Doping creates extra holes and extra electrons to increase the flow of currents. So, conductivity of intrinsic semiconductor increases

- Depending on the type of impurity added, extrinsic Semiconductor are further classified as

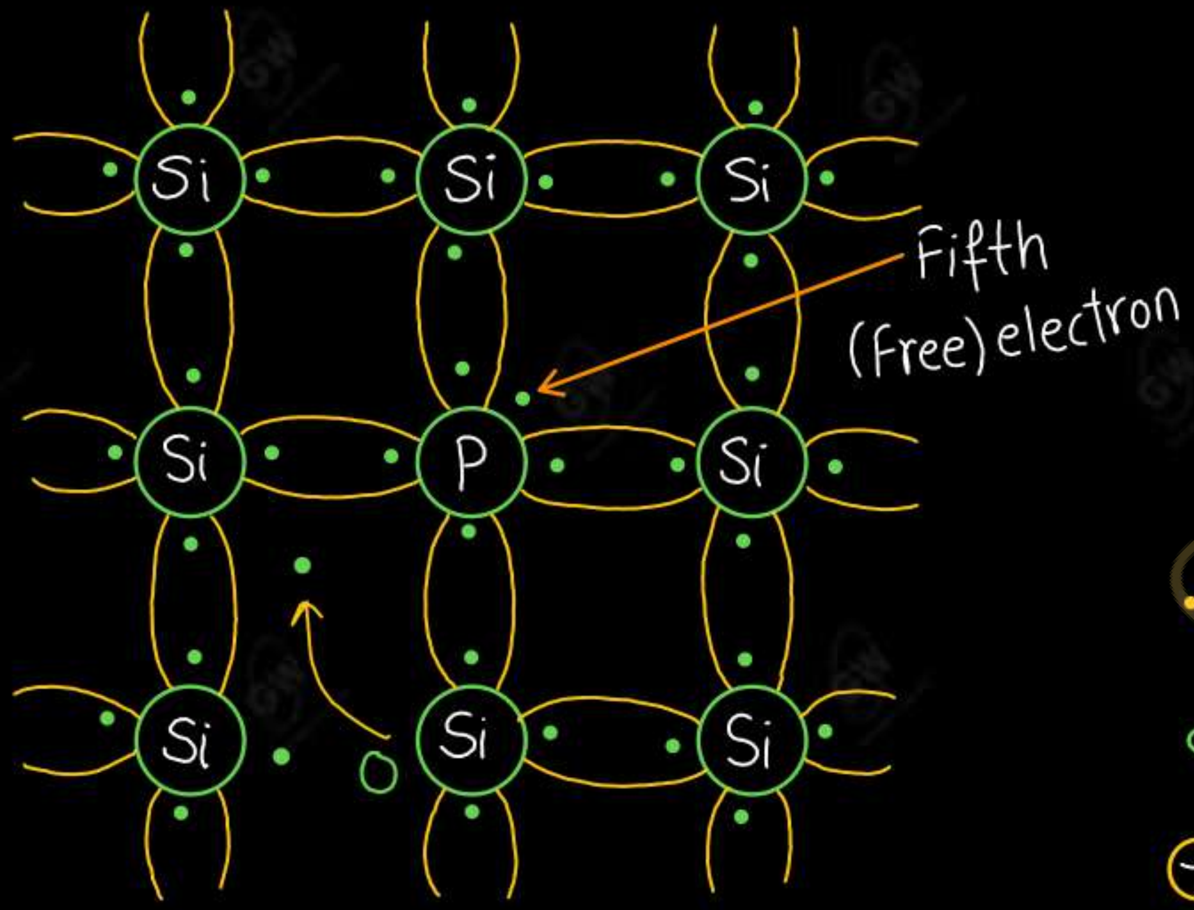
  (i) N - type Semiconductor

  (ii) P – type Semiconductor

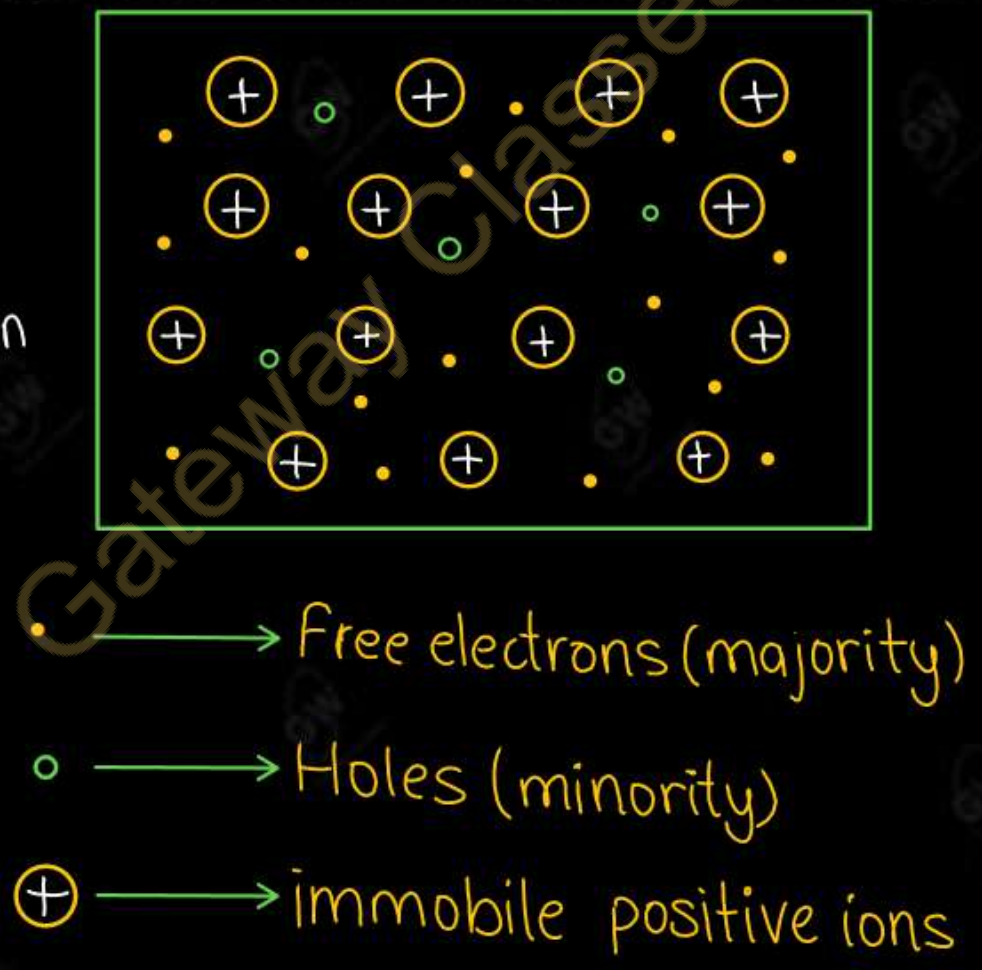
Gateway Classes

(i) N-Type Semiconductor

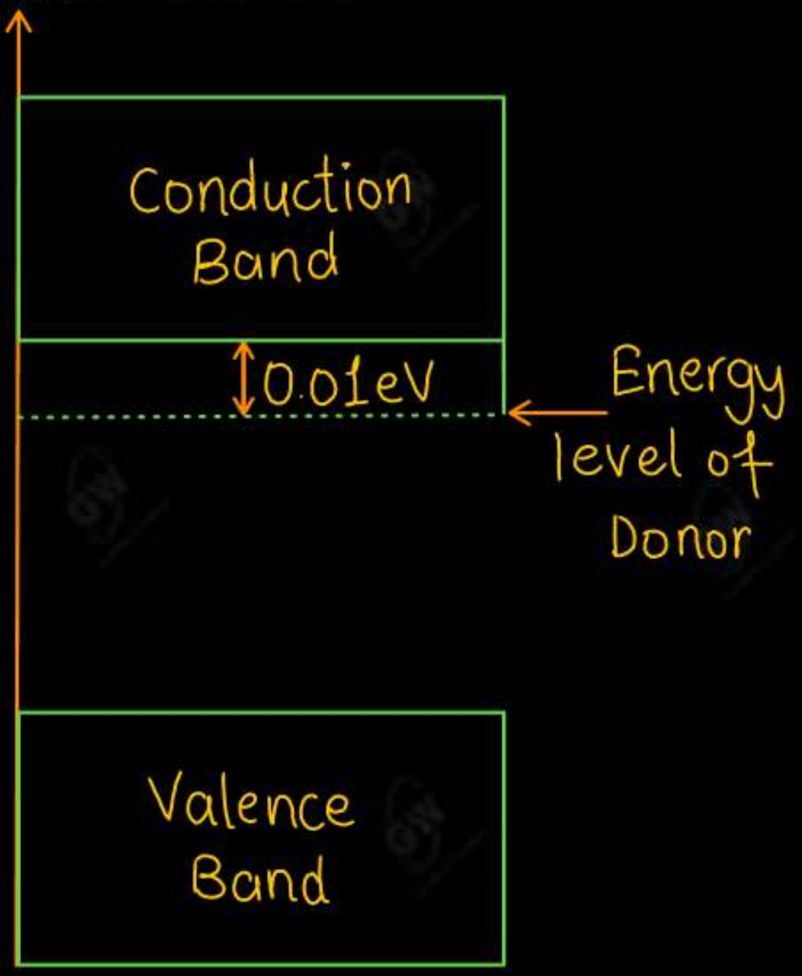
- When a Pentavalent impurity is added in pure semiconductor, the resulting semiconductor is N-type
- The Pentavalent impurity atom is called donor atom because it donates one extra electron for conduction
- Equal number of electrons and holes are also generated by thermal energy
- Hence, electrons are the majority charge carriers and holes are the minority charge carriers.



N-Type Semiconductor



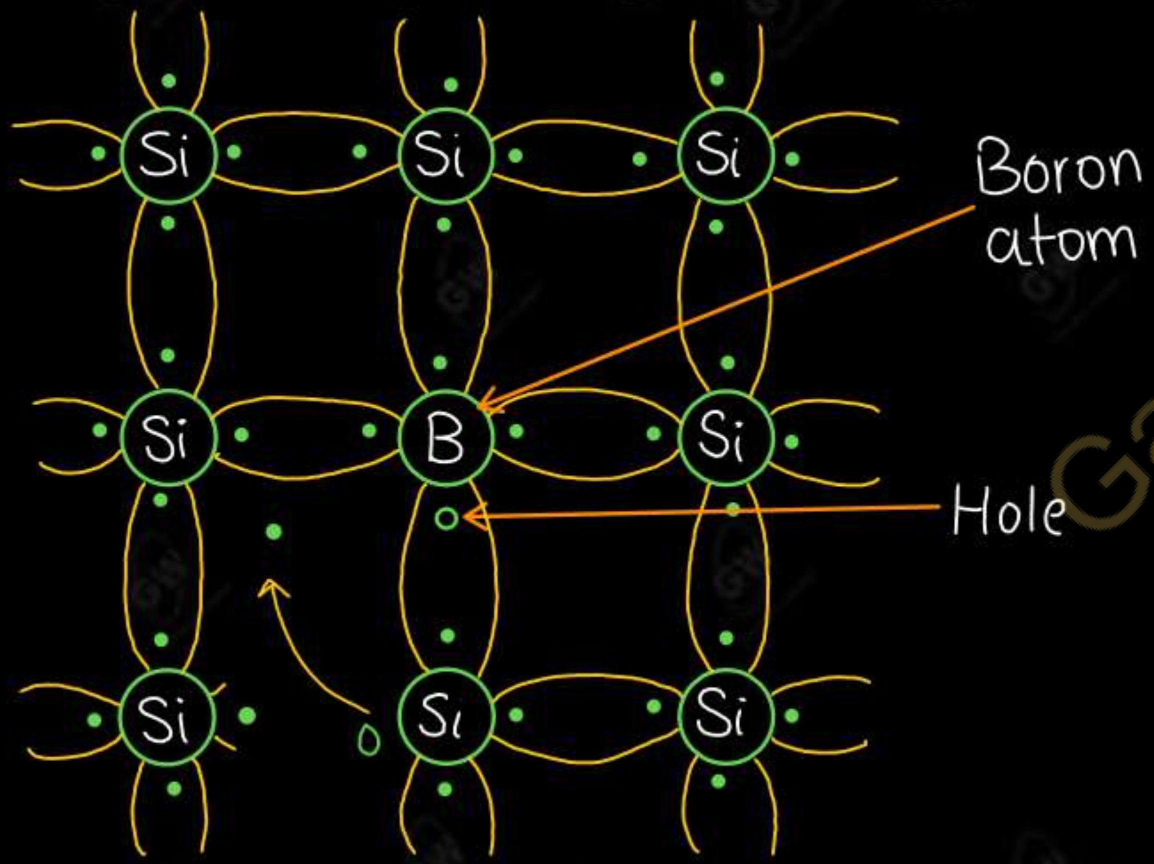
Representation



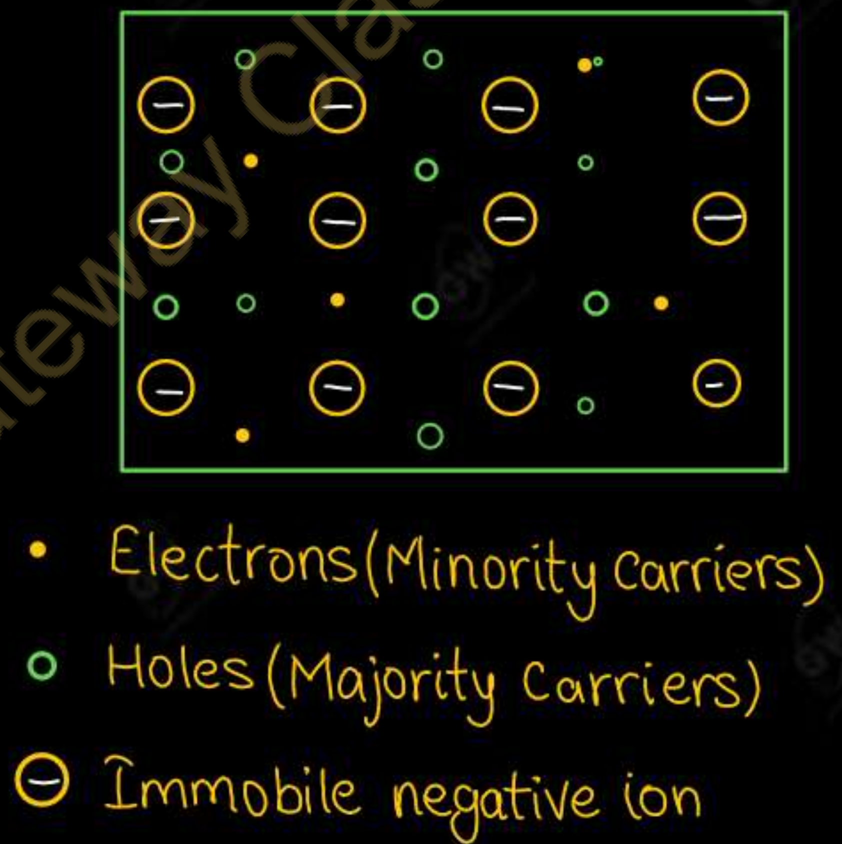
Energy band diagram

(ii) P-type Semiconductor

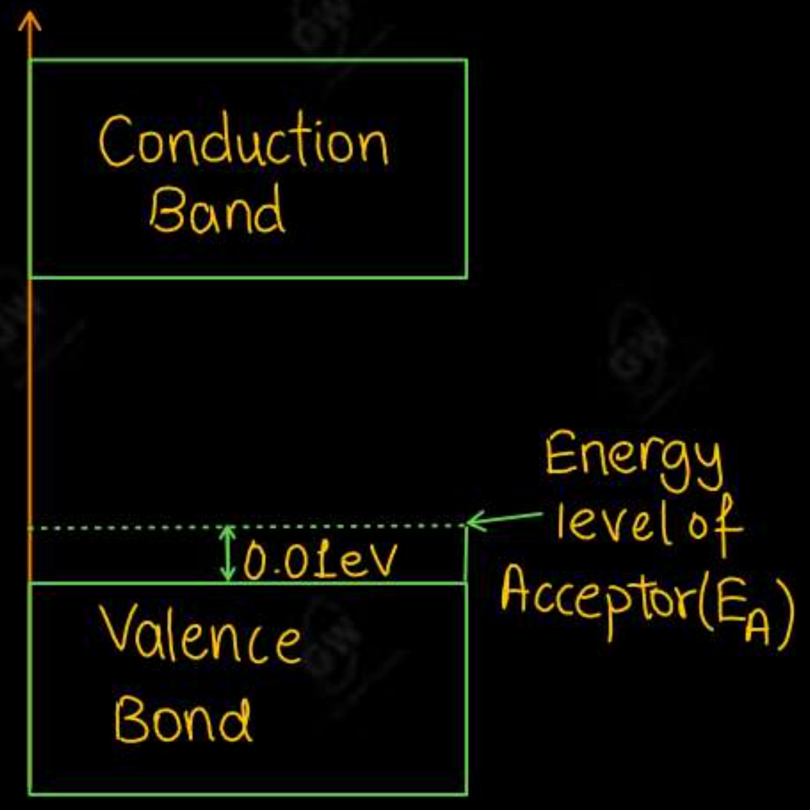
- When a Trivalent impurity is added in pure semiconductor, the resulting semiconductor is P-type
- The trivalent impurity atom is called an acceptor atom because it creates a hole which can accept an electron from the neighbouring bond.
- Equal number of electrons and holes are also generated by thermal energy
- Hence, holes are the majority charge carriers and electrons are the minority charge carriers.



P-Type Semiconductor



Representation



Energy band diagram

Gateway Classes

Differentiate between N-Type and P-type Semiconductor

S.NO	N-type	P-type
1	Pentavalent impurities such as P, As, Sb are added	Trivalent impurities such as B, Al, Ga are added
2	Impurities are donor atoms	Impurities are acceptor atoms
3	Electrons are the majority charge carriers and holes are the minority charge carriers	Holes are the majority charge carriers and electrons are the minority charge carriers.
4	The donor energy level is close to conduction band	The acceptor energy level is close to valence band

Gateway Classes

UNIT - 1 : DPP-1

Topic : Basic Concepts of Semiconductors

Q.1 Classify materials with the help of energy band diagram AKTU-2017

Q.2 Explain the effect of temperature on conductivity of a semiconductor AKTU-2016

Q.3 What do you mean by doping ? Describe its need. AKTU-2022

Q.4 Differentiate between N-type and P-type semiconductor AKTU-2017

Gateway Classes

AKTU

B.Tech : I-Year

GATEWAY CLASSES

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 2

Gateway Classes

Today's Target

- P-N Junction Diode
- Forward Biasing and Reverse Biasing
- V-I characteristic of P-N Junction Diode
- DPP
- PYQs

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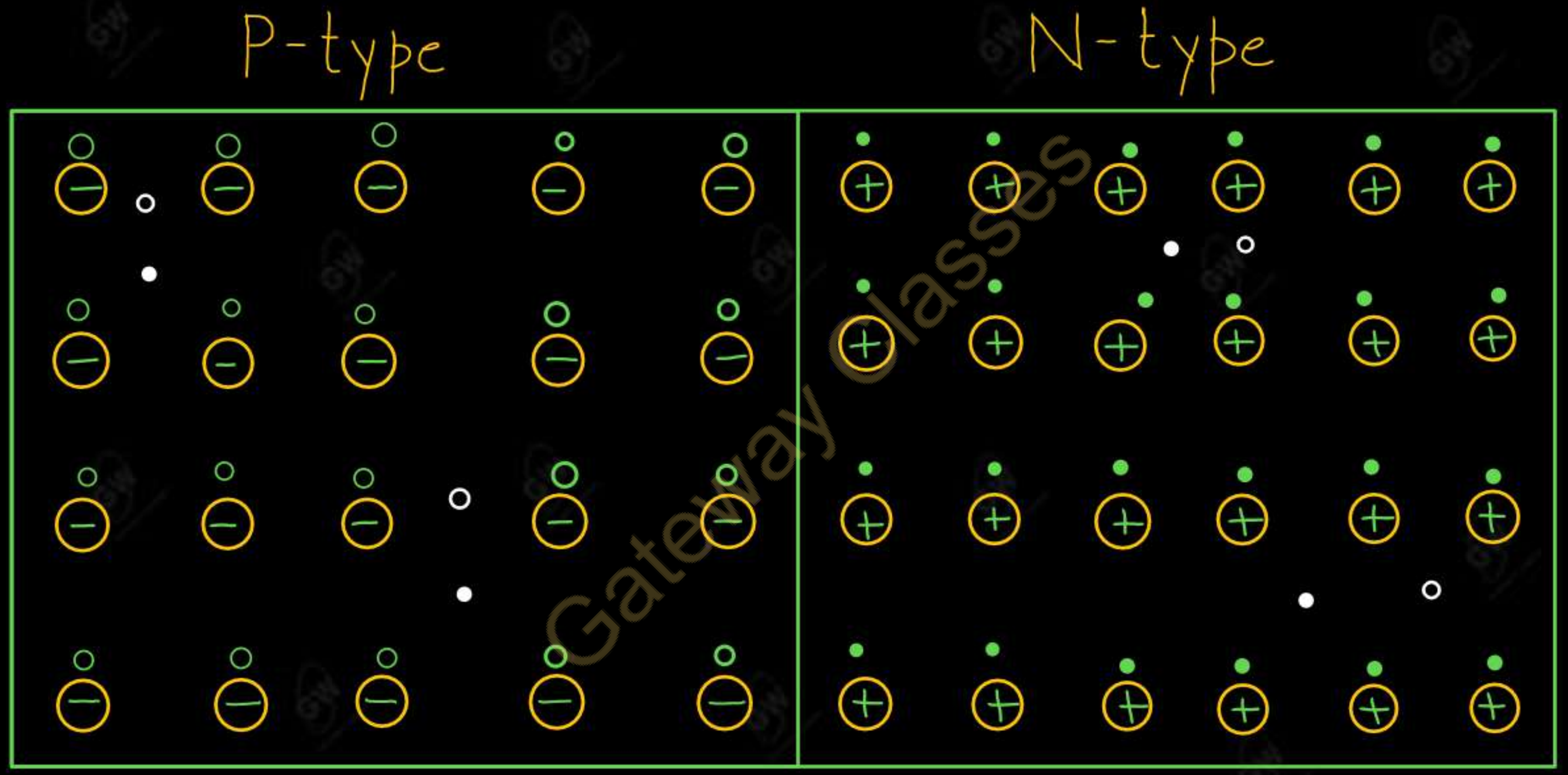
P-N Junction Diode (Unbiased Condition)

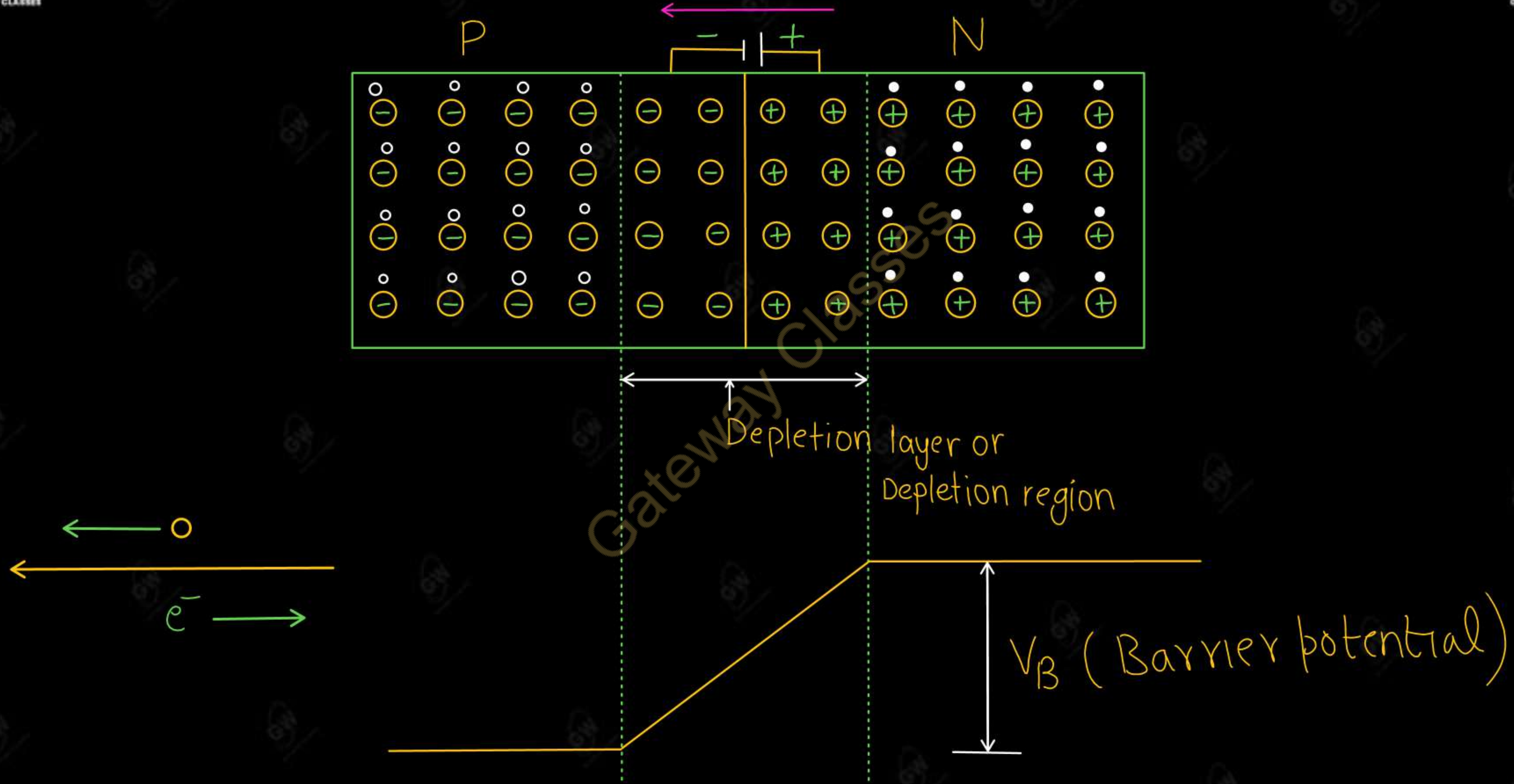
- It is a single crystal of Ge or Si doped in such a manner that one half portion of it acts as a P-type semiconductor and other half as a N-type semiconductor.



- The boundary or region of transition between N-type and P-type semiconductor is called PN junction.

Gateway Classes

➤ Diffusion of electrons leaves positive donor ions on the N-side of the junction.



- Due to the presence of immobile positive and negative ions on opposite side of junction, an electric field is generated at the junction from N-side to P-side.
- Electric field stops further diffusion of electrons and holes.
- The layer of immobile positive and negative ions which have no free electrons and holes called as depletion layer.
- The accumulation of negative charges in the P-region and positive charges in the N-region set up a potential difference across the junction. This act as a barrier and is called barrier potential or junction potential.

(i) Barrier potential for silicon (Si)

$$V_B = 0.7V$$

(ii) Barrier potential for germanium (Ge)

$$V_B = 0.3V$$

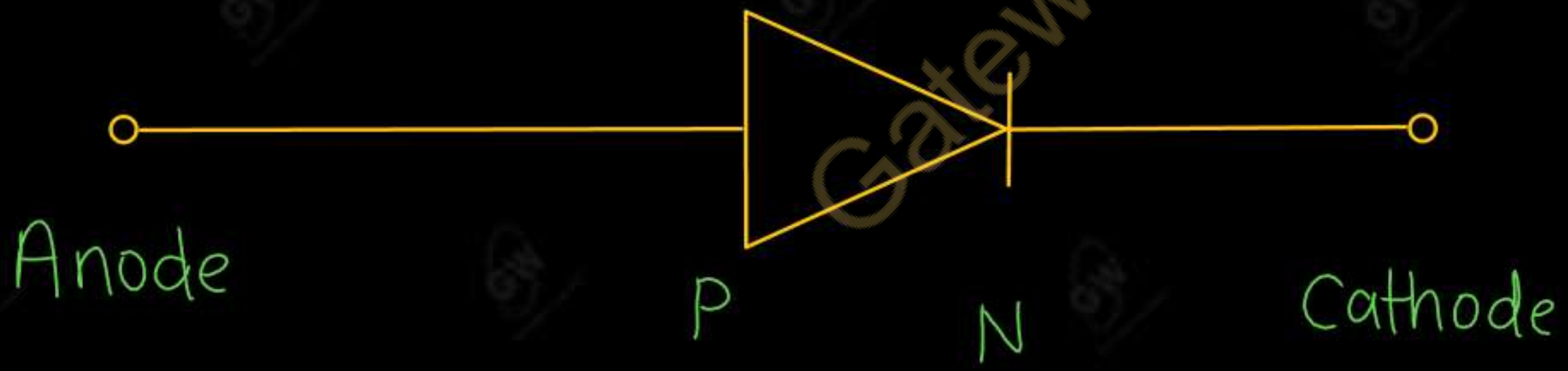
Gateway Classes

NOTE

(1) Electron move from lower to higher potential

(2) Hole move from higher to lower potential

Circuit Symbol for a P-N Junction diode



Gateway Classes

P-type Semiconductor

⊖ ⟶ Acceptor ions (Trivalent impurity)

○ ⟶ Holes (Majority charge carriers)

• ⟶ Electron (Minority charge carriers)

N-type Semiconductor

⊕ ⟶ Donor ions (Pentavalent impurity)

○ ⟶ Holes (Minority charge carriers)

• ⟶ Electron (Majority charge carriers)

Formation of Depletion Region

- When a P-N Junction is formed, the P-side of the junction has a higher concentration of holes while the N-side has a higher concentration of electrons.
- Due to the concentration gradient at the junction

  (i) Holes from the P-region diffuse in to the N-region and recombine with the electrons in the N-region and neutralize each other.

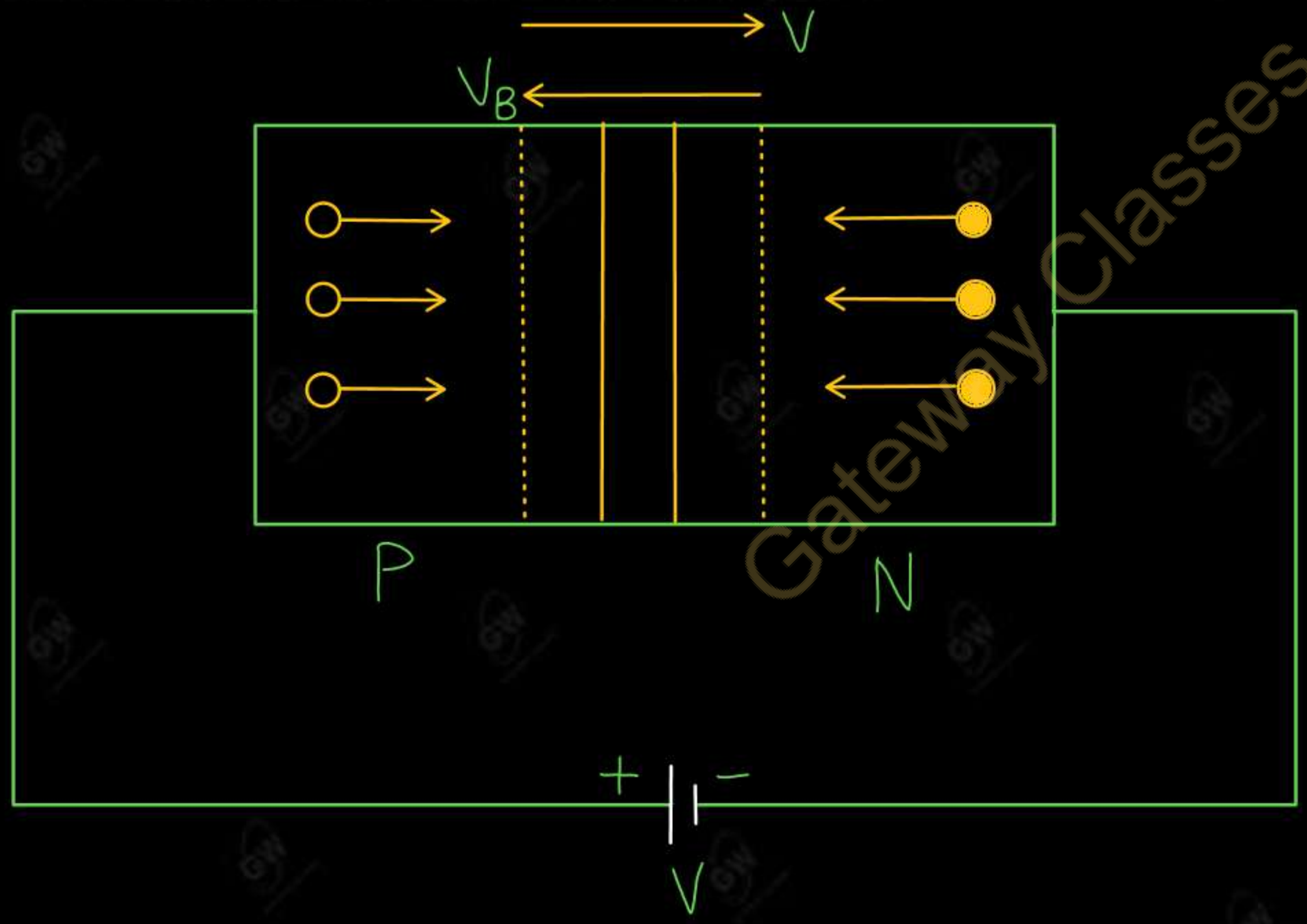
  (ii) Electrons from the N-region diffuse in to the P-region and recombine with the holes in the P-region and neutralize each other.
- Diffusion of holes leaves negative acceptor ions on the P-side of the junction.

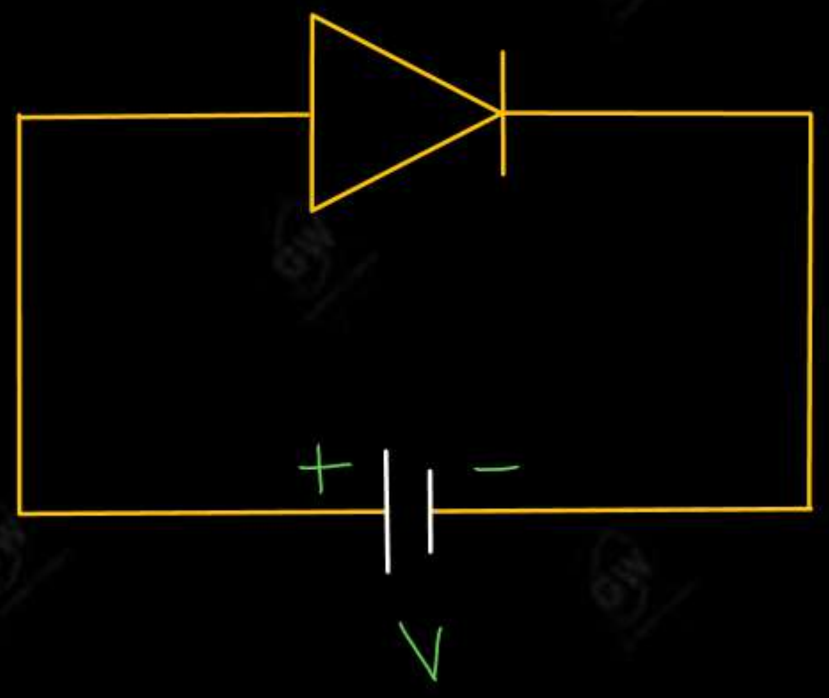
Gateway Classes

Working of P-N Junction under Forward bias and Reverse bias

Forward Biasing

- If the positive terminal of a battery is connected to the P-side and negative terminal to the N-side, then the P-N Junction is said to be forward biased.





➢ In forward bias, majority carriers moves towards the junction and minority carriers moves away from the junction

➢ The diffusion of electrons and holes in to the depletion layer decreases its width.

➢ When V exceeds V_B, the majority charge carriers start flowing easily across the junction and set up a large current (≈ mA) called forward current in the circuit.

➢ The current increases with the increase in applied voltage.

Knee Voltage OR Cut-in voltage

➢ A minimum positive voltage is required to start conduction in a forward biased diode. This minimum positive voltage is called knee voltage or cut-in voltage.

For Germanium (Ge)

$V = 0.3V$

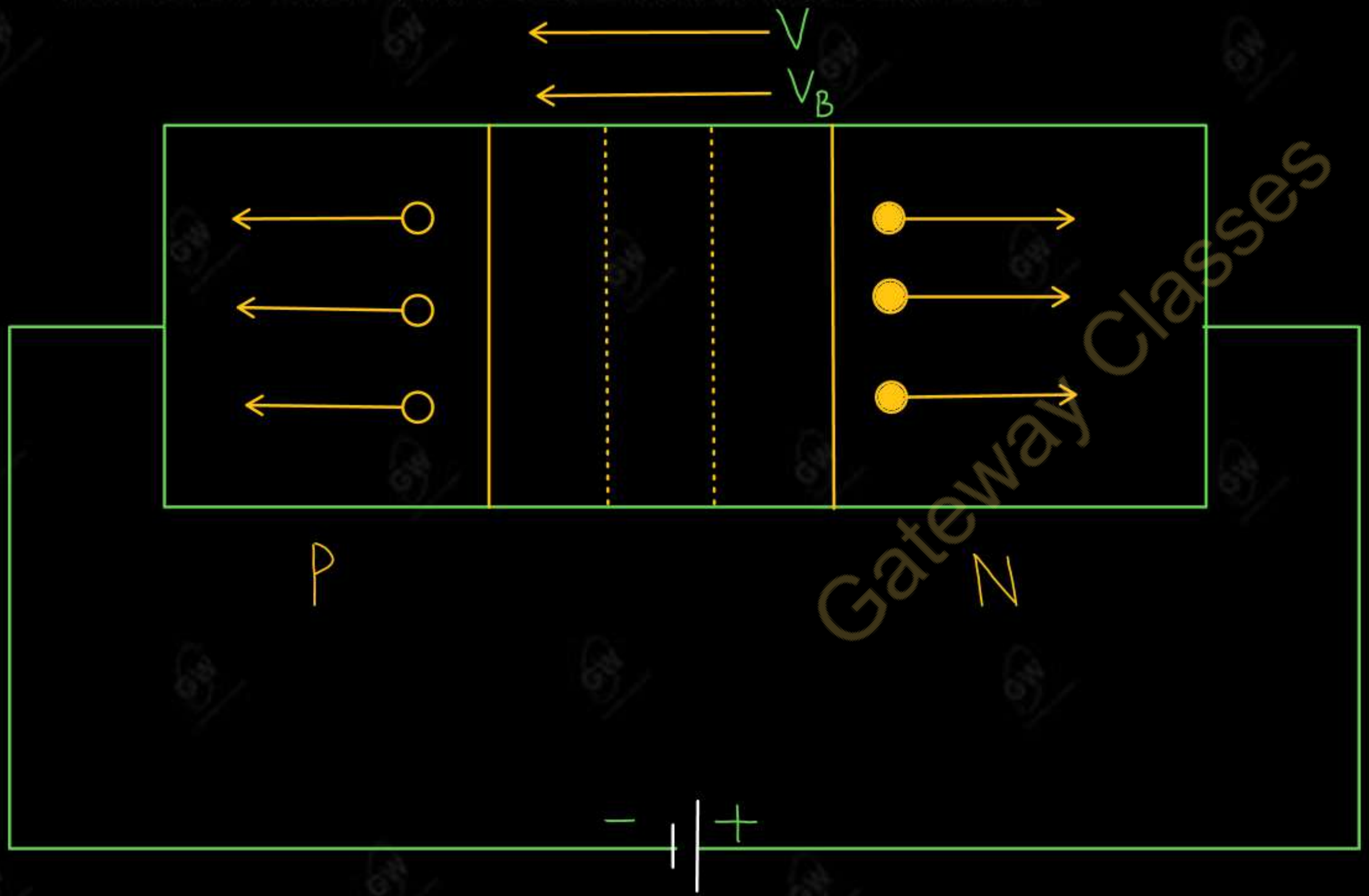
For Silicon (Si)

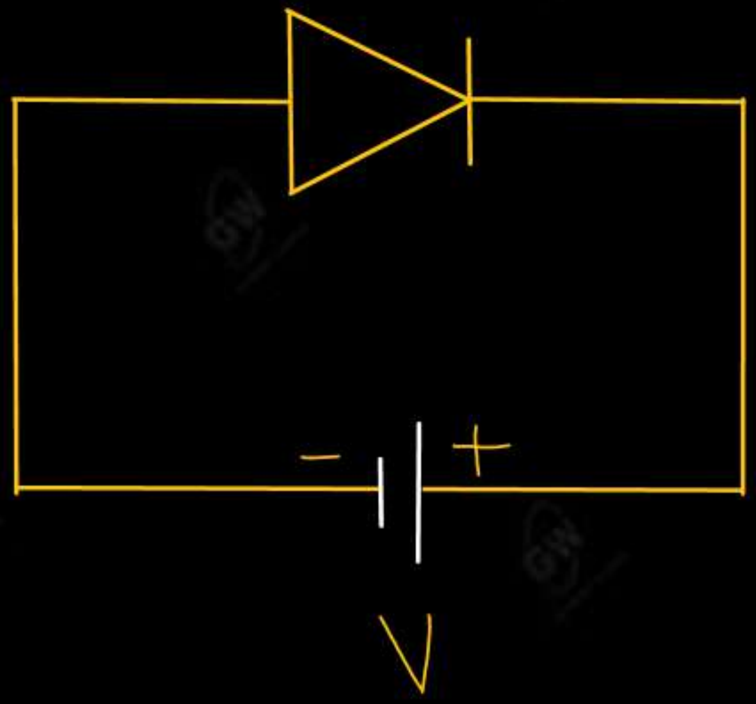
$V = 0.7V$

Gateway Classes

Reverse Biasing

➤ If the positive terminal of a battery is connected to the N-side and negative terminal to the P-side, then the P-N Junction is said to be reverse biased





- In reverse bias, the majority charge carriers move away from the junction, increasing the width of depletion layer.
- No current flow across the junction due to majority charge carriers.
- However, minority charge carriers cross the junction and constitute a small current called reverse saturation current or leakage current.

P-N Junction Diode

P-N junction diode is a semiconductor device, formed by joining P-type and N-type semiconductor materials, that allow current to flow in one direction (forward bias) and block it in the opposite direction (reverse bias). It is also called Semiconductor Diode or Diode

Gateway Classes

Difference between Forward bias and Reverse bias

S.No	Forward Bias	Reverse Bias
(1)	Potential Barrier reduces	Potential Barrier increases
(2)	Width of depletion layer decreases	Width of depletion layer increases
(3)	PN Junction provides very small resistance	PN Junction provides very high resistance
(4)	Forward current flow in the circuit	Reverse saturation current flow in the circuit
(5)	Order of forward current is milli-ampere	Order of current is micro-ampere (Ge) and nano-ampere (Si)
(6)	Knee voltage or cut-off voltage Ge 0.3V Si 0.7V	Break down voltage Ge 25V Si 35V

Gateway Classes

Diode Current Equation

$$I_D = I_o\left(e^{\frac{V}{\eta V_T}} - 1\right)$$

Where

I_D = Diode Current

I_o = Reverse saturation current

V = External voltage applied to the Diode

η = a constant (for Ge, η=1 and for Si, η=2)

V_T = Thermal voltage (26mV at room temperature)

$$V_T = \frac{kT}{q}$$

k = Boltzmann's constant $(1.38 \times 10^{-23}$ J/k)

q = Electron charge$(1.36 \times 10^{-19}C)$

T = Temperature in kelvin

$$V_T = \frac{T}{11600}$$

Gateway Classes

Q.1 The reverse saturation current of Si p-n junction diode is 10 μA at 300K. Determine the forward bias voltage to be applied to obtain diode current of 100mA. **AKTU 2017-18**

Given

$\eta = 2$

$I_0 = 10\,\mu A$

$T = 300K$

$V_T = 26\,mV$

$V = ?$

$I_D = 100\,mA$

$$I_D = I_0\left(e^{\frac{V}{\eta V_T}} - 1\right)$$

$$100\times10^{-3} = 10\times10^{-6}\left(e^{\frac{V}{2\times26\times10^{-3}}} - 1\right)$$

$$e^{\frac{V}{52\times10^{-3}}} - 1 = \frac{10\cancel{0}\times10^{-3}}{\cancel{1}0\times10^{-6}}$$

$$e^{\frac{V}{52\times10^{-3}}} = 1 + 10000$$

$$e^{\frac{V}{52\times10^{-3}}} = 10001$$

Taking log both side

$$\frac{V}{52\times10^{-3}}\log e = \log(10001)$$

$$\frac{V}{52\times10^{-3}} = 9.21$$

$$\boxed{V = 0.4789\,V}$$

Gateway Classes

Q.2 A Ge diode carries a current of 1mA at room temperature when a forward bias of 0.15V is applied. Estimate the reverse saturation current at room temperature. **AKTU 2015-16**

Given

$\eta = 1$

$I_D = 1mA$

$V_T = 26mV$

$V = 0.15V$

$I_0 = ?$

$$I_D = I_0\left(e^{\frac{V}{\eta V_T}} - 1\right)$$

$$1\times10^{-3} = I_0\left(e^{\frac{0.15}{1\times26\times10^{-3}}} - 1\right)$$

$$10^{-3} = I_0\left(e^{\frac{150}{26}} - 1\right)$$

$$10^{-3} = I_0 \times 319.29126$$

$$I_0 = \frac{10^{-3}}{319.29126}$$

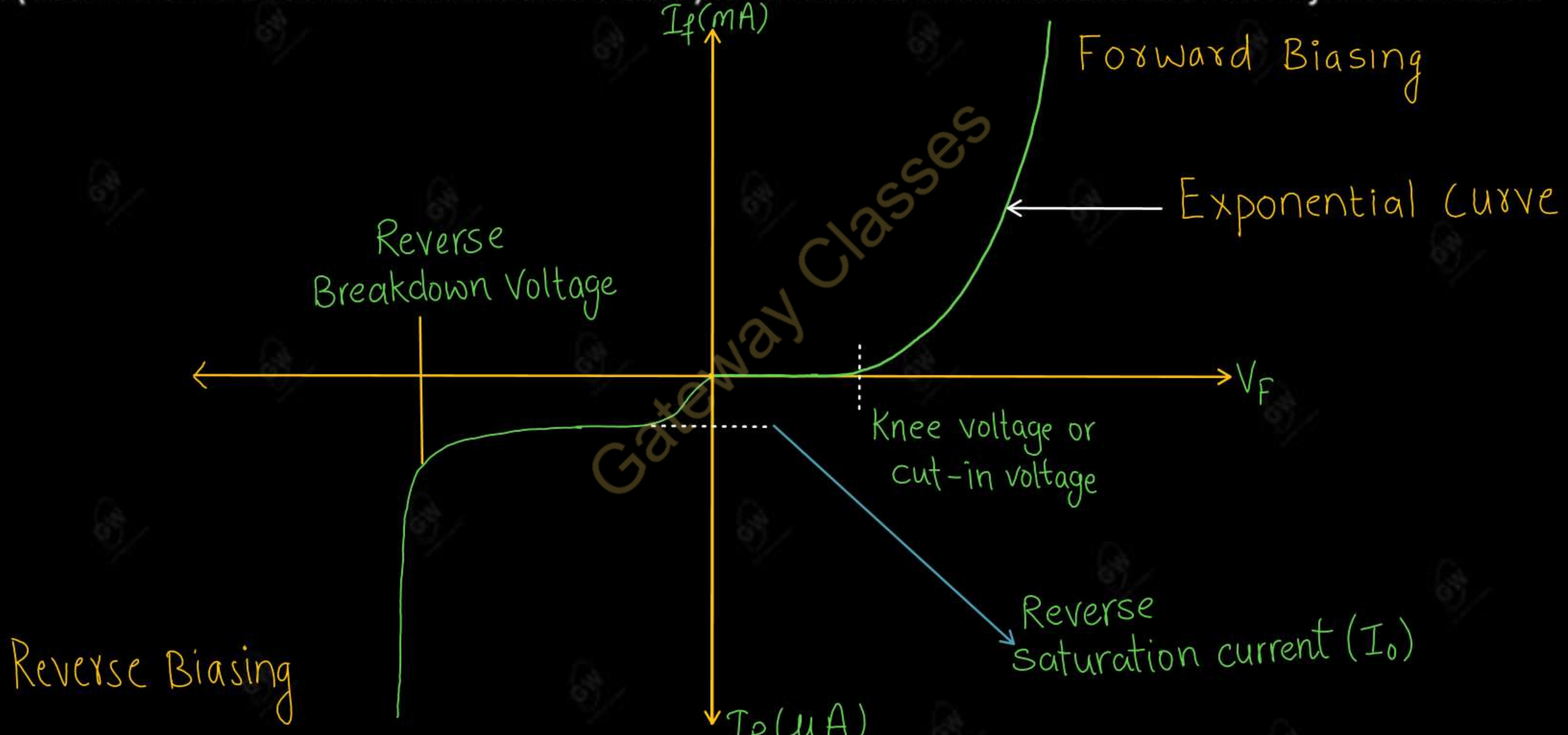
$$I_0 = 3.1319\times10^{-6}\,A$$

$$\boxed{I_0 = 3.1319\,\mu A}$$

Gateway Classes

Volt-Ampere (V-I) Characteristic of a P-N Junction Diode

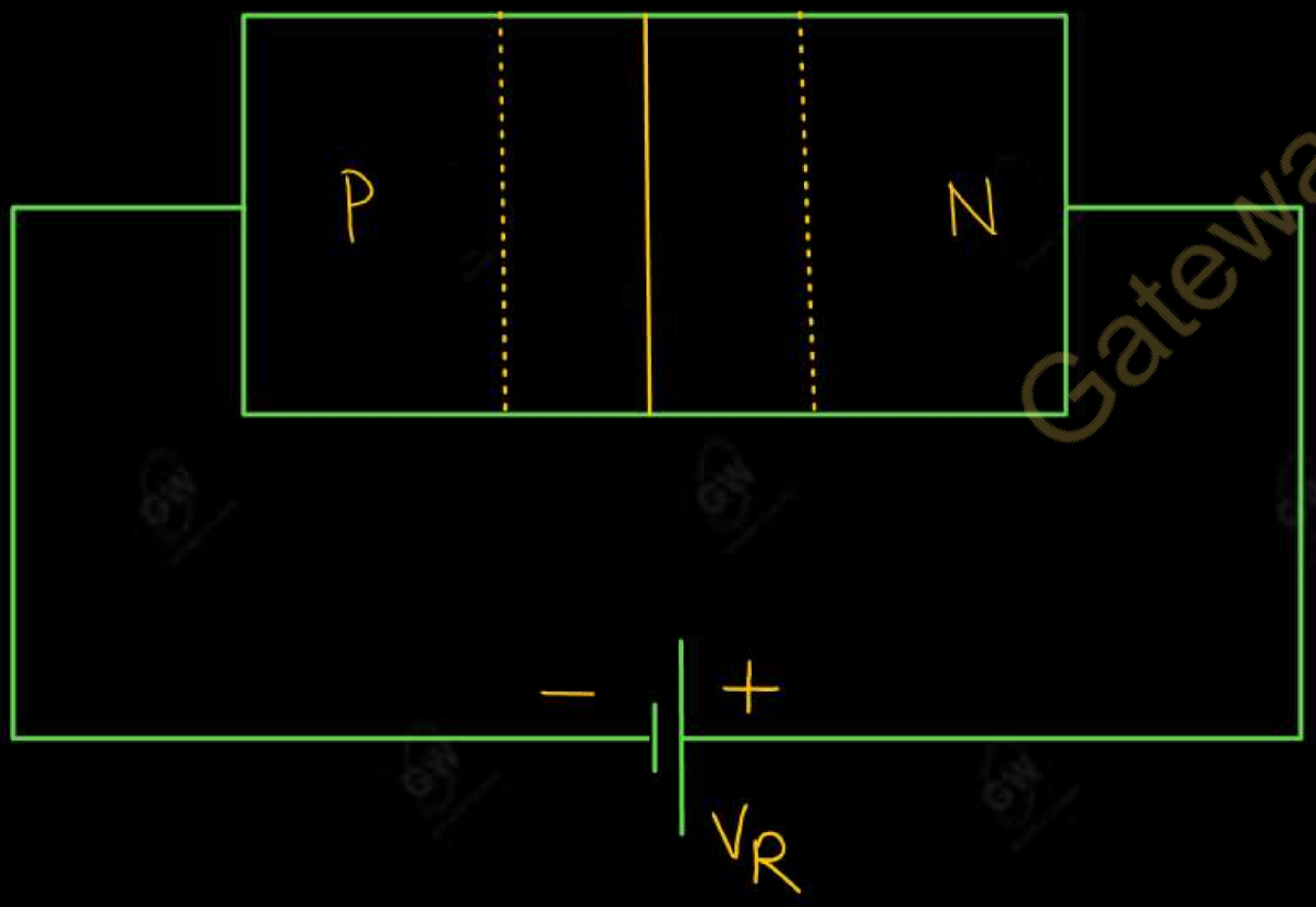
A graph showing the variation of current flowing through a P-N junction with the voltage applied across it (both when it is forward and reverse biased) is called the V-I characteristic of a P-N junction Diode

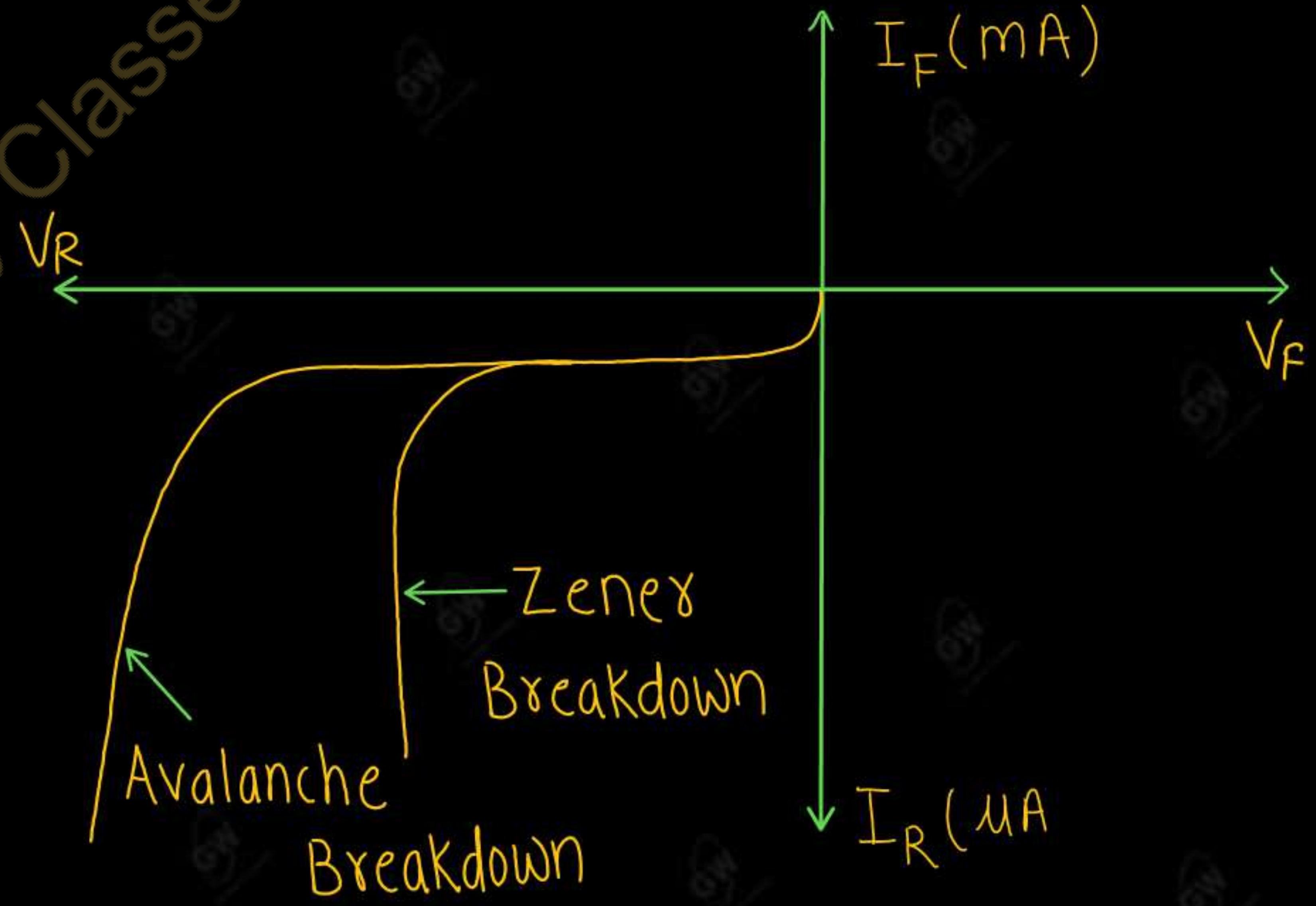


Breakdown Mechanism

- Breakdown mechanism occurs in Reverse biasing of diode
- If applied reverse voltage is continuously increased, a condition come at which current increases at a very rapid rate, the reverse voltage at this condition is called breakdown voltage (Vz)
- The breakdown of a junction diode may occur through two different Mechanism (or processes)

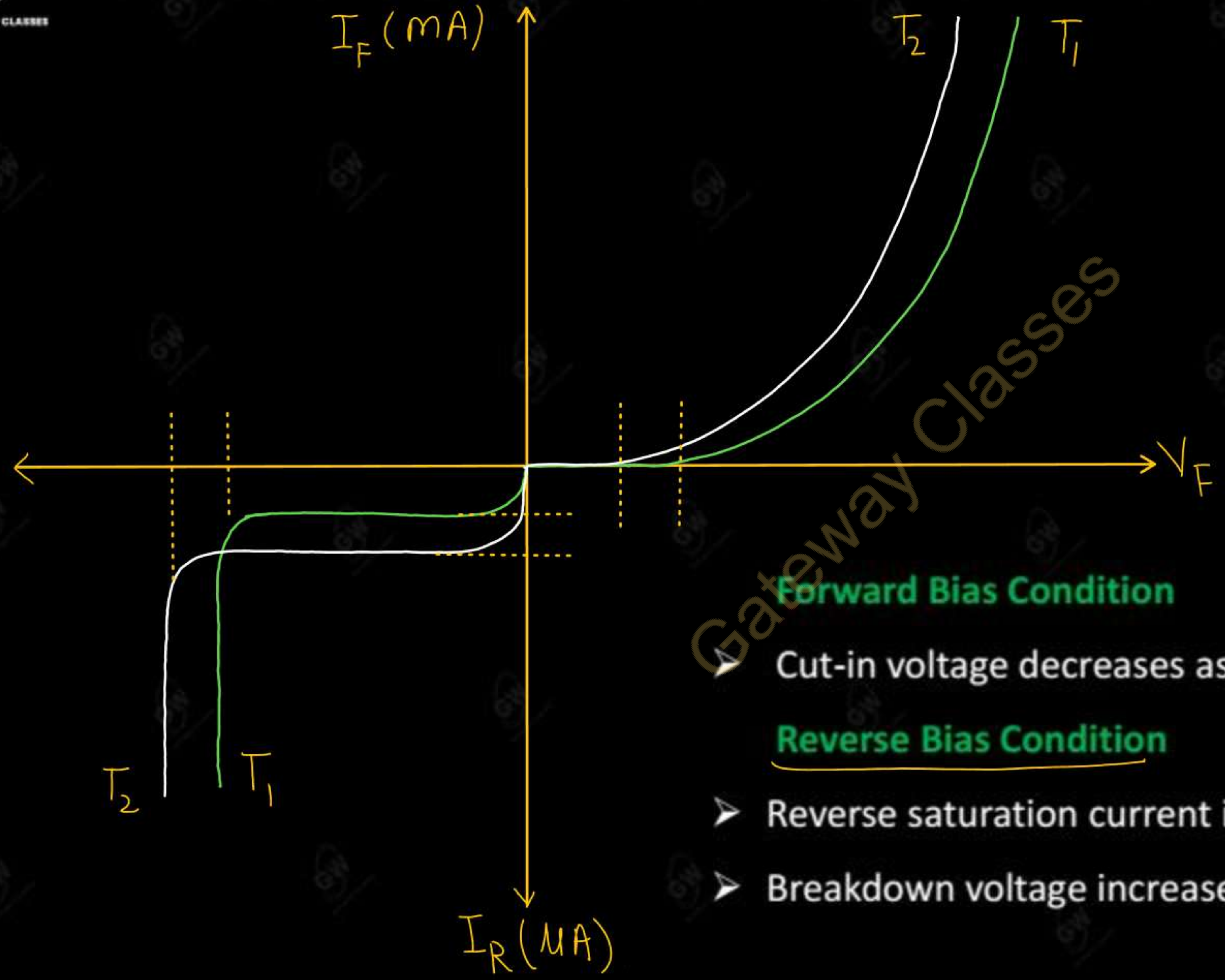
(i) Zener breakdown (ii) Avalanche breakdown





Gateway Classes

Effect of temperature on V-I Characteristic of a P-N Junction Diode



$T_2 > T_1$

Forward Bias Condition

- Cut-in voltage decreases as temperature increases

Reverse Bias Condition

- Reverse saturation current increases as temperature increases
- Breakdown voltage increases as temperature increases

Gateway Classes

UNIT - 1 : DPP - 2

Topic : P-N Junction Diode

Q.1 Discuss the formation of depletion layer in diode AKTU-2022

Q.2 Why does the depletion region width decrease after putting diode in forward bias? AKTU-2025

Q.3 Explain the working of p-n junction diode and draw its V-I characteristics. AKTU-2023

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application
(iii) Special Purpose two terminal Devices

Lecture - 3

Gateway Classes

Today's Target

- Breakdown mechanism
  - (i) Zener Breakdown (ii) Avalanche Breakdown
- Equivalent circuit of a diode
- Diode Configuration
- DPP, PYQs

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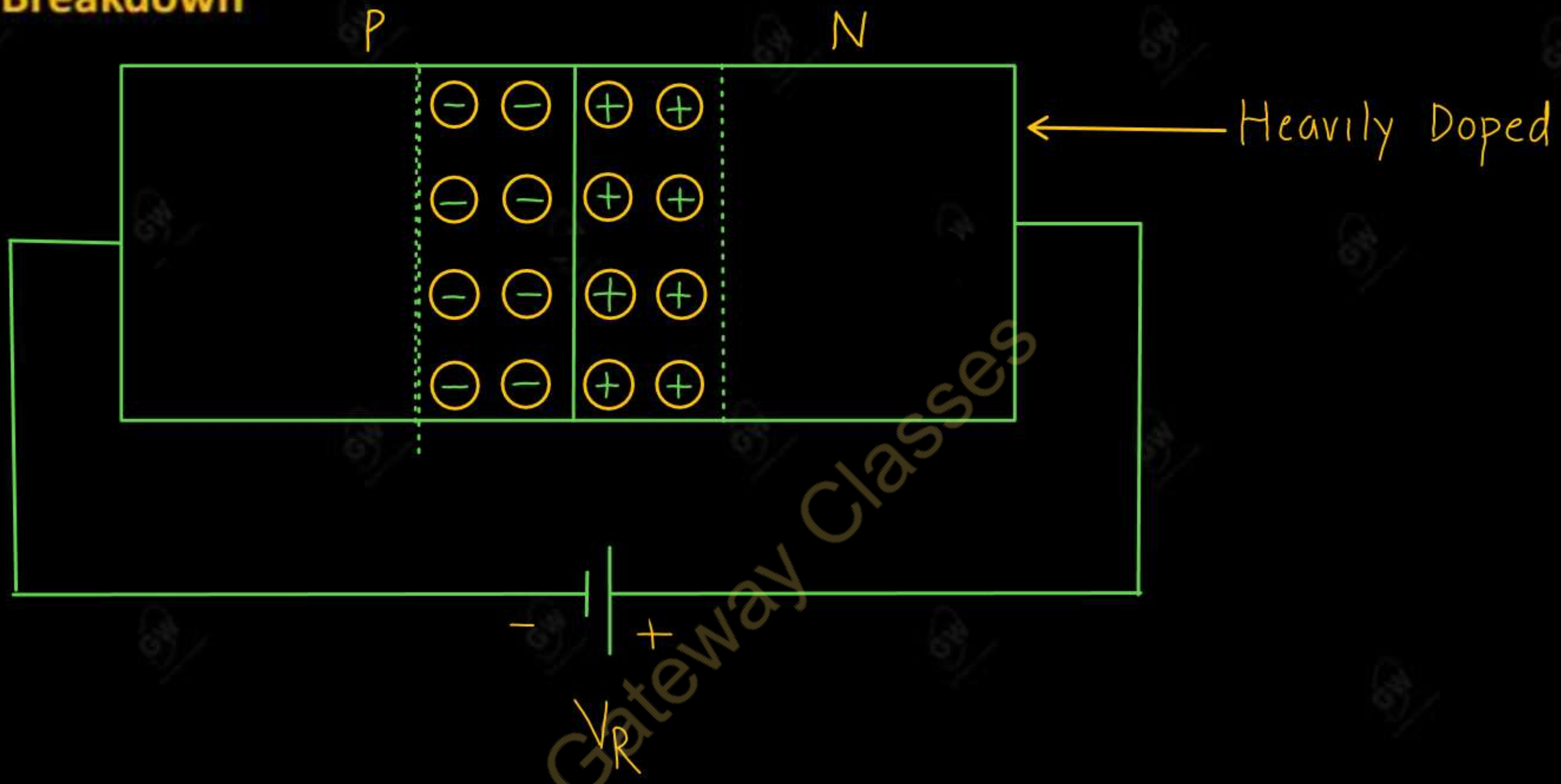
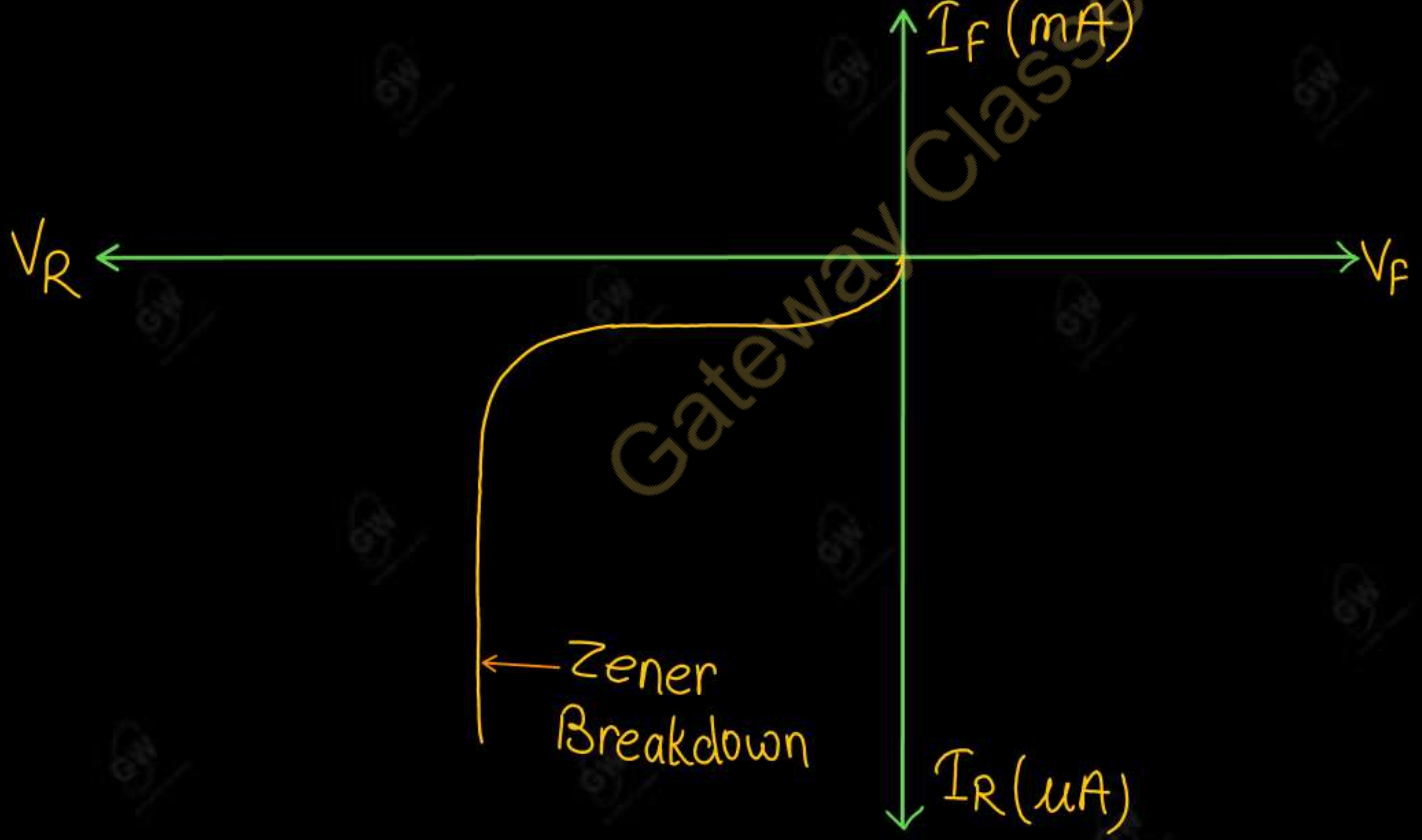
GATEWAY CLASSES

(i) Zener Breakdown

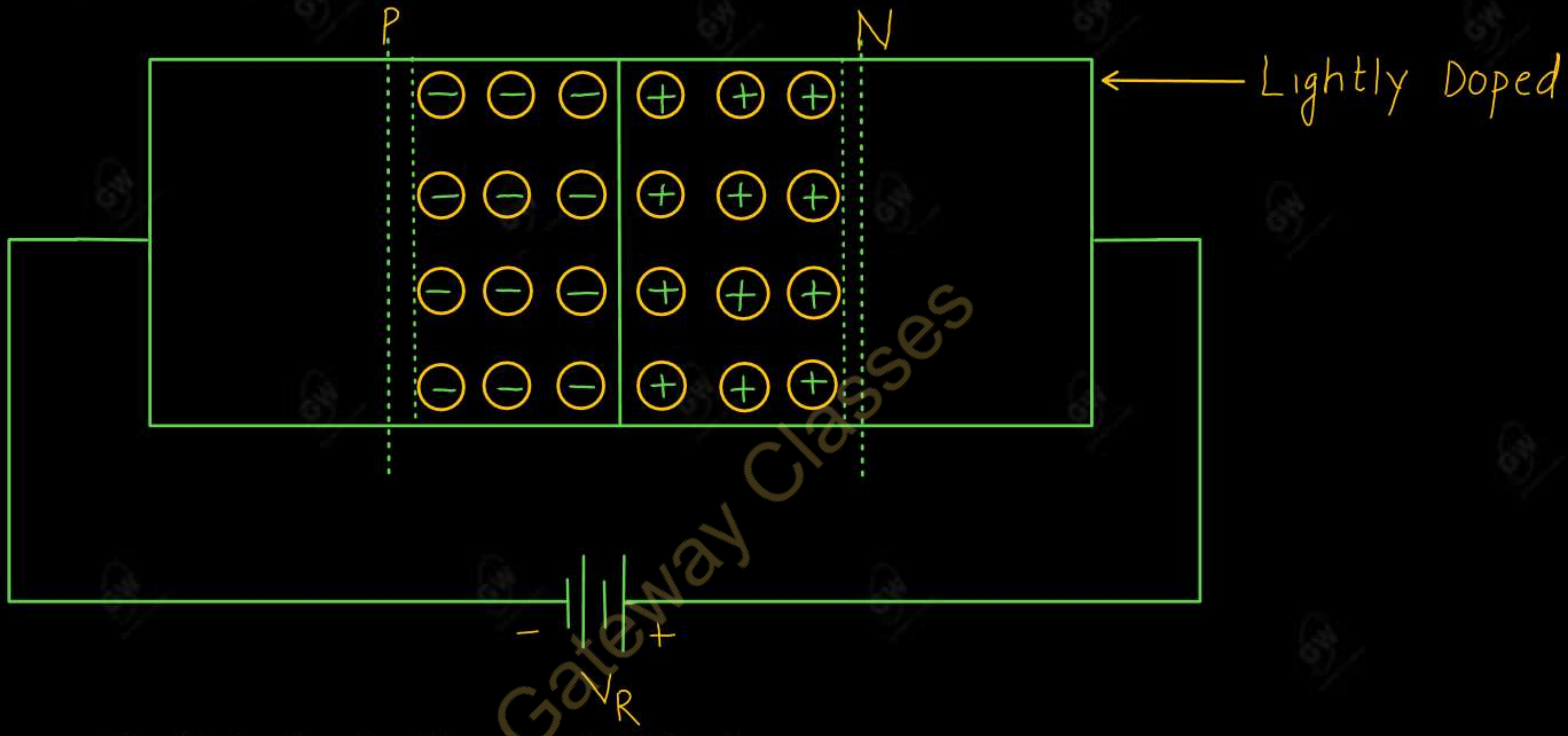


- This process occurs in diodes having high level of doping
- This process occurs for low breakdown voltage (less than 6 V)
- The Breakdown voltage decreases with increase in the junction temperature.

- In heavily doped diode, depletion layer is thin and electric field is very high in the depletion region
- This strong electric field pull the valence electrons in to the conduction band by breaking their covalent bonds
- These electrons then become free electrons which are available for conduction and will constitute a large reverse current. This is called Zener breakdown

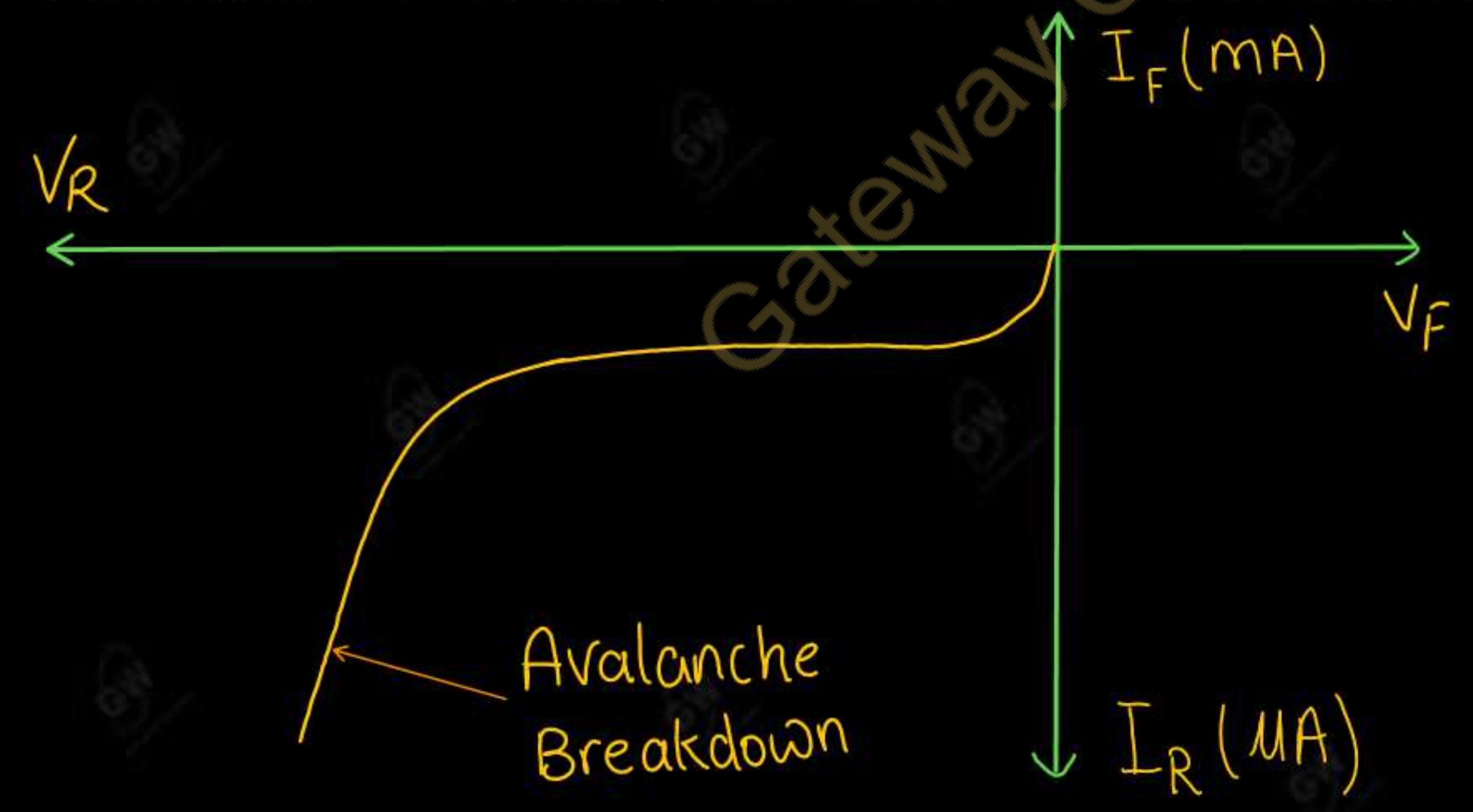


(ii) Avalanche Break down



- This process occur in diodes having low level of doping
- This process occurs for high breakdown voltage (greater than 6 V)
- Breakdown voltage increases with the increase in junction temperature.
- In lightly doped diode, depletion layer is wide and electric field is relatively weak in the depletion region

- This weak electric field can not break covalent bonds
- Due to high reverse voltage, the kinetic energy of minority carriers will increase to a great extent.
- While travelling, the minority carriers will collide with stationary ions and will impart some of the Kinetic energy to the valence electrons present in the covalent bonds.
- Due to additionally acquired energy, these valence electrons will break covalent bonds and jump into the conduction band to become free for conduction. This chain reaction is called Avalanche Breakdown.
- Due to this chain reaction a large current flow through diode and it permanently damage the diode.



Gateway Classes

Comparison of Zener and Avalanche Break down

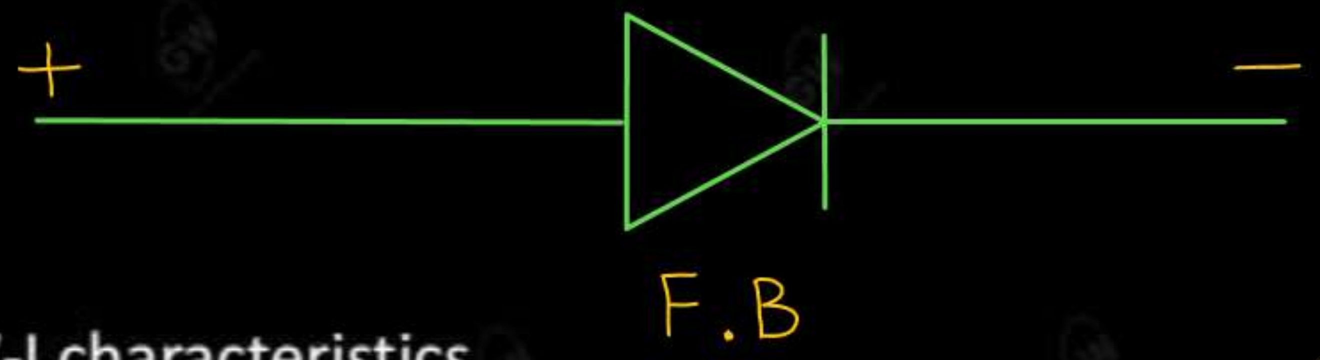
S.NO	Zener Breakdown	Avalanche Breakdown
1	This process occurs in diodes having high level of doping	This process occur in diodes having low level of doping
2	Occurs at low reverse voltages (less than 6 V)	Occurs at high reverse voltages (greater than 6 V)
3	Thin depletion layer	Wide depletion layer
4	The electric field across the junction is Strong	The electric field across the junction is relatively weak
5	The breakdown voltage decreases with increase in temperature.	The breakdown voltage increases with increase in temperature.
6	Zener breakdown does not result in the destruction of the diode.	Avalanche Breakdown destroys the diode.
7	The V-I characteristics has a sharp curve in breakdown region	The V-I characteristics does not have a sharp curve breakdown region

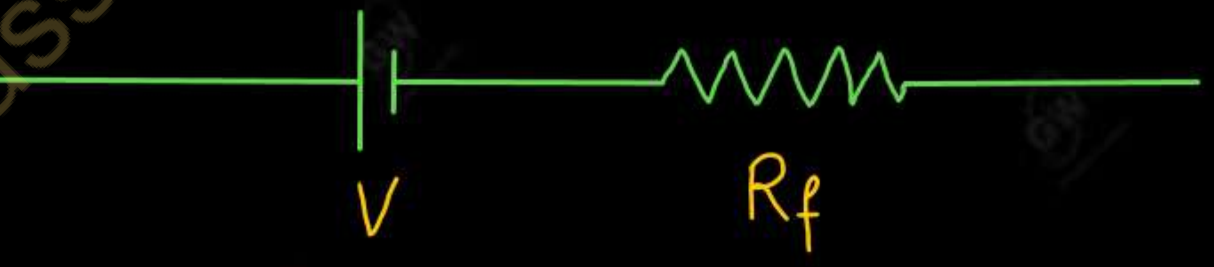
Gateway Classes

Equivalent circuit of a diode

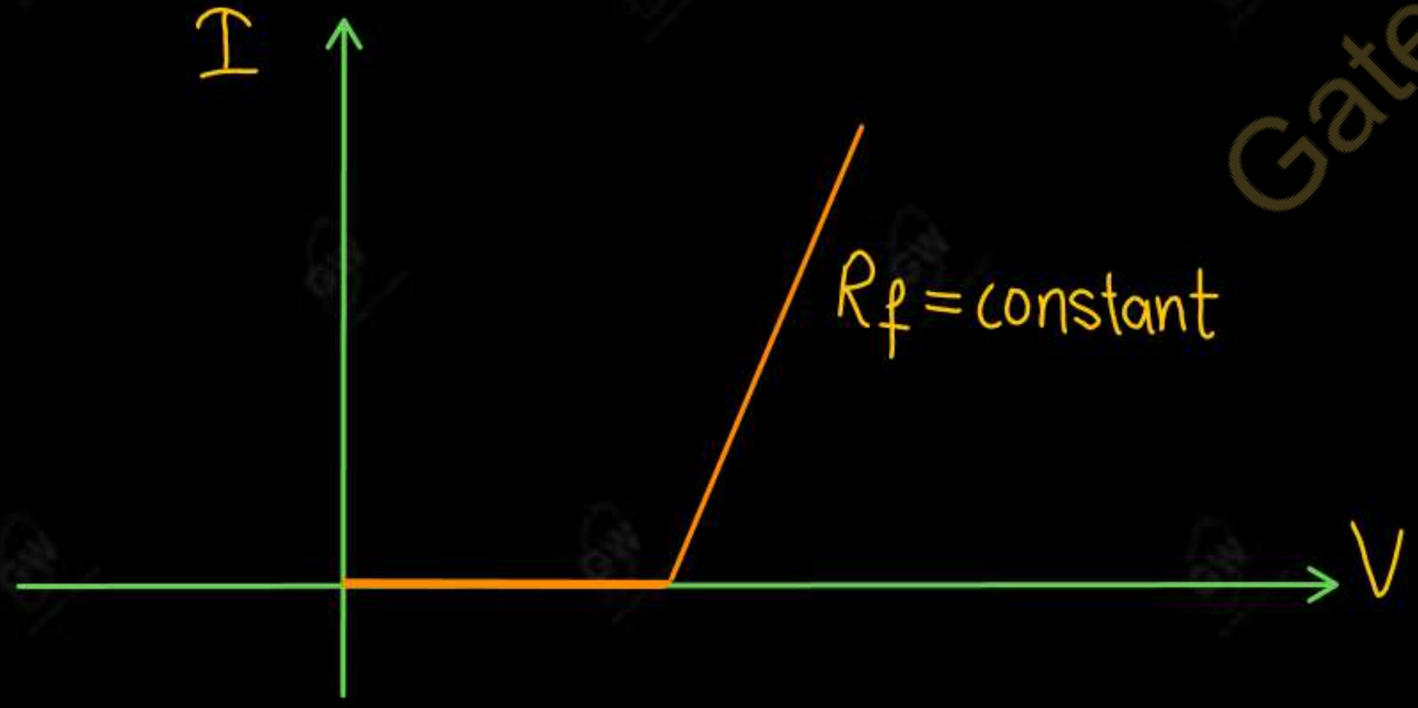
(i) Piecewise-linear equivalent circuits

- Non-linear characteristics of Diode is replaced by a straight line. so, resistance of diode is constant
- Diode is replaced by a battery with resistance in series





- V-I characteristics



Note :

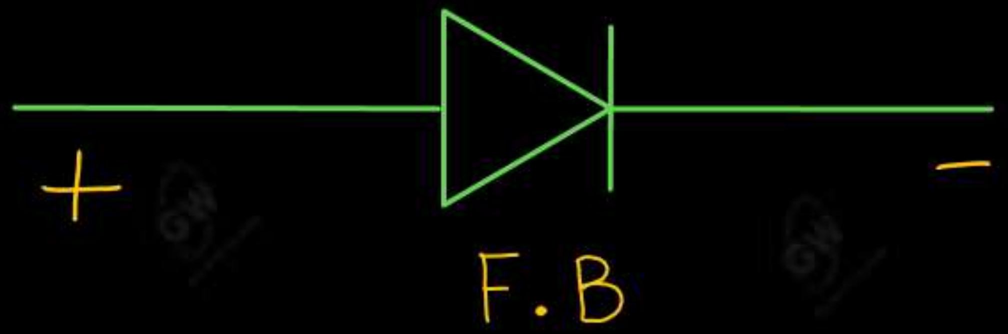
For silicon diode (V = 0.7 V)

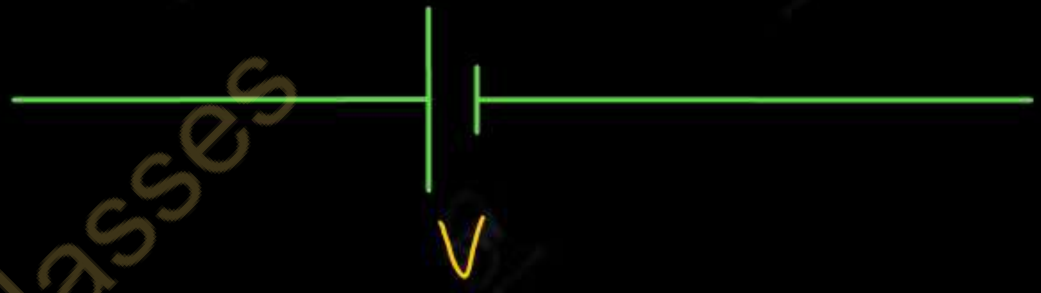
For Germanium diode (V = 0.3 V)

Gateway Classes

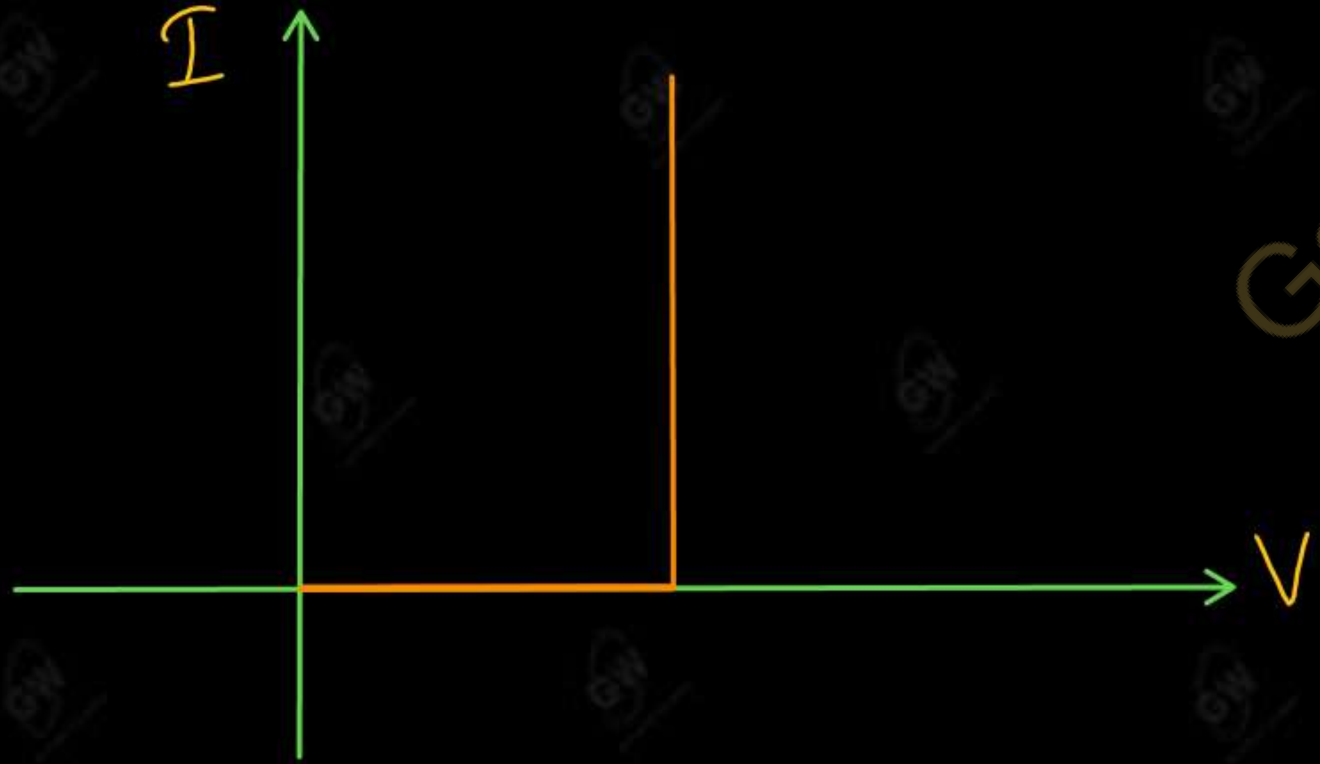
(i) Simplified Equivalent circuit

- Since diode forward resistance is very small, so it can be neglected ($R_f = 0$)
- Diode is replaced by battery





- VI characteristics



Note :

For silicon diode (V = 0.7 V)

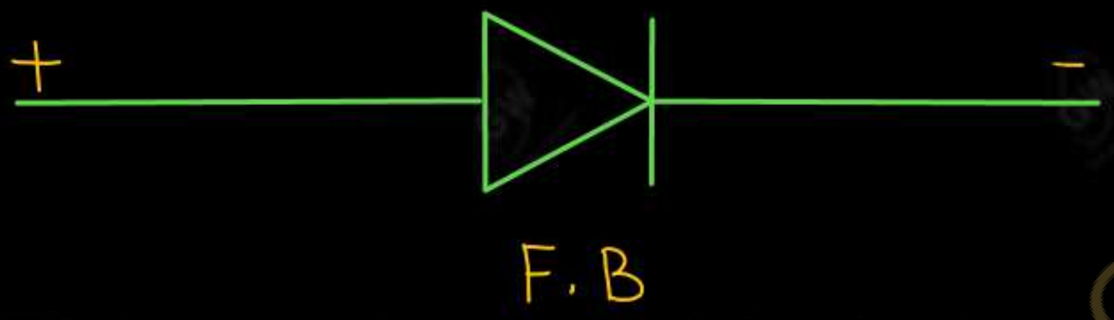
For Germanium diode (V = 0.3 V)

Gateway Classes

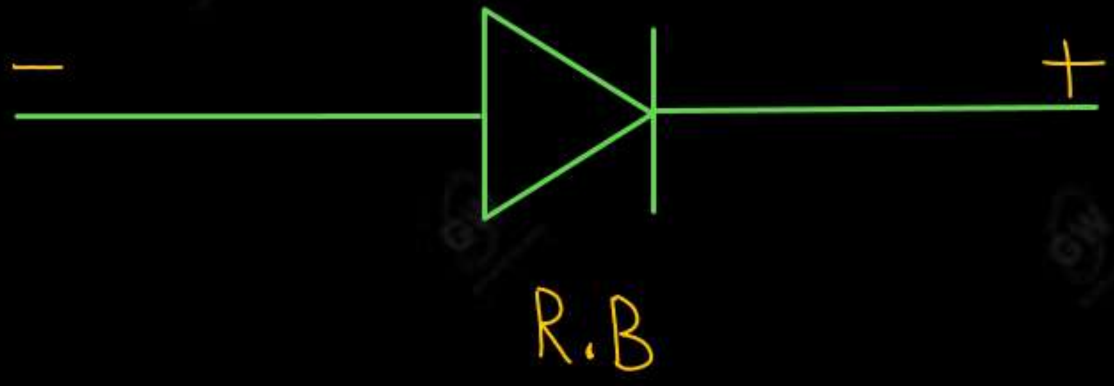
(iii) Equivalent circuits of an ideal diode

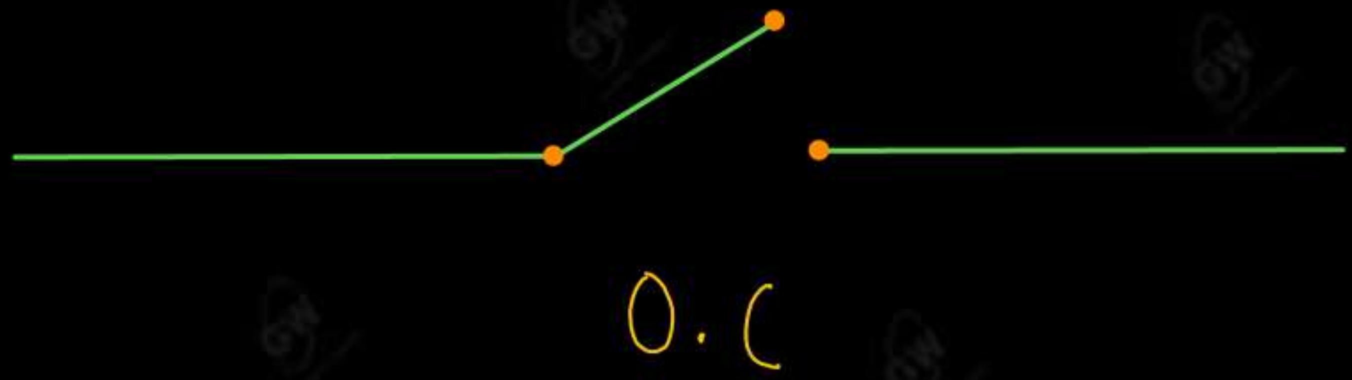
- Diode forward resistance ($R_f = 0$)
- Knee voltage ($V = 0$)
- Reverse saturation current ($I_o = 0$)
- An ideal diode can be used as a perfect switch

  (i) The switch is closed if the diode is forward biased

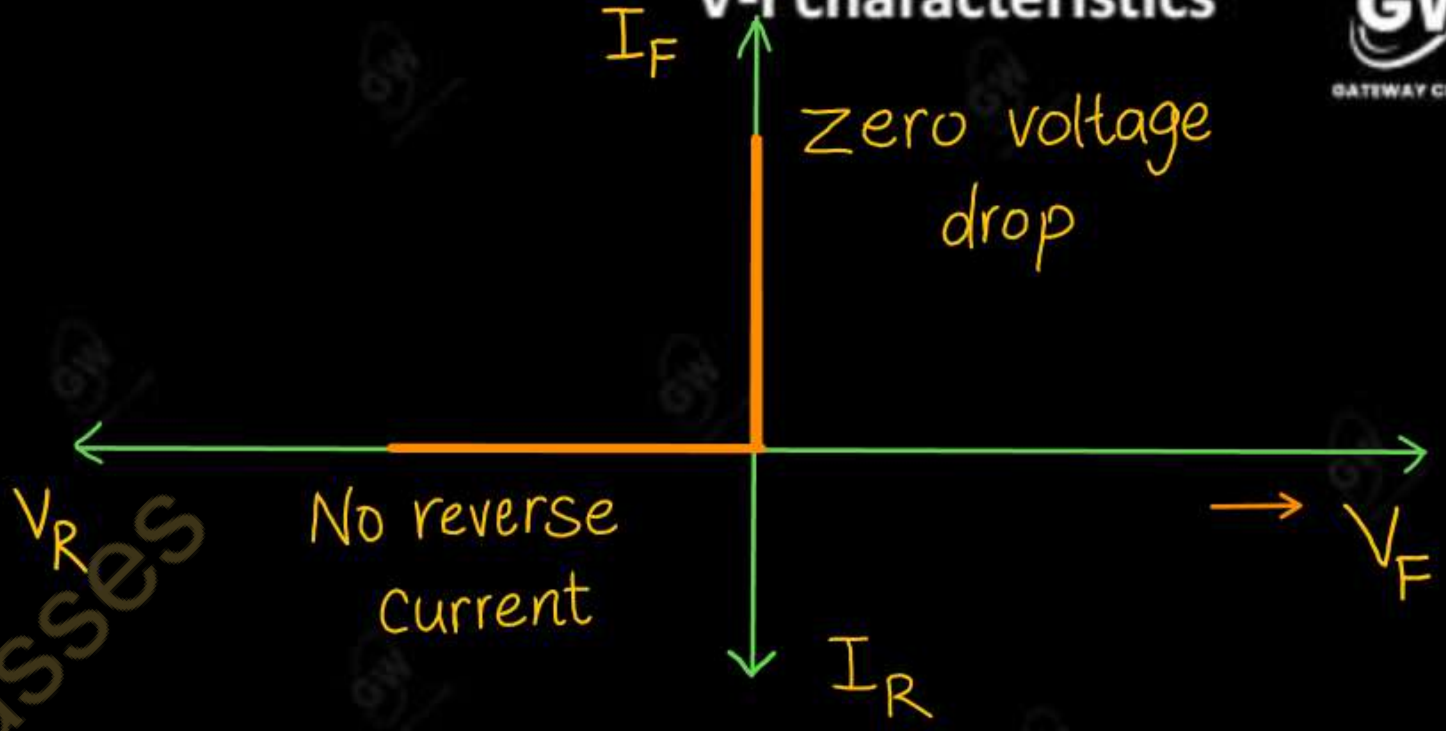


  (ii) The switch is open if the diode is reversed biased.





V-I characteristics



GATEWAY CLASSES
Gateway Classes

Diode Configurations

(i) Series Configurations

- Diodes are connected end-to-end
- Same current flow through all diodes
- Total voltage drop = Sum of voltage drop across each diode

(ii) Parallel Configurations

- Diodes are connected side-by-side
- Current is divided among diodes
- Same voltage drop across each diode

Gateway Classes

Kirchhoff's Law

(1) Kirchhoff's Current Law

At any junction

- Sum of current entering the junction = Sum of current leaving the junction

(2) Kirchhoff's Voltage Law

- Algebraic Sum of Change in potential around a closed loop is Zero

(1) Low potential to high potential Positive

(2) High potential to low potential Negative

Note

(i) Current always flow from high potential to low potential

(ii) Loop can be taken clockwise or anticlockwise

Gateway Classes

Kirchhoff's Current Law

At any junction

Sum of current entering the junction = Sum of current leaving the junction

Kirchhoff's Voltage Law

Algebraic Sum of Change in potential around a closed loop is Zero

(1) Low potential to high potential Positive

(2) High potential to low potential Negative

Note

(i) Current always flow from high potential to low potential
(ii) Loop can be taken clockwise or anticlockwise

Gateway Classes

UNIT - 1 : DPP - 3

Q.1 Give all the equivalent/approximation circuits of a diode AKTU-2017

Q.2 Draw V-I characteristics of an ideal diode in forward and reverse bias conditions. AKTU-2021

Q.3 Differentiate between Avalanche breakdown and Zener breakdown. AKTU-2023

Q.4 Explain the breakdown mechanisms in Zener diodes. AKTU-2024

OR

Describe breakdown mechanisms in diode AKTU-2016

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 4

Gateway Classes

Today's Target

- Half Wave Rectifier
- DPP
- PYQs

By Gulshan sir

GATEWAY CLASSES

UNIT-1

Semiconductor Diode

Depletion layer, V-I characteristics, ideal and practical Diodes, Diode Equivalent Circuits, Zener Diodes breakdown mechanism (Zener and avalanche)

Diode Application

Diode Configuration, Half and Full Wave rectification, Clippers, Clampers, Zener diode as shunt regulator, Voltage-Multiplier Circuits

Special Purpose two terminal Devices

Light-Emitting Diodes, Photo Diodes, Varactor Diodes, Tunnel Diodes.

Gateway Classes

Rectifier

230V, 50Hz — Transformer — Rectifier — Filter — Regulator

A Rectifier is an electronic device which is used to convert a.c. in to pulsating d.c. by using one or more diodes

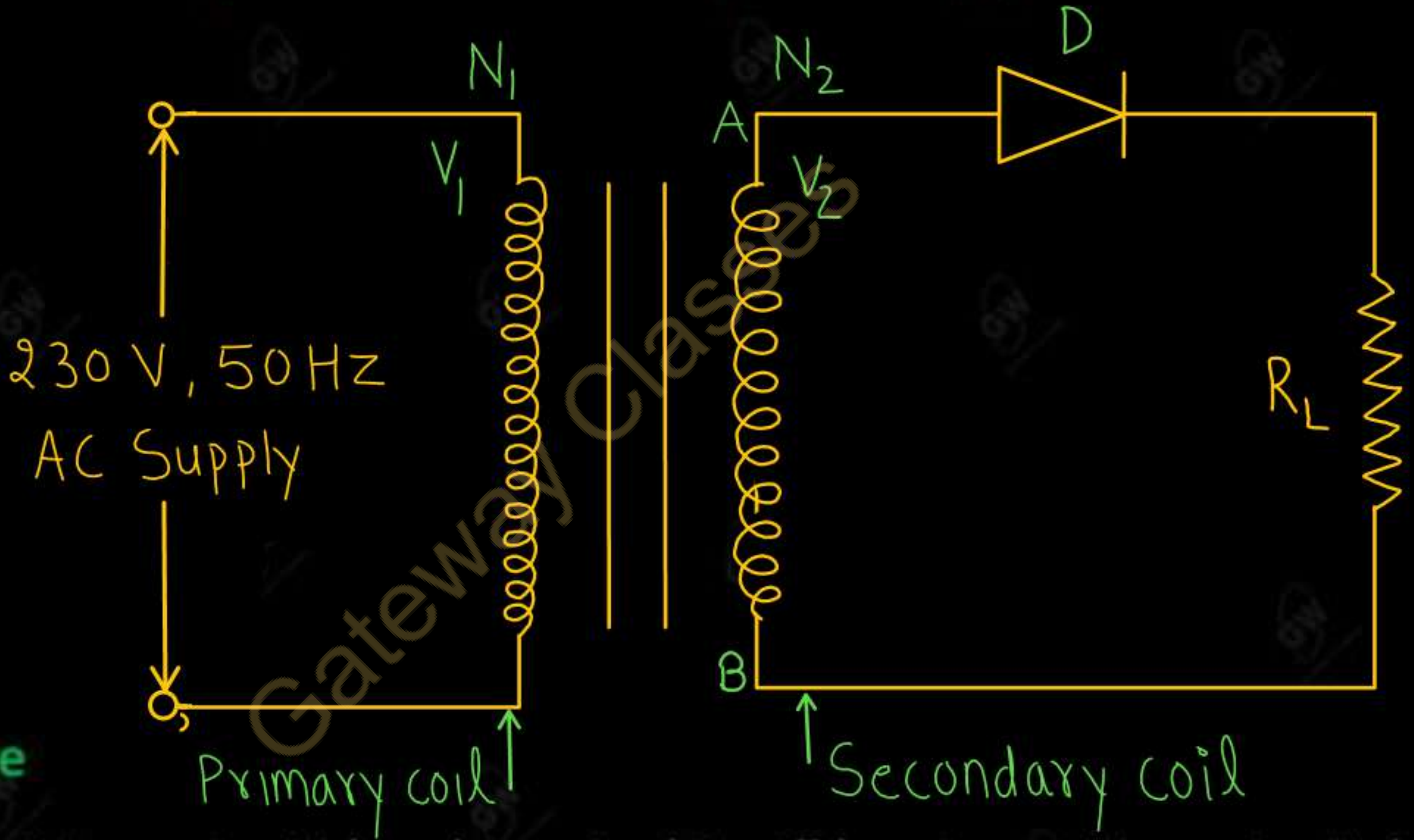
Rectifiers

- Half Wave Rectifier
- Full Wave Rectifier
  - Centre-Tap Full wave rectifier
  - Full wave Bridge Rectifier

Gateway Classes

Half Wave Rectifier

- Half wave Rectifier is a circuit that conduct current only during positive half cycles of input a.c. supply
- A half wave rectifier consists of a **Step–down transformer, a diode (D) and a load resistance (R_L)**

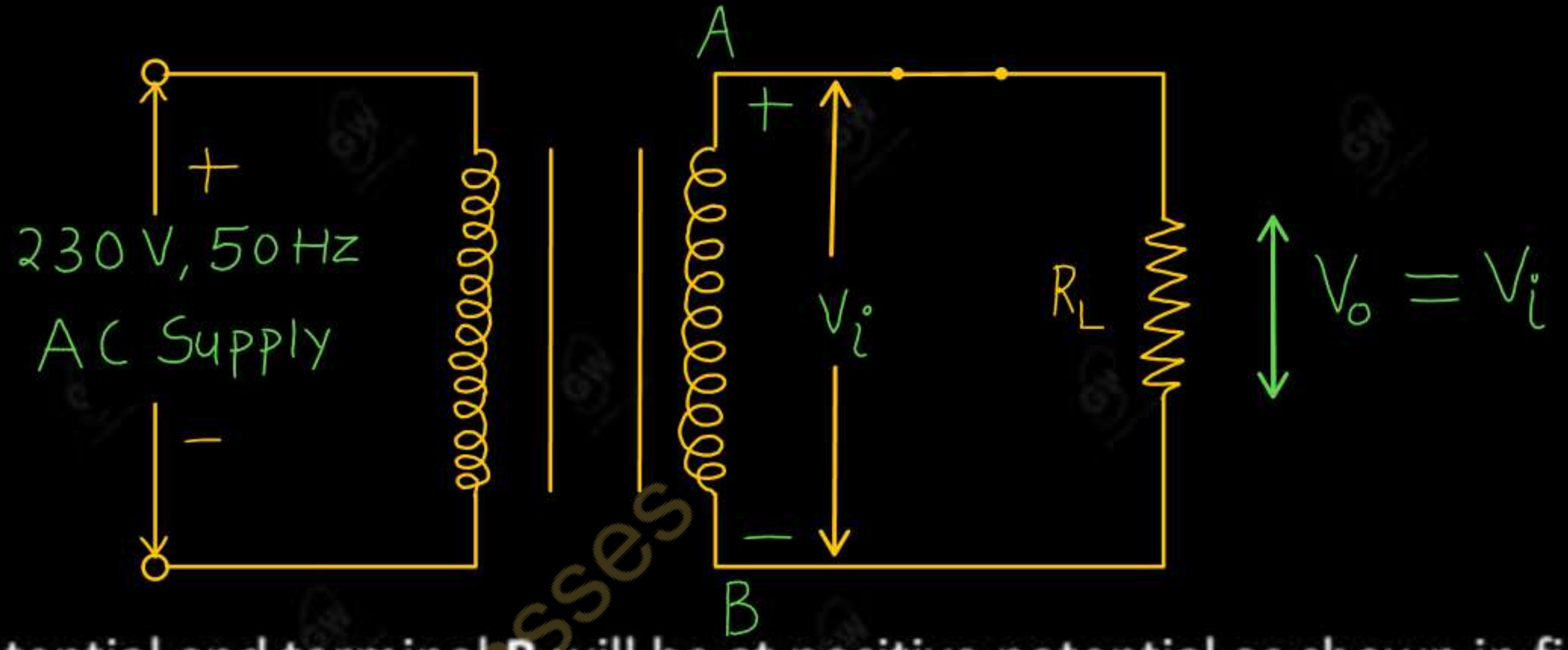


$$\boxed{\frac{N_2}{N_1} = \frac{V_2}{V_1}}$$

Working

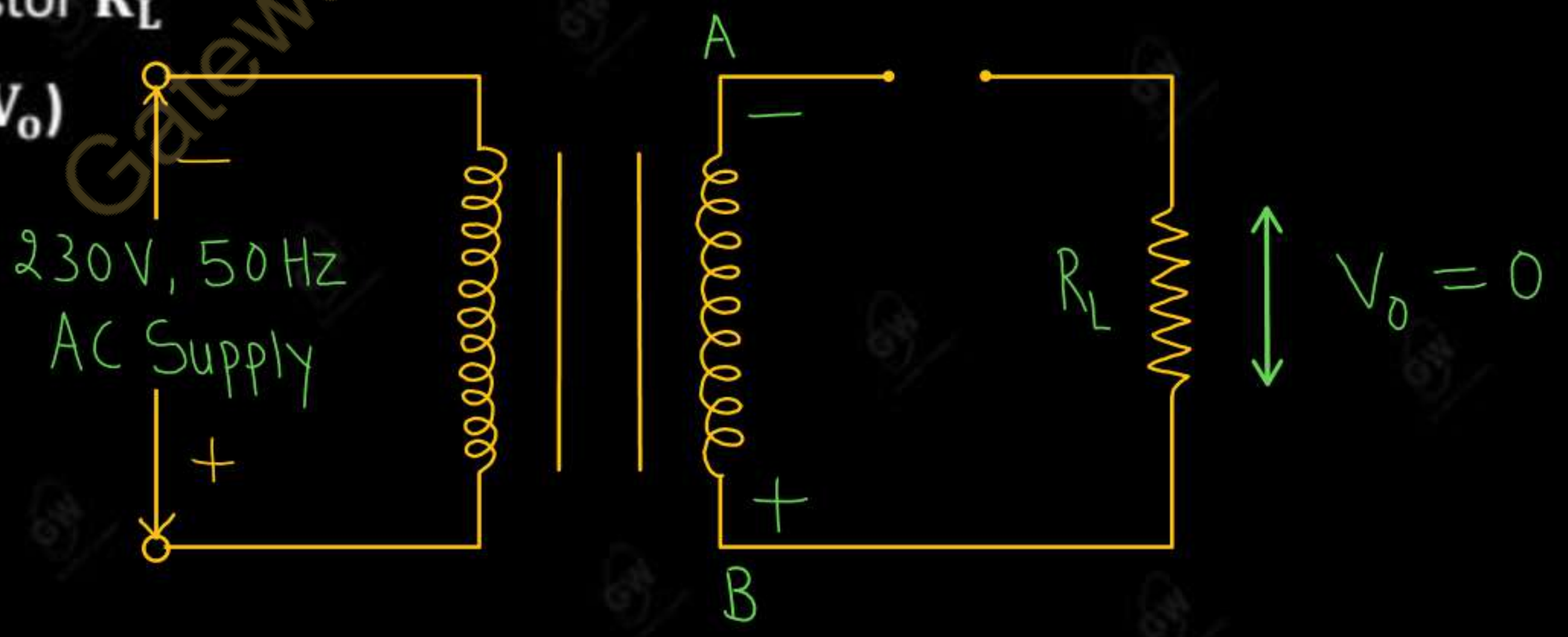
(i) During positive Half cycle

- The terminal **A** will be at positive potential and terminal **B** will be at negative potential as shown in figure
- It will make diode **D** in forward bias and it will act as a short circuit **(ON)**
- Current flow across the output load resistor $\mathbf{R_L}$ and we get output voltage **($\mathbf{V_o}$)**



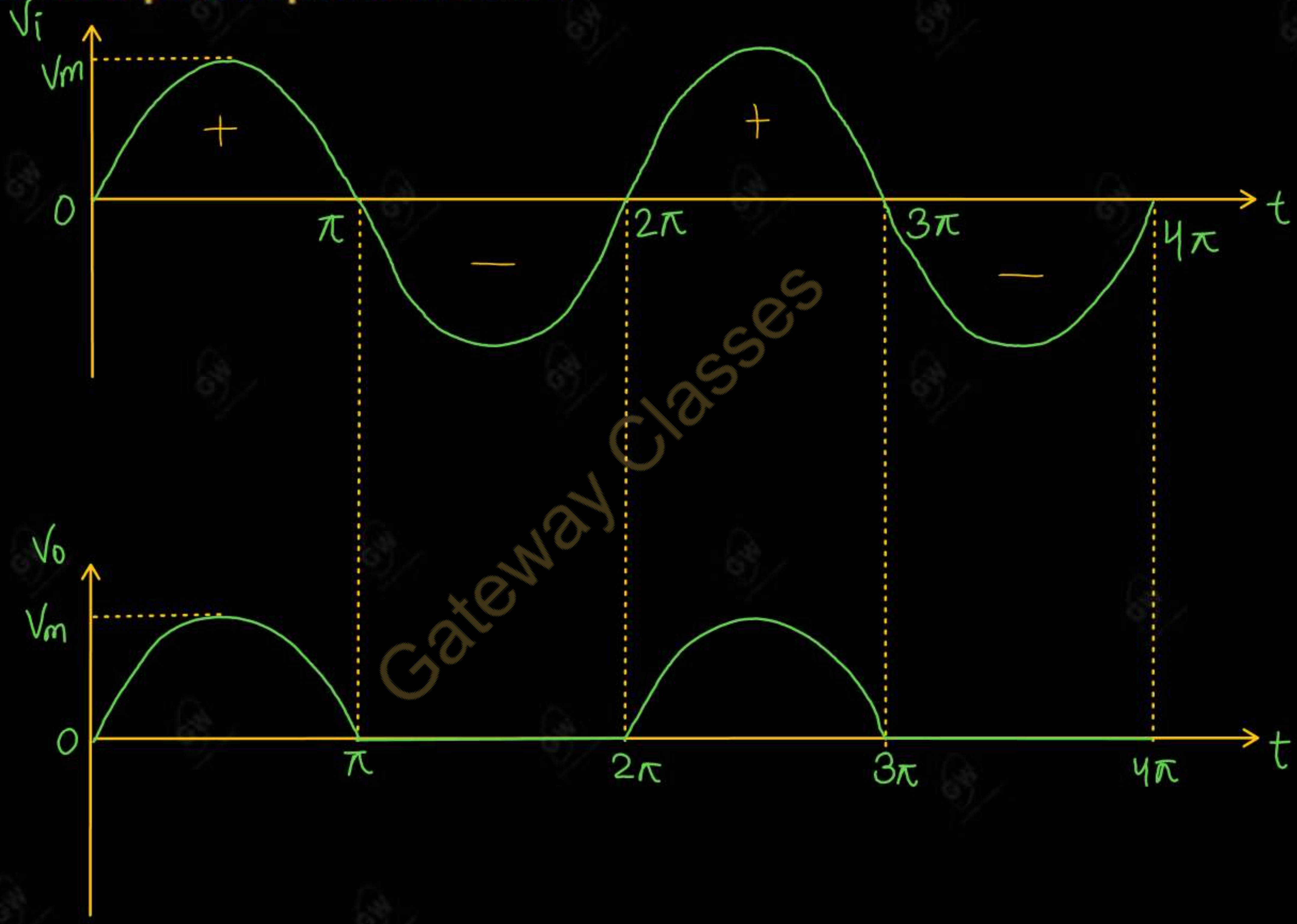
(ii) During Negative Half cycle

- The terminal **A** will be at negative potential and terminal **B** will be at positive potential as shown in figure
- It will make diode **D** in Reverse bias and it will act as an open circuit **(OFF)**
- No Current flow across the load resistor $\mathbf{R_L}$
- Hence, we get Zero output voltage $(\mathbf{V_o})$



Gateway Classes

Half Wave Rectifier Input output waveform



Gateway Classes

Performance Parameters

(1) Average or DC output (Load) current (I_{dc})

$$I_{dc} = \frac{1}{2\pi}\int_0^{2\pi} I_m \sin\theta \, d\theta$$

$$= \frac{I_m}{2\pi}\int_0^{2\pi} \sin\theta \, d\theta$$

$$= \frac{I_m}{2\pi}\left[\int_0^{\pi} \sin\theta \, d\theta + \int_{\pi}^{2\pi} \sin\theta \, d\theta\right]$$

$$= \frac{I_m}{2\pi}\left[\left(-\cos\theta\right)_0^{\pi} + 0\right] = \frac{I_m}{2\pi}(1+1)$$

$$\boxed{I_{dc} = \frac{I_m}{\pi}}$$

(2) Average or DC output (Load) voltage (V_{dc})

$$V_{dc} = I_{dc} \times R_L$$

$$= \frac{I_m}{\pi} \times R_L$$

$$= \frac{V_m}{(R_F + R_L)\pi} \times R_L$$

$$= \frac{V_m}{\pi \cancel{R_L}} \times \cancel{R_L} \qquad \left\{ \because R_L >>> R_F \right.$$

$$\boxed{V_{dc} = \frac{V_m}{\pi}}$$

Gateway Classes

(3) RMS output (Load) current (I_{rms})

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} (I_m \sin\theta)^2 \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi}\int_0^{\pi} \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi}\int_0^{\pi} \frac{1-\cos 2\theta}{2} \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi}\left(\theta - \frac{\sin 2\theta}{2}\right)_0^{\pi}} = \sqrt{\frac{I_m^2}{4\pi} \times \pi}$$

$$\boxed{I_{rms} = \frac{I_m}{2}}$$

(4) RMS output (Load) voltage (V_{rms})

$$V_{rms} = I_{rms} \times R_L$$

$$= \frac{I_m}{2} \times R_L$$

$$= \frac{V_m}{2(R_F + R_L)} \times R_L$$

$$= \frac{V_m}{2 \times R_L} \times R_L \quad \left\{ \because R_L >>> R_F \right.$$

$$\boxed{V_{rms} = \frac{V_m}{2}}$$

Gateway Classes

(5) Ripple factor

- The output of rectifier contains a.c. as well as d.c. component. The ripple factor measures the percentage of a.c. component present in the output.
- Ripple factor is defined as the ratio of rms value of a.c .component to the d.c. component in the rectifier.

$$\text{Ripple factor} = \frac{\text{RMS value of a.c .component of output}}{\text{DC value of output}}$$

$$\gamma = \frac{I_{ac}}{I_{dc}} \quad \text{——} \quad (1)$$

RMS value of a.c component of output

$$I_{ac} = \sqrt{I_{rms}^2 - I_{dc}^2}$$

$$\gamma = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}}$$

$$\gamma = \sqrt{\frac{I_{rms}^2 - I_{dc}^2}{I_{dc}^2}}$$

$$\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$\gamma = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1}$$

$$\boxed{\gamma = 1.21}$$

Gateway Classes

(6) Efficiency of a Rectifier (η)

Efficiency of a Rectifier is defined as the ratio of dc output power to the ac input power

$$\eta = \frac{\text{dc output power}}{\text{ac input power}}$$

$$\eta = \frac{I_{dc}^2 \times R_L}{I_{rms}^2 (R_L + R_f)}$$

$$= \frac{\left(\frac{I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{2}\right)^2 (R_L + R_f)}$$

If $R_f = 0$

$$\eta = \frac{I_m^2 \times R_L \times 4}{I_m^2 \times \pi^2 \times R_L}$$

$$\eta = \frac{4}{\pi^2}$$

$$\boxed{\eta = 40.6\ \%}$$

So, efficiency of half wave Rectifier is low

Gateway Classes

(7) Transformer utilization factor

It may be defined as the ratio of d.c. power delivered to the load and the ac power rating of the transformer secondary

$$\mathbf{TUF} = \frac{\text{d.c power delivered to the load}}{\text{a.c power rating of the transformer secondary}}$$

$$TUF = \frac{V_{dc} \times I_{dc}}{V_{rms} \times I_{rms}}$$

$$\boxed{V_{rms} = \frac{V_m}{\sqrt{2}}}$$

$$= \frac{\frac{V_m}{\pi} \times \frac{I_m}{\pi}}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{2}}$$

$$TUF = \frac{\cancel{V_m} \times \cancel{I_m} \times 2\sqrt{2}}{\pi^2 \times \cancel{V_m} \times \cancel{I_m}}$$

$$TUF = 0.287$$

$$\boxed{TUF = 28.7\%}$$

Gateway Classes

(8) Peak inverse voltage

It is The maximum reverse voltage that can be applied across a diode without damaging it

$$\boxed{PIV = V_m}$$

(9) Ripple frequency (f_r)

It is the frequency of output wave in a rectifier. It is called output frequency

$$\boxed{f_r = f_i}$$

Gateway Classes

Q.1 An ac supply of 230 V is applied to a half wave rectifier circuit through a transformer of turn ratio 10 : 1. Find (i) the output dc voltage and (ii) the peak inverse voltage. Assume the diode to be ideal

AKTU 2025

Given

$V_1 = 230$

$\frac{N_1}{N_2} = \frac{10}{1}$

$\frac{V_2}{V_1} = \frac{N_2}{N_1}$

$V_2 = \frac{N_2}{N_1} V_1$

$= \frac{1}{10} \times 230$

$\boxed{V_2 = 23 V}$

$V_m = V_2 \sqrt{2}$

$V_m = 23\sqrt{2}$

$V_m = 32.5 V$

(i) $V_{dc} = \frac{V_m}{\pi} = \frac{32.5}{\pi}$

$\boxed{V_{dc} = 10.3 V}$

(ii) $PIV = V_m$

$\boxed{PIV = 32.5 V}$

Gateway Classes

UNIT - 1 : DPP - 4

Q.1 For a half wave rectifier derive an expression for ripple factor **AKTU-2016**

Q.2 A.C voltage of 200 V is applied to a half wave rectifier circuit through transformer of turns ratio 4 : 1. the load resistance value is 300 Ω . Assuming the diode as ideal one, determine the following

(i) dc output voltage (ii) PIV (iii) maximum value of power delivered to the load (iv) average power delivered to the load

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 5

Today's Target

- Full Wave Rectifier
- DPP
- PYQs

Gateway Classes

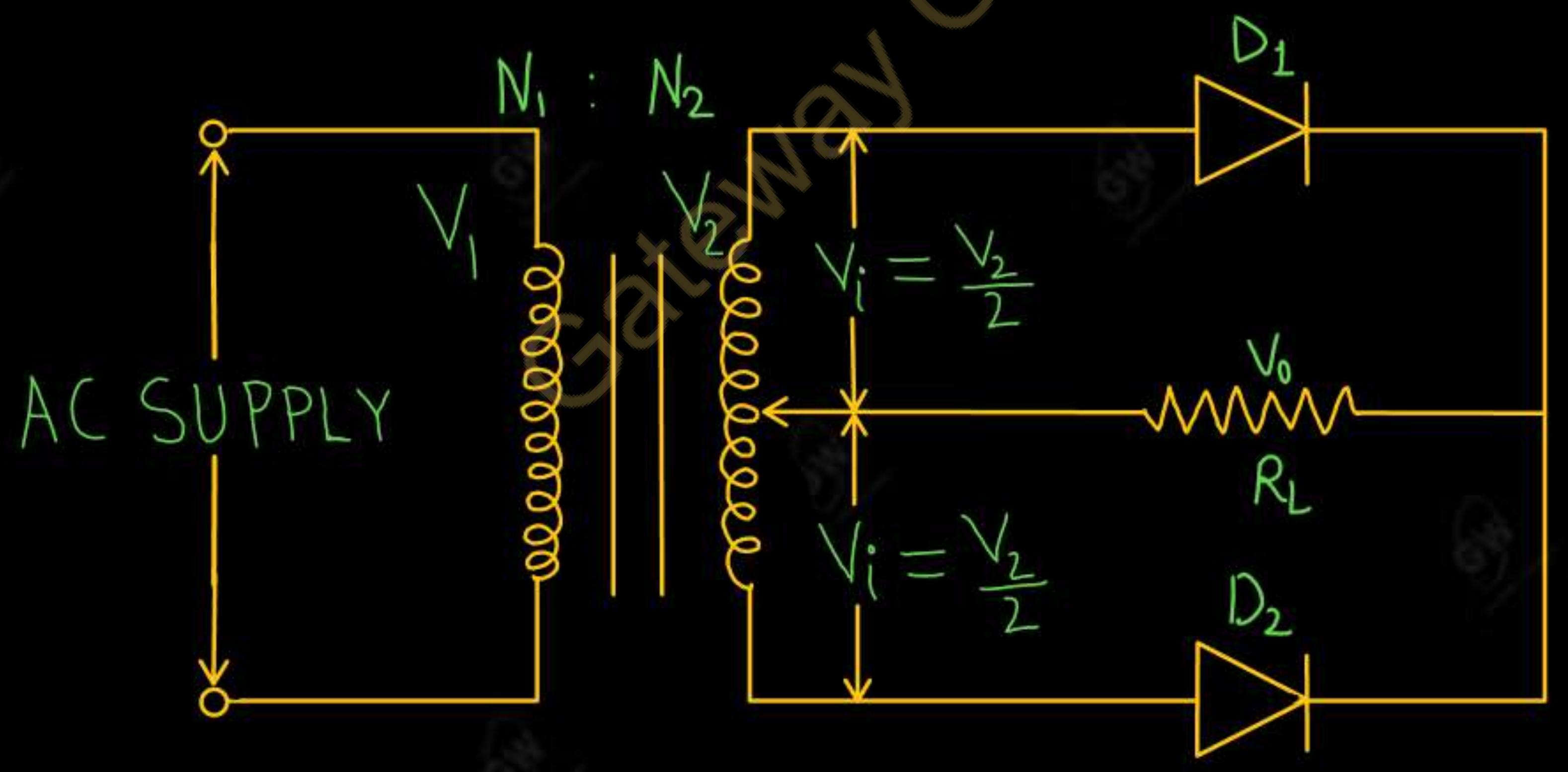
By Gulshan sir

Full Wave Rectifier

- Full wave Rectifier is a circuit that conduct current during both the positive and negative half cycles of input a.c. supply
- Full wave rectifier are of two types (i) Centre Tapped Full wave rectifier (i) Full wave Bridge Rectifier

Centre Tapped Full Wave Rectifier

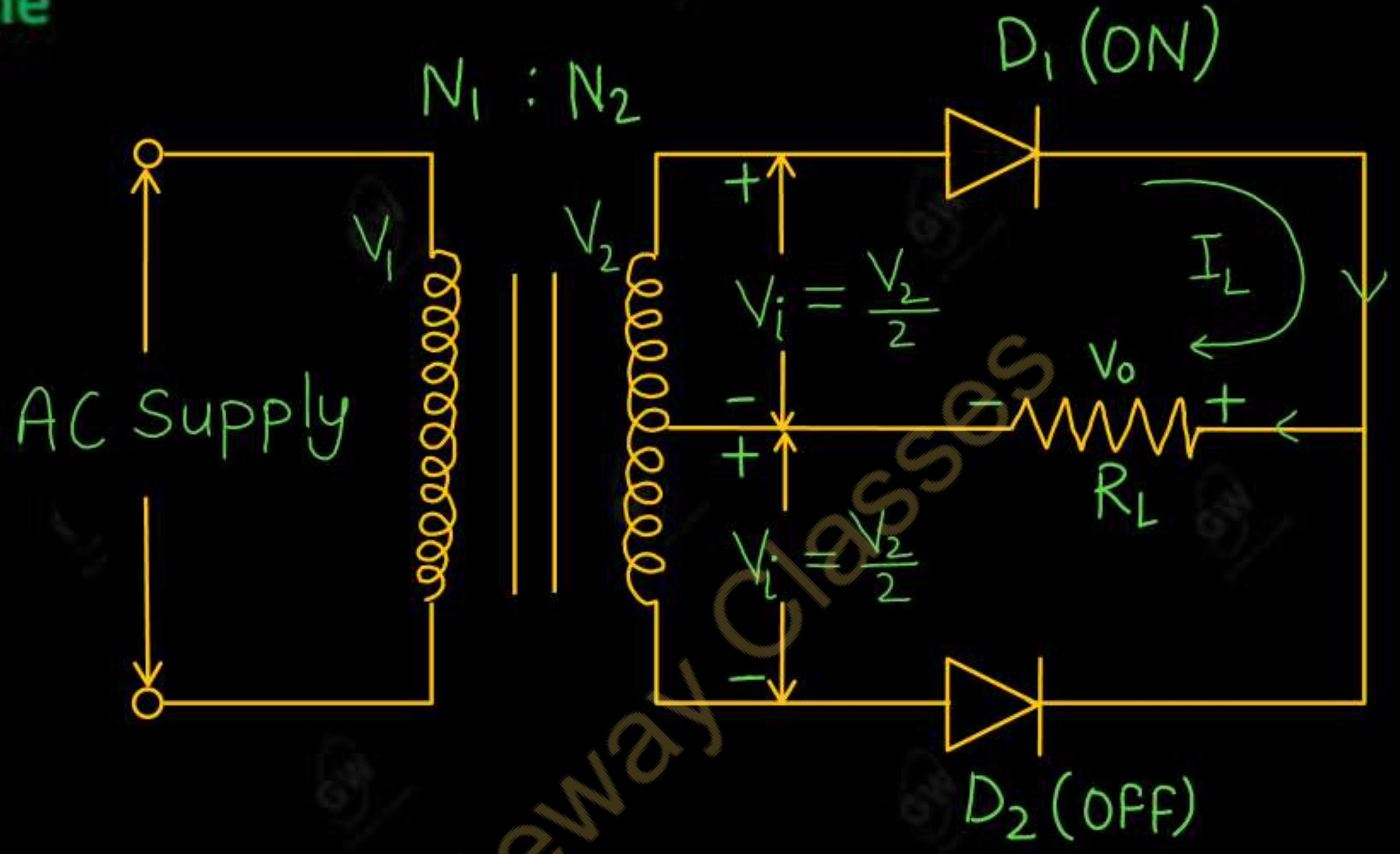
- Centre Tapped full wave rectifier have a Centre tap step-down transformer, two diodes D_1 , D_2 and load resister R_L .In Centre tapped transformer secondary winding is divided in to two equal parts



Gateway Classes

Working

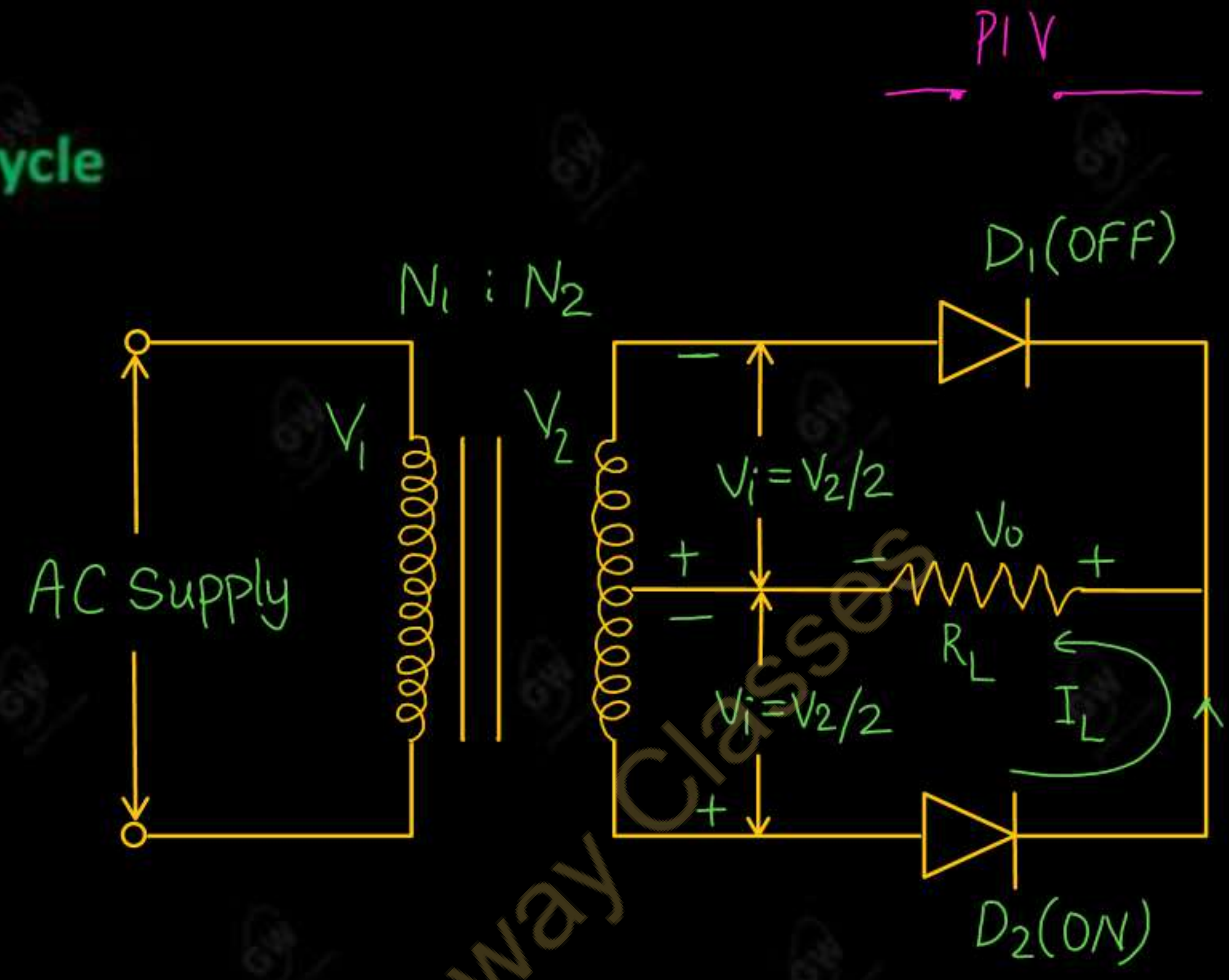
(i) During positive Half cycle



- Diode $\mathbf{D_1}$ is forward biased and $\mathbf{D_2}$ is reversed biased.
- So diode $\mathbf{D_1}$ is ON and $\mathbf{D_2}$ is OFF
- Current flow across the load resistor $\mathbf{R_L}$ in upper half of secondary winding
- Hence, we get output voltage $(\mathbf{V_o})$ across load resistor $\mathbf{R_L}$

Gateway Classes

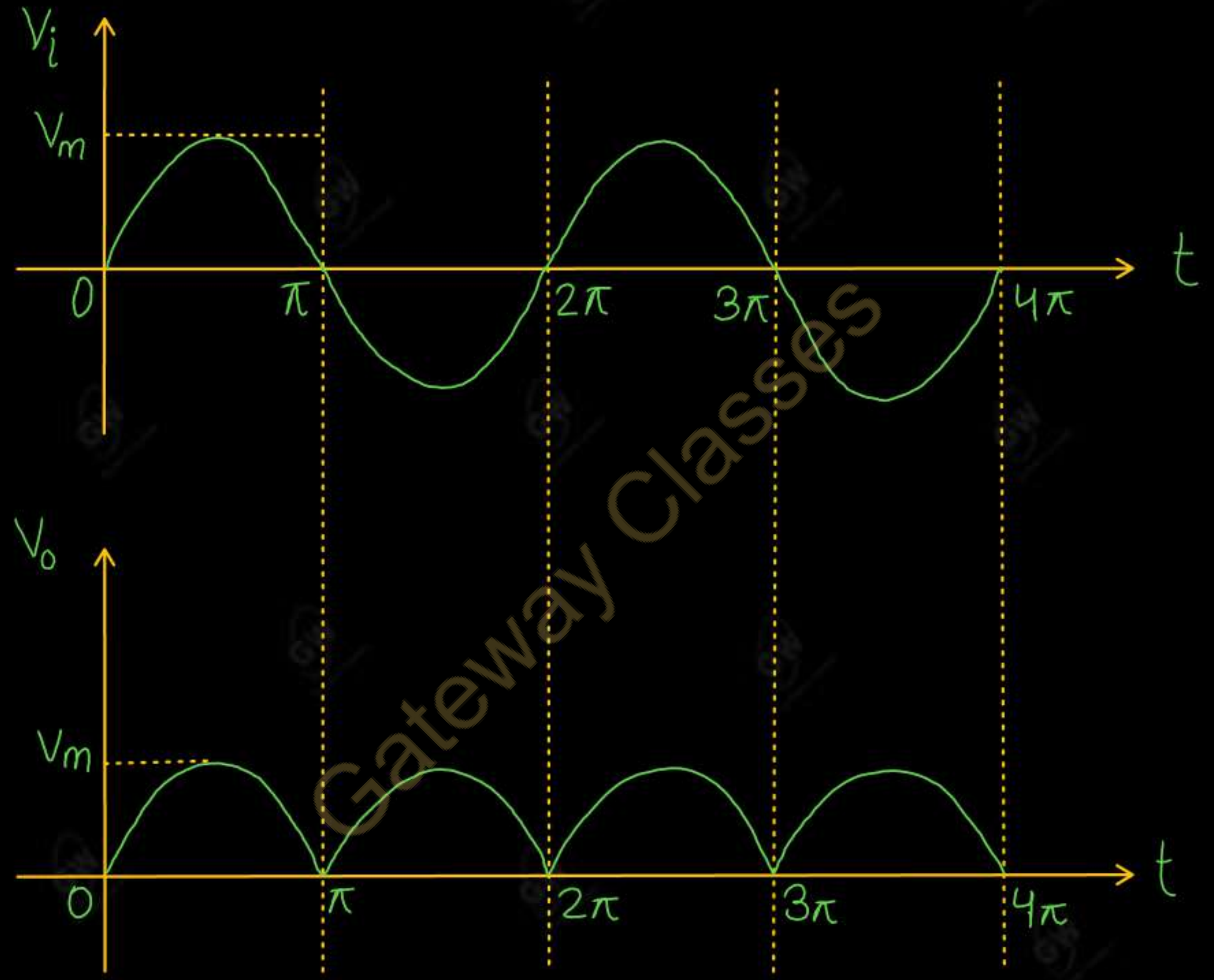
(ii) During Negative Half cycle



- Diode $\mathbf{D_1}$ is reversed biased and $\mathbf{D_2}$ is forward biased.
- So diode $\mathbf{D_1}$ is **OFF** and $\mathbf{D_2}$ is **ON**
- Current flow across the load resistor $\mathbf{R_L}$ in Lower half of secondary winding
- Hence, we get output voltage $(\mathbf{V_o})$ across load resistor $\mathbf{R_L}$

Gateway Classes

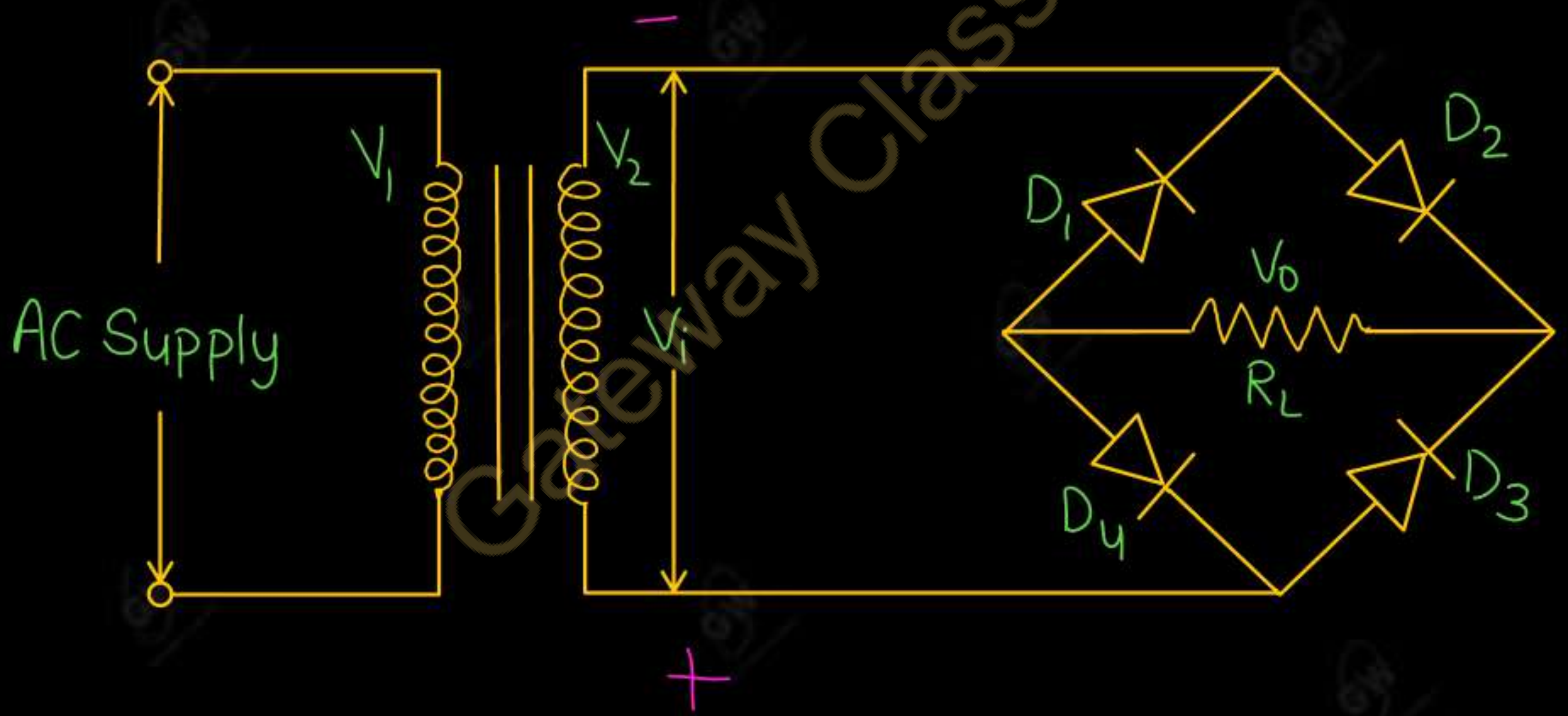
Input output waveform of Centre Tapped Full wave Rectifier



Gateway Classes

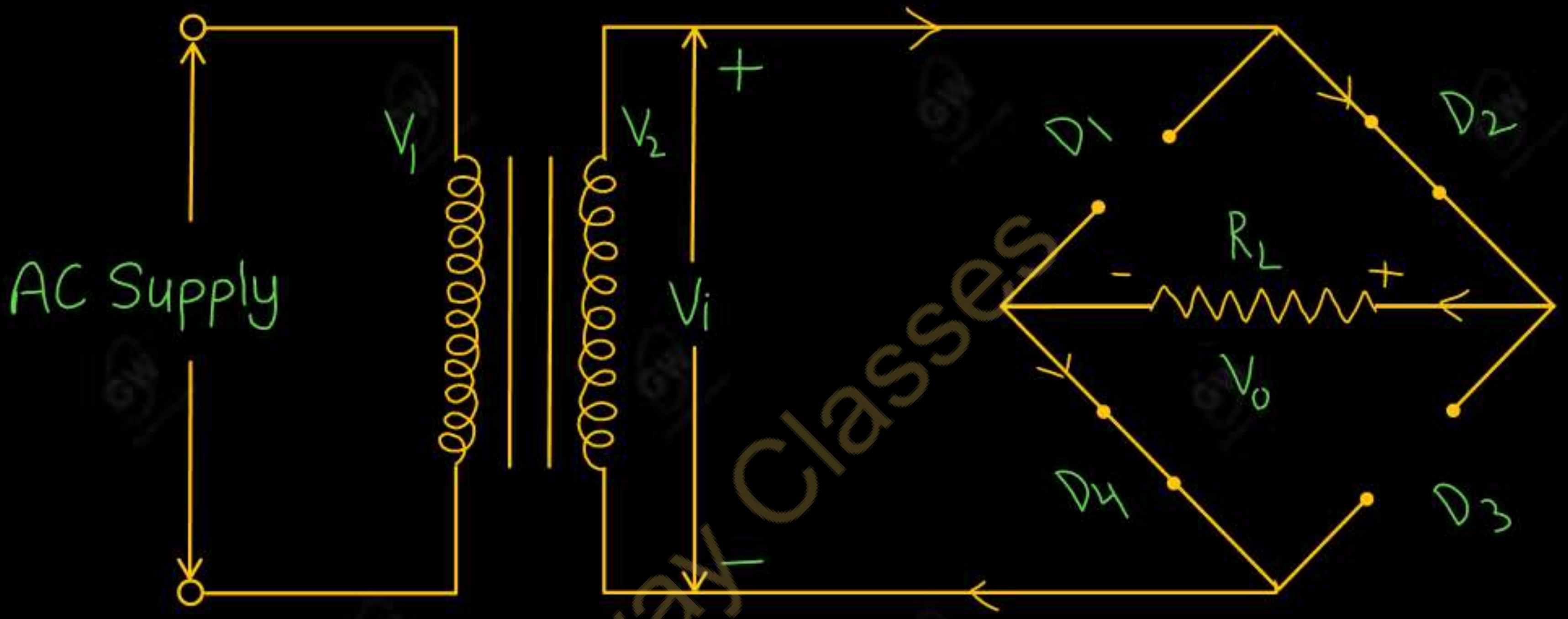
Full wave Bridge Rectifier

- Full wave Bridge Rectifier have a Step-down transformer, four diodes D_1 , D_2 , D_3 , D_4 and a load resister R_L as shown in figure
- Bridge Rectifier avoids the need of a Centre-tapped transformer



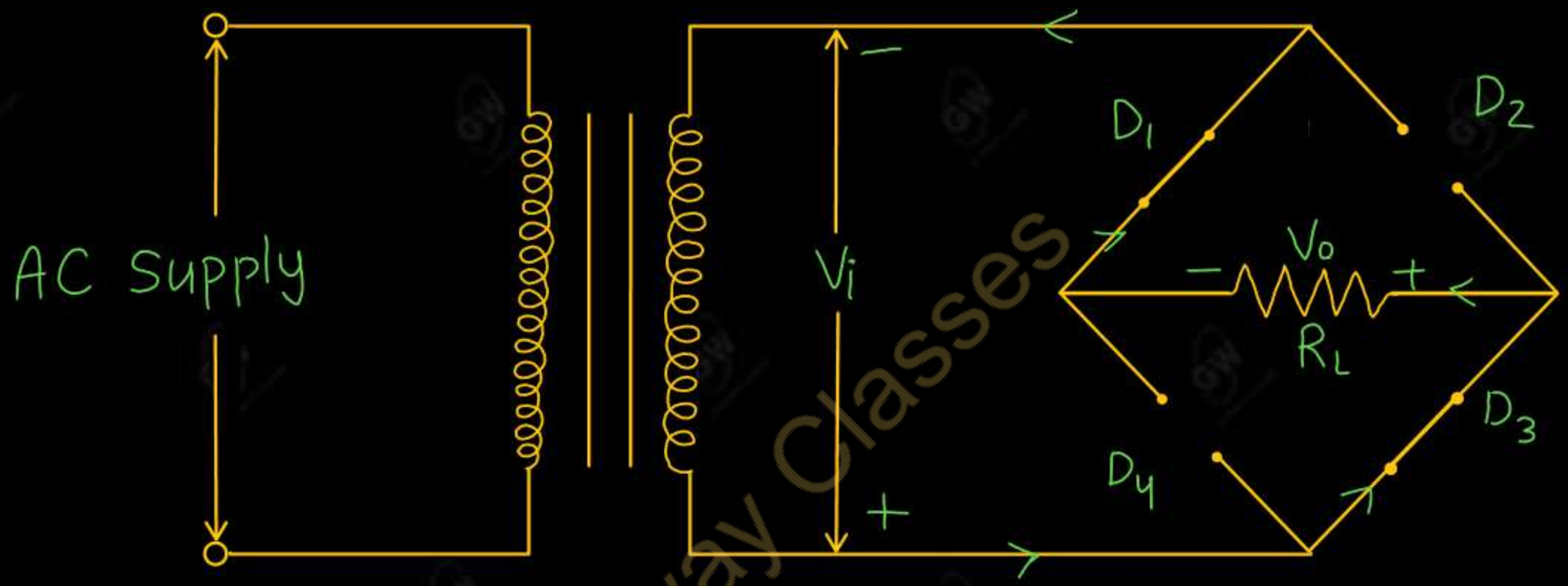
Working

(i) During positive Half cycle



- Diode D_2 , D_4 are forward biased and D_1 , D_3 are reversed biased.
- So diode D_2 , D_4 are ON and D_1 , D_3 are OFF
- Current flow through D_2 , D_4 and load resistor R_L
- Hence, we get output voltage (V_0) across load resistor R_L

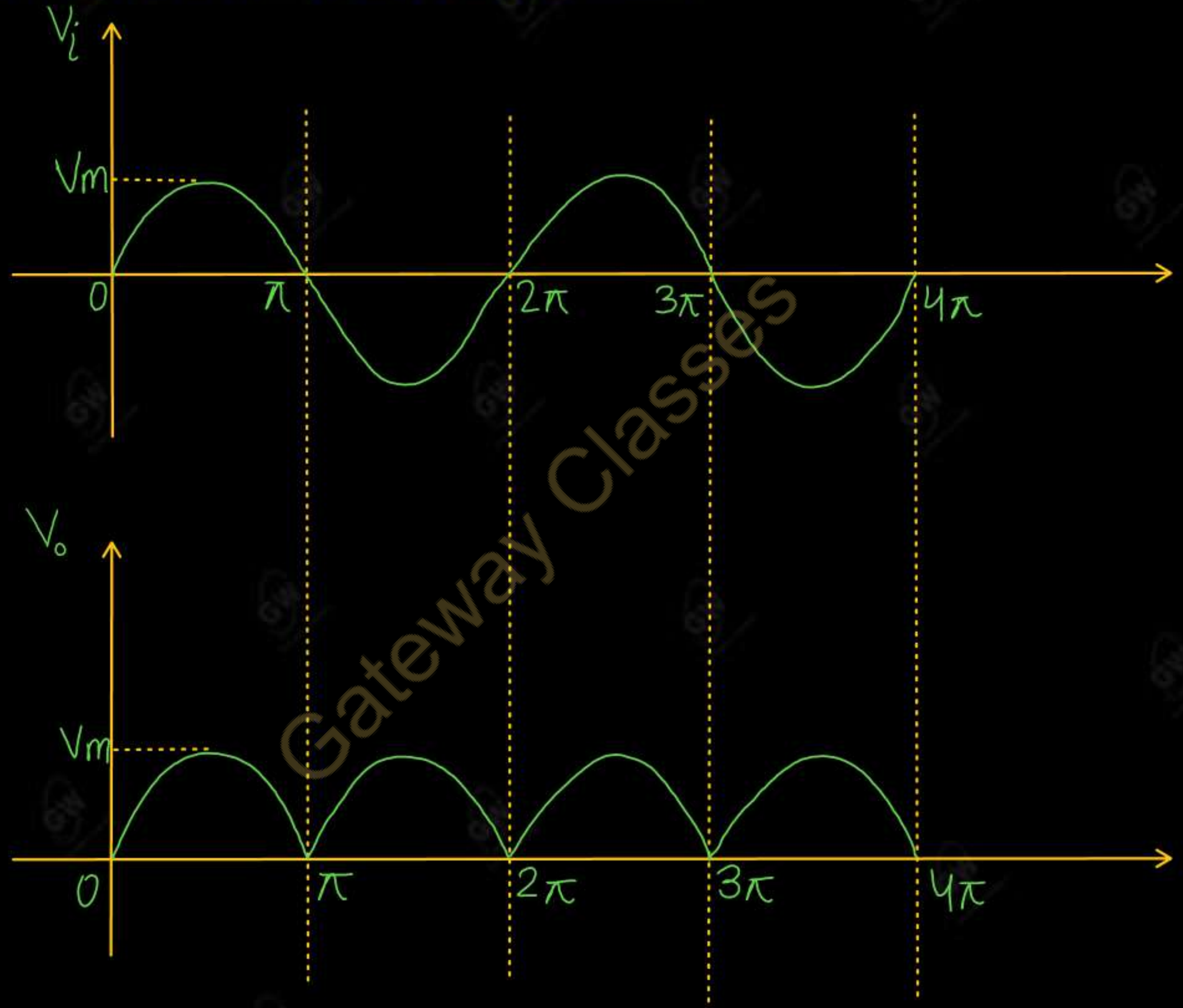
(ii) During Negative Half cycle



- Diode D_2 , D_4 are reversed biased and D_1 , D_3 are forward biased.
- So diode D_2 , D_4 are OFF and D_1 , D_3 are ON
- Current flow through D_3 , D_1 and load resistor R_L
- Hence, we get output voltage (V_o) across load resistor R_L

Gateway Classes

Input output waveform of Full Wave Bridge Rectifier



Gateway Classes

UNIT - 1 : DPP - 3

Q.1 Draw and explain the working of Centre Tapped full wave rectifier with input output waveform

Q.2 Draw and explain the Full wave Bridge rectifier with input output waveform

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 6

Today's Target

- Performance Parameters of Full Wave Rectifier
- DPP
- PYQs

By Gulshan sir

Gateway Classes

Performance Parameters

(1) Half Wave Rectifier

(2) Centre Tapped Full Wave Rectifier

(3) Full Wave Bridge Rectifier

$$I_{av} = \frac{1}{T}\int_0^T I \, d\theta$$

Let $v = V_m \sin\theta$ be the voltage across the secondary winding

Let $I = I_m \sin\theta$ be the current across the secondary winding

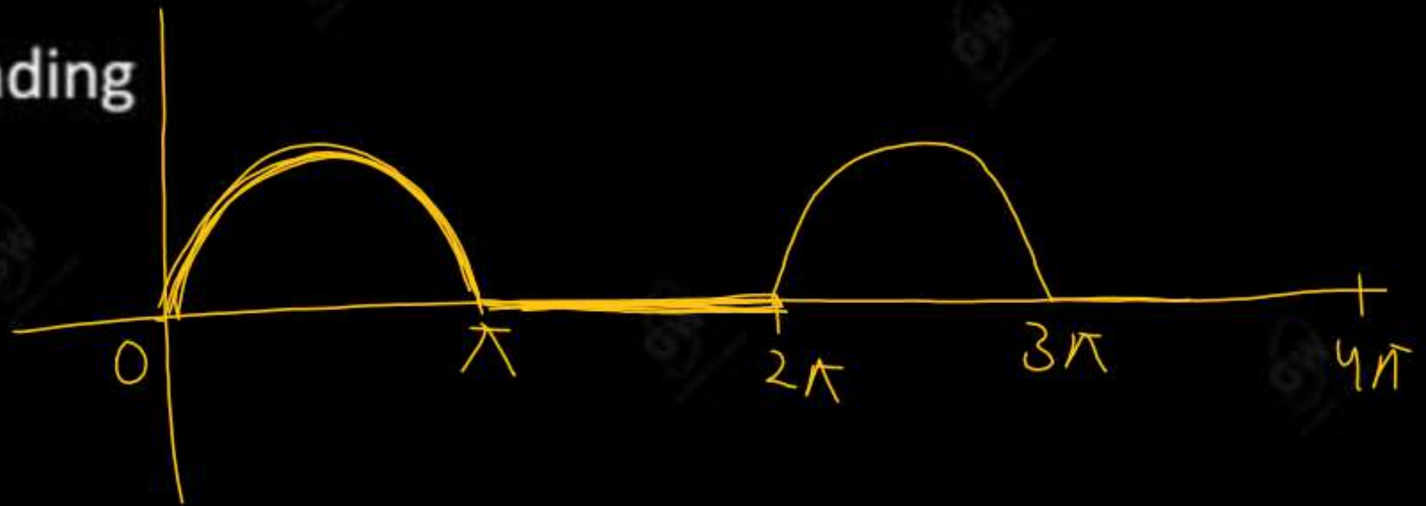
Where

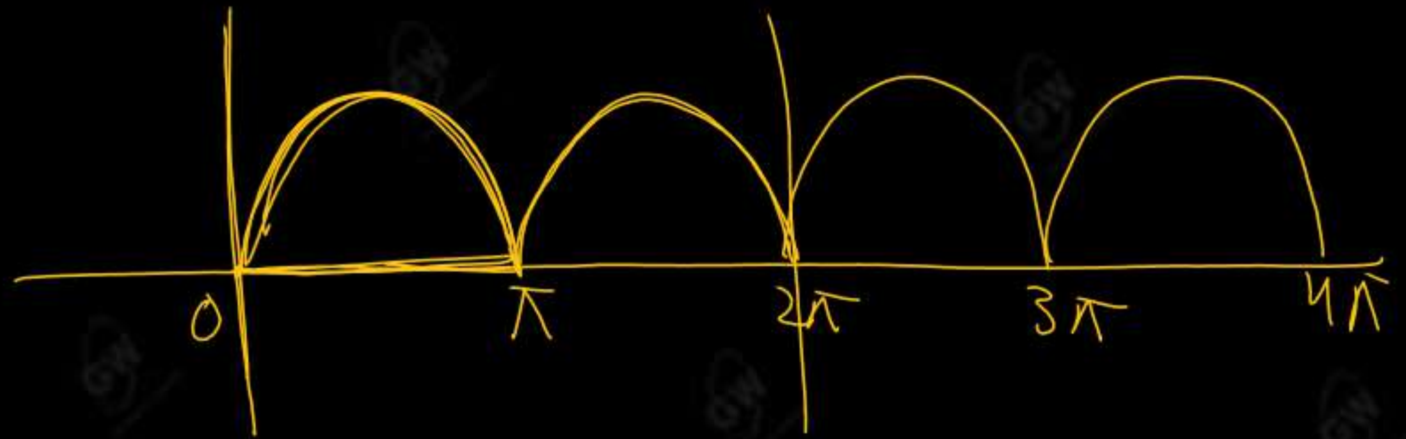
$$I_m = \frac{V_m}{R_L + R_f}$$

R_L = Load resistance

R_f = Diode resistance

$\theta = \omega t$





Gateway Classes

Performance Parameters

(1) Average or DC output (Load) current (I_{dc})

$$I_{dc} = \frac{1}{T}\int_0^T I\,d\theta = \frac{1}{\pi}\int_0^\pi I_m \sin\theta\,d\theta$$

$$= \frac{I_m}{\pi}\int_0^\pi \sin\theta\,d\theta$$

$$= \frac{I_m}{\pi}\left(-\cos\theta\right)_0^\pi$$

$$= \frac{I_m}{\pi}\left(-\cos\pi + \cos 0\right)$$

$$= \frac{I_m}{\pi}(1+1)$$

$$\boxed{I_{dc} = \frac{2I_m}{\pi}}$$

(2) Average or DC output (Load) voltage (V_{dc})

$$V_{dc} = I_{dc} \times R_L$$

$$= \frac{2I_m}{\pi} \times R_L$$

$$= \frac{2V_m}{\pi(R_L + R_f)} \times R_L$$

$$= \frac{2V_m}{\pi \cancel{R_L}} \times \cancel{R_L} \qquad \{\text{If } R_f = 0\}$$

$$\boxed{V_{dc} = \frac{2V_m}{\pi}}$$

Gateway Classes

(3) RMS output (Load) current (I_{rms})

$$I_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} I^2 \, d\theta}$$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi} (I_m \sin\theta)^2 \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{\pi}\int_0^{\pi} \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{\pi}\int_0^{\pi} \frac{1-\cos 2\theta}{2} \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi}\left(\theta - \frac{\sin 2\theta}{2}\right)_0^{\pi}}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi}(\pi - 0)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\cancel{\pi}}\cancel{\pi}}$$

$$\boxed{I_{rms} = \frac{I_m}{\sqrt{2}}}$$

(4) RMS output (Load) voltage (V_{rms})

$$V_{rms} = I_{rms} \times R_L$$

$$= \frac{I_m}{\sqrt{2}} \times R_L$$

$$= \frac{V_m}{\sqrt{2}(R_L + R_f)} \times R_L$$

$$= \frac{V_m}{\sqrt{2} \times \cancel{R_L}} \times \cancel{R_L} \quad \{\text{If } R_f = 0\}$$

$$\boxed{V_{rms} = \frac{V_m}{\sqrt{2}}}$$

Gateway Classes

(5) Ripple factor (r)

- The output of rectifier contains a.c. as well as d.c. component. The ripple factor measures the percentage of a.c. component present in the output.
- The effectiveness of a rectifier depends on the magnitude of ripple factor. Smaller the ripple factor, more effective is the rectifier
- Ripple factor is defined as the ratio of rms value of a.c .component to the d.c. component in the rectifier.

$$\text{Ripple factor} = \frac{\text{RMS value of a.c .component of output}}{\text{DC value of output}}$$

$$\gamma = \frac{I_{ac}}{I_{dc}}$$

RMS value of a.c. component of output

$$I_{ac} = \sqrt{I_{rms}^2 - I_{dc}^2}$$

$$\gamma = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}}$$

$$\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - \left(\frac{I_{dc}}{I_{dc}}\right)^2}$$

$$\boxed{\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}}$$

$$\gamma = \sqrt{\left(\frac{\cancel{I_m}}{\sqrt{2}} \times \frac{\pi}{2\cancel{I_m}}\right)^2 - 1}$$

$$\gamma = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1}$$

$$\boxed{\gamma = 0.482}$$

Gateway Classes

(6) Efficiency of a Rectifier (η)

Efficiency of a Rectifier is defined as the ratio of dc output power to the ac input power

$$\eta = \frac{\text{dc output power}}{\text{ac input power}}$$

$$\boxed{\eta = \frac{I_{dc}^2 \times R_L}{I_{rms}^2 (R_L + R_f)}}$$

$$\eta = \frac{\left(\frac{2 I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 \times (R_L + R_f)}$$

$$\eta = \frac{4 \cancel{I_m^2} \times R_L \times 2}{\cancel{I_m^2} \times (R_L + R_f) \times \pi^2} = \boxed{\frac{8}{\pi^2} \times \frac{R_L}{R_L + R_f}}$$

$$= \frac{8 \cancel{R_L}}{\pi^2 \times \cancel{R_L}} \quad \{\text{If } R_f = 0$$

$$= \frac{8}{\pi^2}$$

$$\boxed{\eta = 81.2\%}$$

Gateway Classes

(7) Peak inverse voltage

It is The maximum reverse voltage that can be applied across a diode without damaging it

(i) For Centre Tapped full wave rectifier

$$PIV = 2V_m$$

(ii) For Full Wave Bridge Rectifier

$$PIV = 2V_m$$

(8) Ripple frequency (f_r)

It is the frequency of output wave in a rectifier. It is called output frequency

$$f_r = 2f_i$$

Gateway Classes

Comparison of Half Wave Rectifier and Full Wave Rectifier

S.NO	Parameter of comparison	Half Wave Rectifier	Centre tapped Full Wave Rectifier	Full Wave Bridge Rectifier
1	Output	Positive Half Cycle	Both Half Cycle	Both Half Cycle
2	Number of Diodes	1	2	4
3	Efficiency	40.6 %	81.2 %	81.2 %
4	Ripple Factor	1.21	0.48	0.48
5	PIV	V_m	$2V_m$	V_m
6	DC Load current (I_{dc})	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
7	RMS Load Current(I_{rms})	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
8	Ripple Frequency	f_i	$2f_i$	$2f_i$

Gateway Classes

Advantages and disadvantages of Bridge rectifier over Centre Tapped Rectifier

Advantages

- Bridge Rectifier does not require Centre Tapped secondary winding
- The diode in a Bridge Rectifier is required to have PIV rating of only V_m
- The diodes used in a Centre Tapped Rectifier are costlier than those used in Bridge Rectifier

Disadvantages

- The only disadvantage of bridge rectifier is that this needs four diodes

NOTE : For low voltage application, the Centre Tapped Rectifiers are preferred

Gateway Classes

Q.1 A Centre tapped full wave rectifier has a load resistance $R_L = 1000\Omega$. The forward resistance R_f of each diode is $20\ \Omega$. The voltage across half of the secondary winding is given by the equation $v = 200 \sin 314t$. Determine : (i) The peak value of current (ii) The average or dc value of current (iii) The RMS value of current (iv) The Ripple factor (v) The Rectifier efficiency

Given

$R_L = 1000\Omega$

$R_f = 20\Omega$

$V = 200 \sin 314t$

$\Rightarrow V_m = 200V$

(i) $I_m = \dfrac{V_m}{R_L + R_f}$

$= \dfrac{200}{1000+20}$

$\boxed{I_m \approx 0.2A}$

(ii) $I_{dc} = \dfrac{2I_m}{\pi}$

$= \dfrac{2 \times 0.2}{3.14}$

$\boxed{I_{dc} = 0.127A}$

Gateway Classes

(III) $I_{rms} = \frac{I_m}{\sqrt{2}}$

$= \frac{0.2}{\sqrt{2}}$

$\boxed{I_{rms} = 0.141\ A}$

(IV) $\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

$= \sqrt{\left(\frac{0.141}{0.127}\right)^2 - 1}$

$\boxed{\gamma = 0.482}$

(V) $\eta = \frac{8}{\pi^2} \times \frac{R_L}{R_L + R_f}$

$= \frac{8}{\pi^2} \times \frac{1000}{1000 + 20}$

$\boxed{\eta = 79.5\%}$

GW GATEWAY CLASSES
Gateway Classes

Q.2 In the Bridge rectifier circuit, the secondary voltage $V_s = 100 \sin 50t$ and load resistance is 1 kΩ. Calculate (i) DC current (ii) RMS value of current (iii) Efficiency (iv) Ripple factor

AKTU 2022

Given

$V_s = 100 \sin 50t$

$V_m = 100\,V$

$R_L = 1000\,\Omega$

$R_f = 0$ (say)

(i) $I_{dc} = \frac{2I_m}{\pi}$

$= \frac{2V_m}{\pi(R_L+R_f)}$

$= \frac{2\times 100}{\pi \times 1000}$

$I_{dc} = 0.06366\,A$

$\boxed{I_{dc} = 63.66\,mA}$

(ii) $I_{rms} = \frac{I_m}{\sqrt{2}}$

$= \frac{V_m}{(R_L+R_f)\sqrt{2}}$

$= \frac{100}{1000\sqrt{2}}$

$= 0.07071\,A$

$\boxed{I_{rms} = 70.71\,mA}$

Gateway Classes

(III) $\boxed{\eta = 81.2\%}$

(IV) $\boxed{\gamma = 0.482}$

$$\boxed{\eta = \frac{8}{\pi^2} \times \frac{R_L}{R_L + R_f}}$$

$$\boxed{\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}}$$

GW GATEWAY CLASSES
Gateway Classes

UNIT - 1 : DPP - 6

Q.1 How does a half wave rectifier differ from a full wave rectifier **AKTU 2024**

Q.2 Draw and explain the working of Centre tapped full wave rectifier. Also calculate its Efficiency and Ripple factor. **AKTU 2023**

Q.3 Why bridge type full wave rectifier is preferred over centre tapped full wave rectifier. State two reasons **AKTU 2021**

Q.4 Draw a neat circuit diagram of the bridge rectifier and explain its operation with output waveforms. Derive the average value of current and voltage **AKTU 2020**

Gateway Classes

AKTU

B.Tech : I-Year

GATEWAY CLASSES

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 7

Gateway Classes

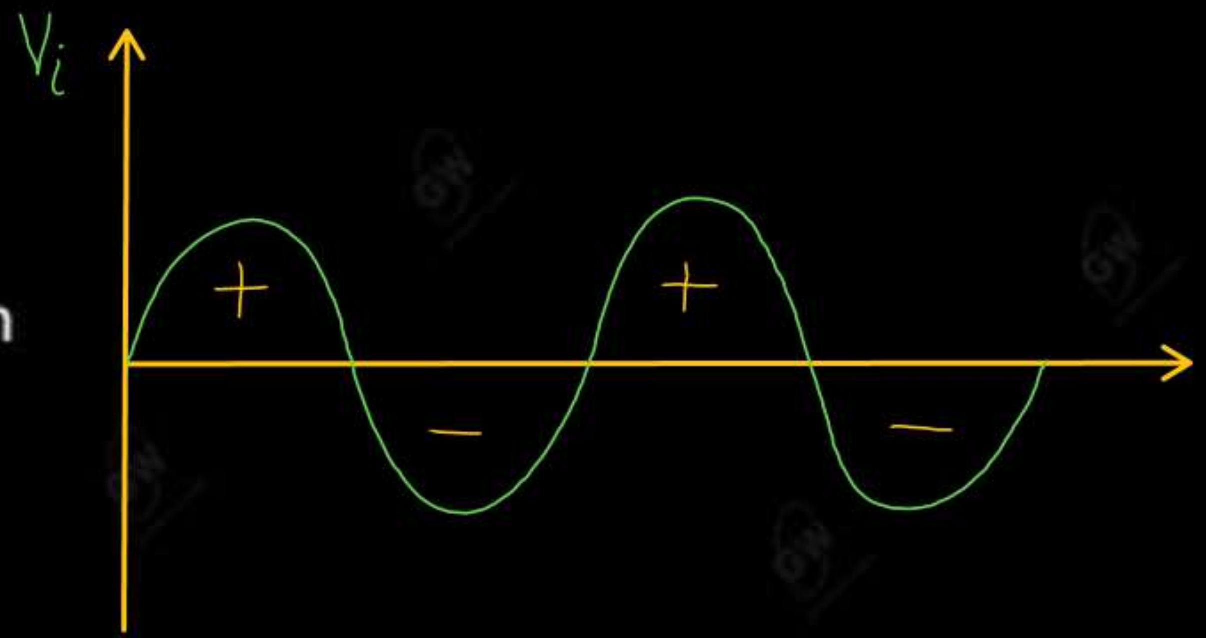
Today's Target

- Clipper Circuits
- DPP
- PYQs

By Gulshan sir

Clipper

Clipper is a circuit which is used to remove some portion of input waveform without distorting the remaining part.



Types of Clipper circuits

(1) Positive clipper : Remove Positive Half cycle of input waveform

(i) Series

(ii) Shunt

(2) Negative clipper : Remove Negative Half cycle of input waveform

(i) Series

(ii) Shunt

(3) Biased clipper : Remove some portion of Positive Half cycle or Negative Half cycle or both

(a) Biased Positive clipper

(i) Series

(ii) Shunt

Gateway Classes

(b) Biased Positive clipper with reverse polarity of Battery

(i) Series

(ii) Shunt

(c) Biased Negative clipper

(i) Series

(ii) Shunt

(d) Biased Negative clipper with reverse polarity of Battery

(i) Series

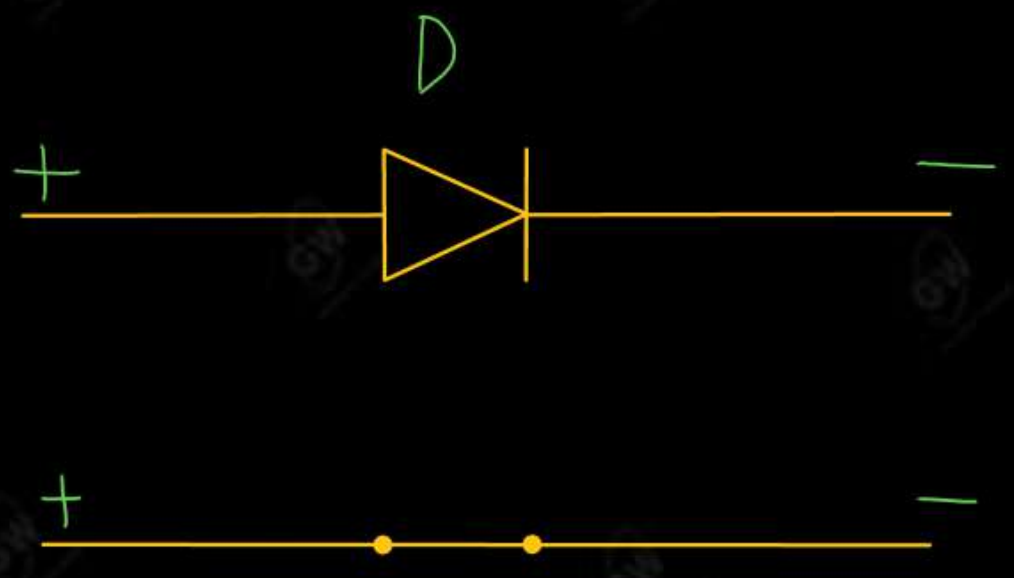
(ii) Shunt

(4) Combinational clipper : Remove some portion of Positive Half cycle and some portion of Negative Half cycle

Gateway Classes

Properties of ideal Diode

(1)

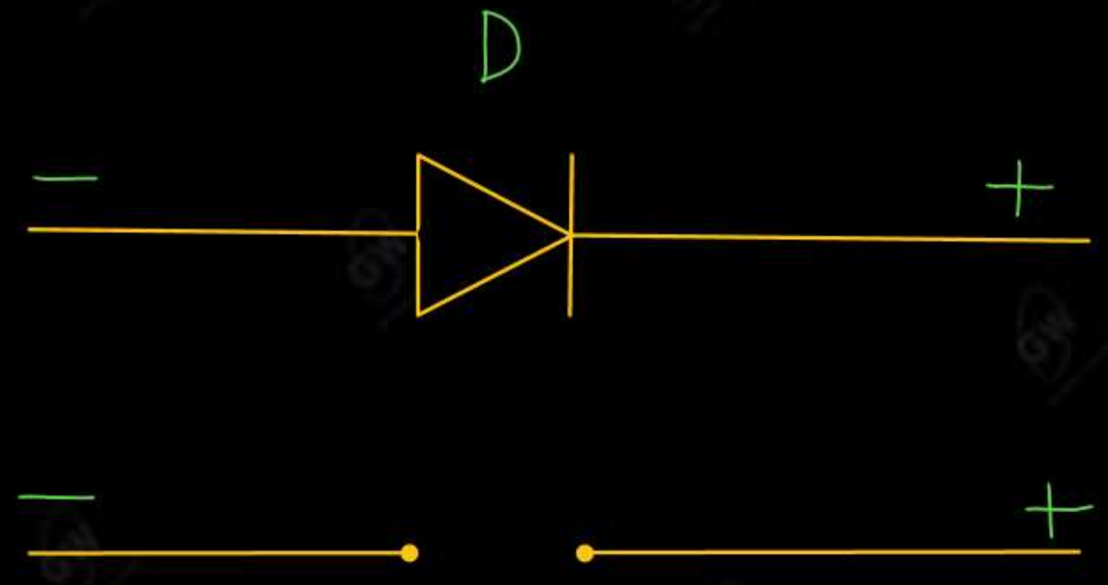


(i) FB / ON / SC

(ii) Current flow

(iii) No Voltage drop

(2)



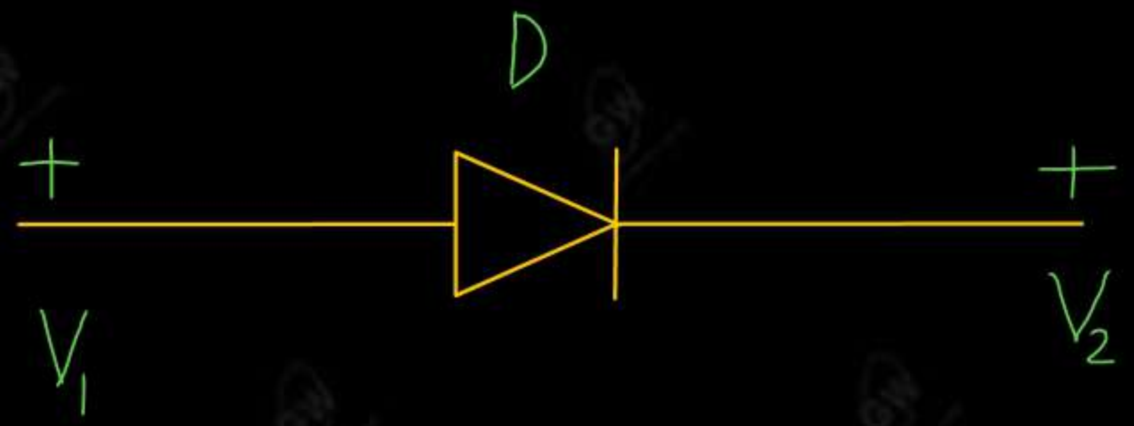
(i) RB / OFF / OC

(ii) No current flow

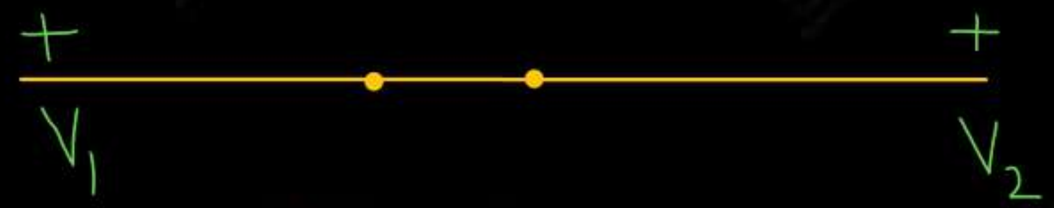
(iii) Voltage drop

Gateway Classes

(3)



CASE-1 : When $V_1 > V_2$

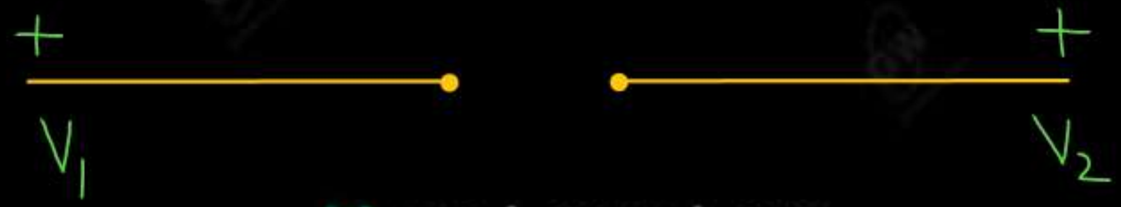


(i) FB | ON | SC

(ii) Current flow

(iii) No Voltage drop

CASE-1 : When $V_1 < V_2$



(i) RB | OFF | OC

(ii) No current flow

(iii) Voltage drop

Gateway Classes

Positive clipper

A positive clipper removes the positive half cycle of the input a.c voltage waveform.

Working

(i) Series positive clipper

- **During positive cycle**

Diode D is in Reversed biased

$$V_o = 0$$

- **During Negative cycle**

Diode D is in Forward biased

$$V_o = V_i$$

(ii) Shunt positive clipper

- **During positive cycle**

Diode D is in Forward biased

$$V_o = 0$$

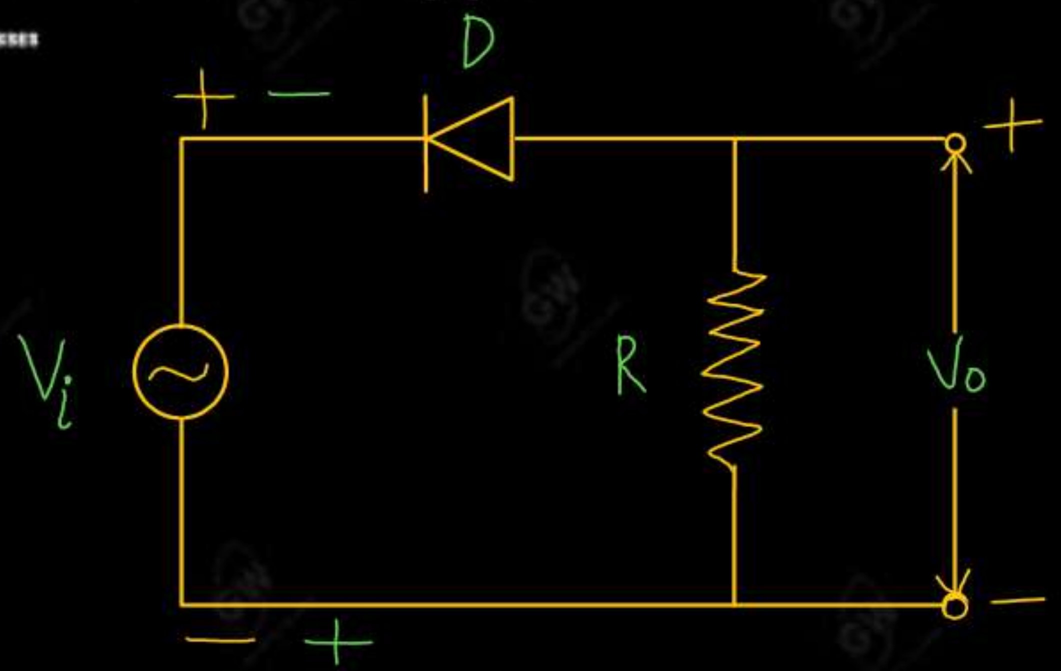
- **During Negative cycle**

Diode D is in Reversed biased

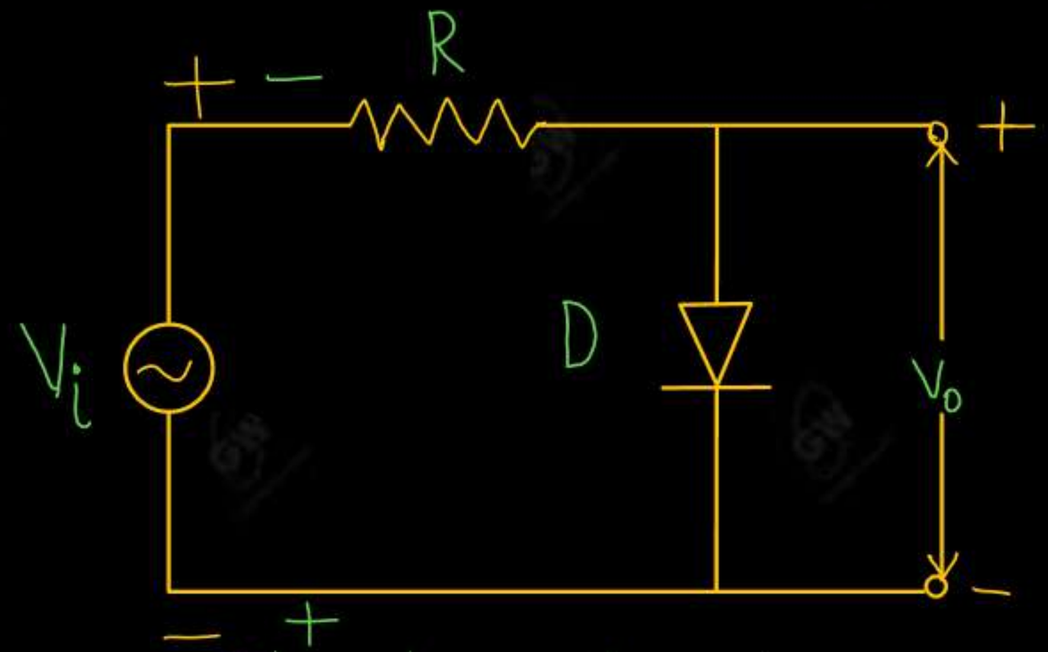
$$V_o = V_i$$

Gateway Classes

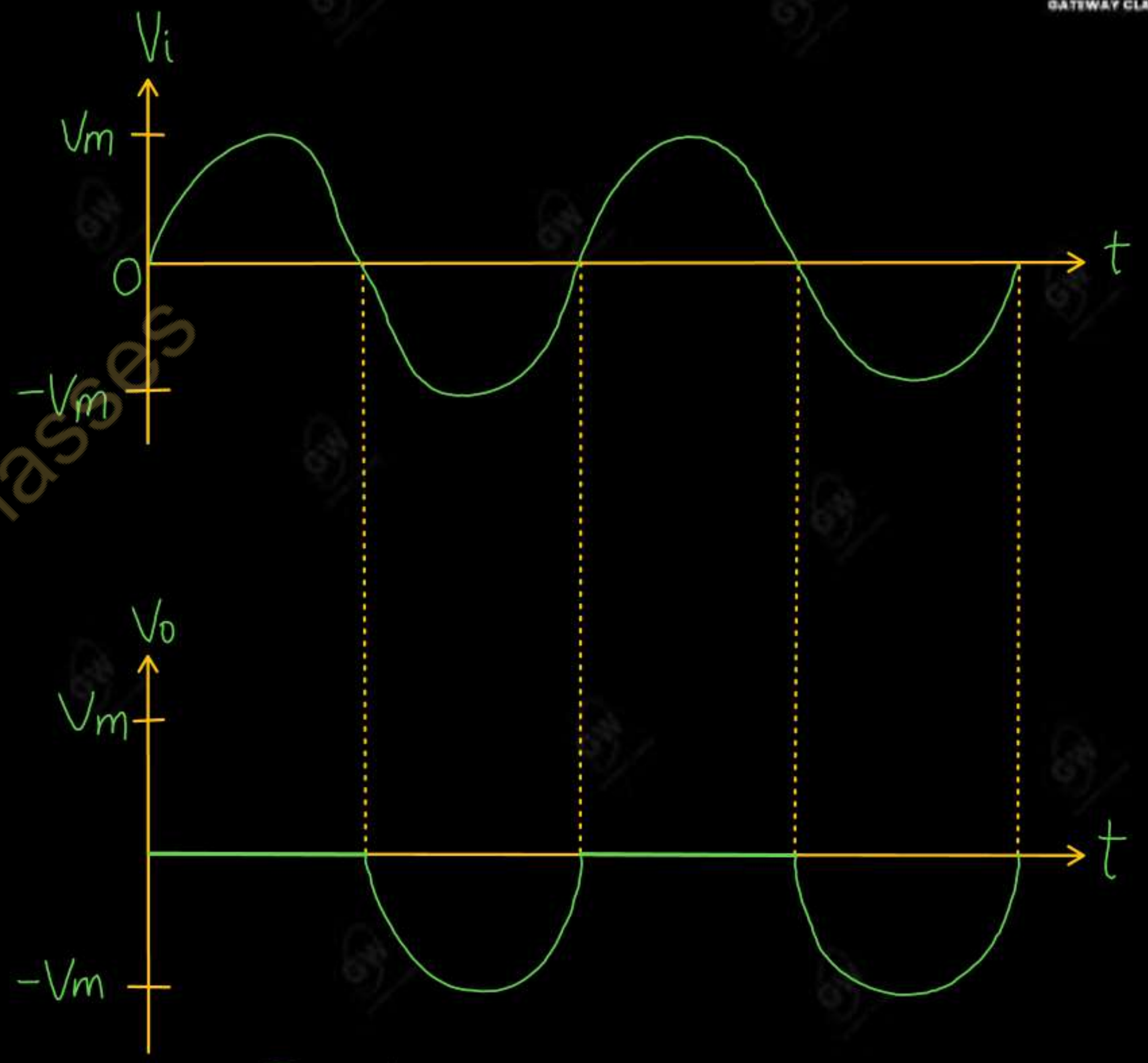
Positive clipper



Series Positive Clipper



Shunt positive Clipper



Input Output Waveform

Gateway Classes

Negative Clipper

A Negative Clipper removes the negative half cycle of the input a.c. voltage waveform.

Working

(i) Series negative clipper

- **During positive cycle**

Diode D is in Forward biased

$$V_o = V_i$$

- **During Negative cycle**

Diode D is in Reversed biased

$$V_o = 0$$

(ii) Shunt negative clipper

- **During positive cycle**

Diode D is in Reversed biased

$$V_o = V_i$$

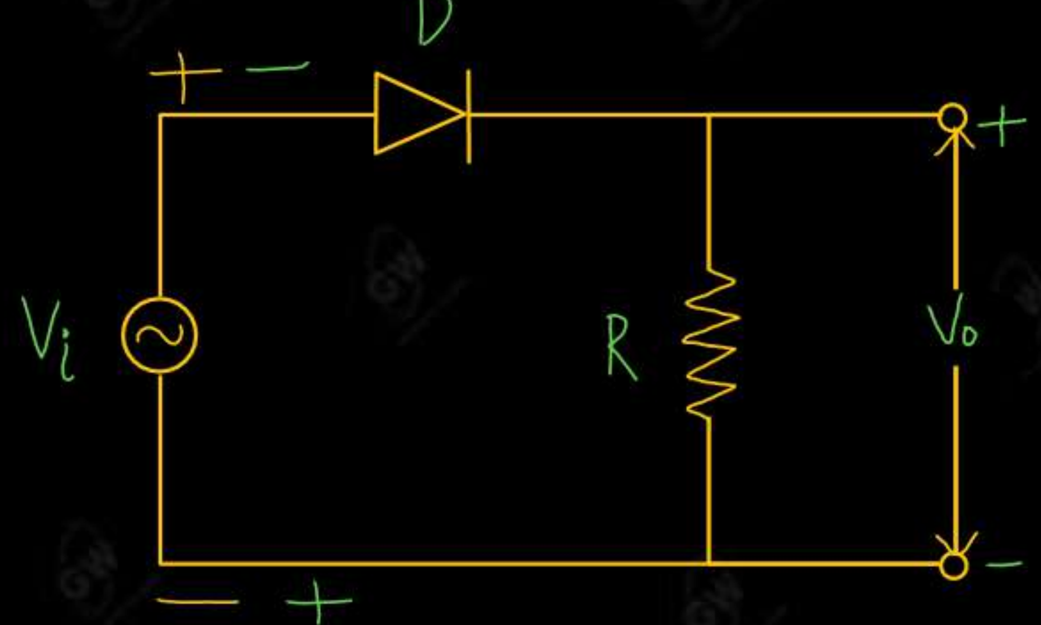
- **During Negative cycle**

Diode D is in Forward biased

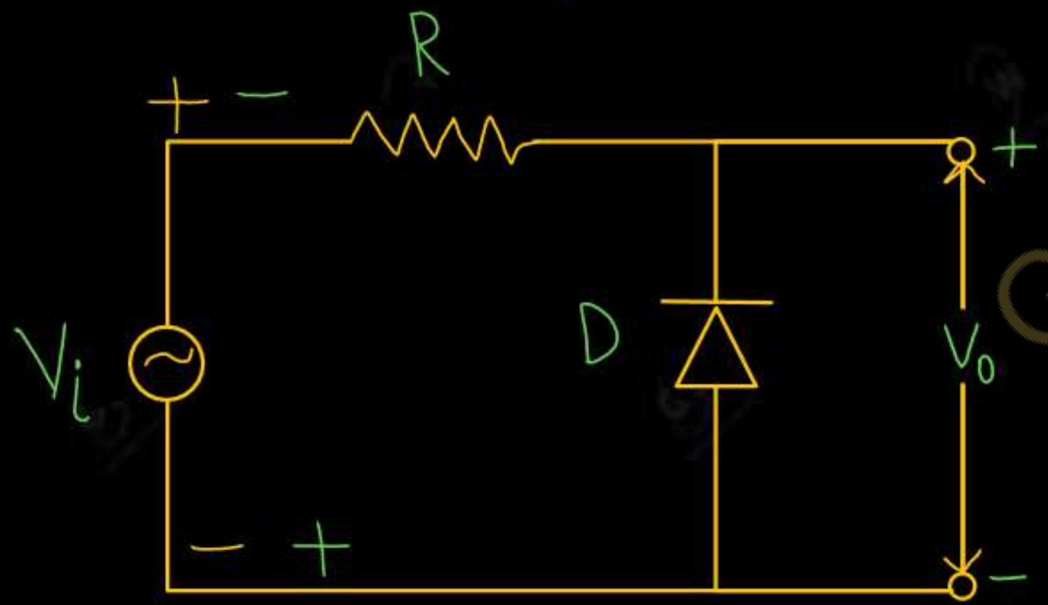
$$V_o = 0$$

Gateway Classes

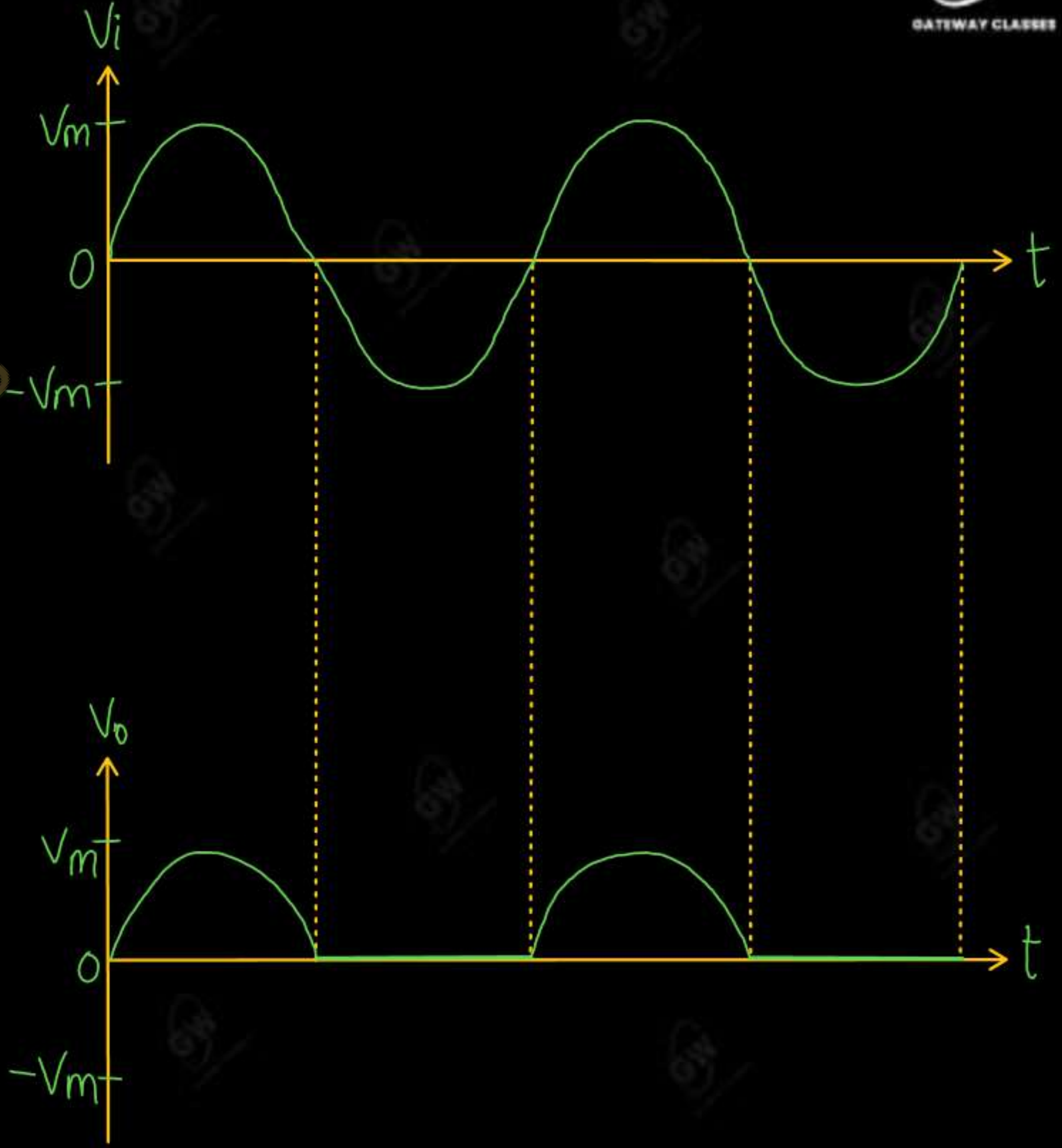
Negative Clipper



Series Negative clipper



Shunt negative clipper



Input output Waveform

Gateway Classes

Biased clipper : Biased clipper is used to remove a small portion of Positive Half cycle or Negative Half cycle or both. This is achieved by adding a battery in series with Diode or Resister

(a) Biased positive Clipper

Working

(i) Series

- **During positive cycle**

If $V_i < V_b$

Diode D is in Forward biased

$$V_o = V_i$$

If $V_i > V_b$

Diode D is in Reversed biased

$$V_o = V_b$$

- **During Negative cycle**

Diode D is in Forward biased

$$V_o = V_i$$

(ii) Shunt

- **During positive cycle**

If $V_i < V_b$

Diode D is in Reversed biased

$$V_o = V_i$$

If $V_i > V_b$

Diode D is in Forward biased

$$V_o = V_b$$

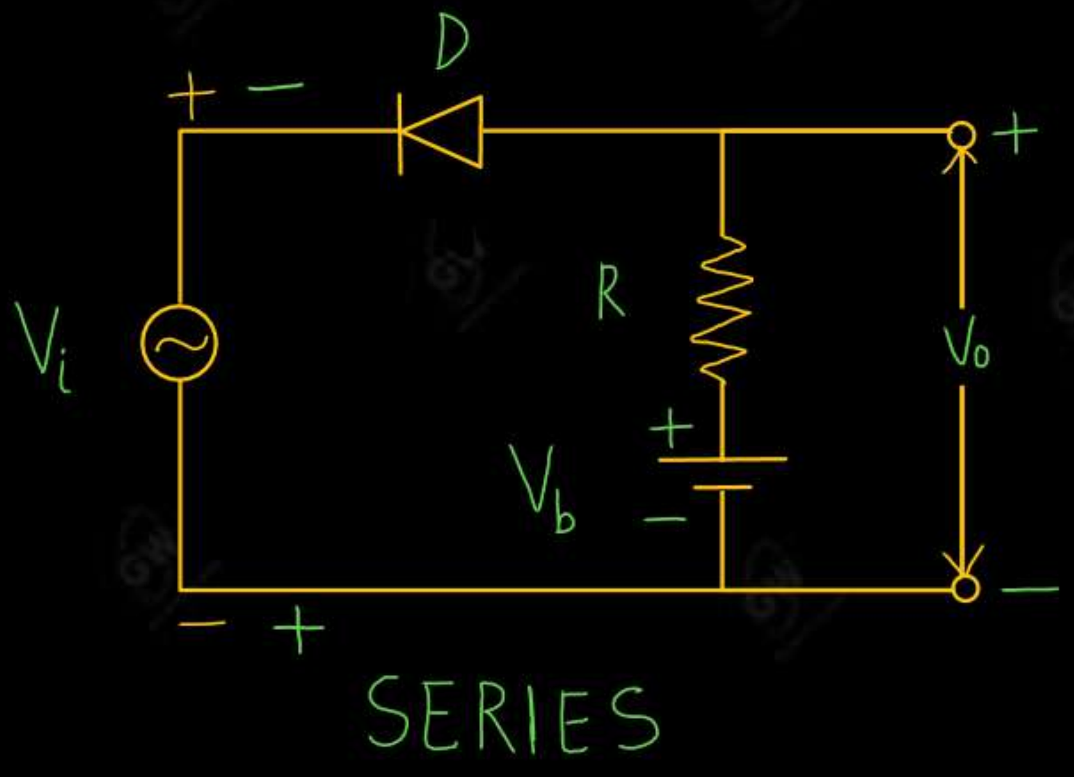
- **During Negative cycle**

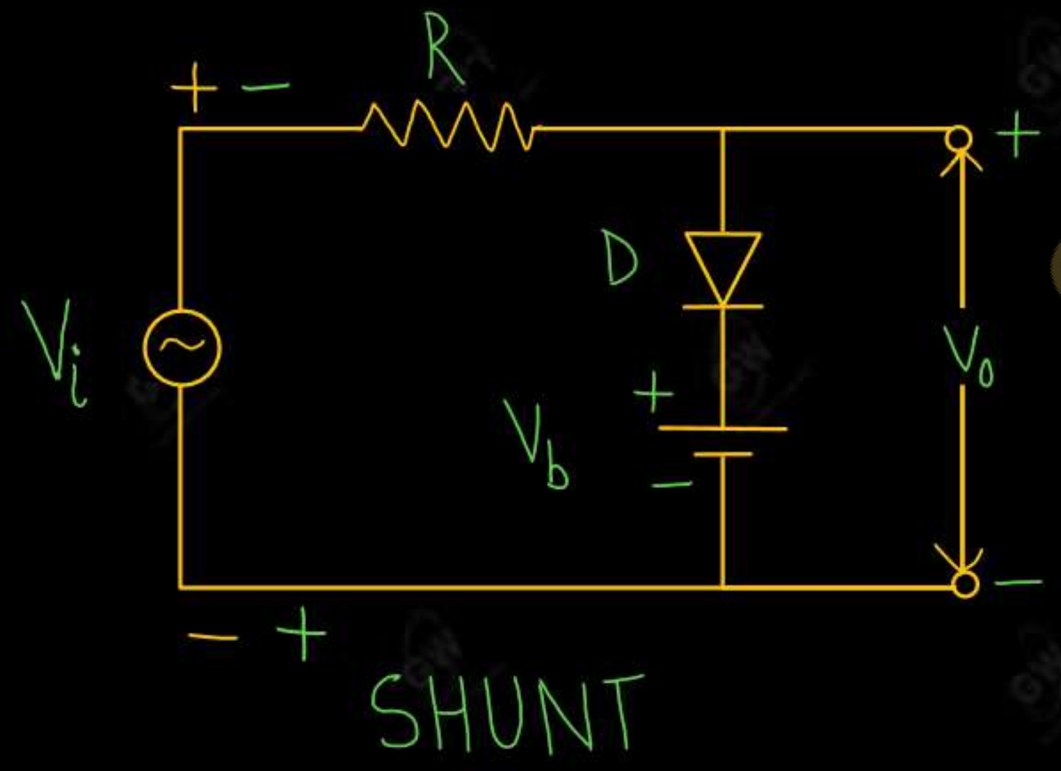
Diode D is in Reversed biased

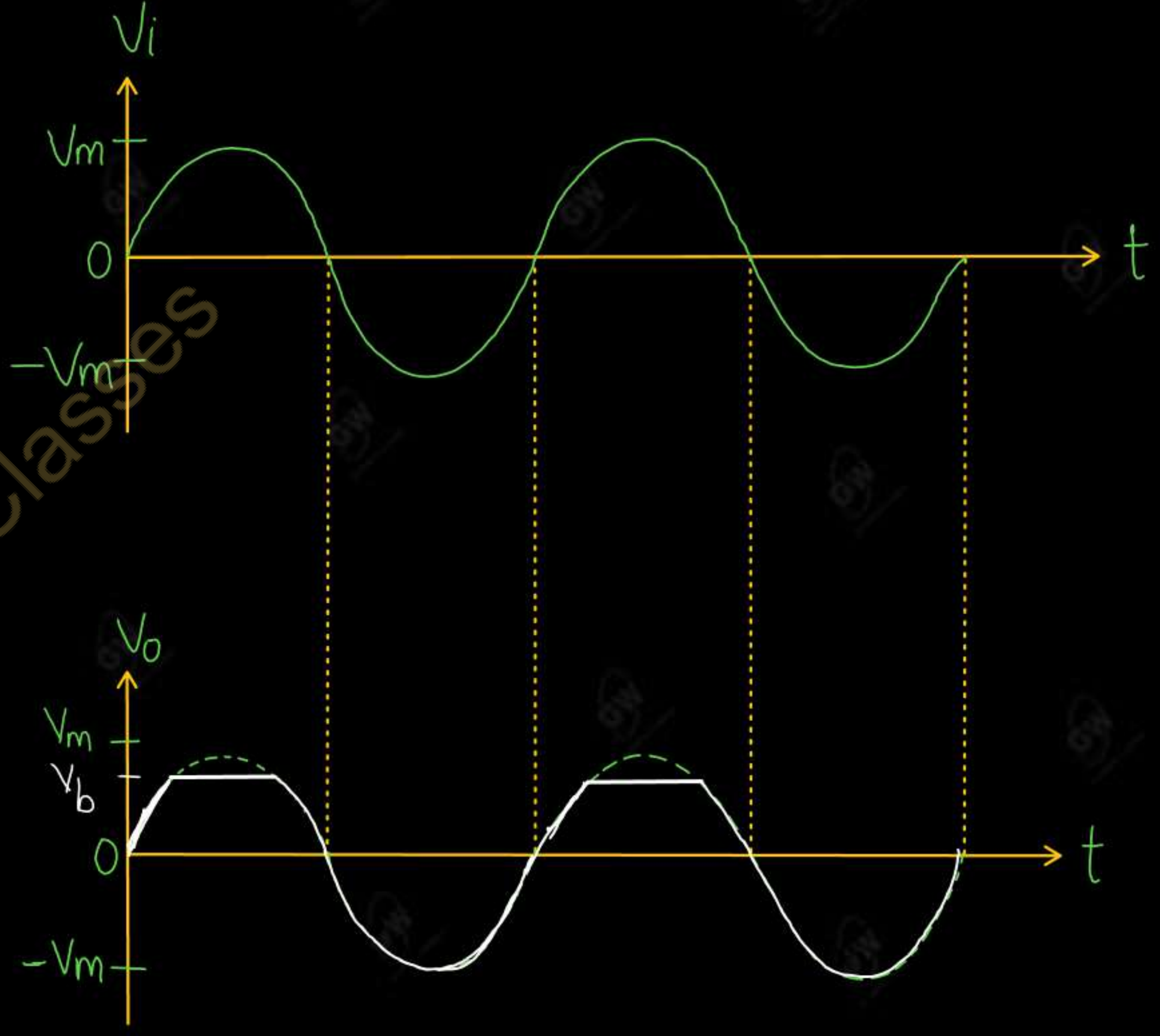
$$V_o = V_i$$

Gateway Classes

Biased positive Clipper







Gateway Classes

(a) Biased positive Clipper With reverse polarity of Battery

Working

(i) Series

- **During positive cycle**

Diode D is in Reversed biased

$$V_o = -V_b$$

- **During Negative cycle**

If $V_i < V_b$

Diode D is in Reversed biased

$$V_o = -V_b$$

If $V_i > V_b$

Diode D is in Forward biased

$$V_o = V_i$$

(ii) Shunt

- **During positive cycle**

Diode D is in Forward biased

$$V_o = -V_b$$

- **During Negative cycle**

If $V_i < V_b$

Diode D is in Forward biased

$$V_o = -V_b$$

If $V_i > V_b$

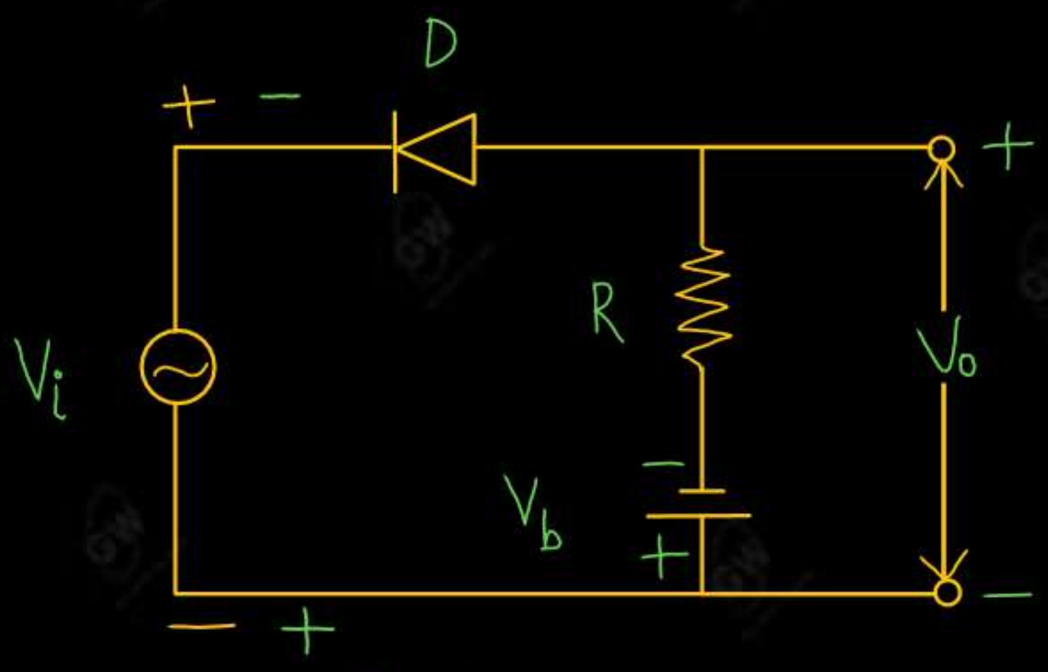
Diode D is in Reversed biased

$$V_o = V_i$$

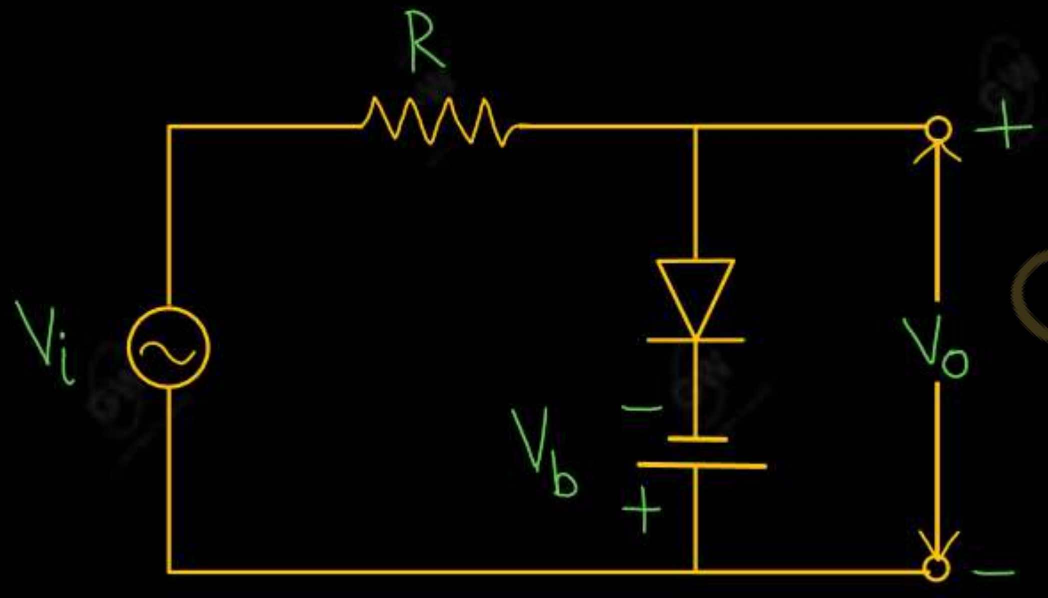
Gateway Classes

Biased positive Clipper With reverse polarity of Battery

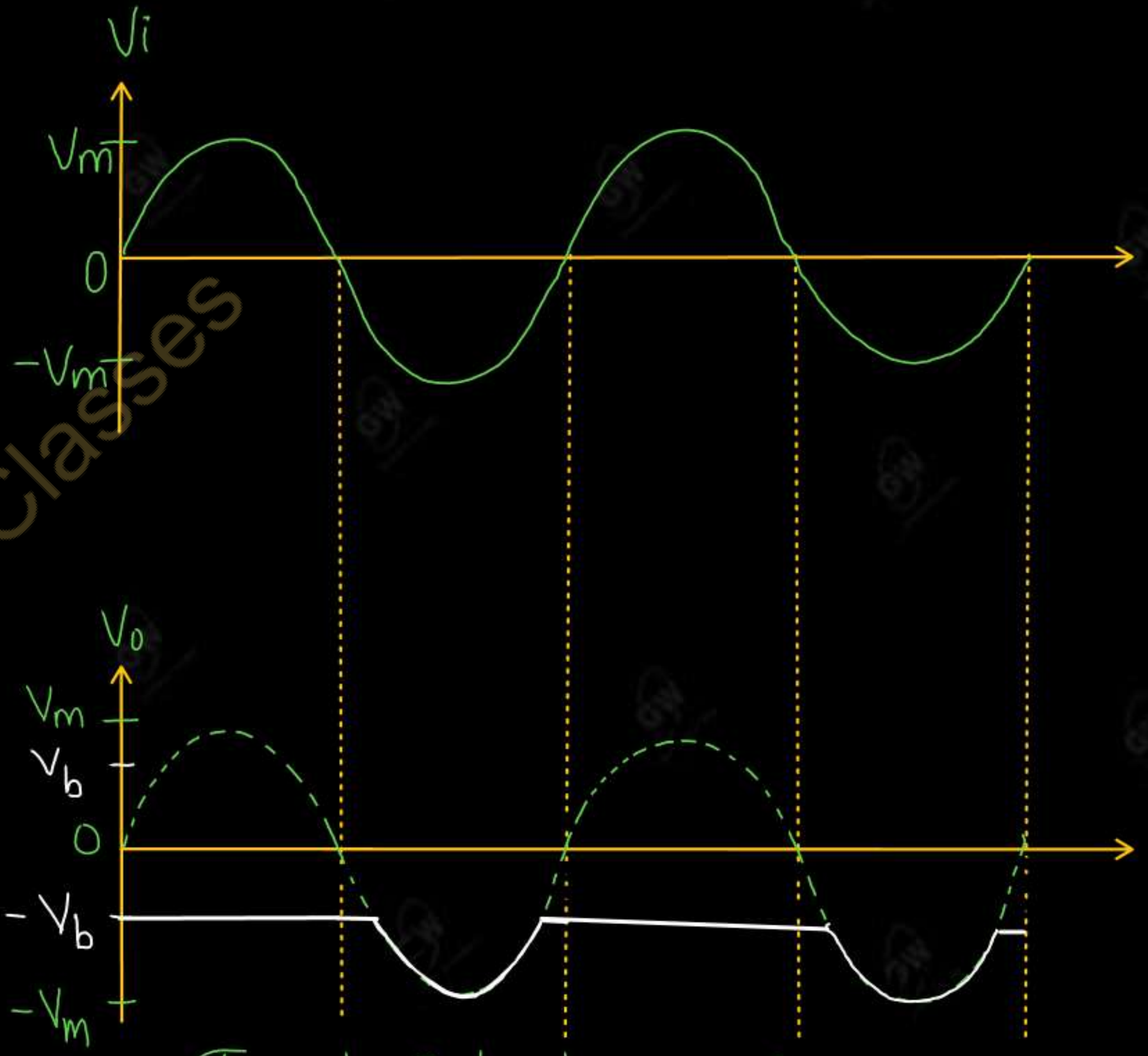
$$V_0 = -V_b$$



Series



Shunt



Input Output Waveform

Gateway Classes

(a) Biased Negative Clipper

Working

(i) Series

- **During positive cycle**

Diode D is in Forward biased

$$V_o = V_i$$

- **During Negative cycle**

If $V_i < V_b$

Diode D is in Forward biased

$$V_o = V_i$$

If $V_i > V_b$

Diode D is in Reversed biased

$$V_o = -V_b$$

(ii) Shunt

- **During positive cycle**

Diode D is in Forward biased

$$V_o = V_i$$

- **During Negative cycle**

If $V_i < V_b$

Diode D is in Reversed biased

$$V_o = V_i$$

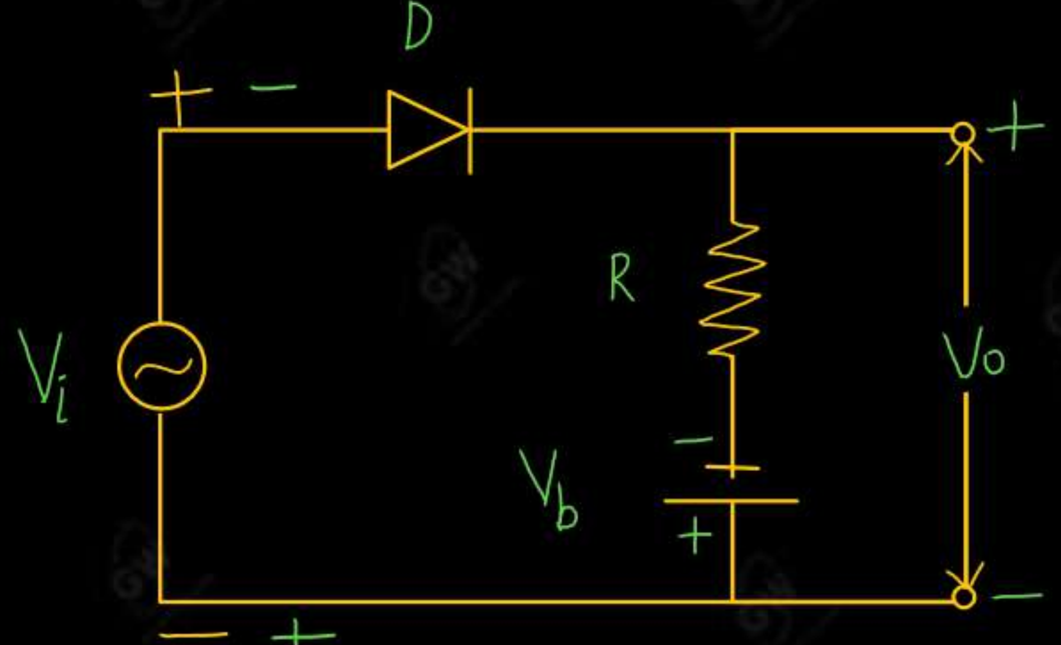
If $V_i > V_b$

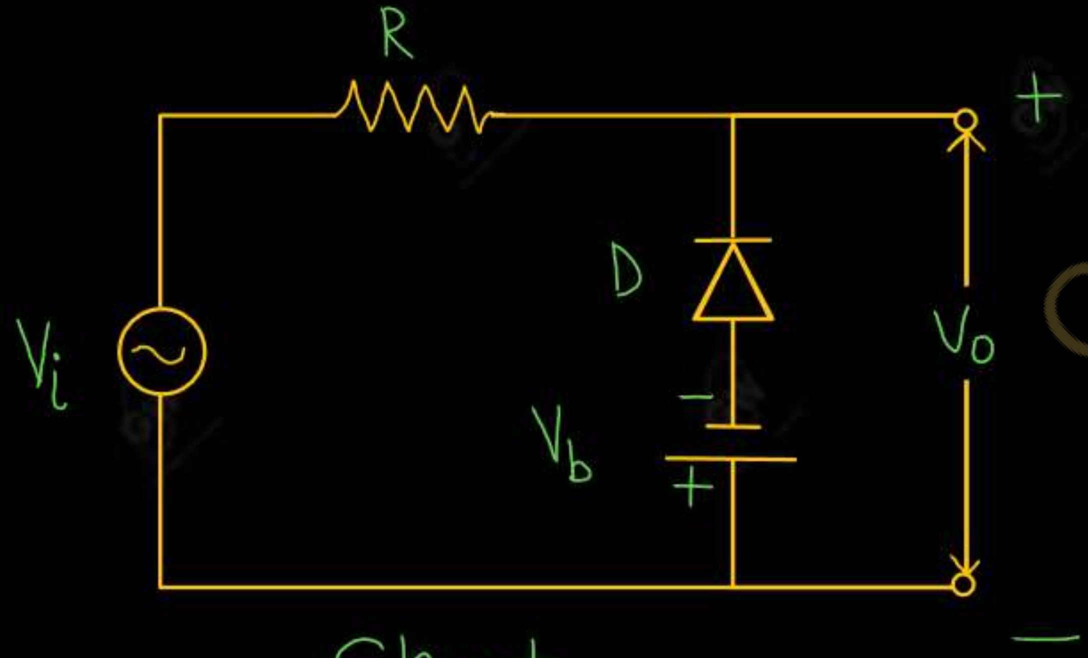
Diode D is in Forward biased

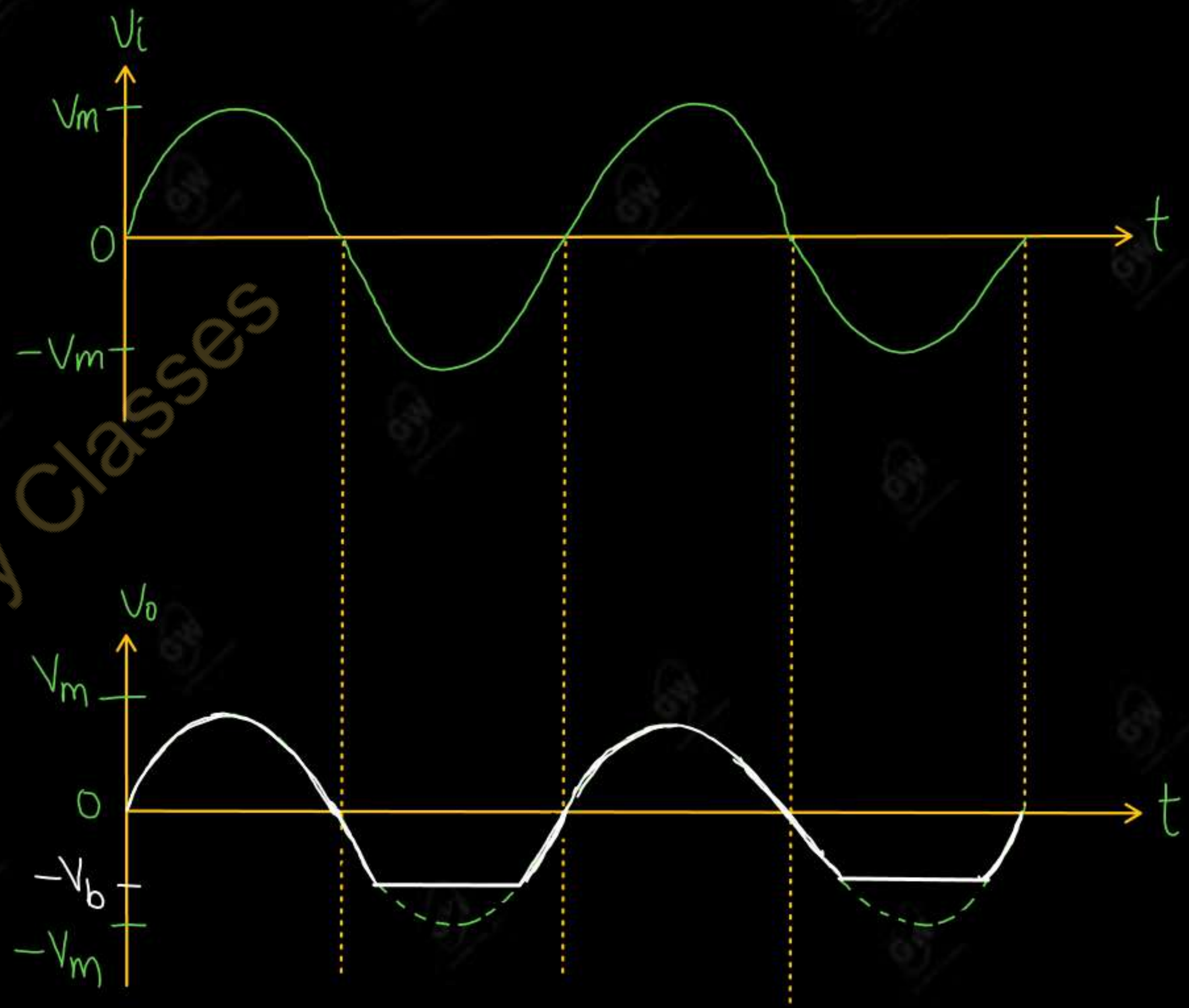
$$V_o = -V_b$$

Gateway Classes

Biased Negative Clipper







Gateway Classes

(a) Biased negative Clipper With reverse polarity of Battery

Working

(i) Series

➢ **During positive cycle**

If $V_i < V_b$

Diode D is in Reversed biased

$$V_o = V_b$$

If $V_i > V_b$

Diode D is in Forward biased

$$V_o = V_i$$

➢ **During Negative cycle**

Diode D is in Reversed biased

$$V_o = V_b$$

(ii) Shunt

➢ **During positive cycle**

If $V_i < V_b$

Diode D is in Forward biased

$$V_o = V_b$$

If $V_i > V_b$

Diode D is in Reversed biased

$$V_o = V_i$$

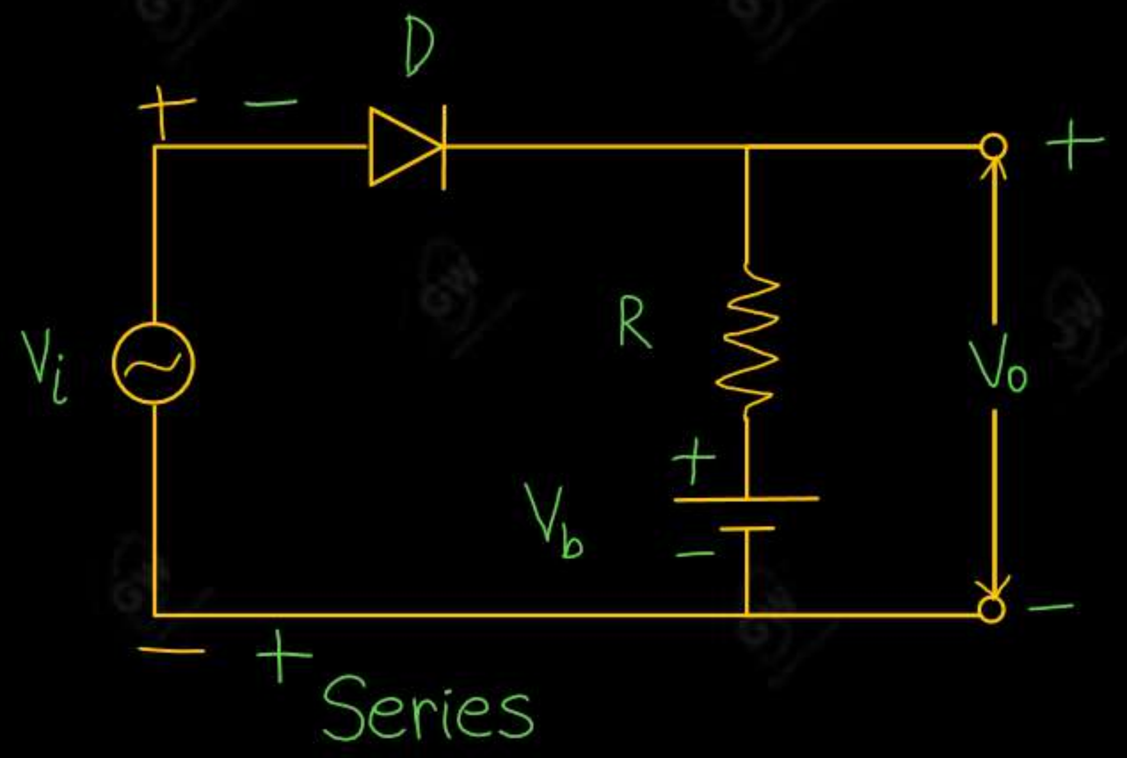
➢ **During Negative cycle**

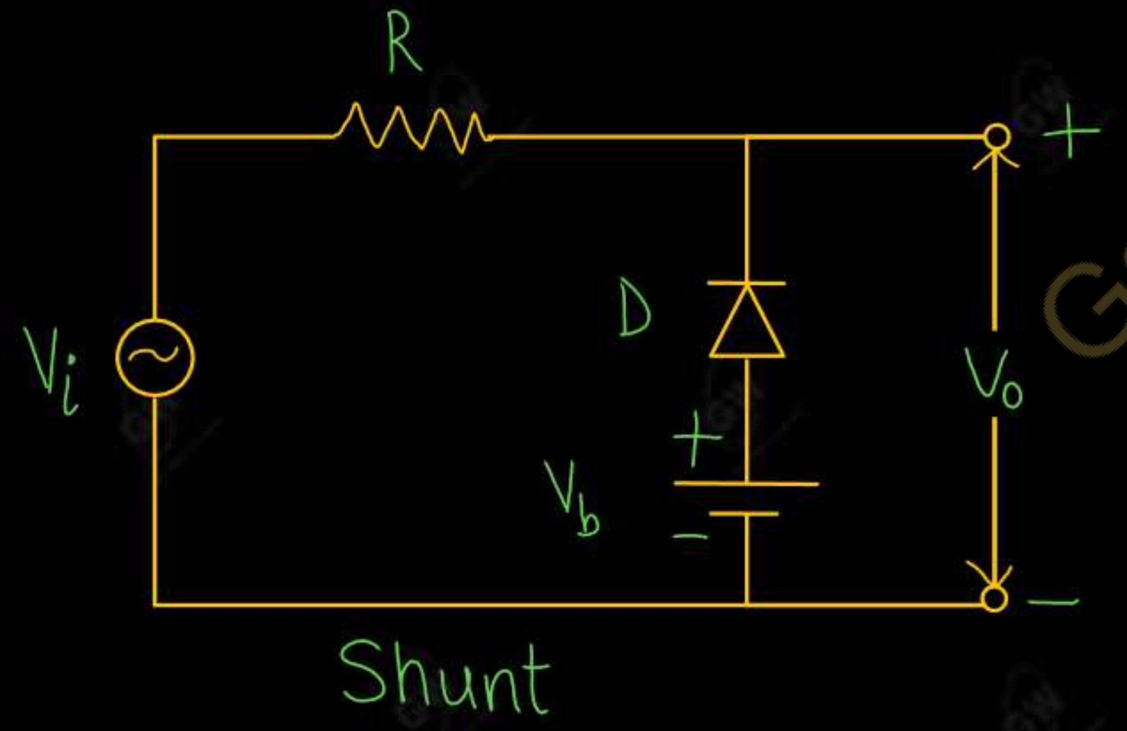
Diode D is in Forward biased

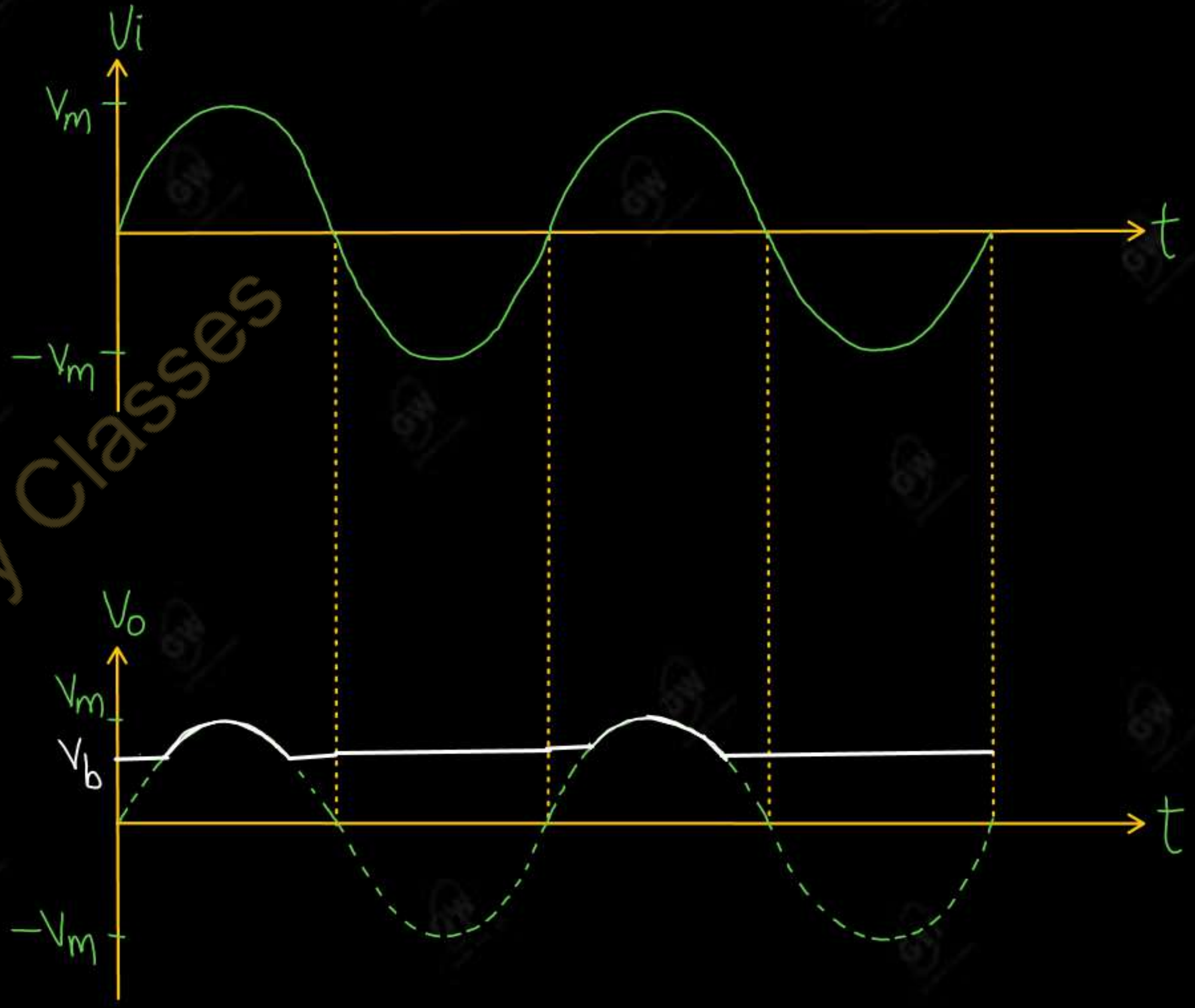
$$V_o = V_b$$

Gateway Classes

Biased negative Clipper With reverse polarity of Battery







Gateway Classes

(a) Combinational Clipper

- Combinational clipper is a combination of a positive biased clipper and a negative biased clipper
- Remove some portion of Positive Half cycle and some portion of Negative Half cycle

Working

- **During positive cycle**

Diode D_2 is in Reversed biased

If $V_i < V_b$

Diode D_1 is also in Reversed biased

$$V_o = V_i$$

If $V_i > V_b$

Diode D_1 is in forward biased and

$$V_o = V_b$$

- **During negative cycle**

Diode D_1 is in Reversed biased

If $V_i < V_b$

D_2 is also in Reversed biased

$$V_o = V_i$$

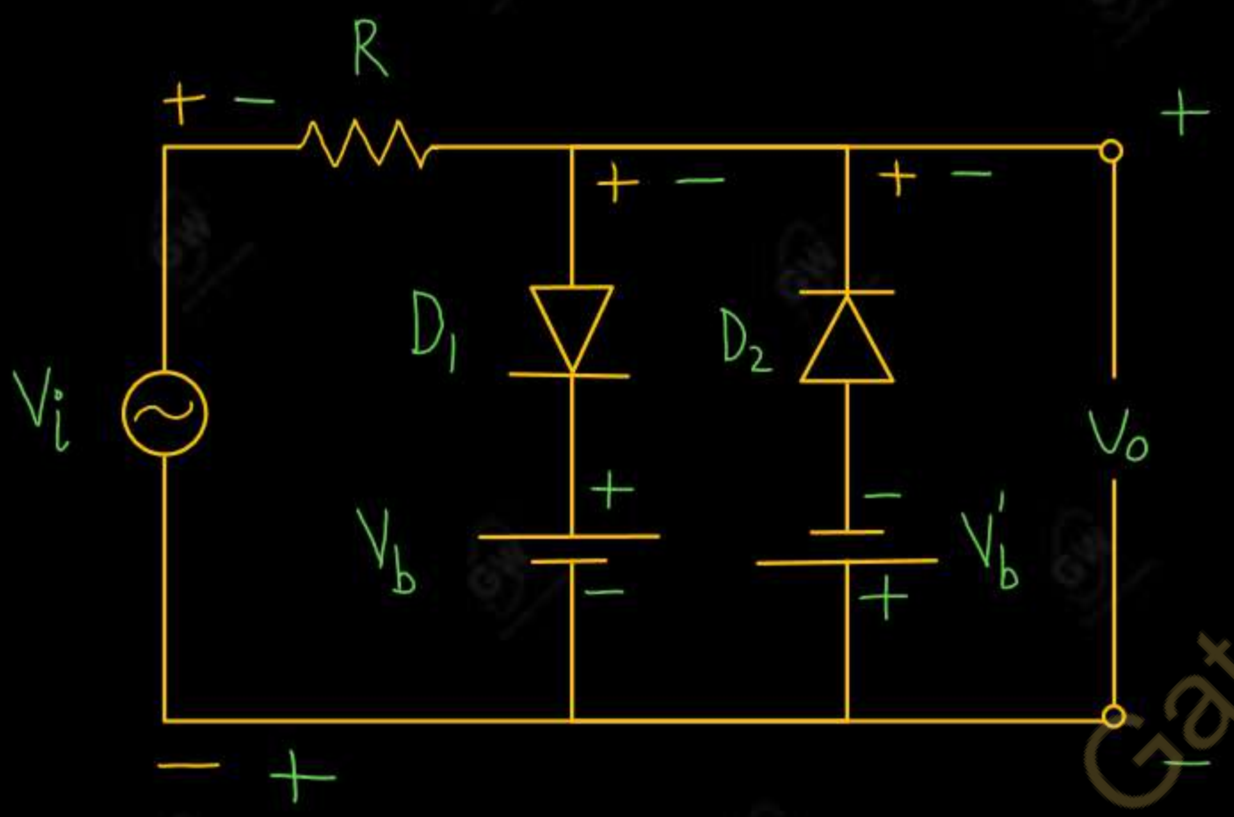
If $V_i > V_b$

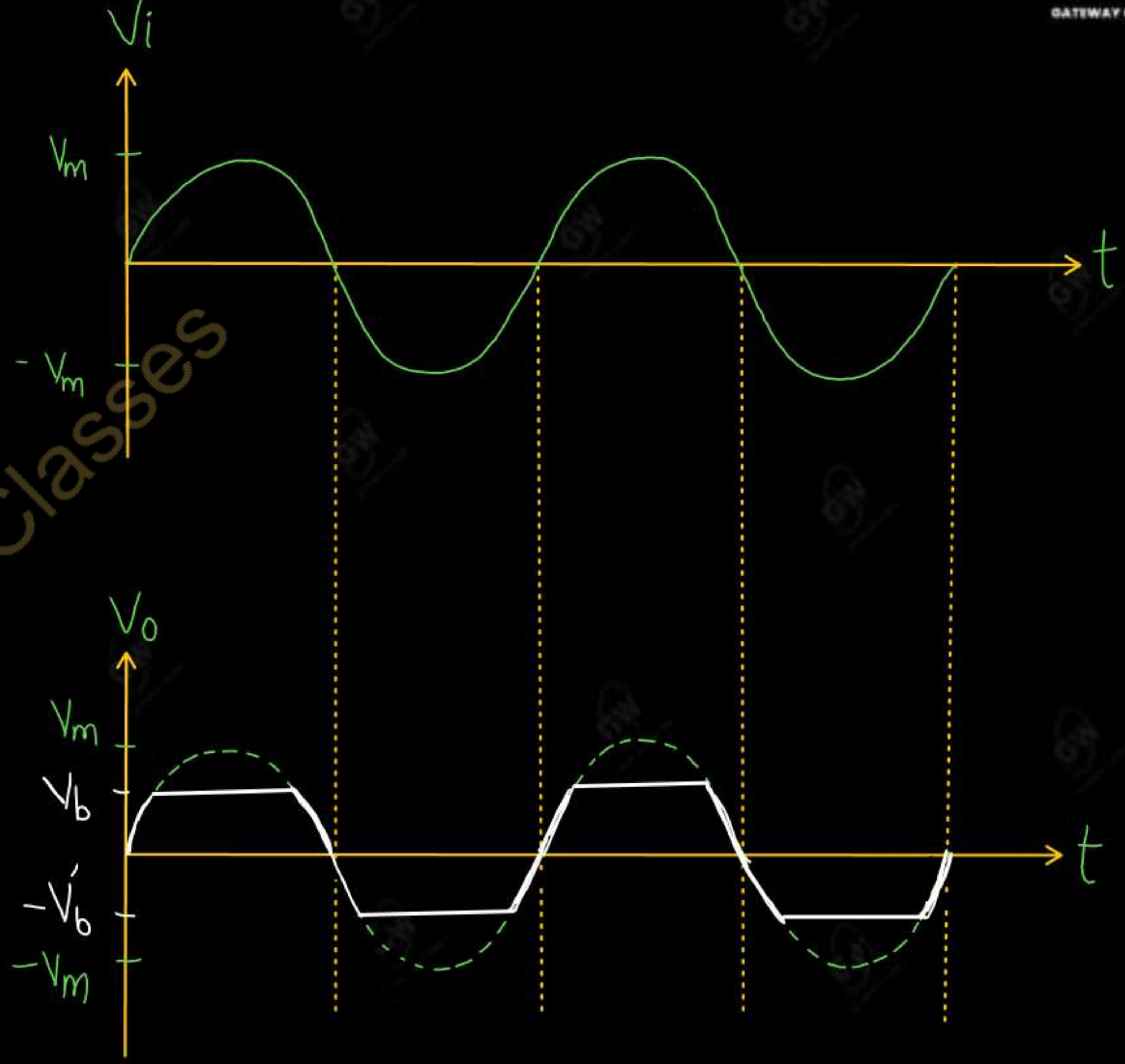
Diode D_2 is in Forward biased

$$V_o = -V_b$$

Gateway Classes

Combinational Clipper





Gateway Classes

UNIT - 1 : DPP - 7

Understand the working of all the Clipper circuits for numerical problems

Gateway Classes

AKTU

B.Tech : I-Year

GATEWAY CLASSES

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 8

Gateway Classes

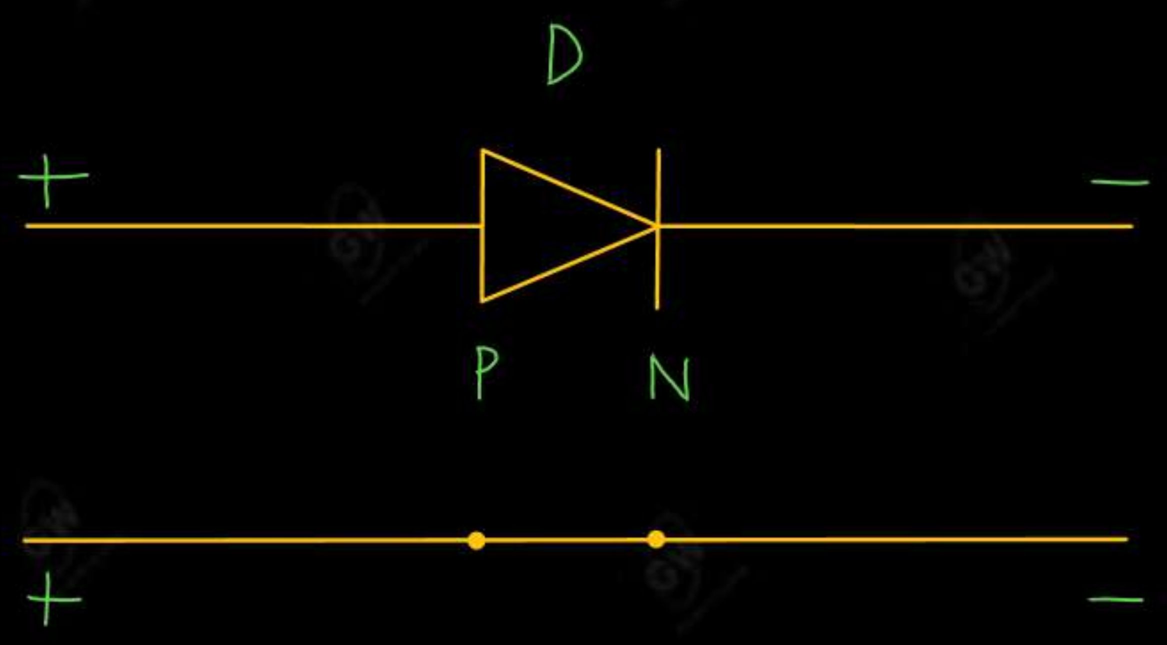
Today's Target

- Numericals on Clipper Circuits
- DPP
- PYQs

By Gulshan sir

Properties of ideal Diode

(1)

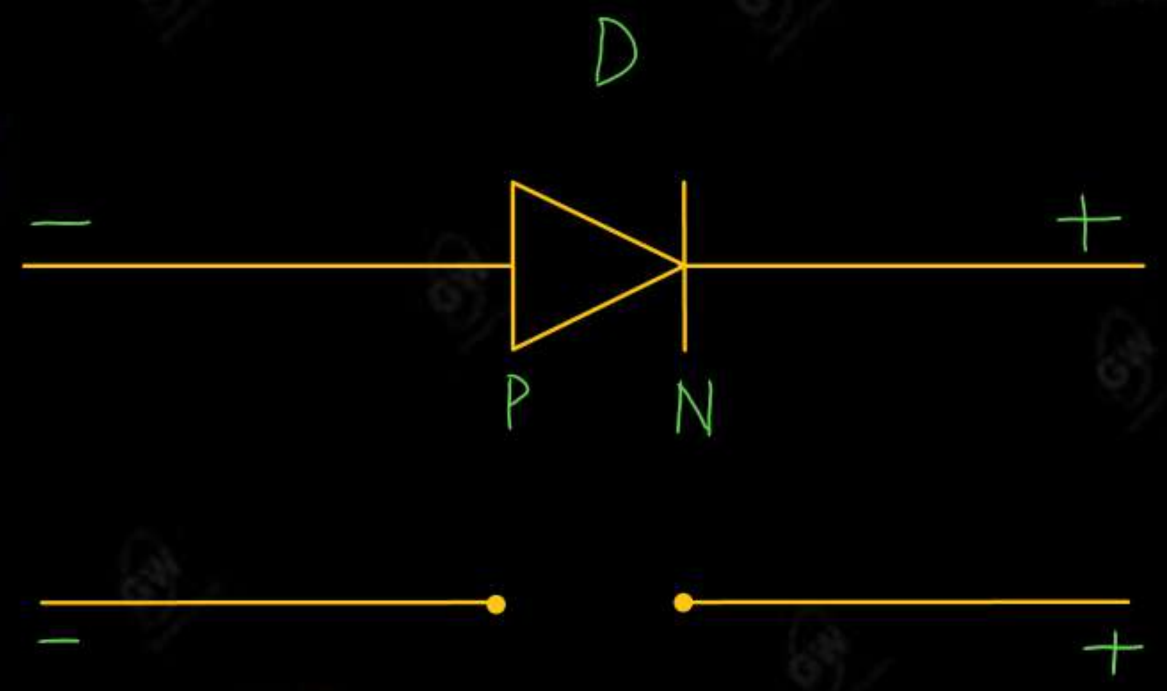


(i) FB / ON / SC

(ii) Current flow

(iii) No Voltage drop

(2)



(i) RB / OFF / OC

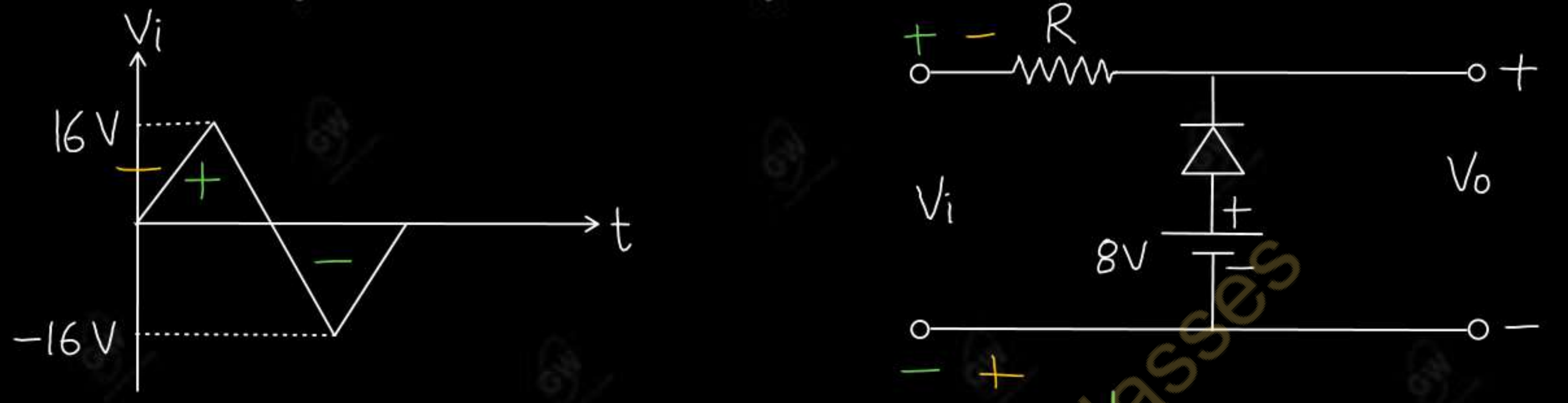
(ii) No current flow

(iii) Voltage drop

Gateway Classes

Q.1 What are the advantages of clipper circuits? For the given input waveform to the given circuit, what is the peak value of the output waveform

AKTU 2025



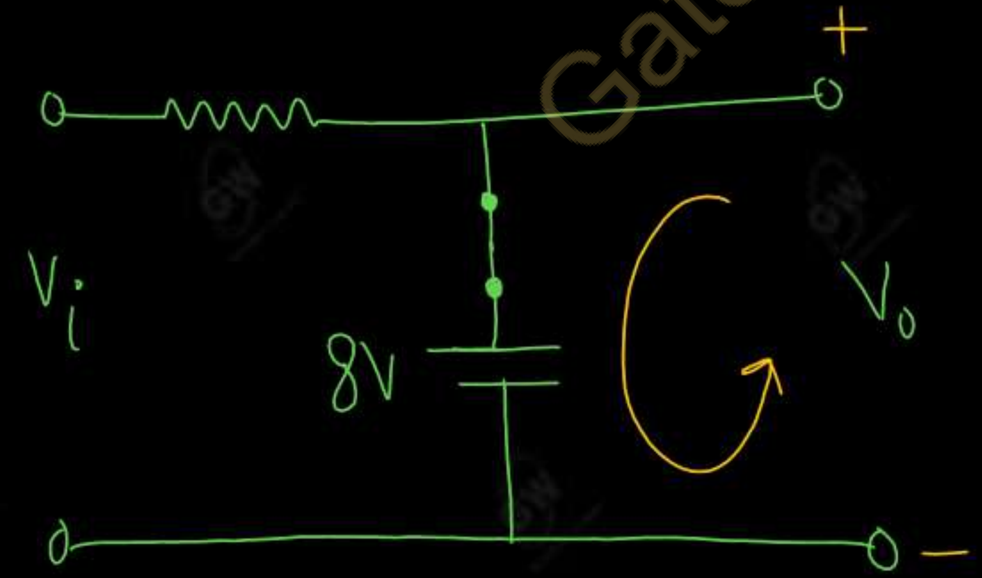
SOLUTION

During positive Half cycle

If $V_i < 8V$

Diode is ON

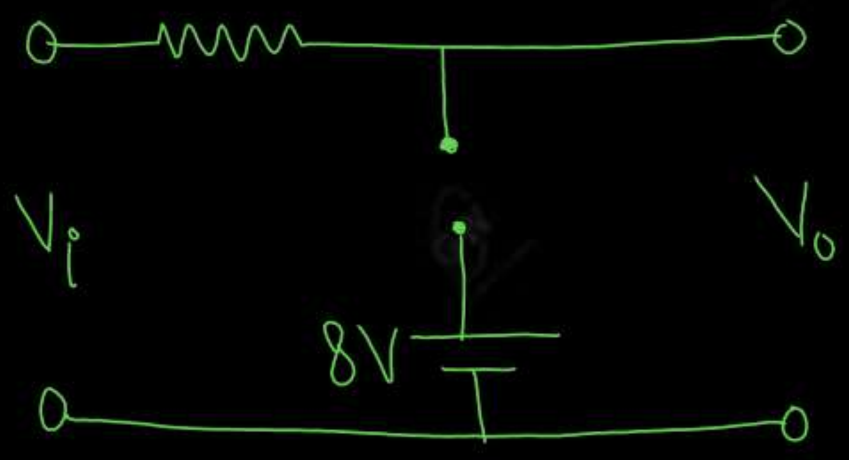
$V_o = 8V$



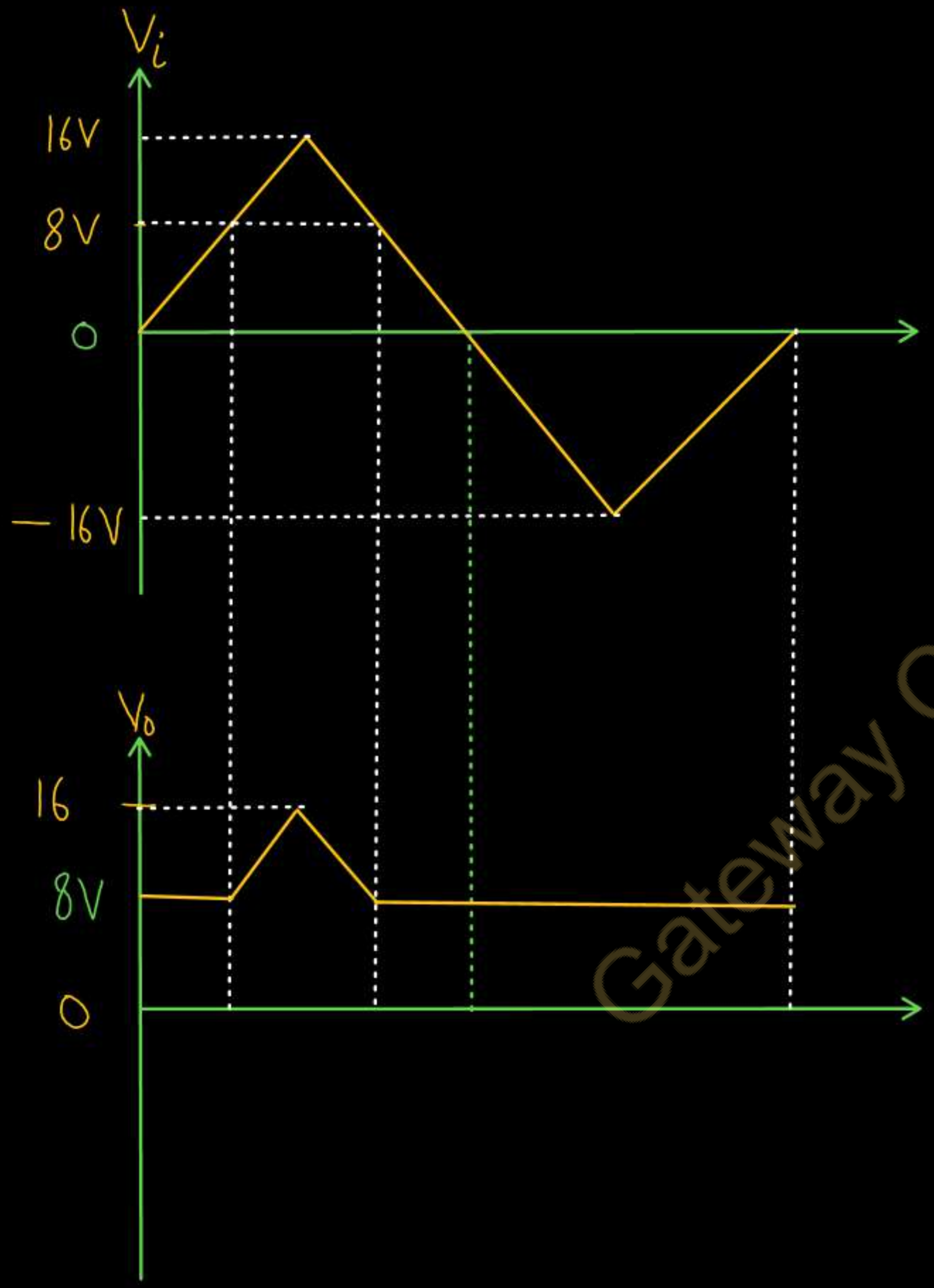
If $V_i > 8V$

Diode is OFF

$V_o = V_i$

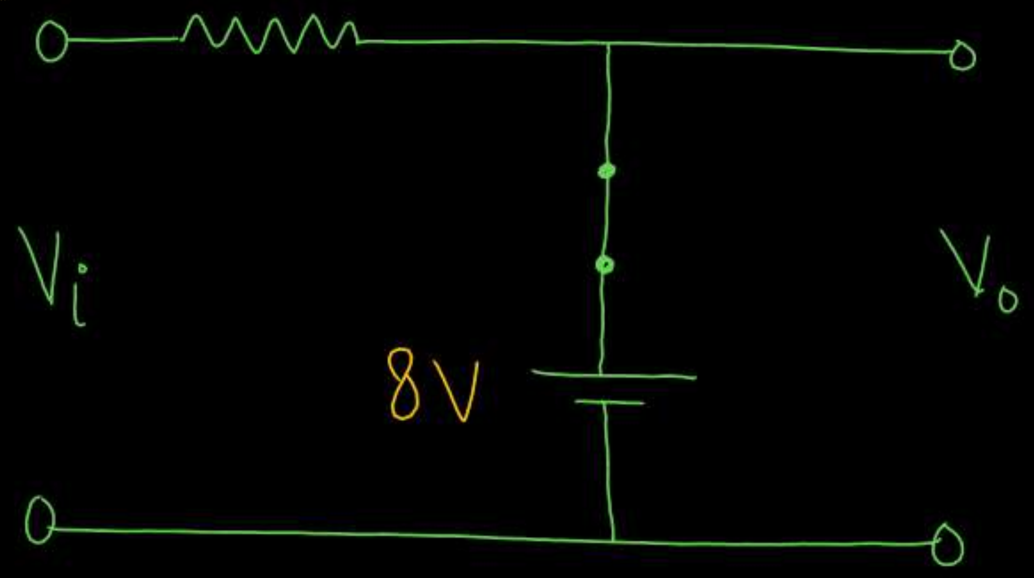


Gateway Classes



During Negative Half cycle

Diode is ON

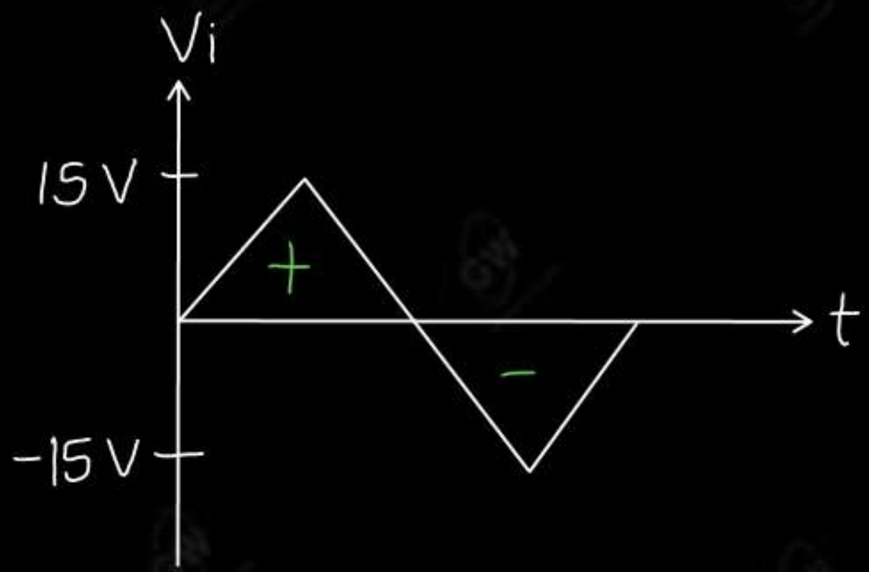


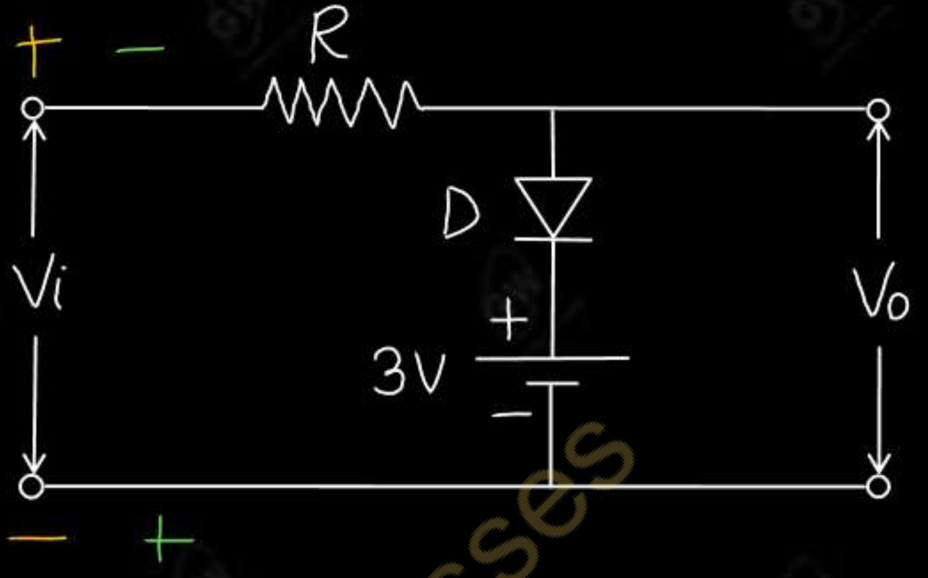
$V_o = 8V$

$\boxed{\text{Peak value} = 16\,V}$

Gateway Classes

GW **Q.2** Sketch the output waveform for the circuit and given input waveform





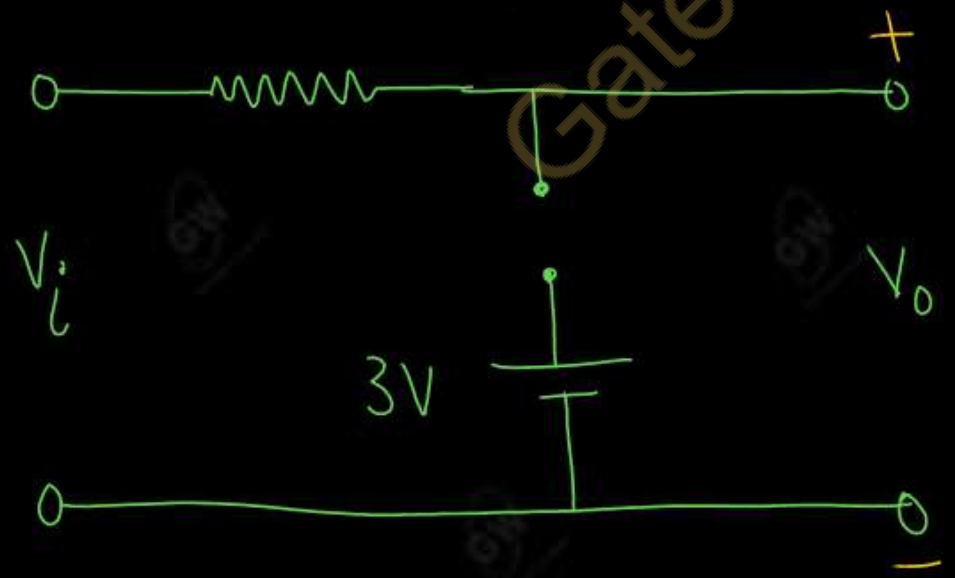
SOLUTION

During positive half cycle

If $V_i < 3V$

Diode is OFF

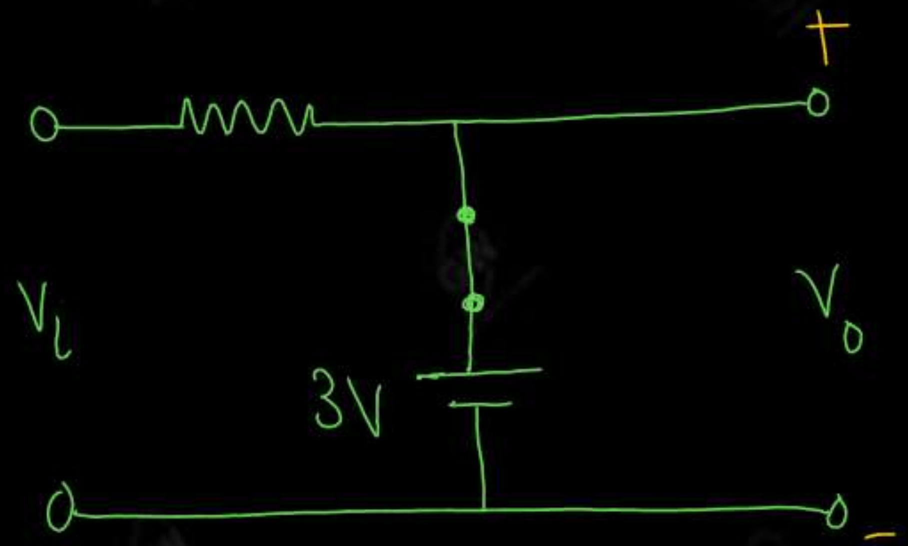
$V_0 = V_i$



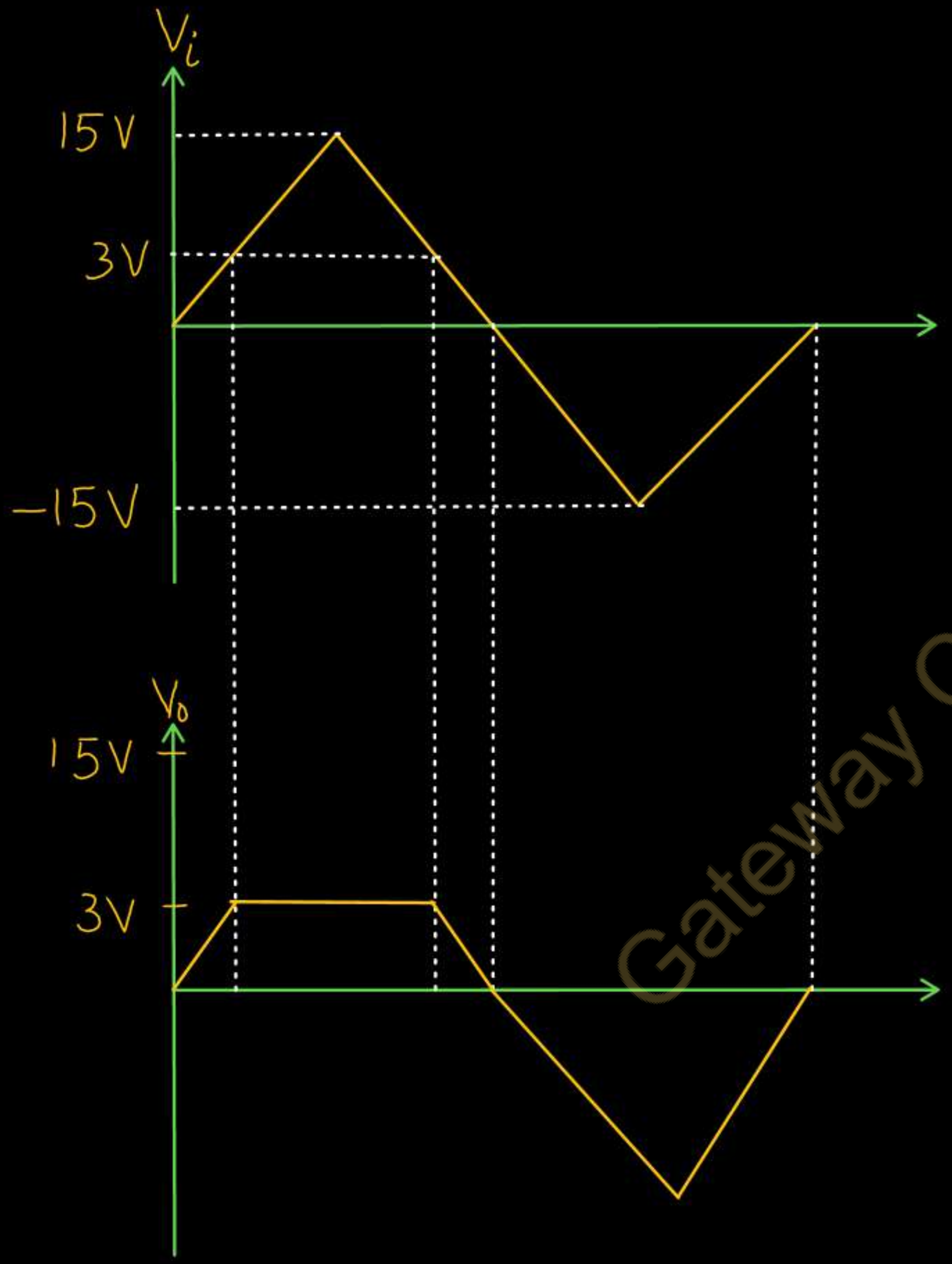
If $V_i > 3V$

Diode is ON

$V_0 = 3V$



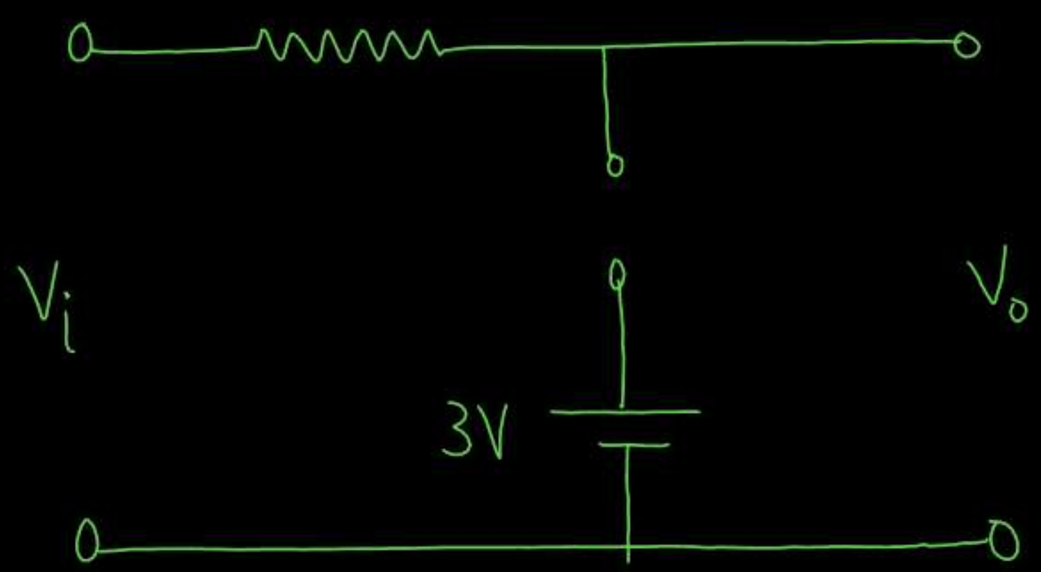
Gateway Classes



During Negative Half cycle

Diode is OFF

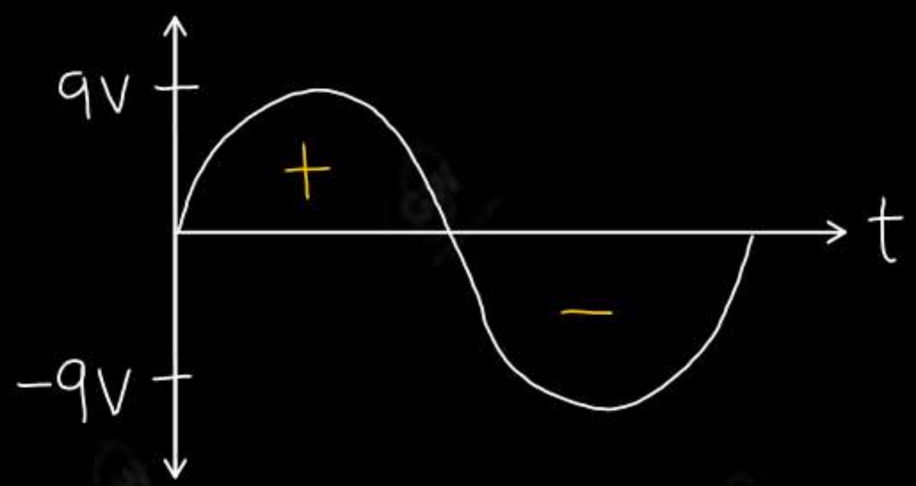
$$V_o = V_i$$

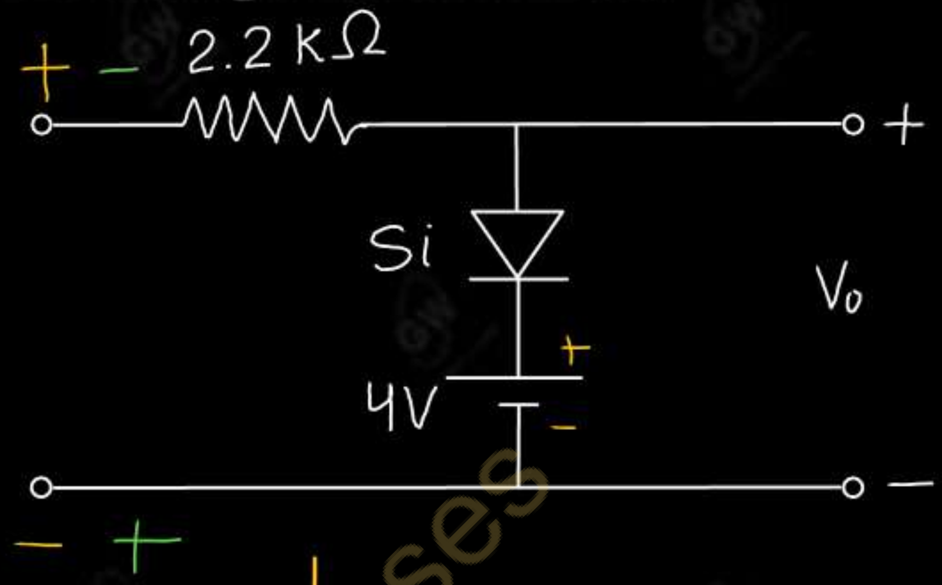


Gateway Classes

Q.3 Determine and draw the output voltage of the given network

AKTU 2022





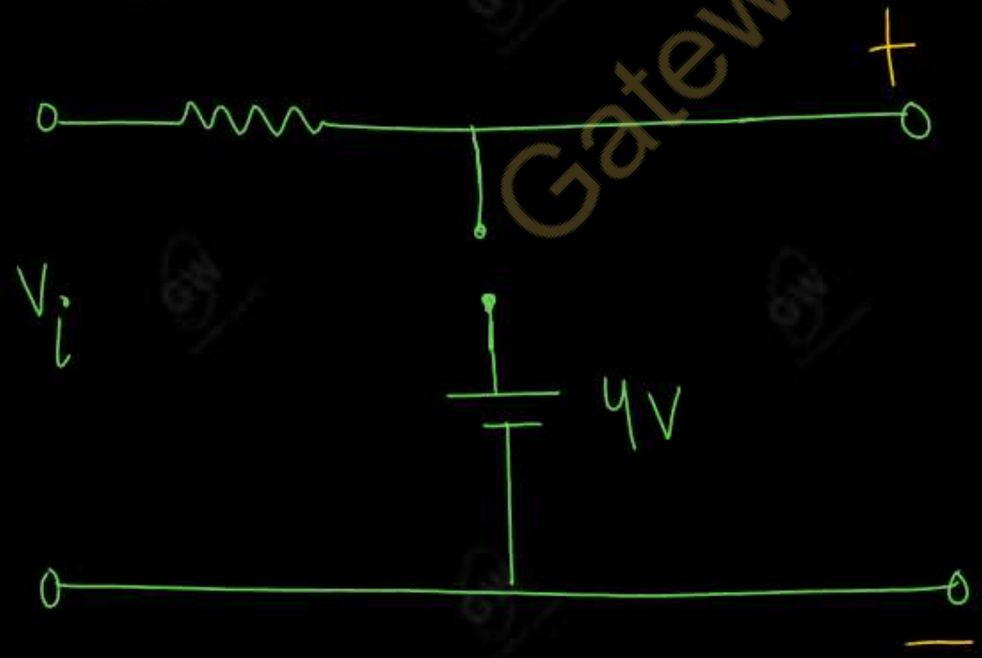
SOLUTION

During Positive half cycle

If $V_i < 4.7V$

Diode is OFF

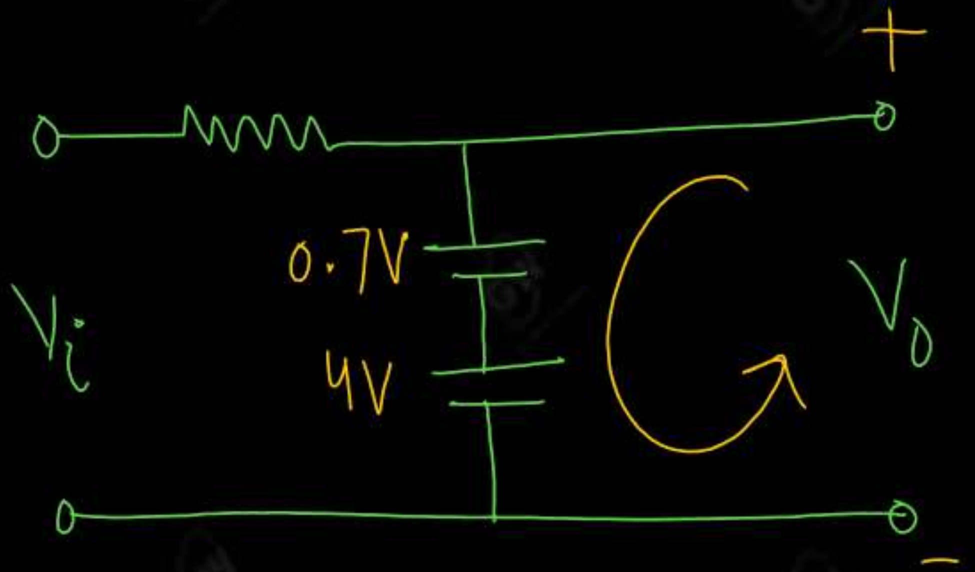
$V_o = V_i$



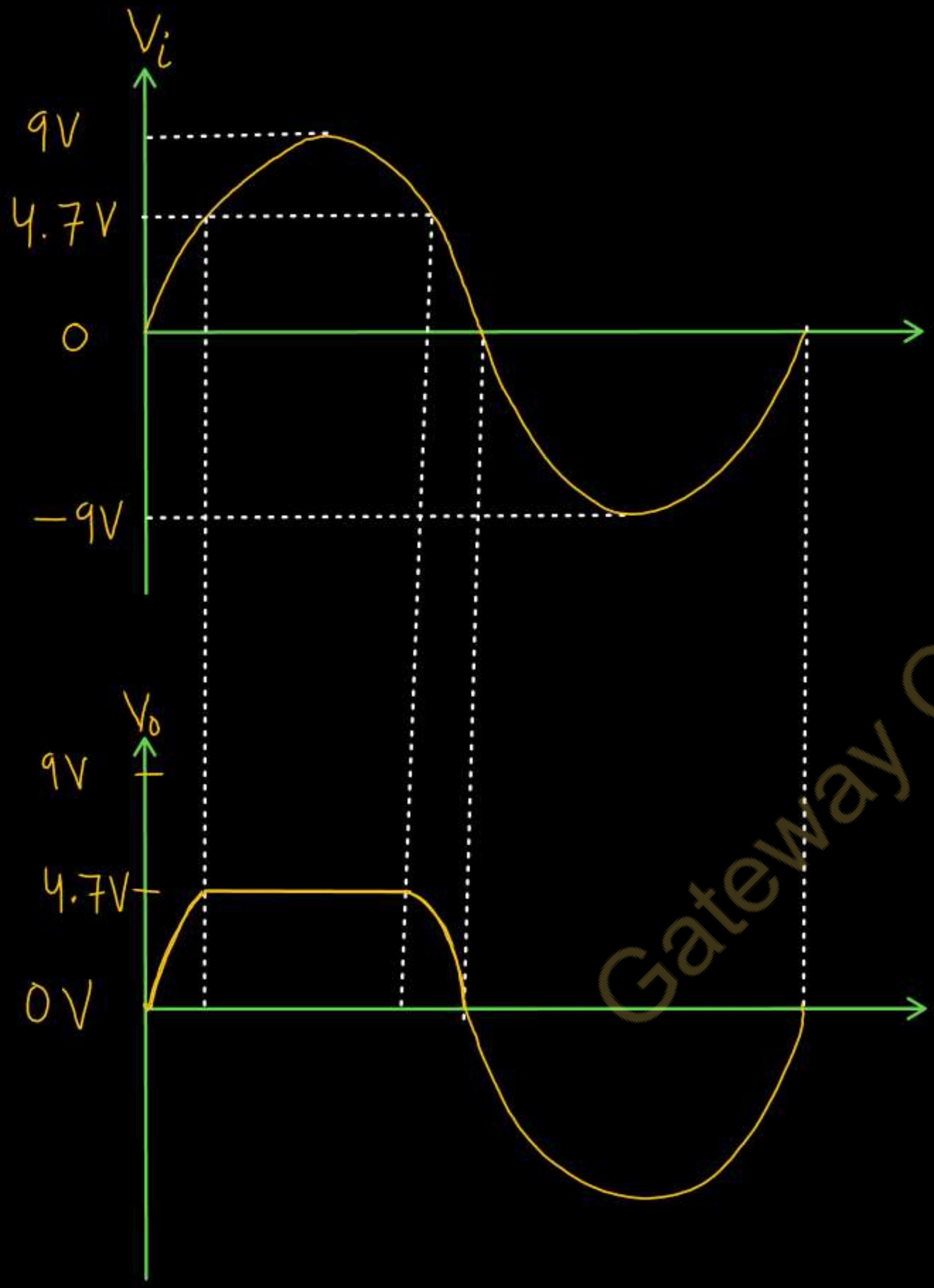
If $V_i > 4.7V$

Diode is ON

$V_o = 4.7V$



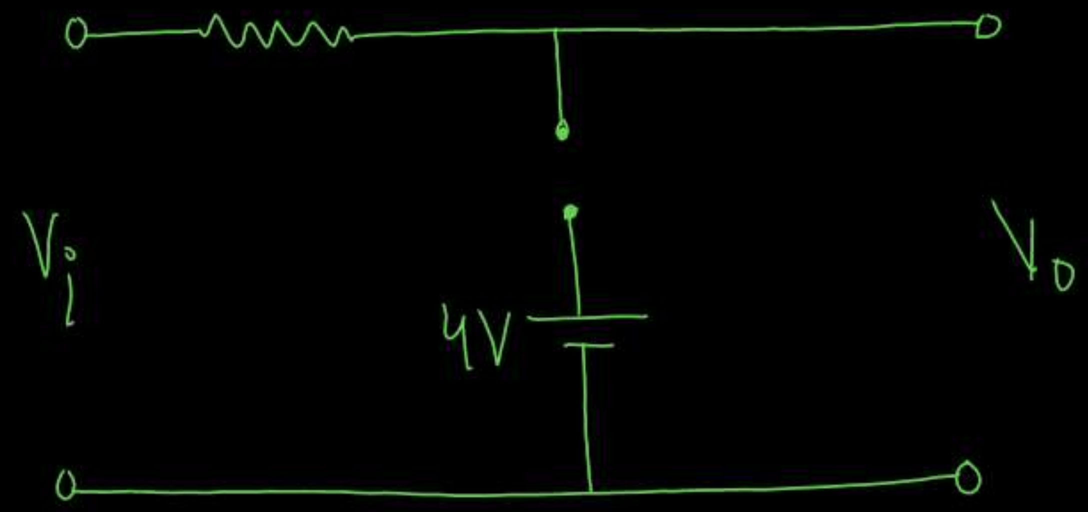
Gateway Classes



During Negative Half cycle

Diode is OFF

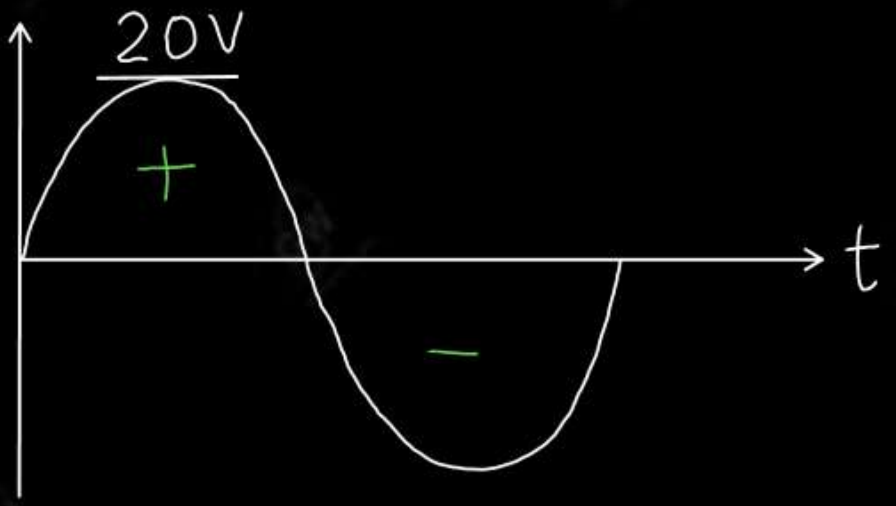
$$V_o = V_i$$

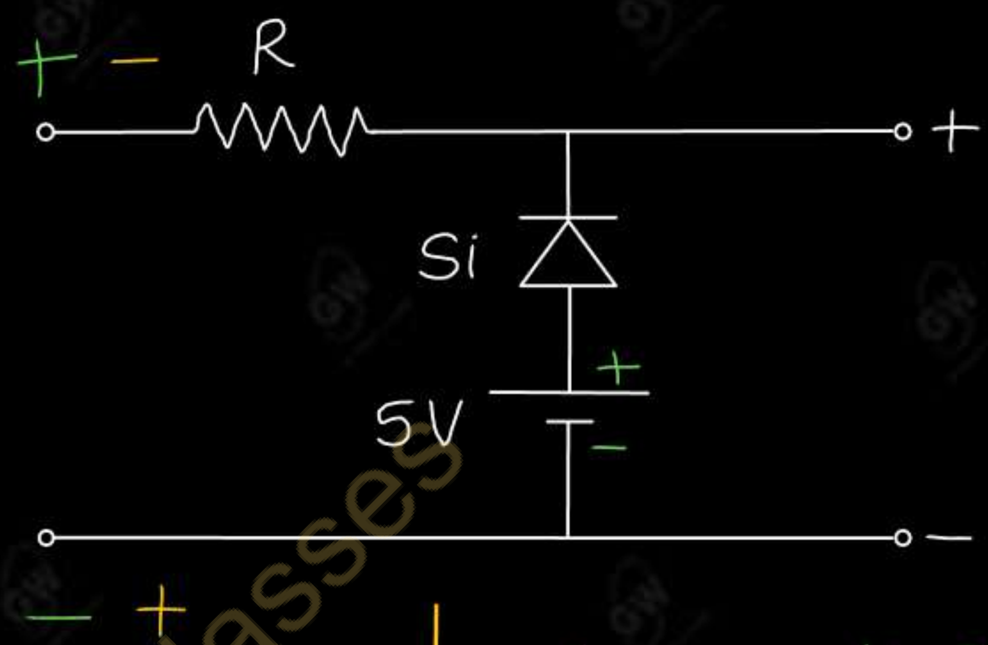


Gateway Classes

Q.4 Determine and draw the output voltage of the given network

AKTU 2024





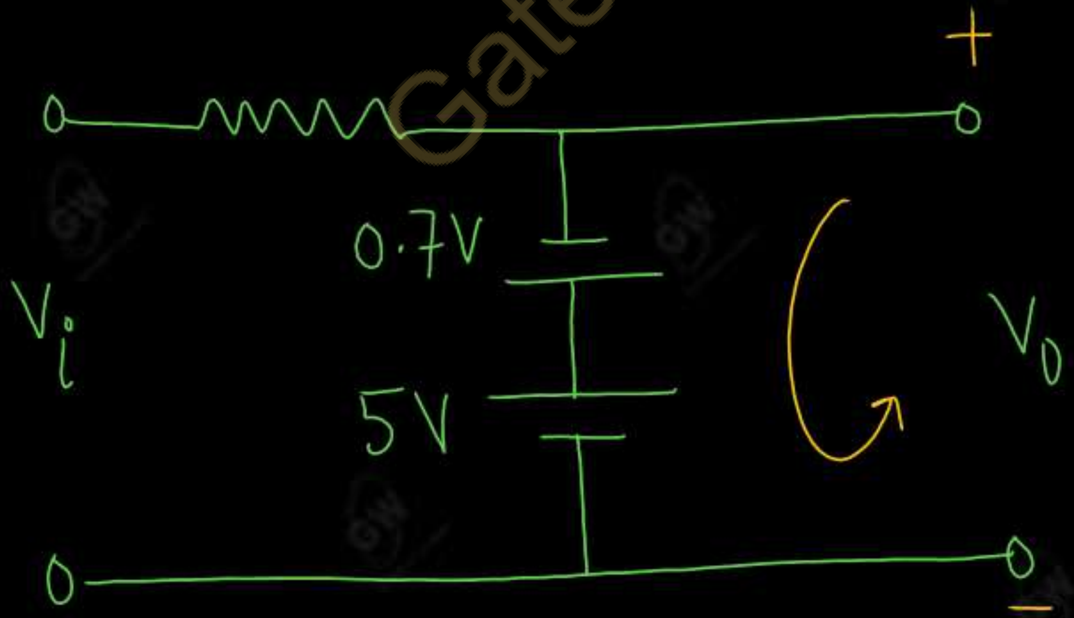
SOLUTION

During positive Half cycle

If $V_i < 4.3V$

Diode is ON

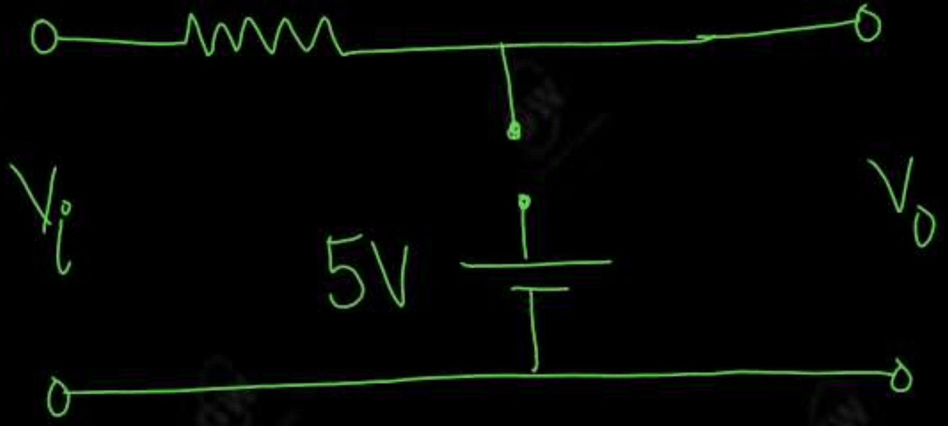
$V_0 = 4.3V$



If $V_i > 4.3V$

Diode is OFF

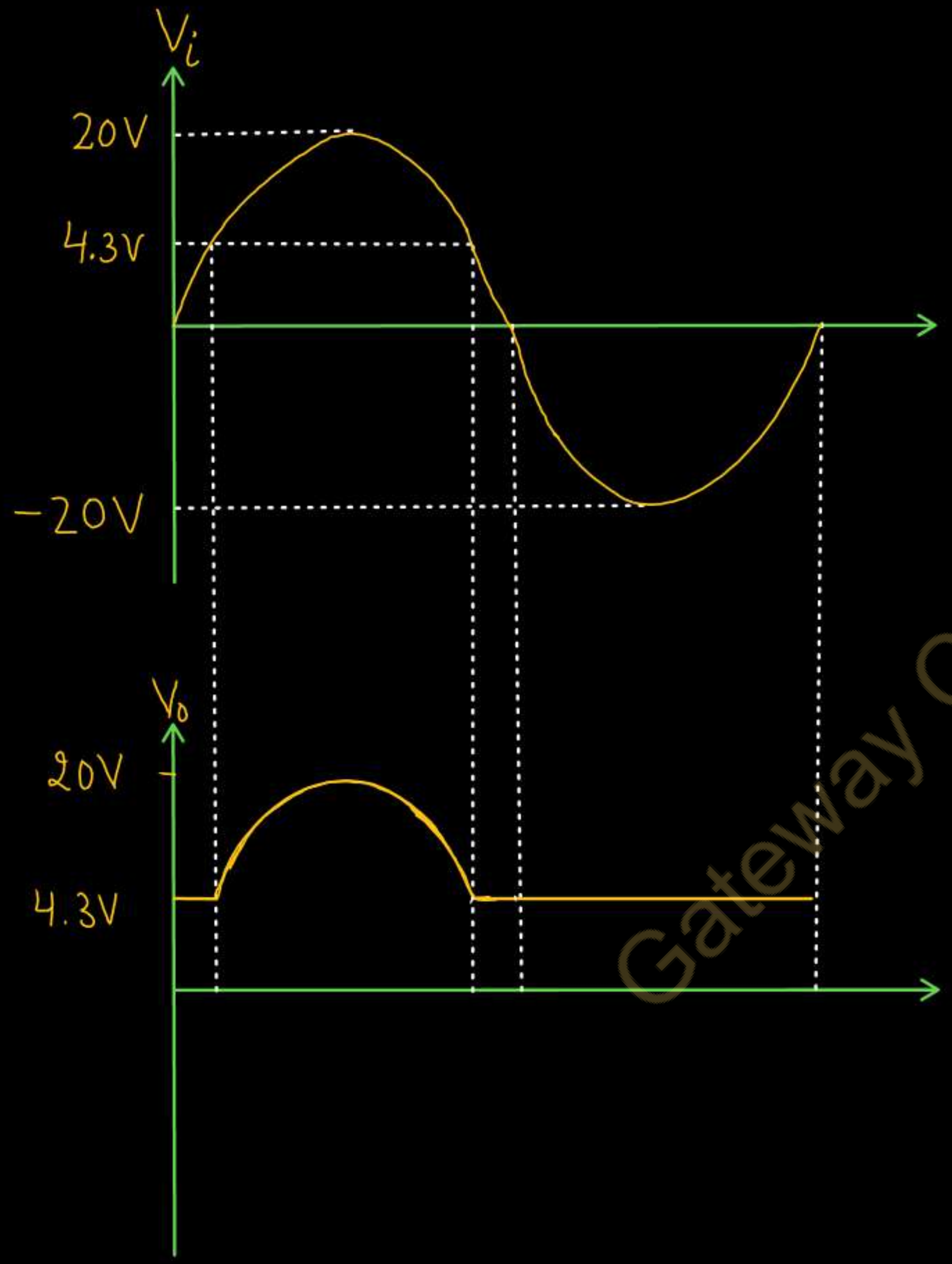
$V_0 = V_i$

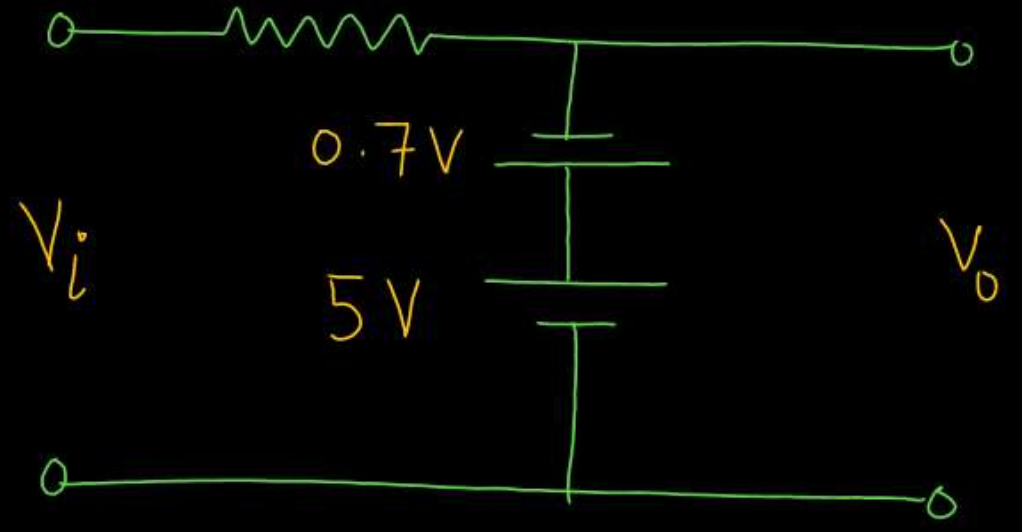


Gateway Classes

During Negative half cycle

Diode is ON

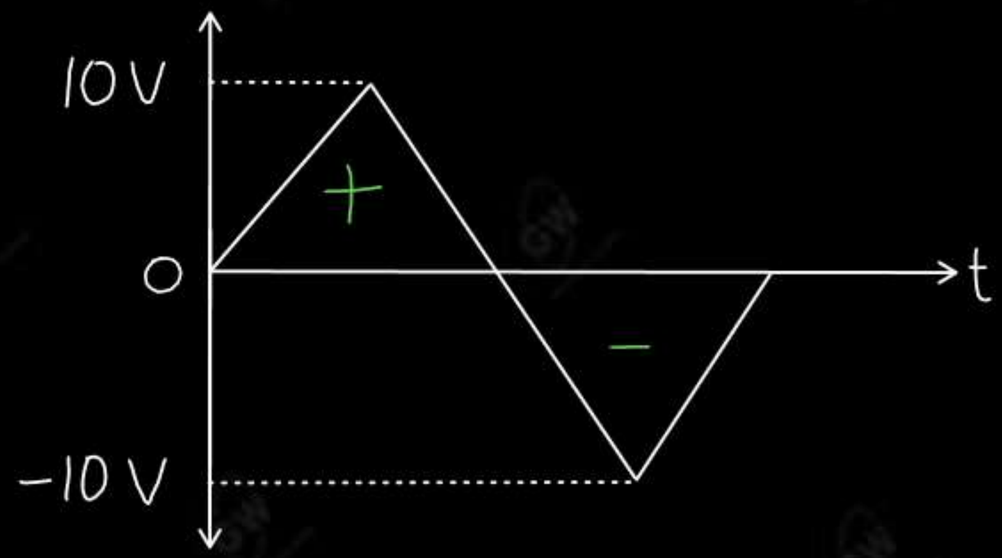


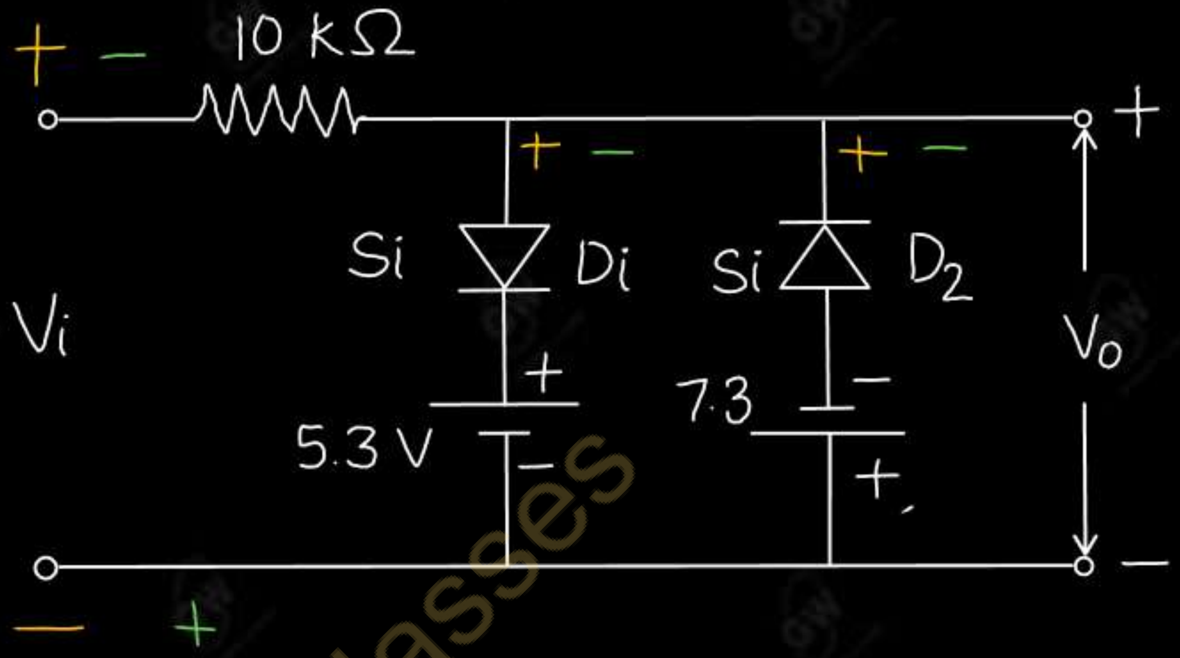


$V_o = 4.3V$

Gateway Classes

Q. 5 Draw the output waveform of the given circuit and input waveform

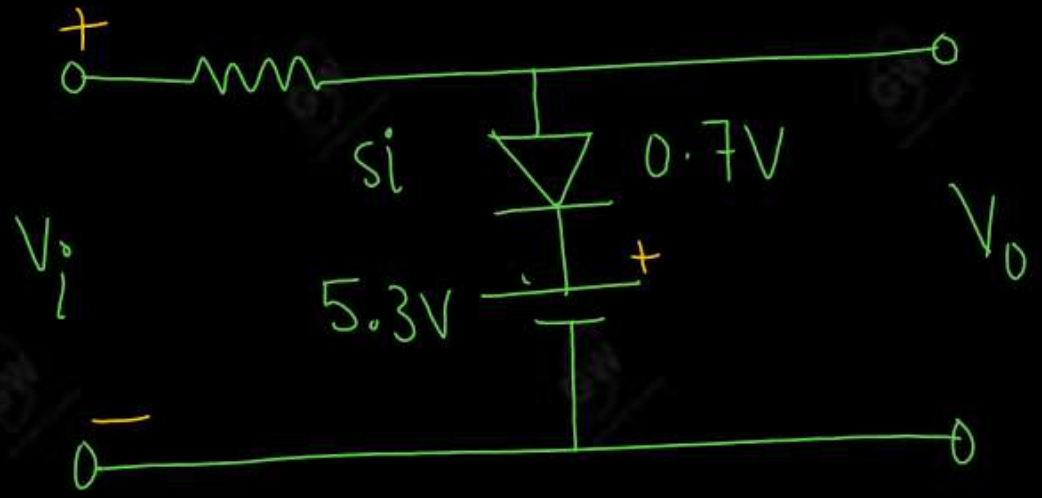




SOLUTION

During positive Half cycle

Diode D_2 is OFF



If $V_i < 6V$

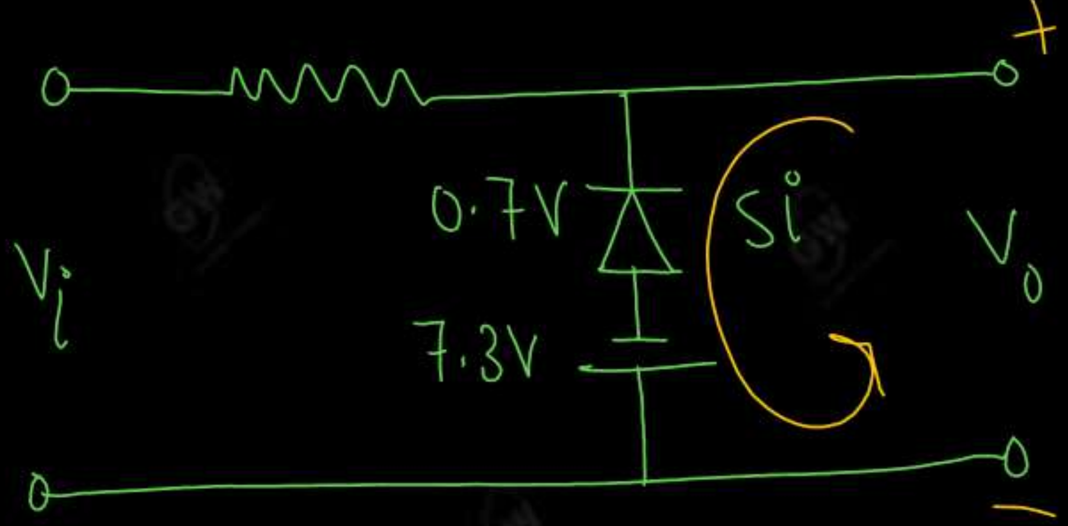
$V_o = V_i$

If $V_i > 6V$

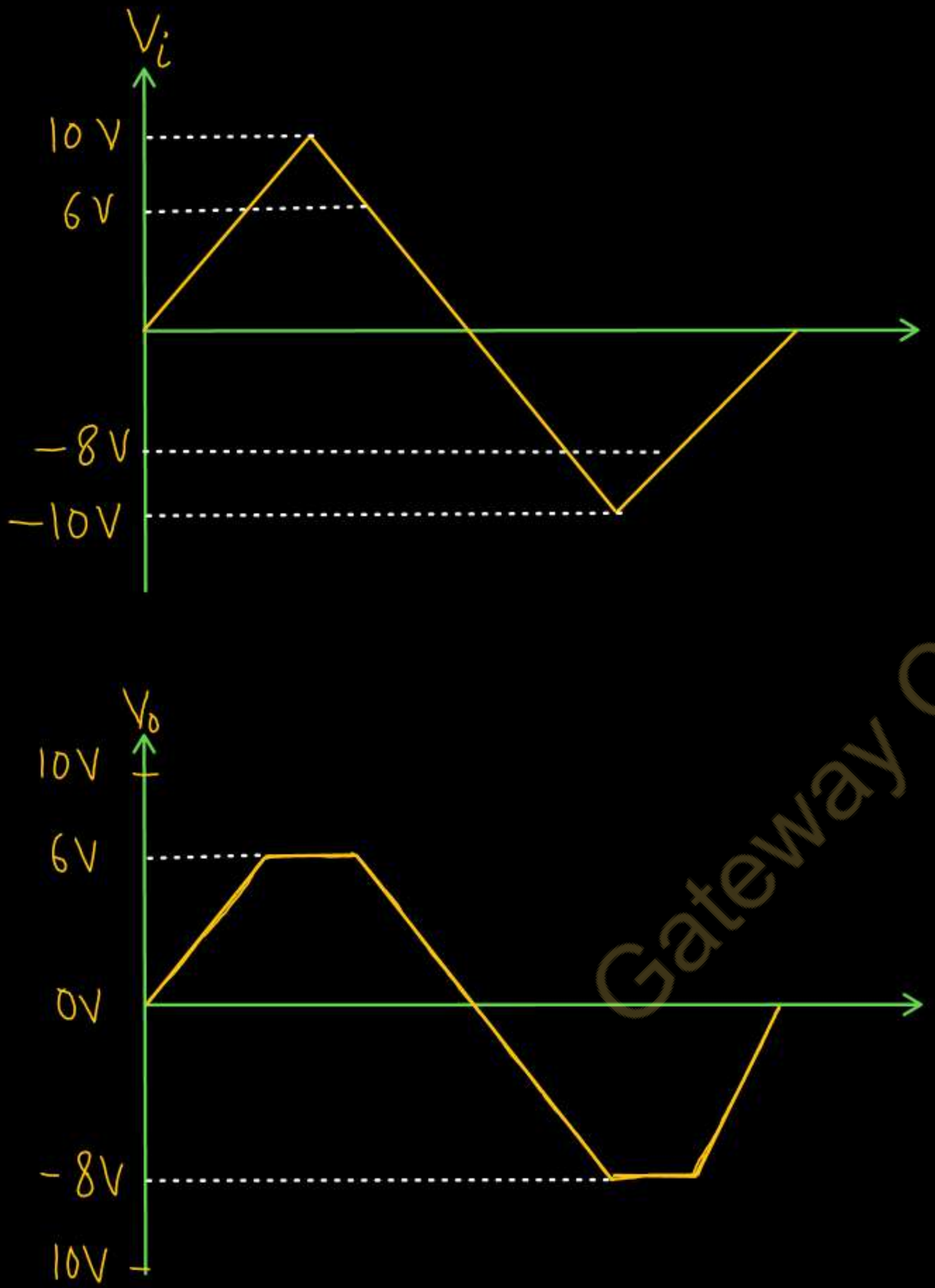
$V_o = 6V$

During Negative Half cycle

Diode D_1 is OFF



Gateway Classes



Gateway Classes

If $V_i < 8V$

Diode is OFF

$V_o = V_i$

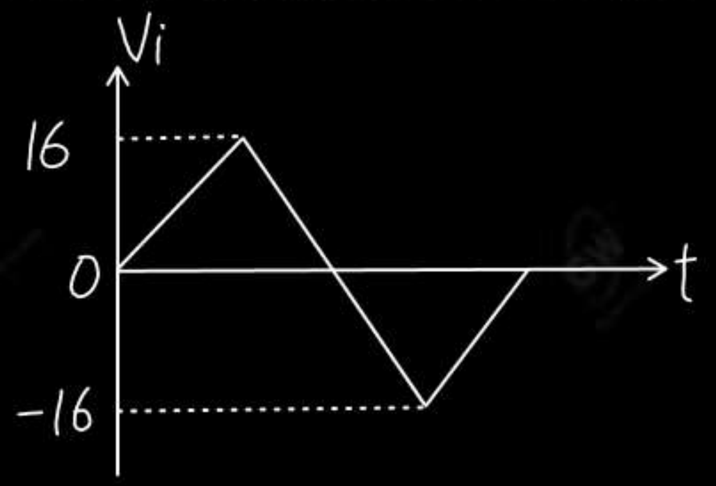
If $V_i > 8V$

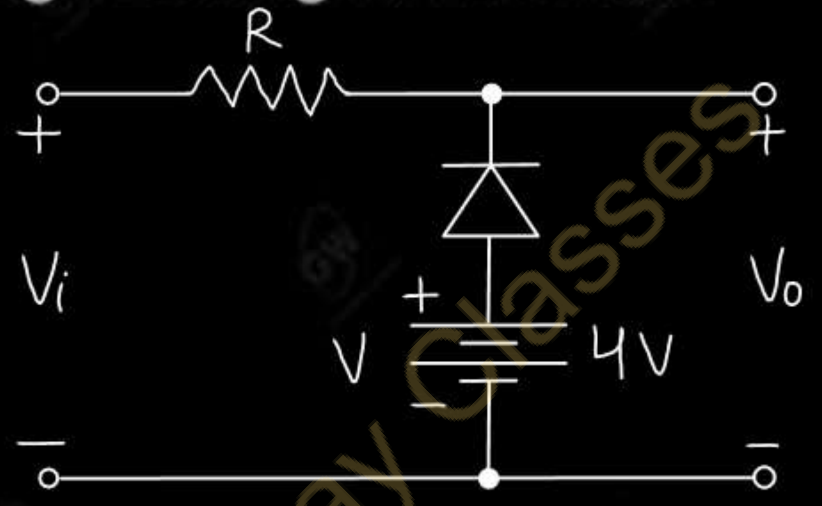
Diode is ON

$V_o = -8V$

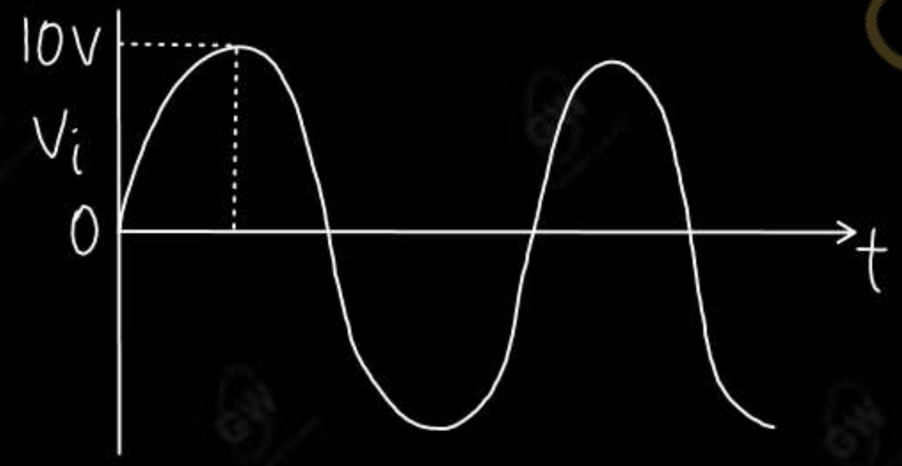
UNIT - 1 : DPP - 8

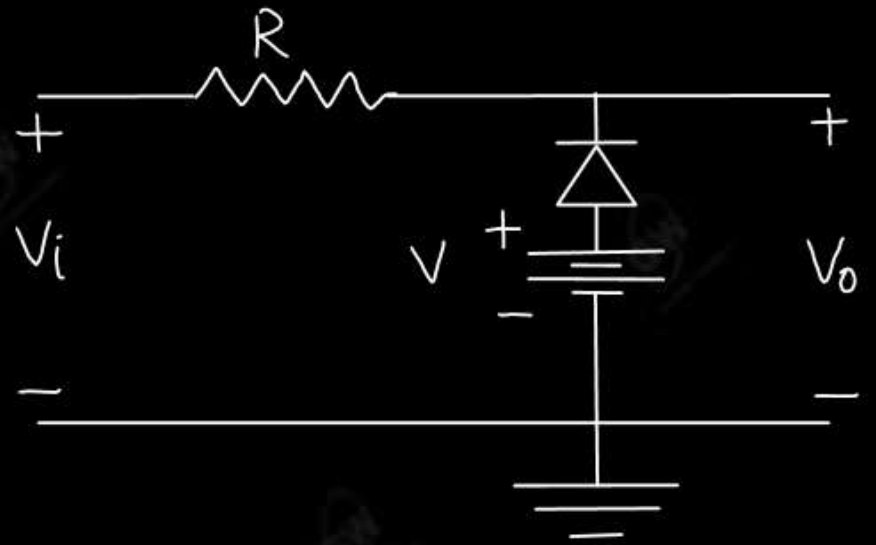
Q.1 Determine and draw the output voltage of the given network — AKTU 2023





Q.2 What do you mean by clipper? Draw the output waveform of the given circuit — AKTU 2022

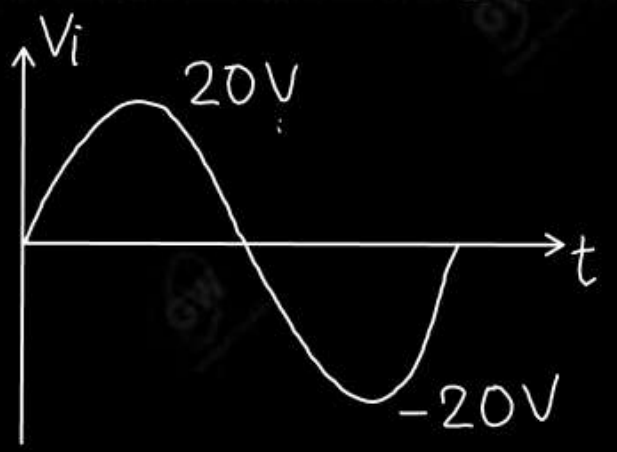


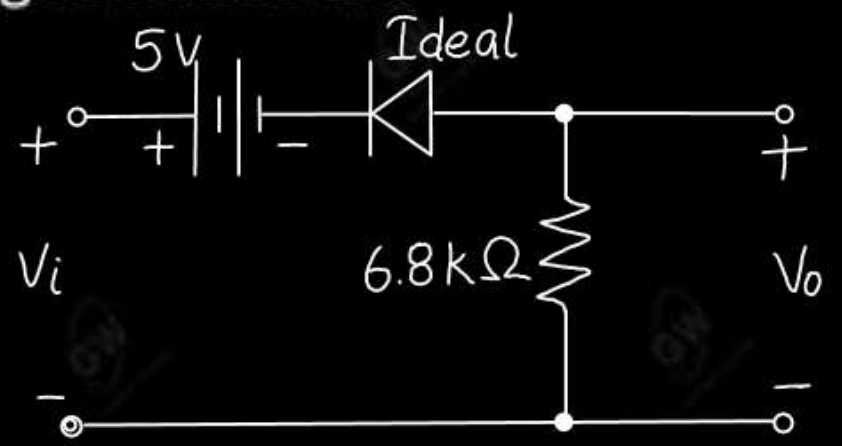


Gateway Classes

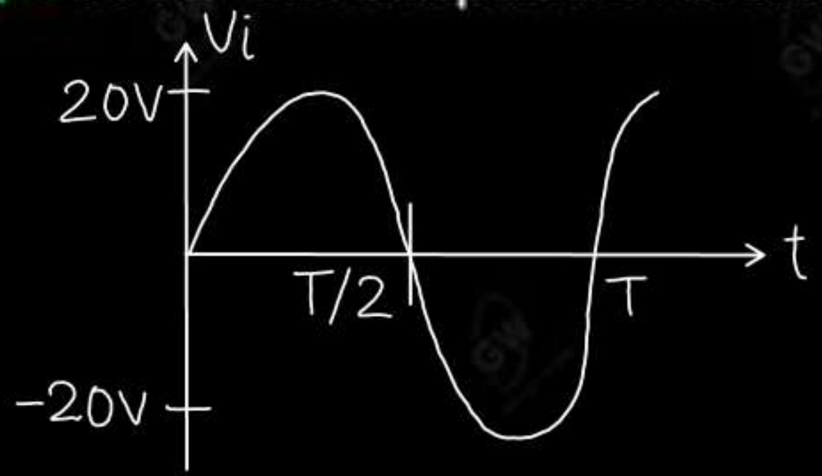
GATEWAY CLASSES

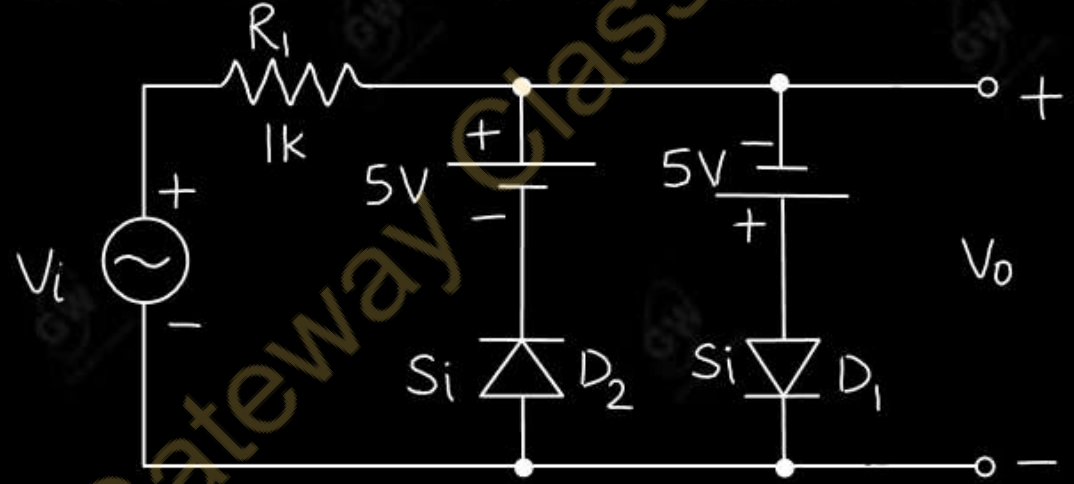
Q.3 Determine and Sketch V_o for the given network



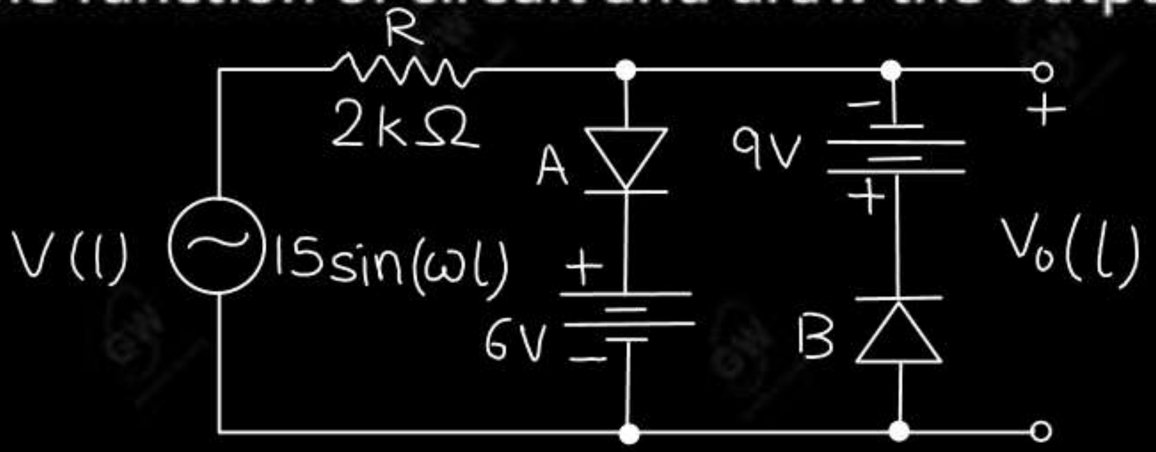


Q.4 Draw the output waveform of the given circuit and input waveform



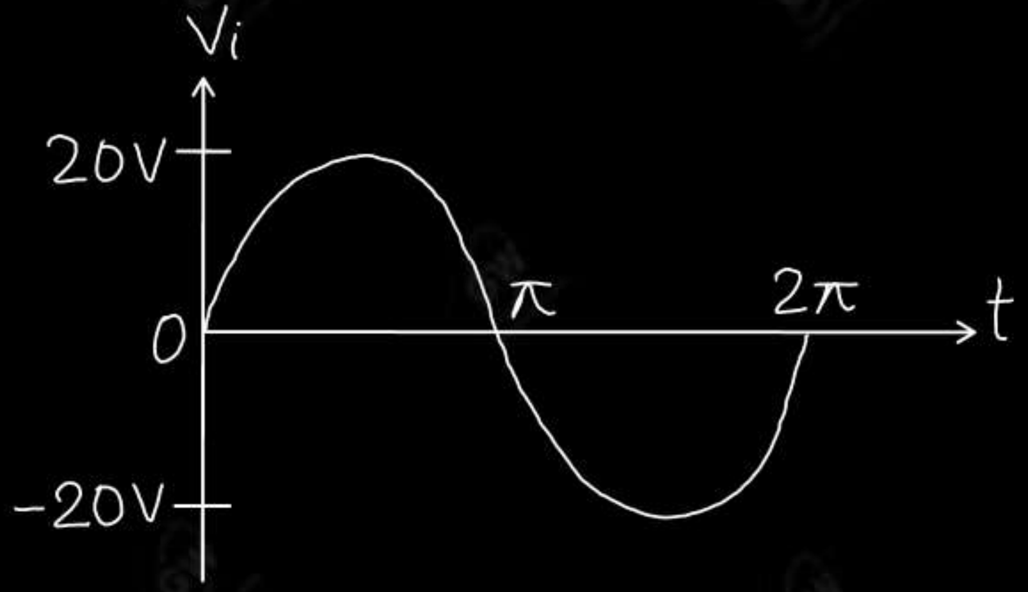


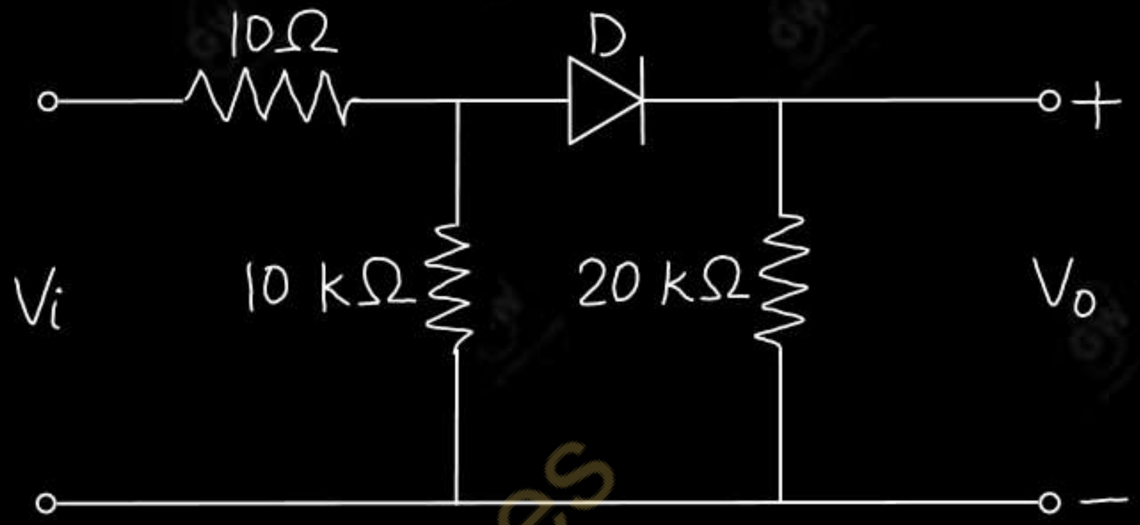
Q.5 Explain the function of circuit and draw the output waveform



Gateway Classes

Q.6 Draw the output waveform of the given circuit and input waveform





Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 9

Today's Target

- Clamper Circuits
- DPP
- PYQs

By Gulshan sir

Gateway Classes

➢ Clamper is a circuit consists of a diode, a resistor and a capacitor that shifts a waveform to a desired DC level without changing the actual appearance of the applied signal

➢ Additional shifts can also be obtained by introducing a DC supply to the basic structure

➢ Clamper circuits are of three types

(1) Positive Clamper

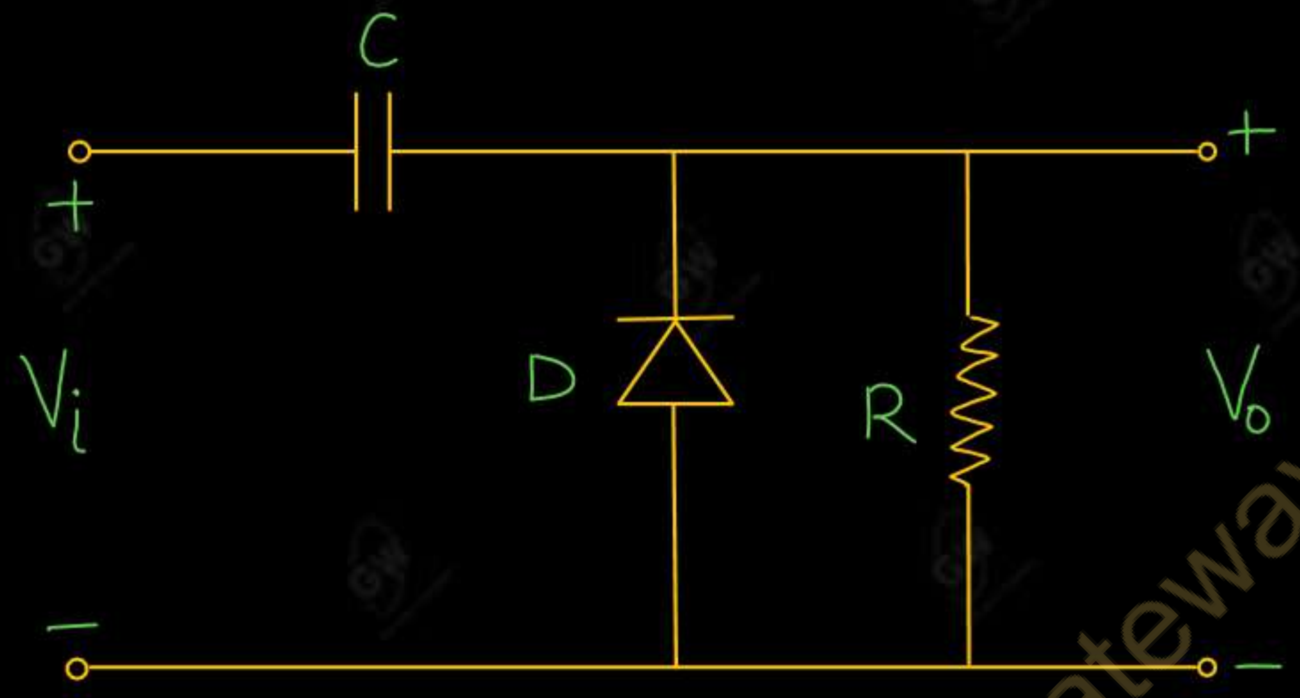
(2) Negative Clamper

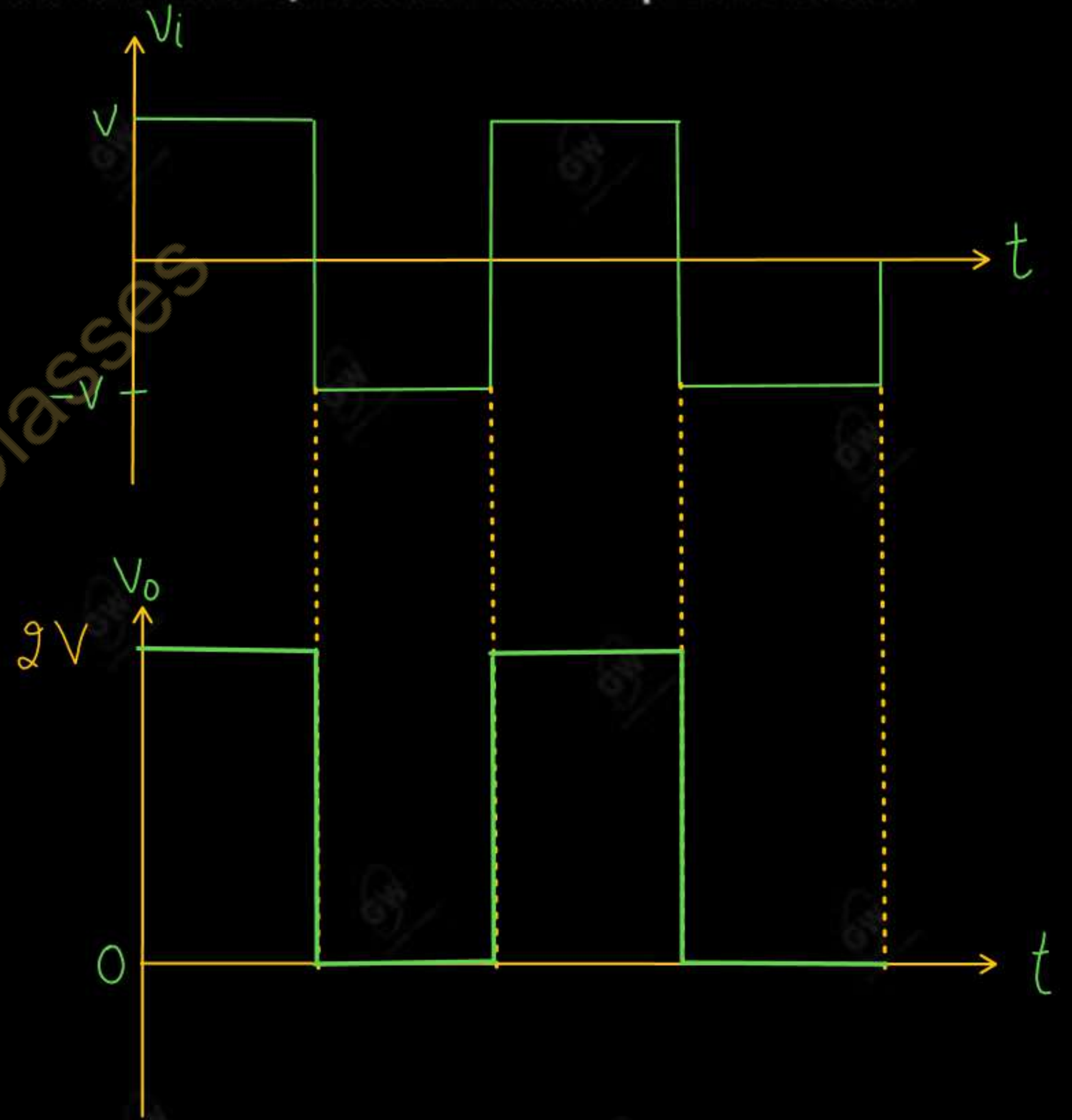
(3) Biased Clamper

Gateway Classes

Positive Clamper

- When the clamper circuit shifts AC signal in the upward direction, then the clamper is called positive clamper



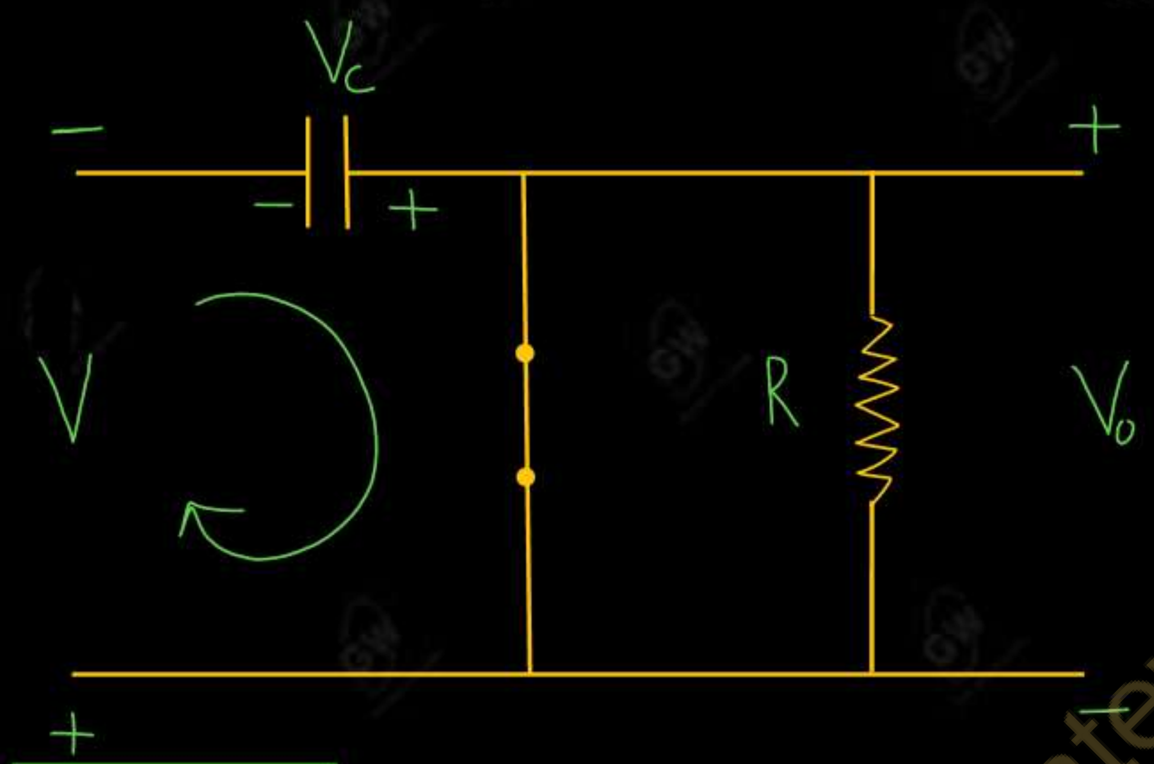


Gateway Classes

GW GATEWAY CLASSES

During Negative cycle

Diode D is ON and Capacitor starts to charge



$$\boxed{V_0 = 0}$$

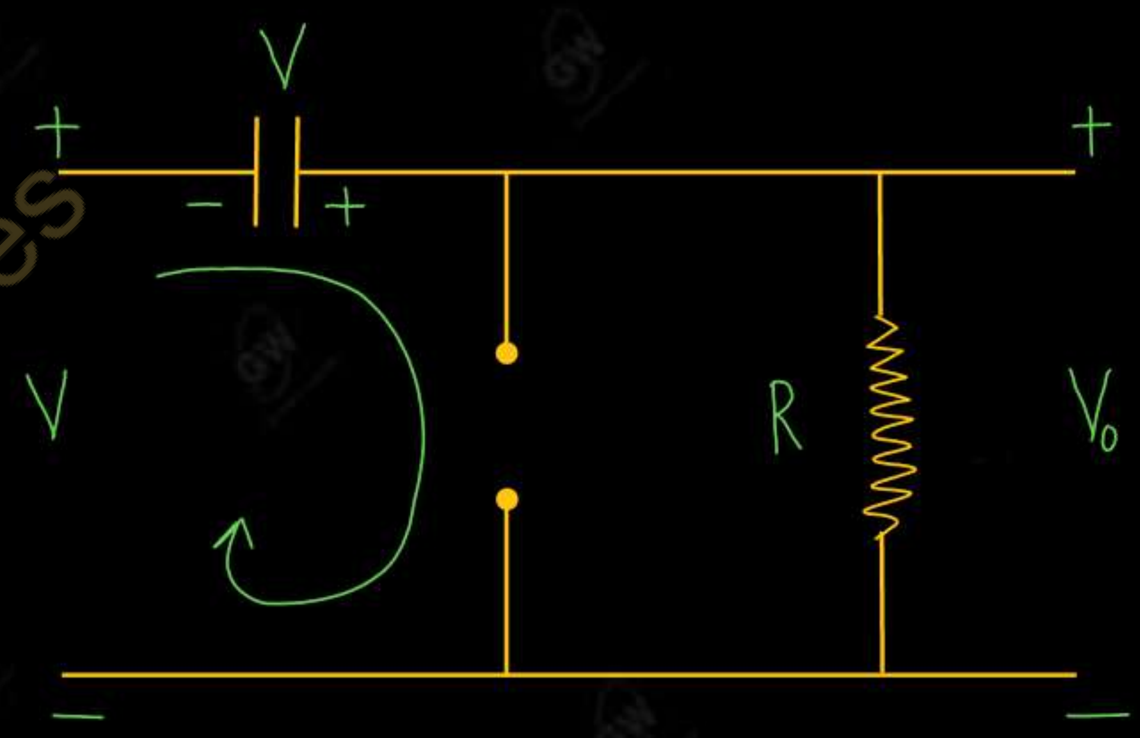
Apply KVL

$$V_c - V = 0$$

$$\boxed{V_c = V}$$

During positive cycle

Diode D is OFF



Apply KVL

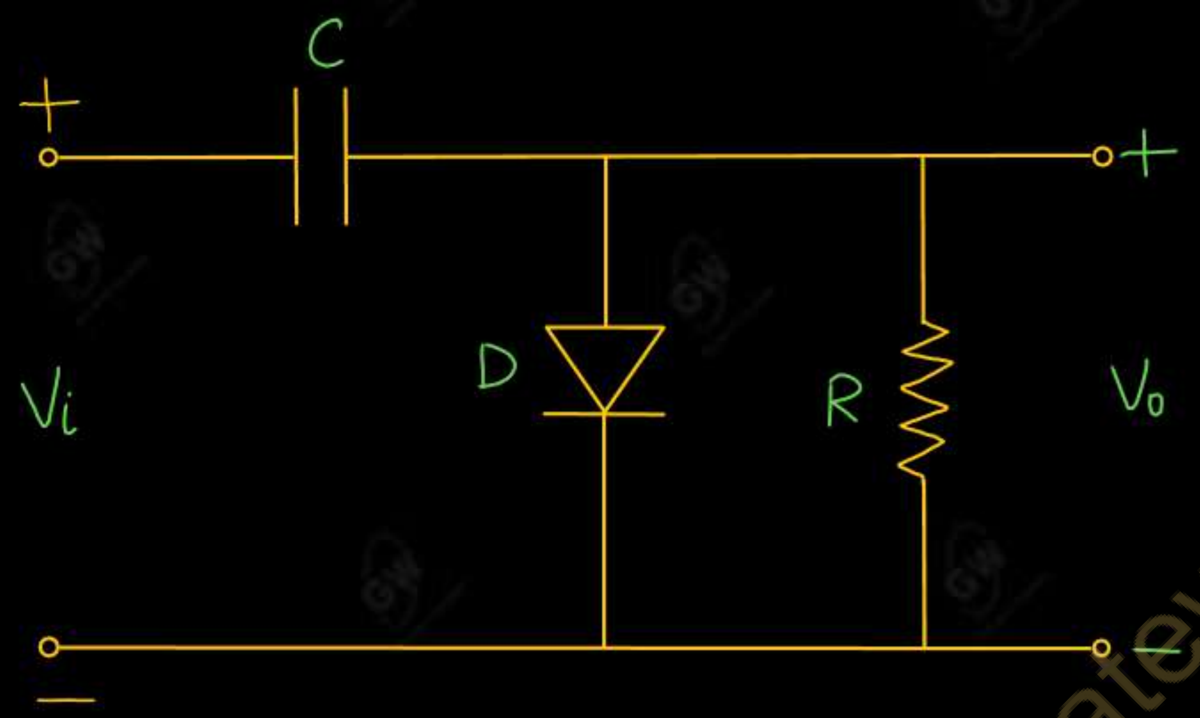
$$V - V_0 + V = 0$$

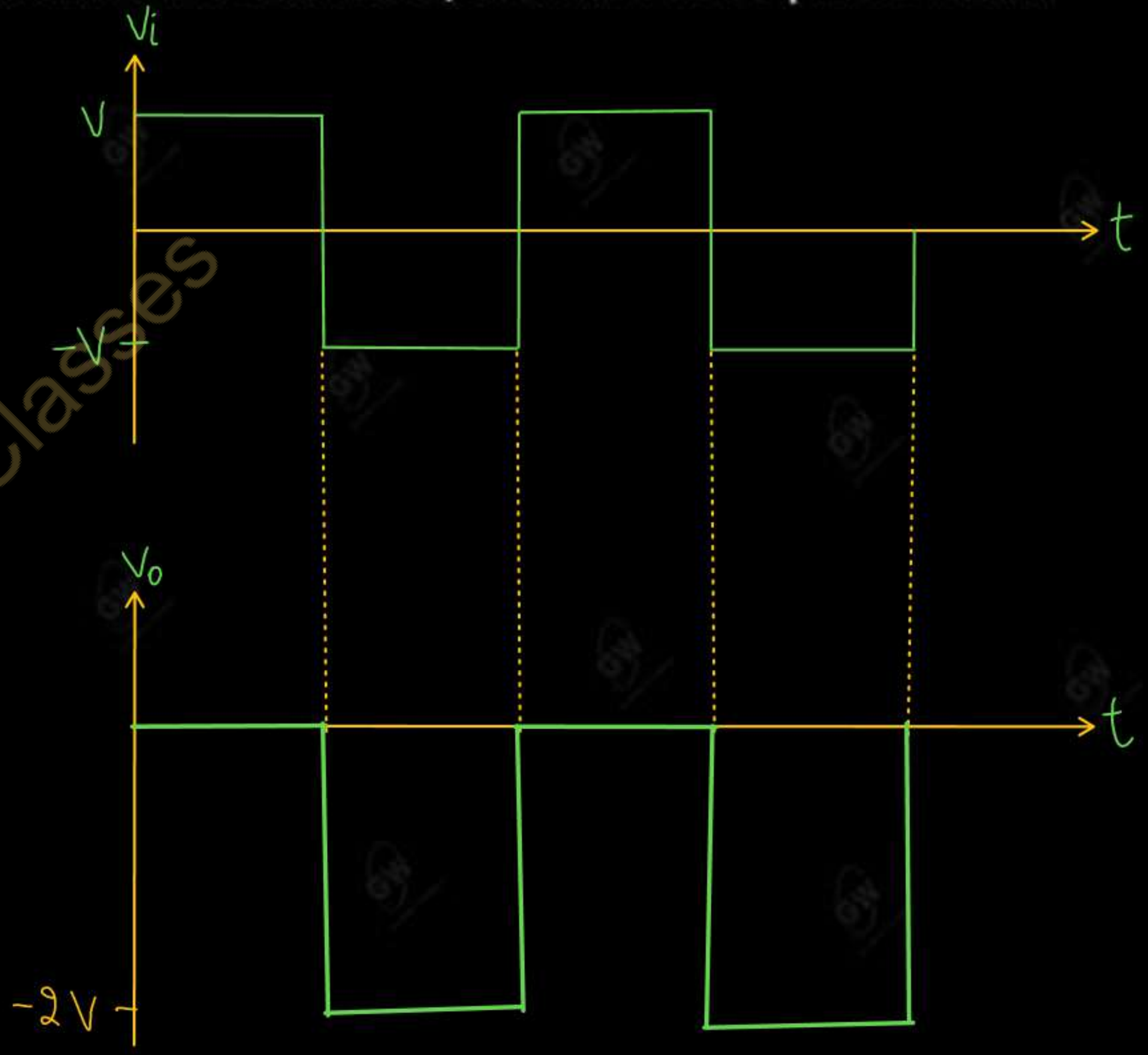
$$\boxed{V_0 = 2V}$$

Gateway Classes

Negative Clamper

- When the clamper circuit shift the AC signal in the downward direction, then the clamper is called negative clamper



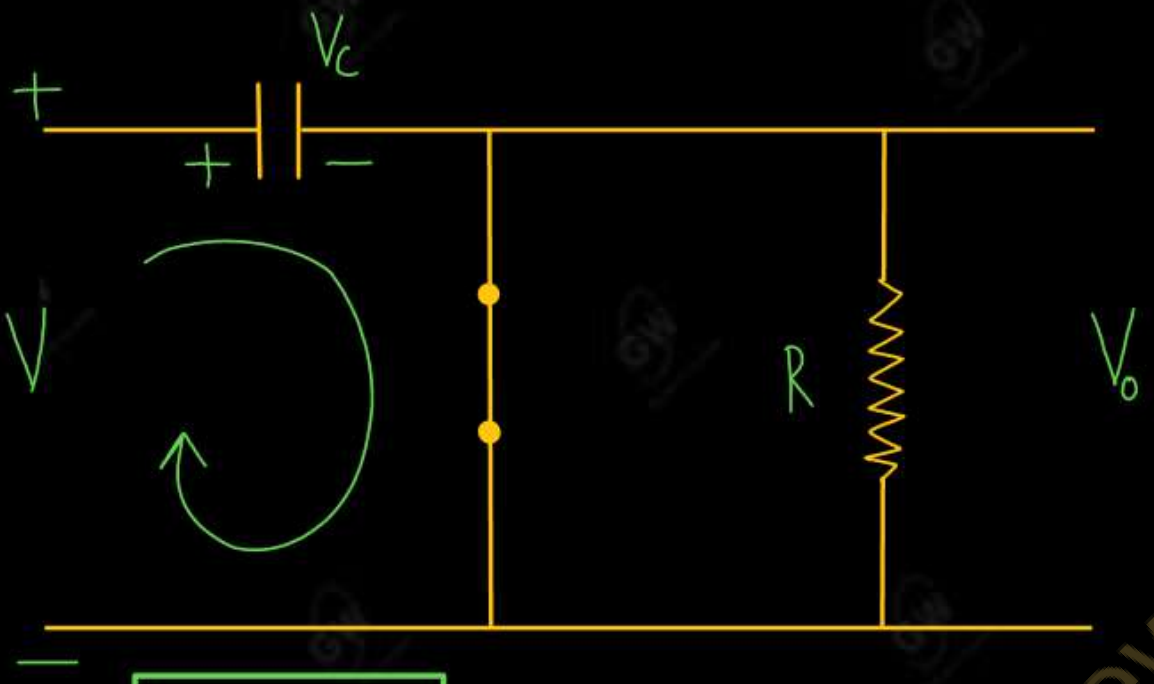


Gateway Classes

Working

During positive cycle

Diode D is ON (Capacitor starts to charge)



$\boxed{V_o = 0}$

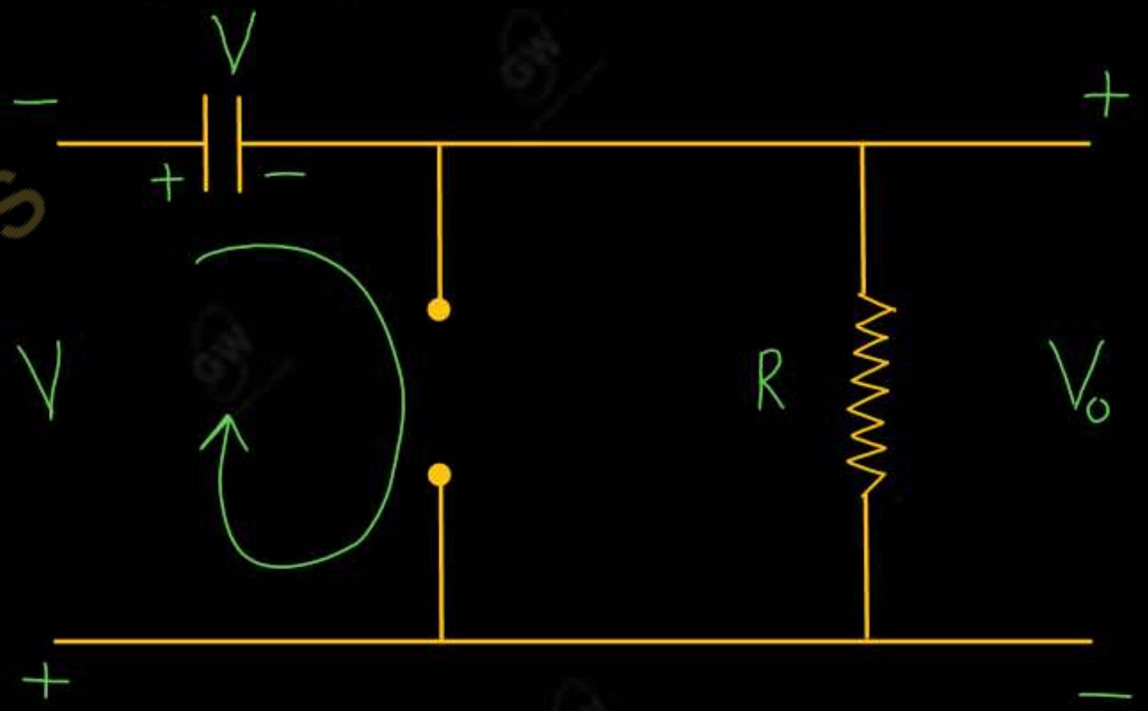
Apply KVL

$-V_c + V = 0$

$\boxed{V_c = V}$

During Negative cycle

Diode D is OFF



Apply KVL

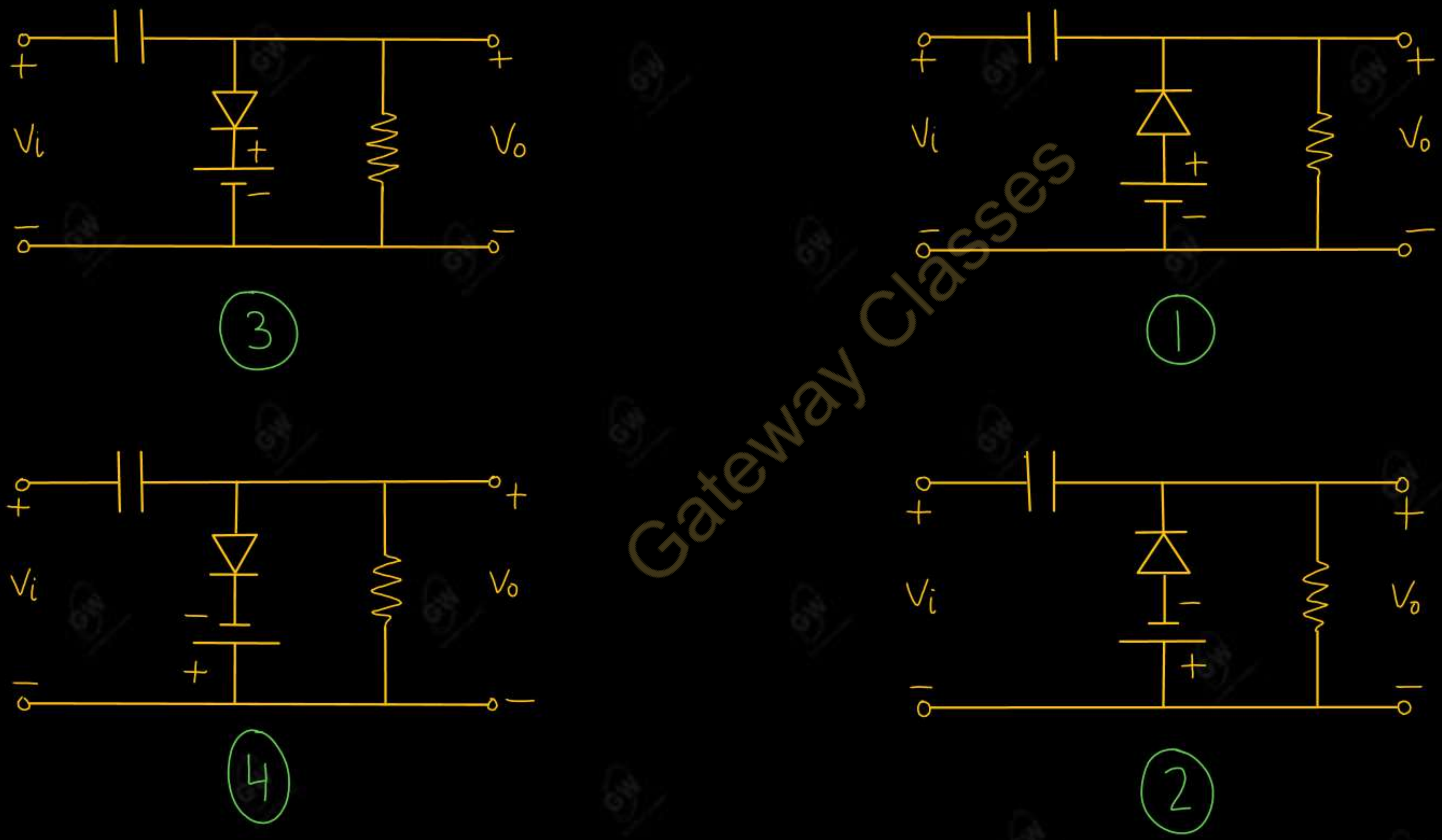
$-V - V_o - V = 0$

$\boxed{V_o = -2V}$

Gateway Classes

Biased Clamper

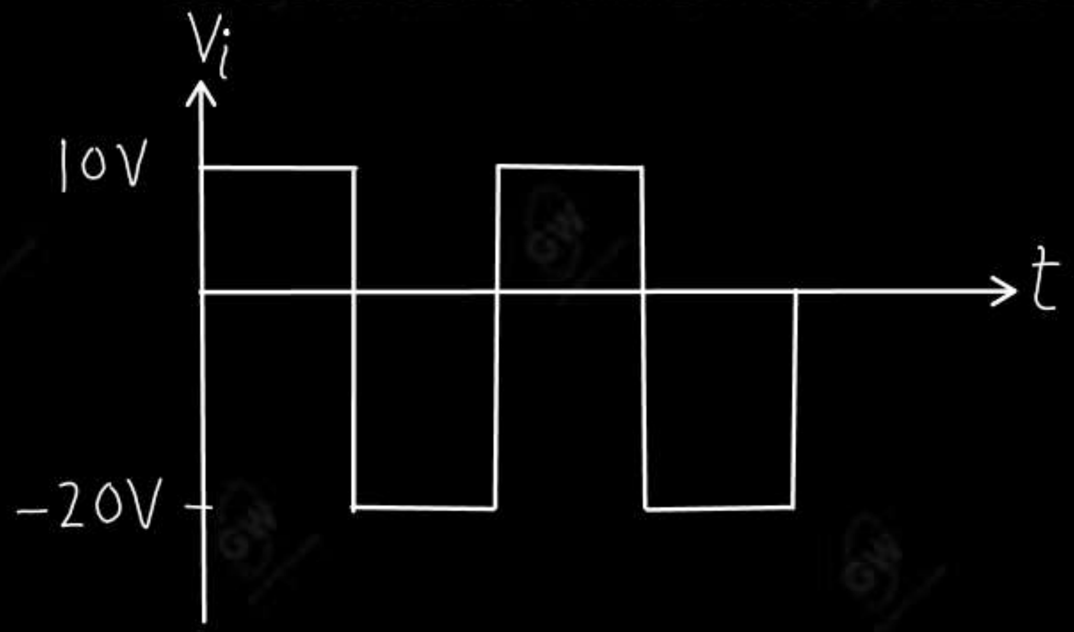
If DC supply is also present in the clamper circuit the such type of clamper is called biased clamper

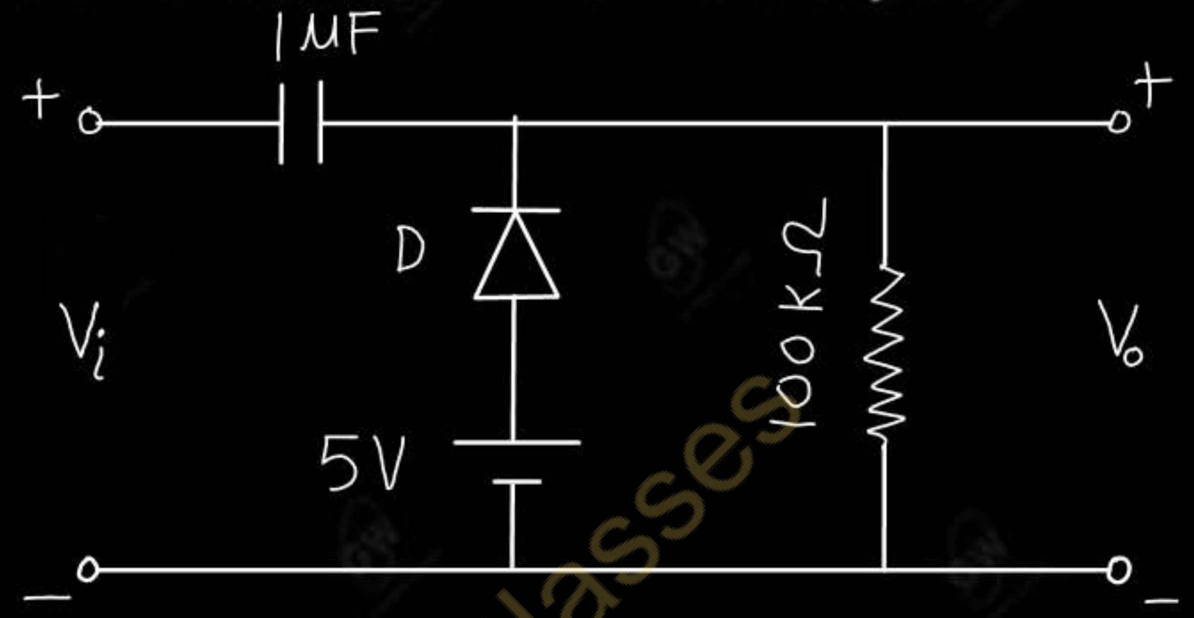


Gateway Classes

Q.1 Determine the output waveform of the following circuit by presenting all the necessary calculations which have been done to determine the output

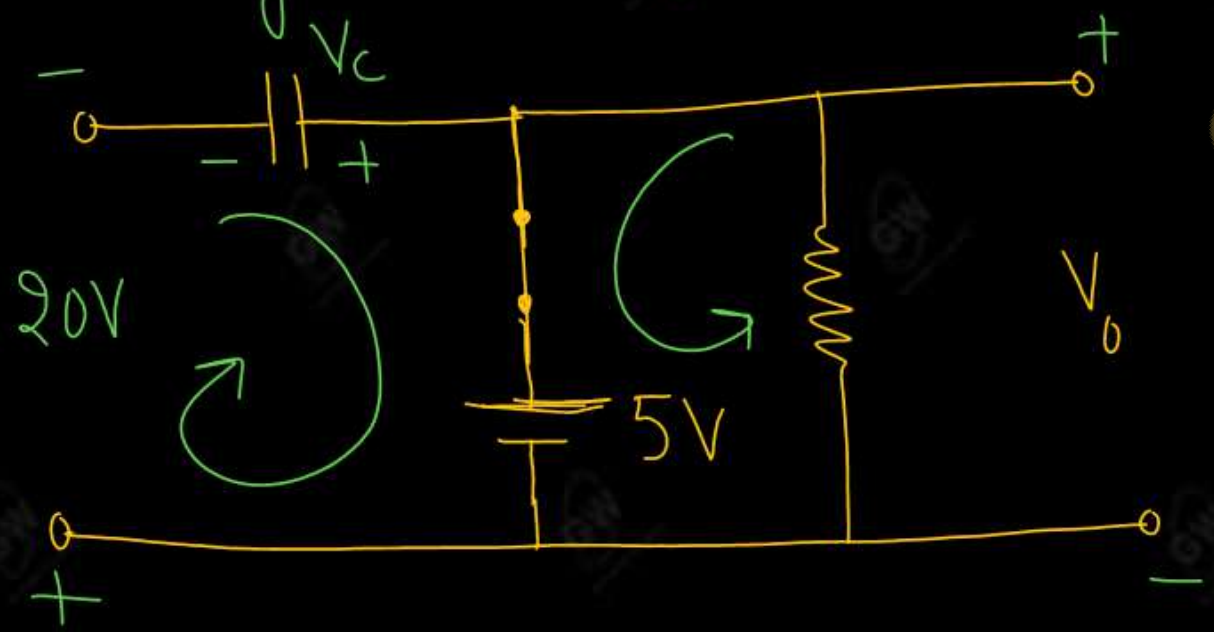
AKTU 2021





Solution

During −ve Half cycle



Apply KVL

$$V_c - 5 - 20 = 0$$

$$V_c = 25V$$

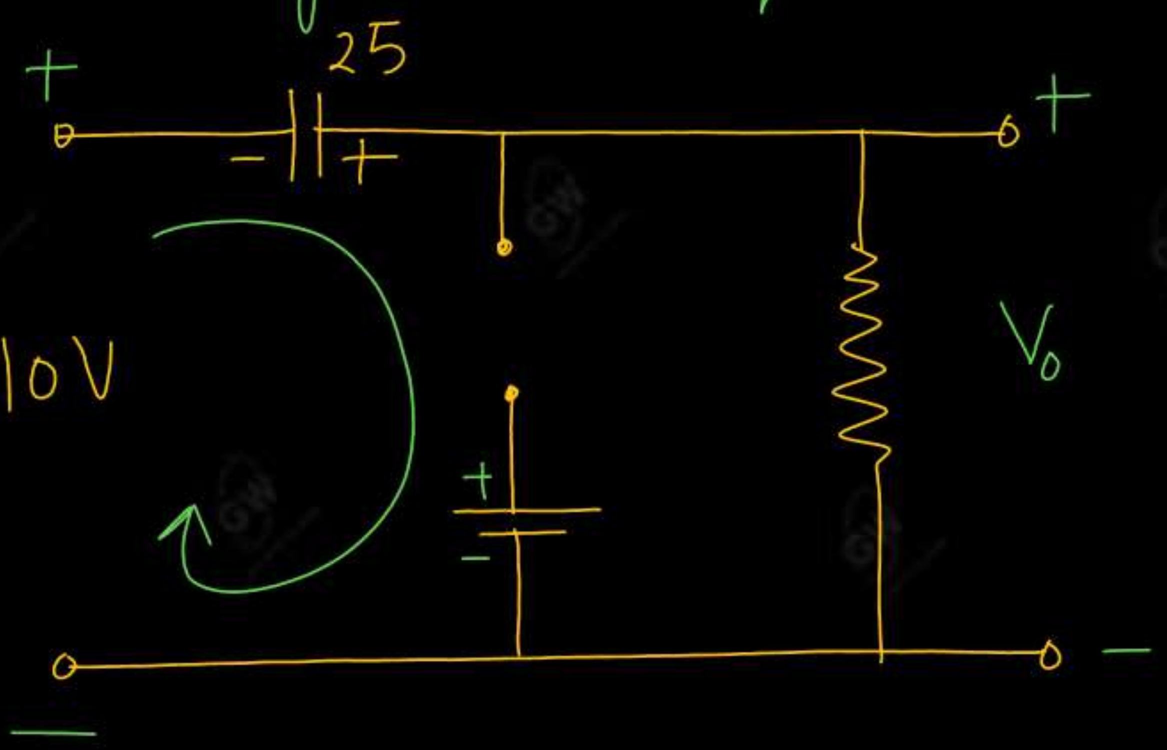
Apply KVL

$$-5 + V_0 = 0$$

$$V_0 = 5V$$

Gateway Classes

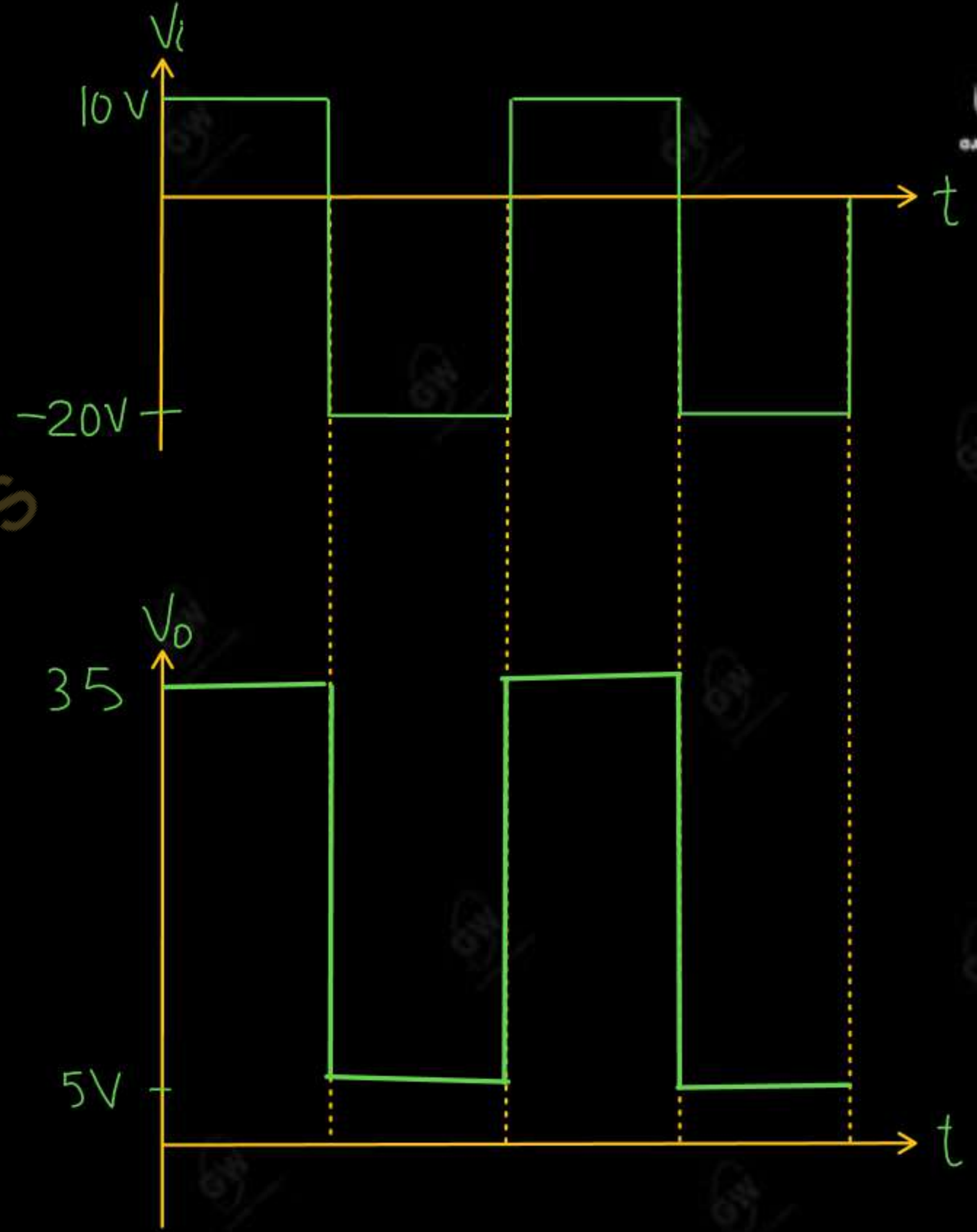
During +ve Half cycle



Apply KVL

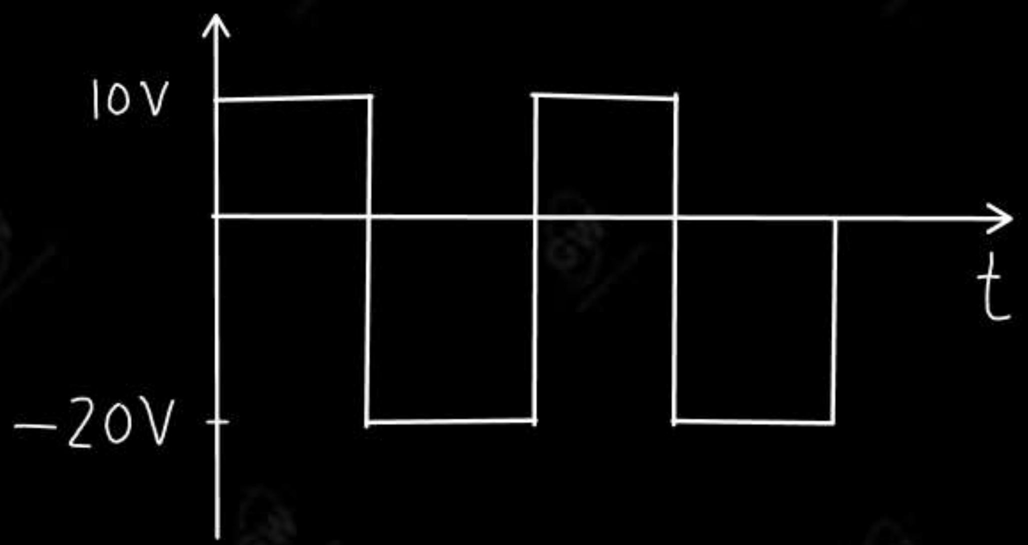
$+25 - V_0 + 10 = 0$

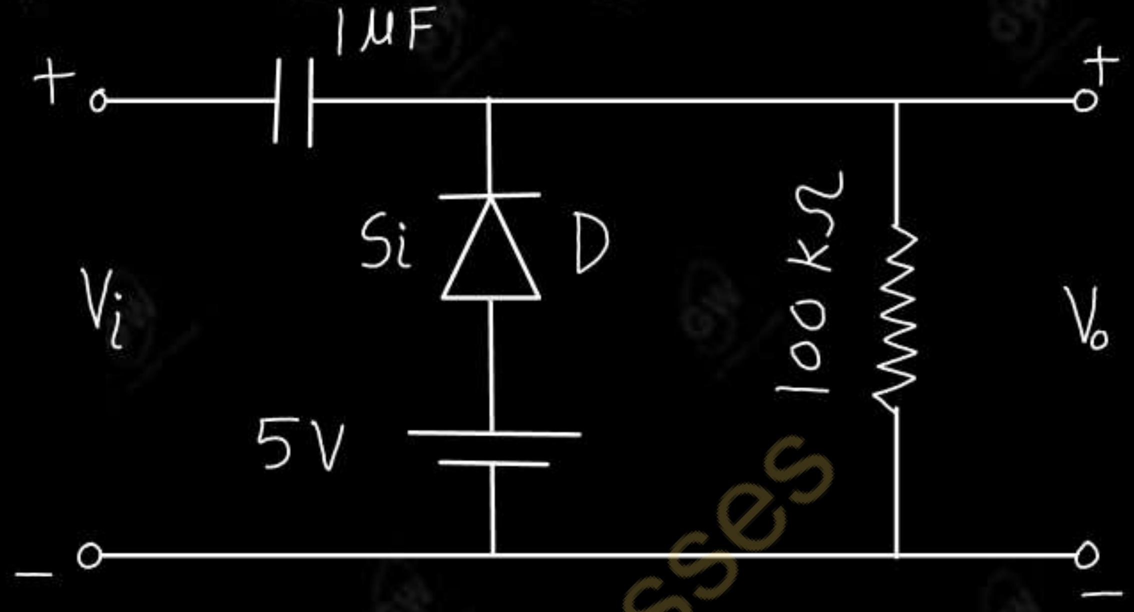
$$\boxed{V_0 = 35V}$$



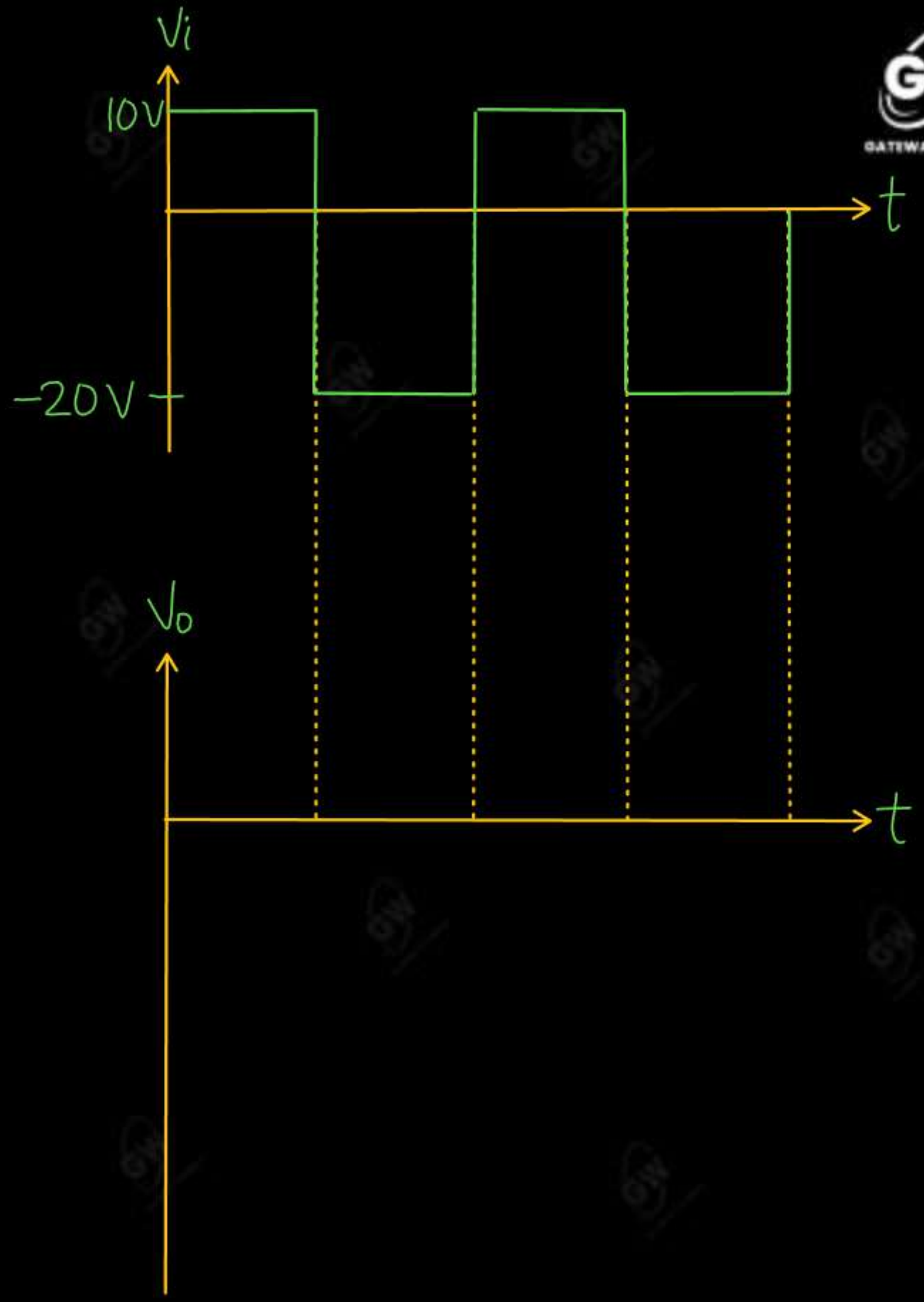
Gateway Classes

Q.2 Determine and draw the output voltage of the given network



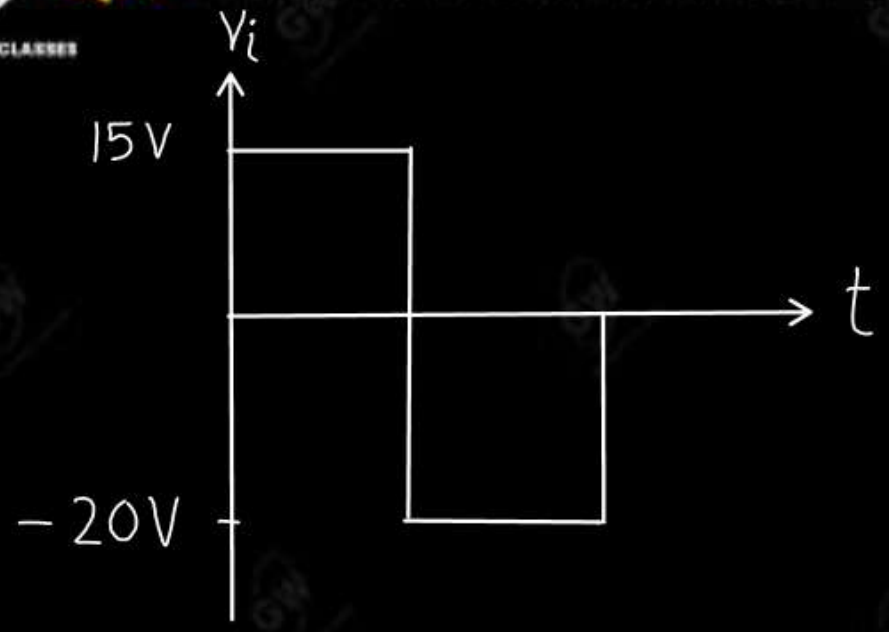


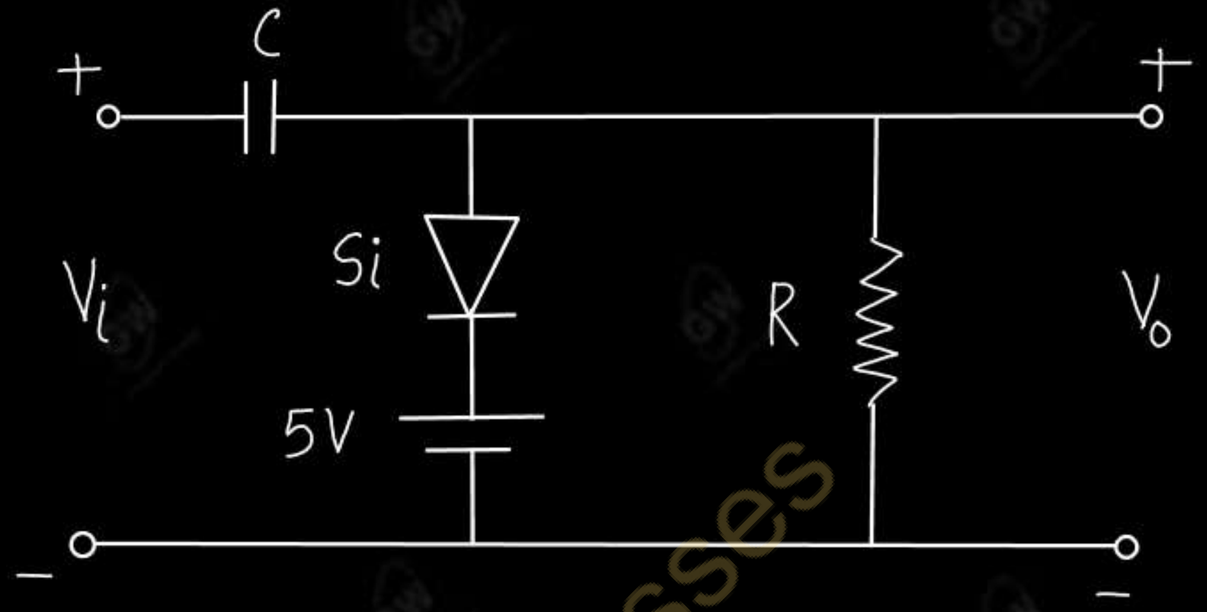
Solution



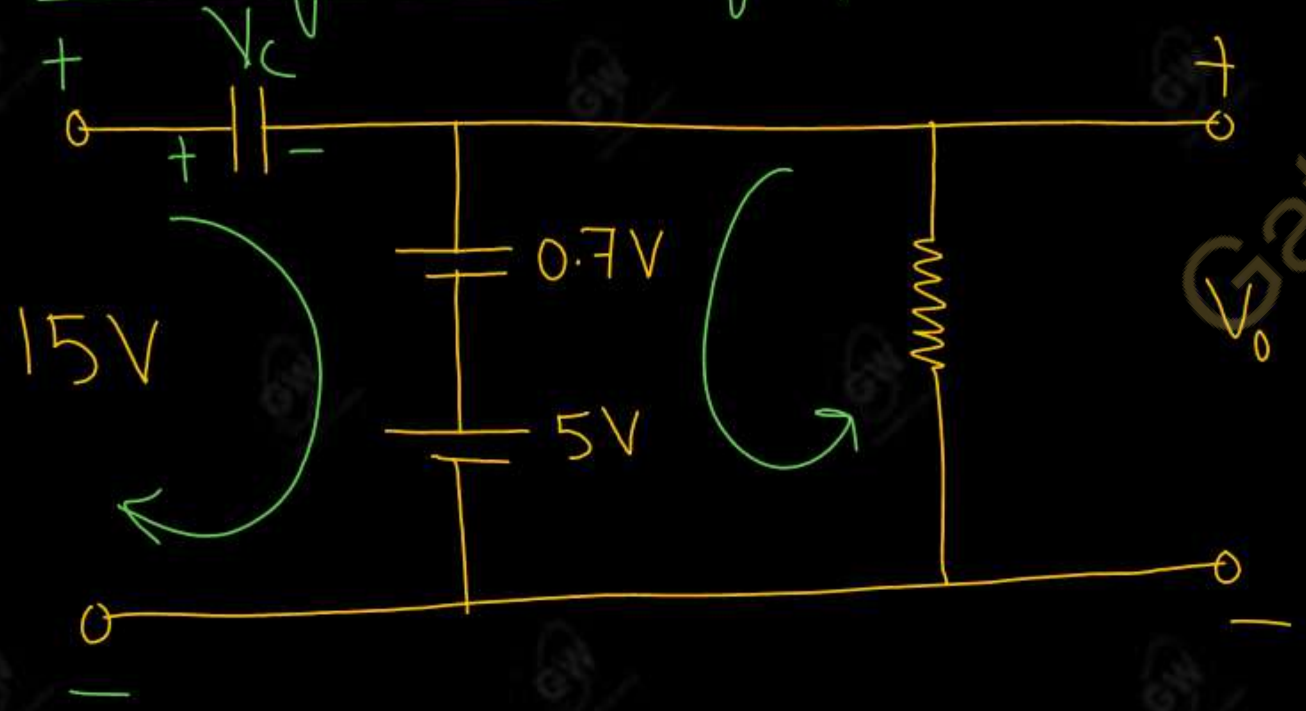
Gateway Classes

Q.3 Determine and draw the output voltage of the given network





During +ve Half cycle



Apply KVL in output section

$$-0.7-5+V_0=0$$

$$\boxed{V_0 = 5.7V}$$

Apply KVL in input section

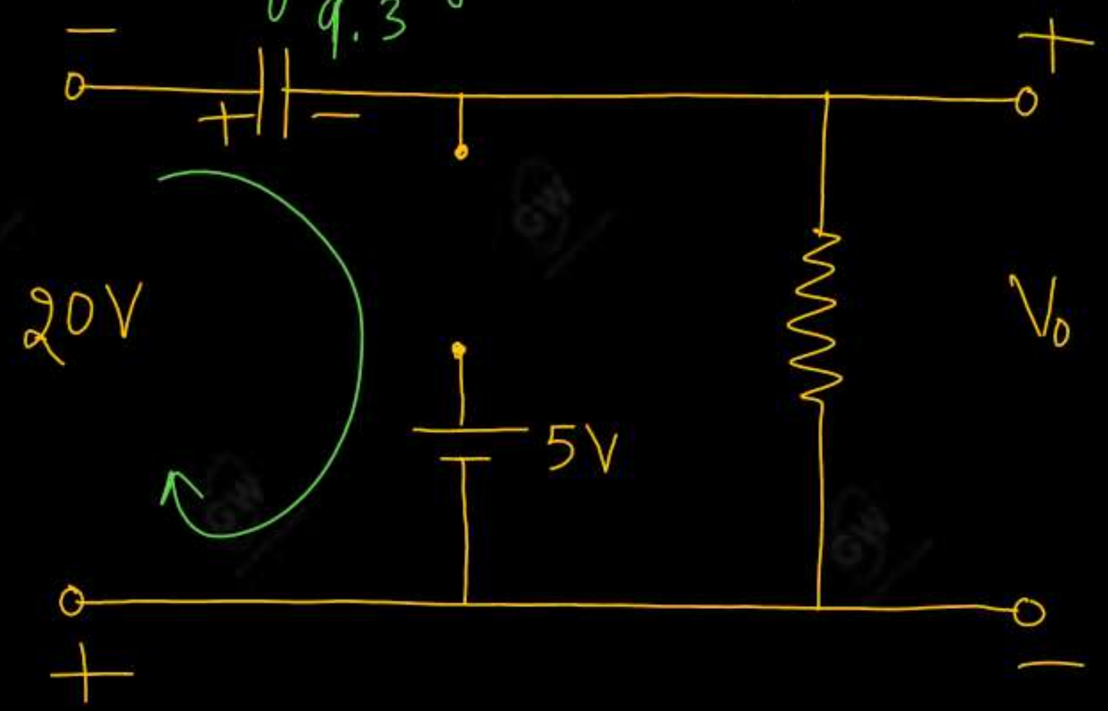
$$-V_C-0.7-5+15=0$$

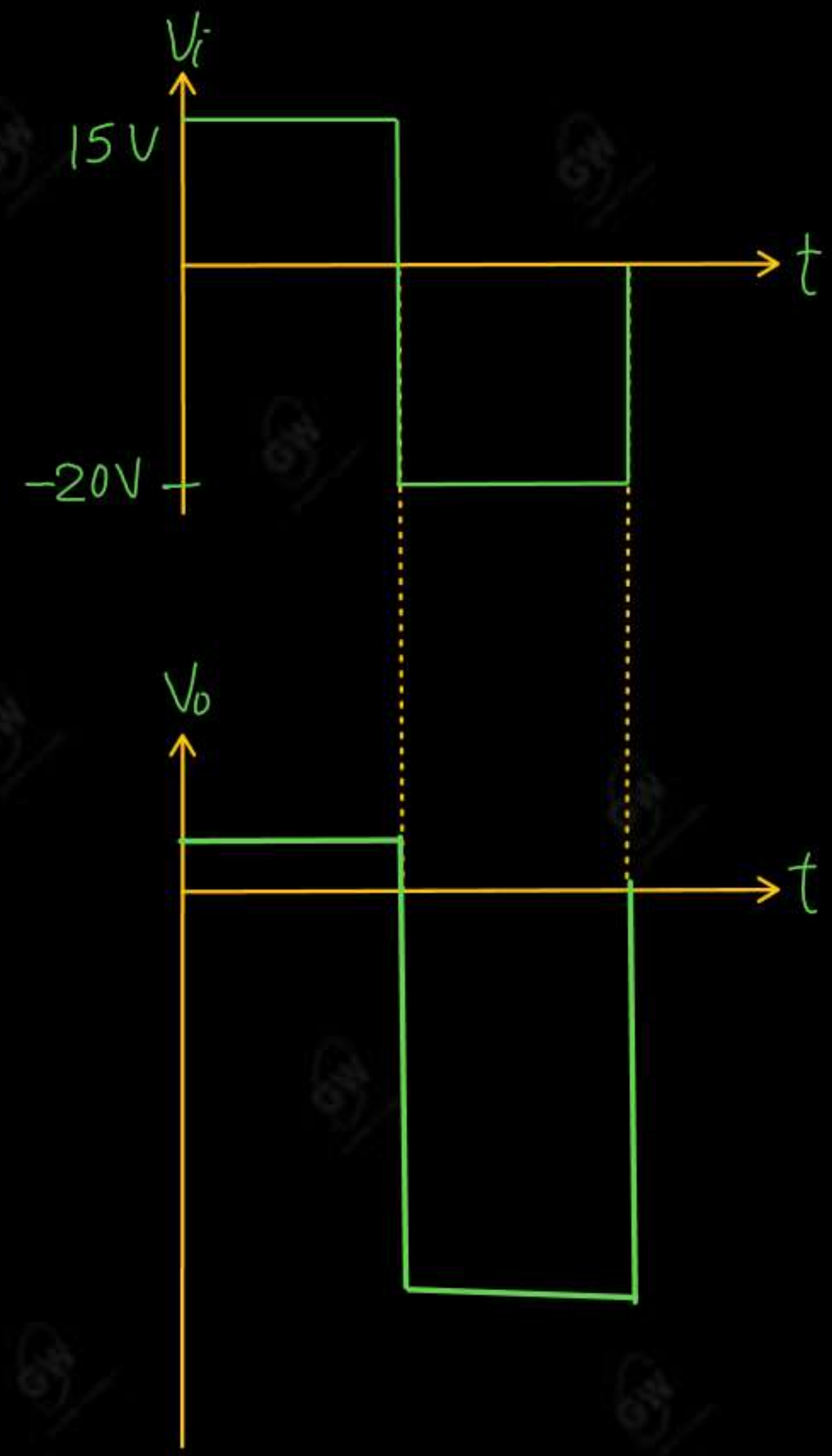
$$V_C = 15-5.7$$

$$\boxed{V_C = 9.3V}$$

Gateway Classes

During negative Half cycle





Apply KVL

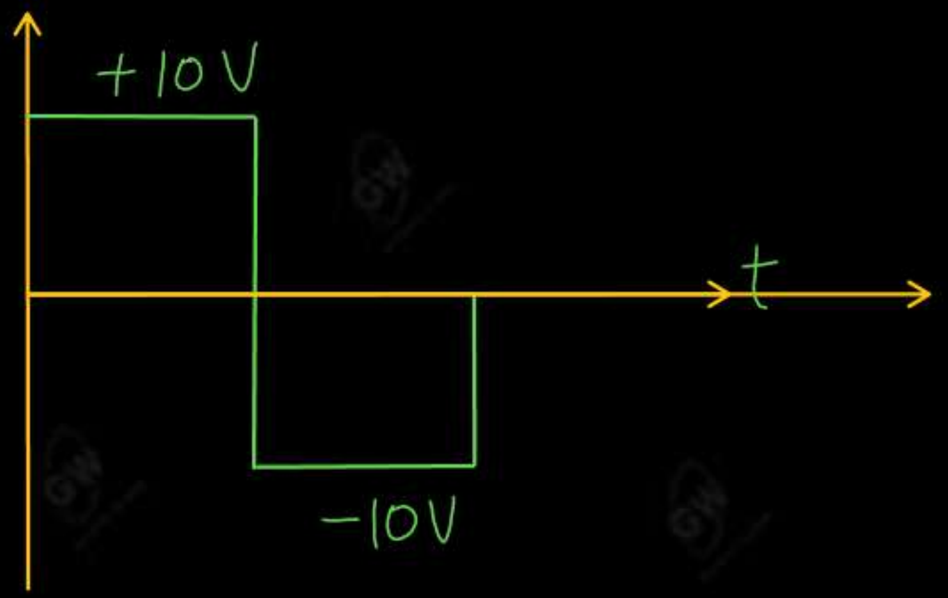
$$-9.3 - V_0 - 20 = 0$$

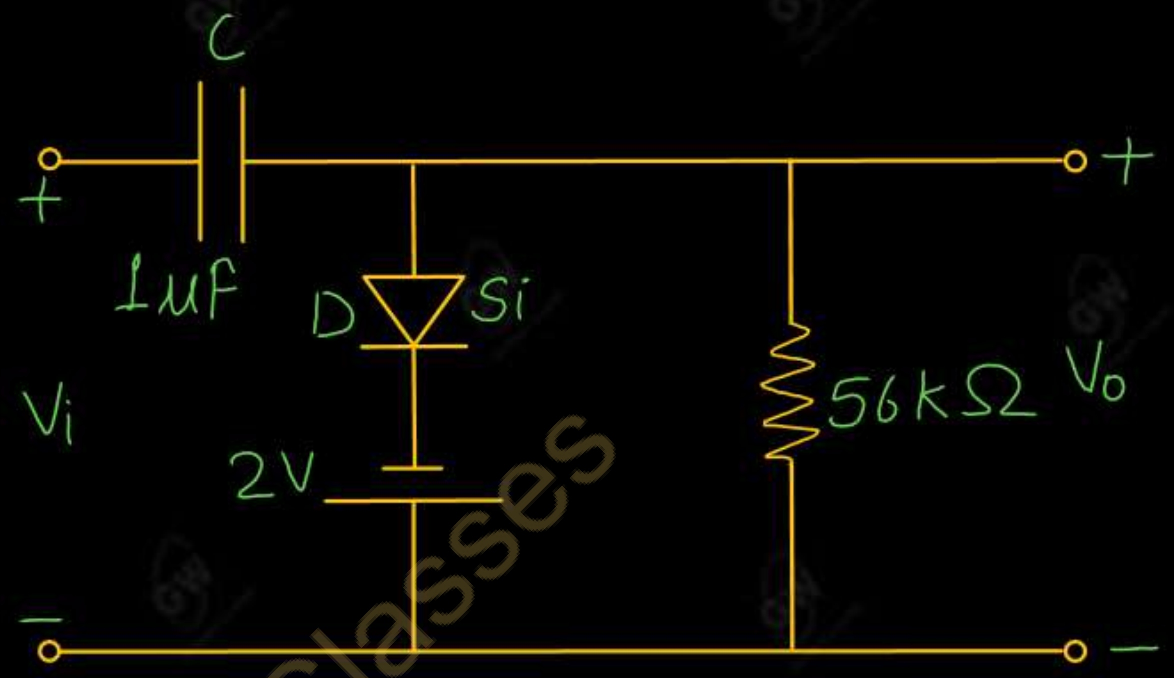
$$\boxed{V_0 = -29.3V}$$

Gateway Classes

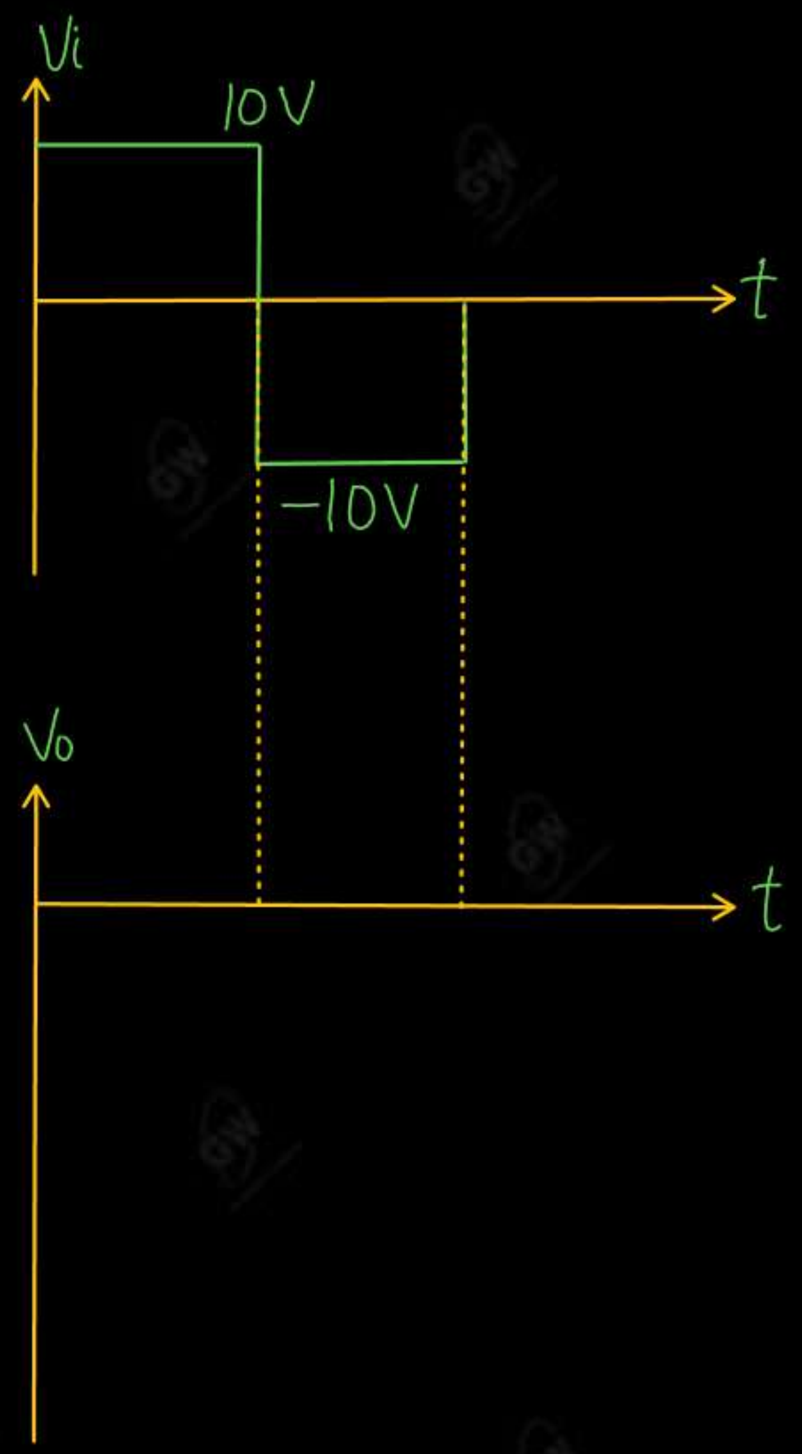
Q.4 Determine V_o for the following figure

AKTU 2017



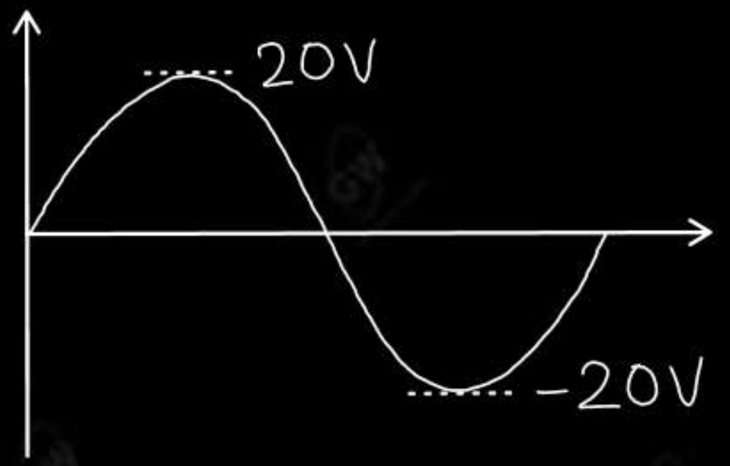


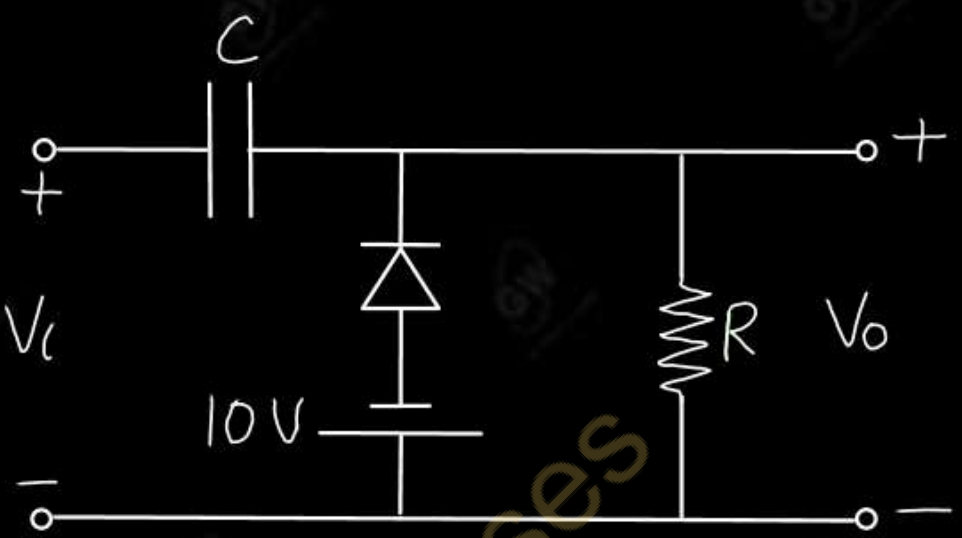
Gateway Classes



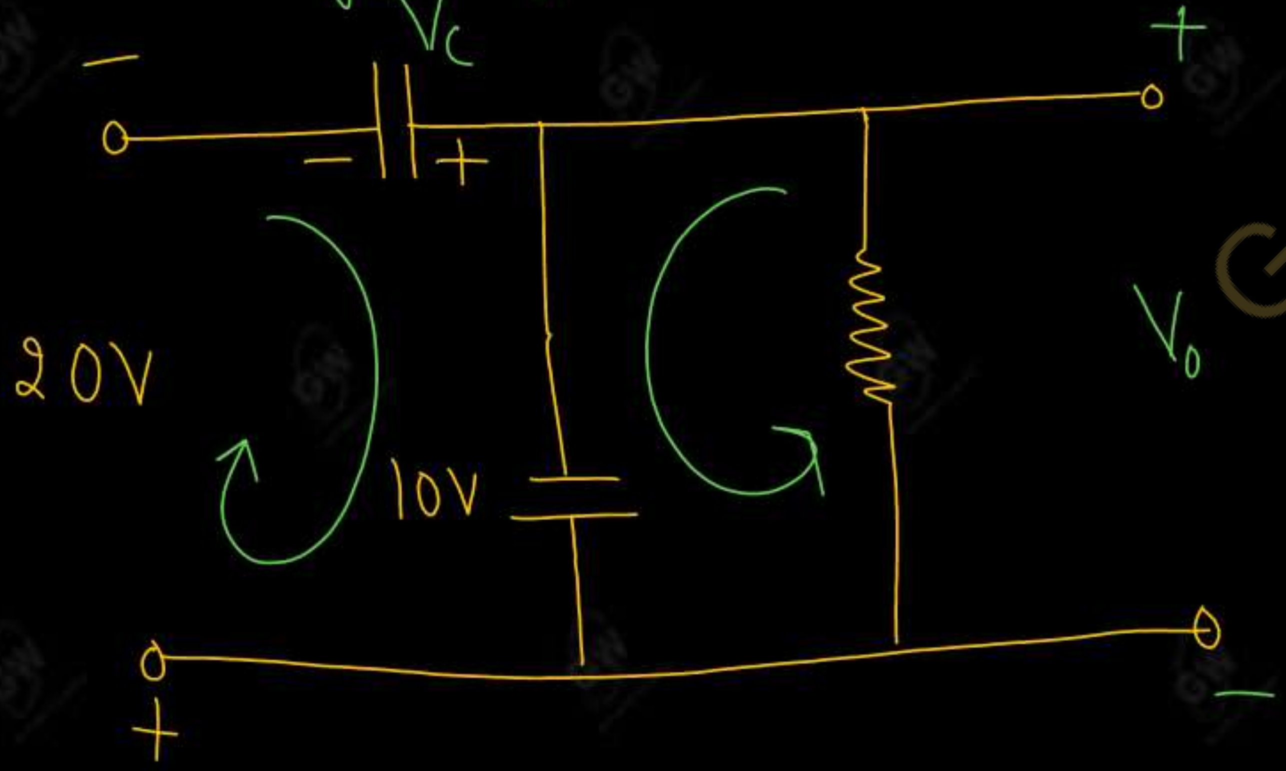
Gateway Classes

Q.5 Draw the output waveform





During negative Half cycle



Apply KVL

$$10 + V_0 = 0$$

$$\boxed{V_0 = -10V}$$

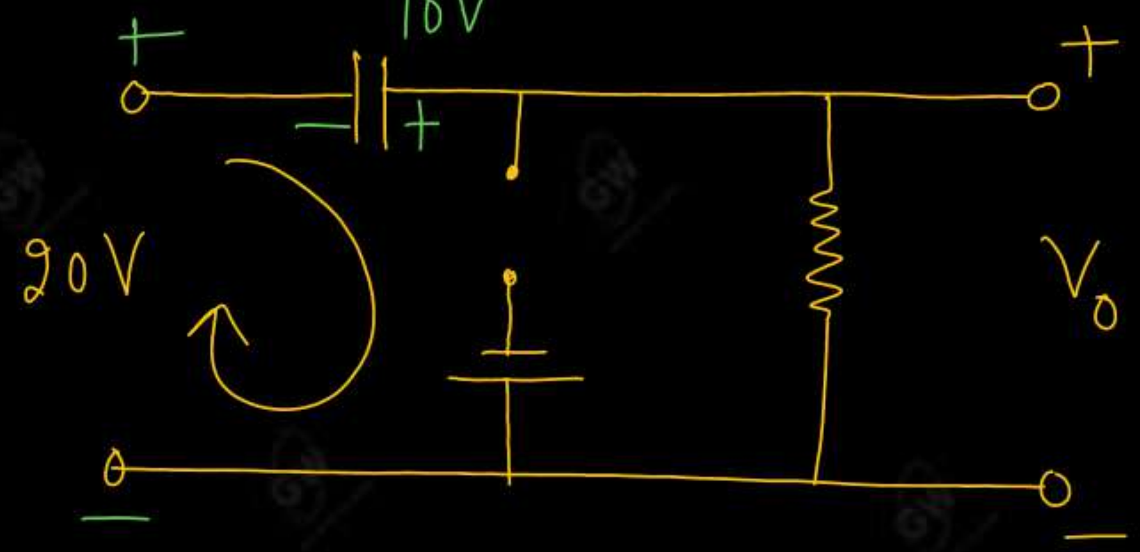
Apply KVL

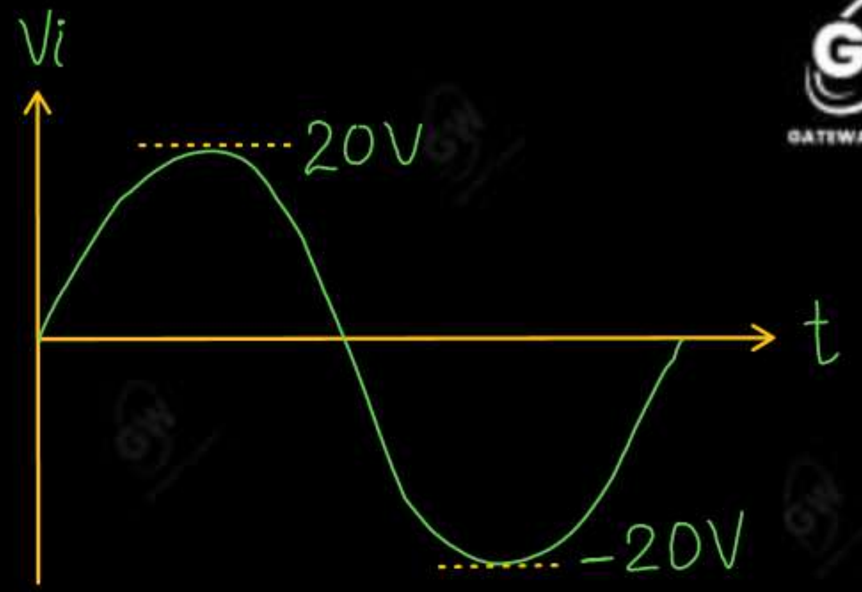
$$V_c + 10 - 20 = 0$$

$$V_c = 10$$

Gateway Classes

During positive cycle





Apply KVL

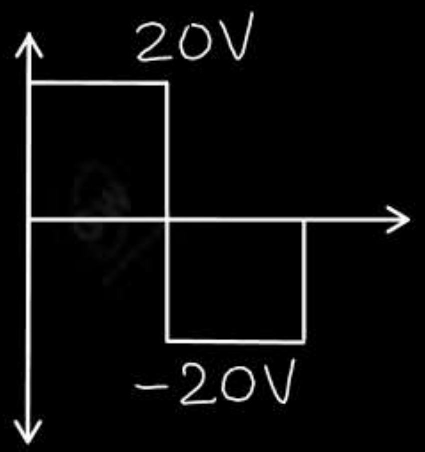
$$10 - V_0 + 20 = 0$$

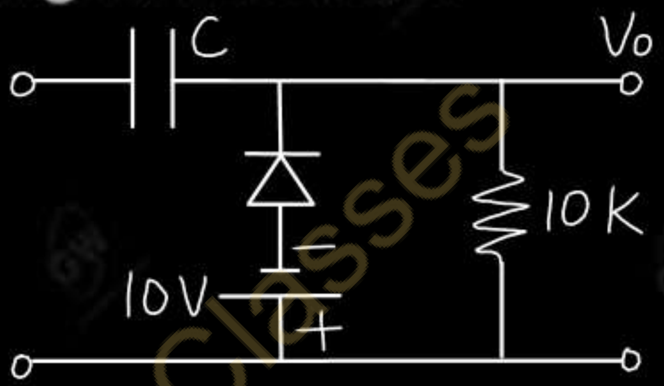
$$\boxed{V_0 = 30V}$$

Gateway Classes

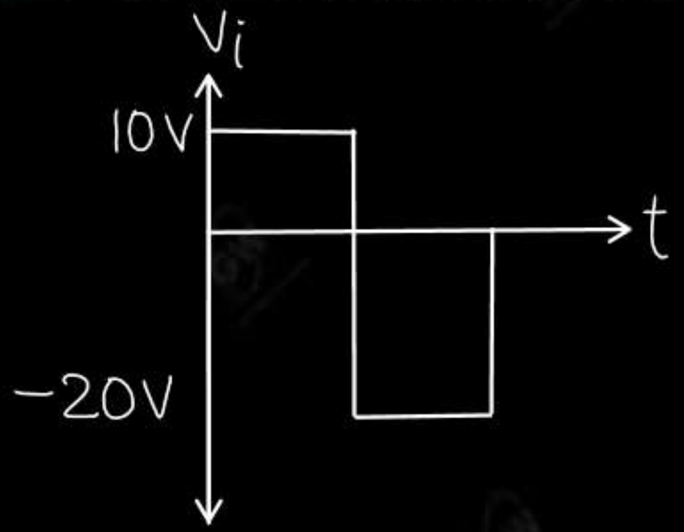
UNIT - 1 : DPP - 9

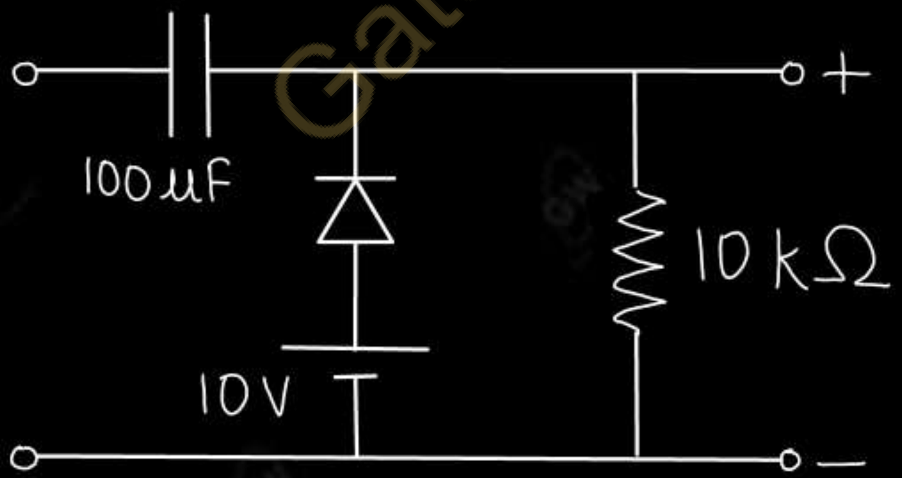
Q.1 Determine and draw the output voltage of the given network





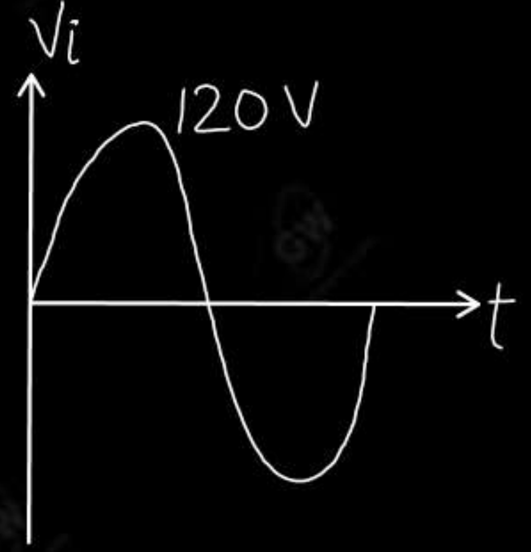
Q.2 Define Clamper. Determine output for the given network **AKTU 2022**

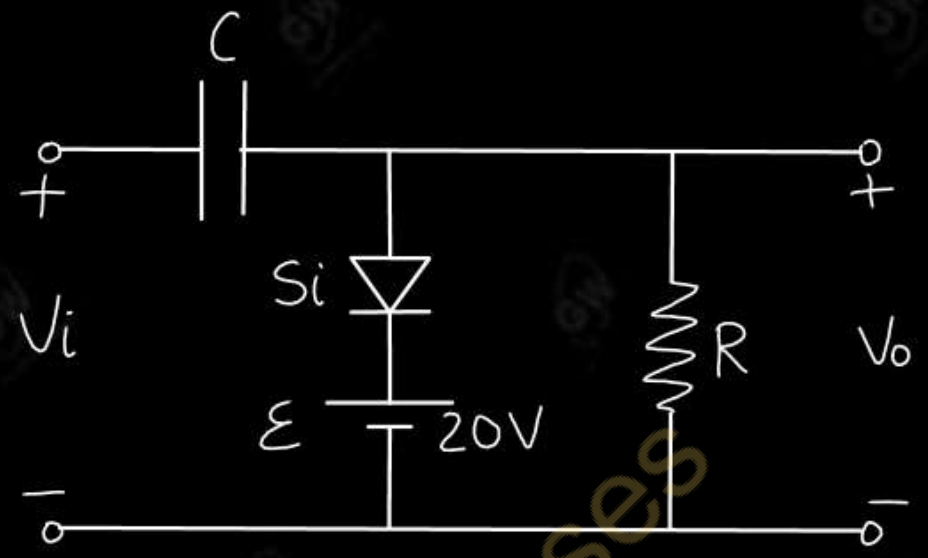




Gateway Classes

Q.3 Sketch the output for the given clamper circuit





Gateway Classes

AKTU

B.Tech : I-Year

GATEWAY CLASSES

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 10

Today's Target

- Voltage Multiplier Circuits
- DPP
- PYQs

Gateway Classes

By Gulshan sir

Voltage Multiplier

- A circuit that produces an output dc voltage equal to a multiple of the peak input a.c. voltage (V_m) is called a voltage multiplier.
- If peak input ac voltage be V_m then the output dc voltage be $2V_m$, $3V_m$, $4V_m$ ....
- On the basis of multiplying factor voltage multiplier can be classified as

  (1) Voltage Doubler ($V_o = 2\,V_m$)

  (2) Voltage Tripler ($V_o = 3V_m$)

  (3) Voltage Quadrupler ($V_o = 4V_m$)

$$\boxed{V_i = V_m \sin wt}$$

Voltage Doubler

- A circuit that produces an output dc voltage equal to two times of the peak input ac voltage (V_m) is called a voltage Doubler.
- Two Diode and two capacitor are needed for Voltage Doubler circuit

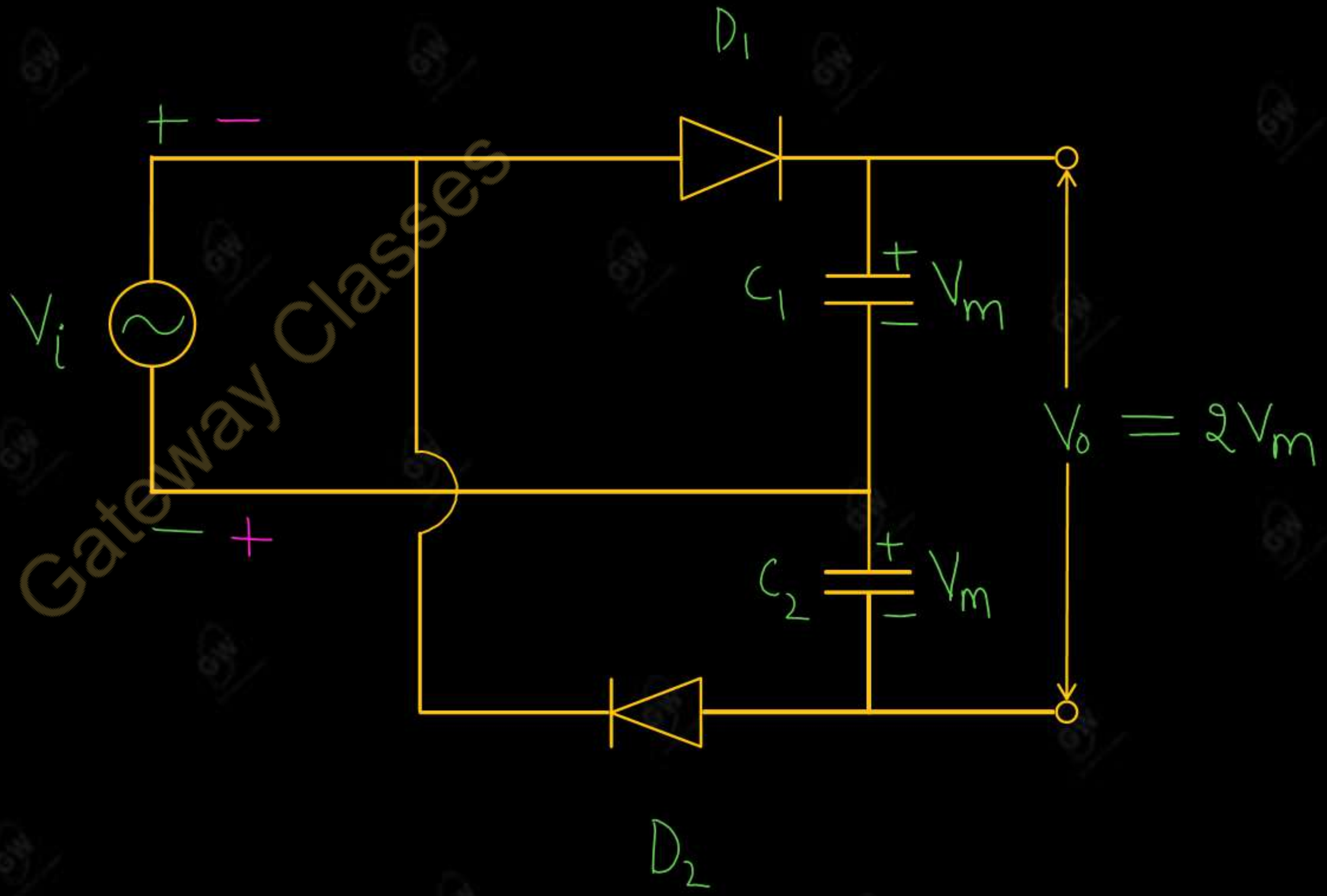
Gateway Classes

➤ A Voltage Doubler circuit may be classified as

(i) Full Wave Voltage Doubler

(ii) Half Wave Voltage Doubler

Full Wave Voltage Doubler



Gateway Classes

Working

(i) During positive Half cycle

- Diode D_1 is forward biased (ON)
- Diode D_2 is reversed biased (OFF)
- So, capacitor C_1 is charged up to voltage V_m

(ii) During negative Half cycle

- Diode D_1 is reversed biased (OFF)
- Diode D_2 is forward biased (ON)
- So, capacitor C_2 is charged up to voltage V_m

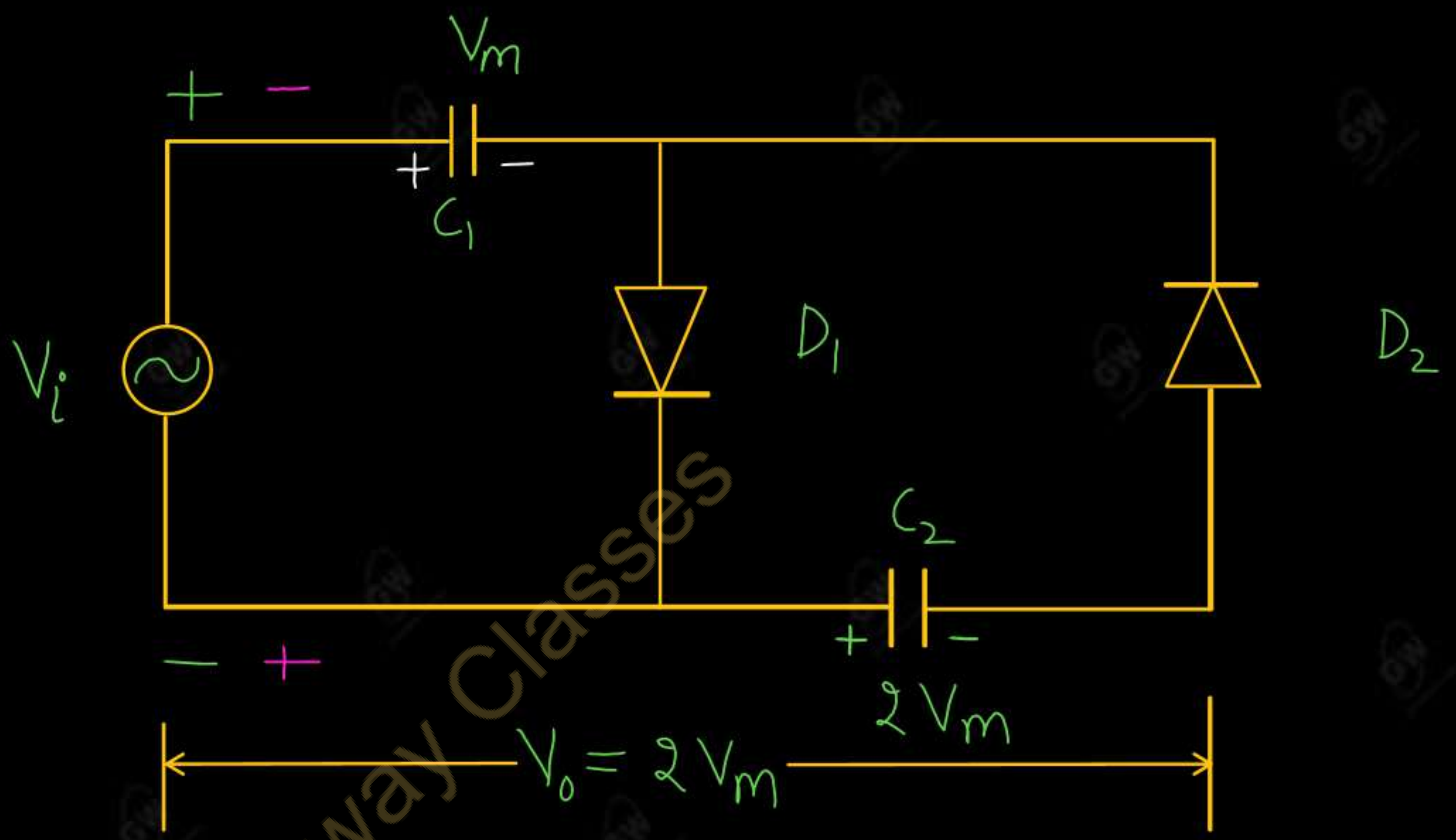
Final output voltage

$$V_o = V_m + V_m$$

$$V_o = 2V_m$$

Gateway Classes

Half Wave Voltage Doubler



Working

(ii) During positive Half cycle

- Diode D_1 is forward biased (ON)
- Diode D_2 is reversed biased (OFF)
- So, capacitor C_1 is charged up to voltage V_m

$$V_{c_1} = V_m$$

Gateway Classes

(ii) During negative Half cycle

- Diode D_1 is reversed biased (OFF)
- Diode D_2 is forward biased (ON)
- So, capacitor C_2 is charged up to voltage $2V_m$

$$V_{c_2} = 2V_m$$

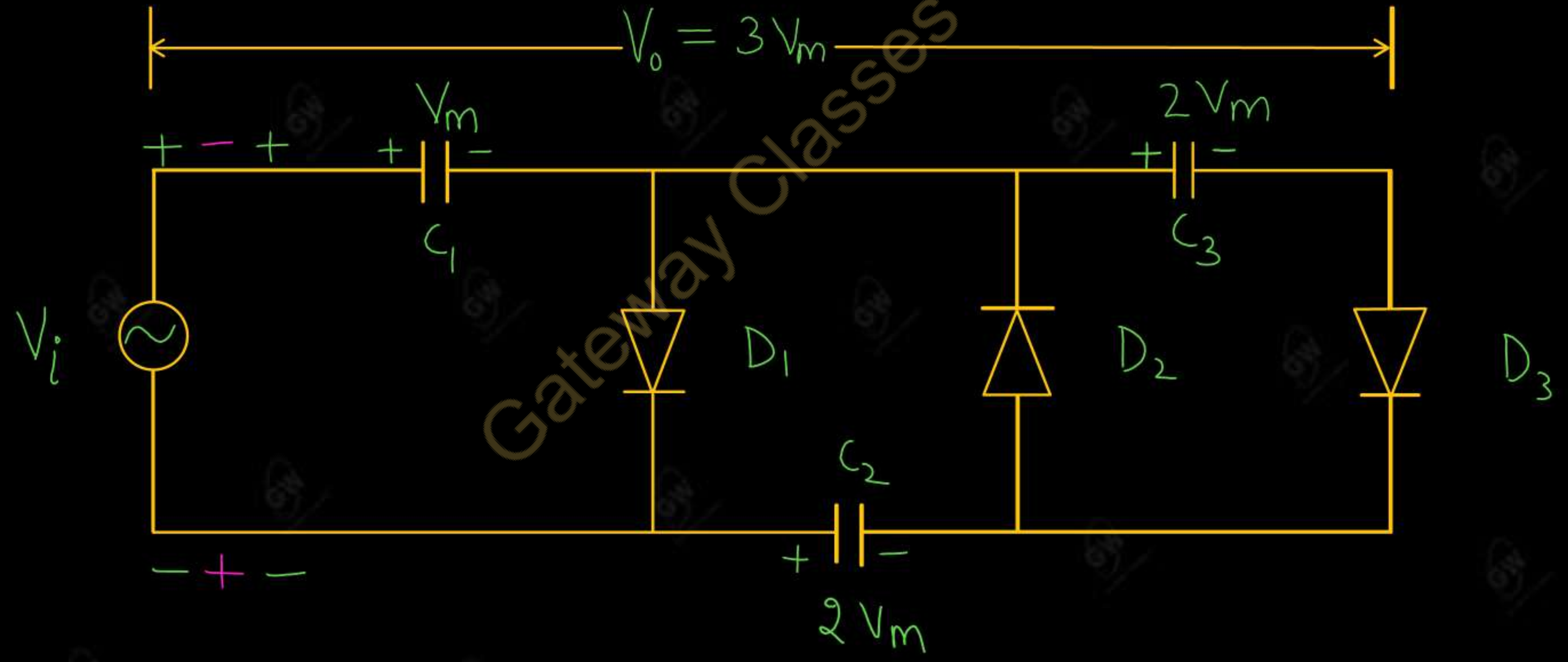
Final output voltage

$$V_o = 2V_m$$

Gateway Classes

Voltage Tripler

- A circuit that produces an output dc voltage equal to three times of the peak input ac voltage (V_m) is called voltage tripler.
- Three Diode and Three capacitors are needed for Voltage Tripler Circuit



Gateway Classes

Working

(ii) During positive Half cycle

- Diode D_1 and D_3 are forward biased (ON)
- Diode D_2 is reversed biased (OFF)
- So, capacitor C_1 is charged up to voltage V_m

$$V_{c_1} = V_m$$

(ii) During negative Half cycle

- Diode D_1 and D_3 are reversed biased (OFF)
- Diode D_2 is forward biased (ON)
- So, capacitor C_2 is charges up to voltage $2V_m$

$$V_{c_2} = 2V_m$$

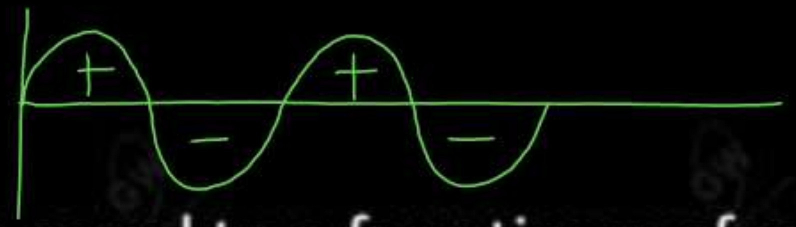
(ii) During second positive Half cycle

- Diode D_1 and D_3 are forward biased (ON)
- Diode D_2 is reversed biased (OFF)
- So the capacitor C_3 is charges up to voltage $2V_m$

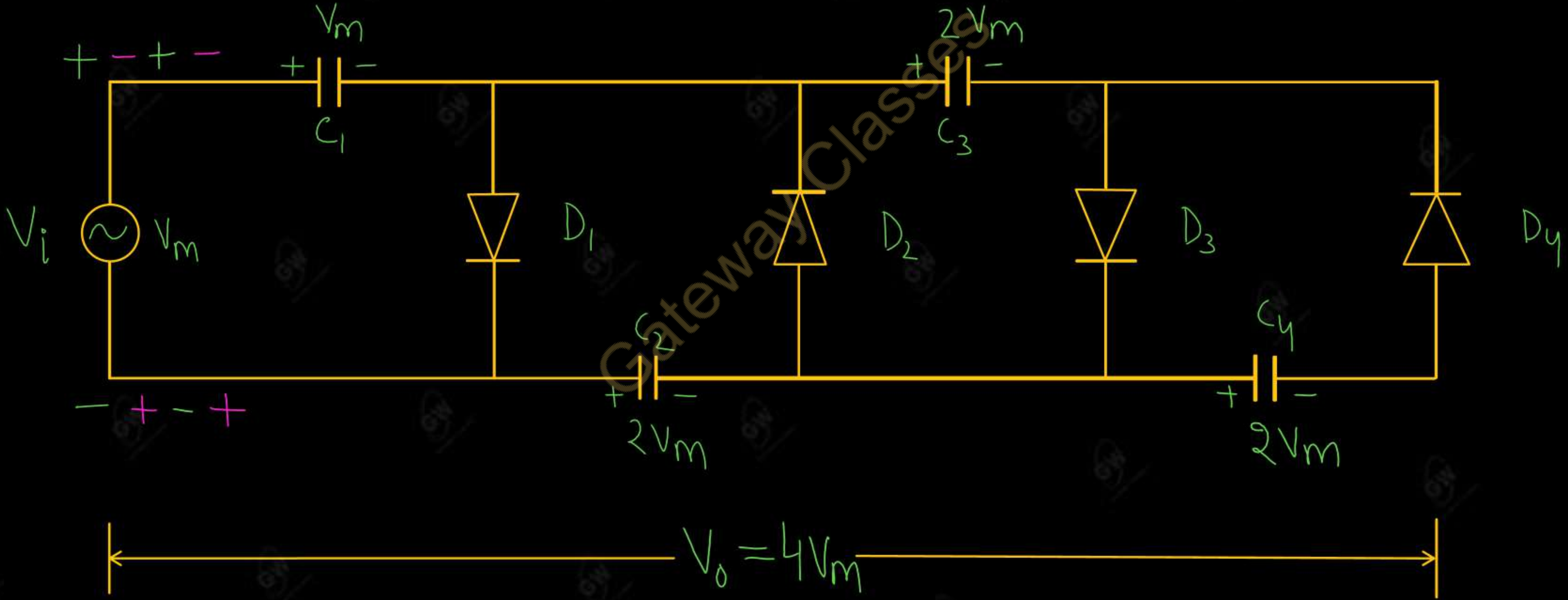
$$V_{c_3} = 2V_m$$

Gateway Classes

Voltage Quadrupler



- A circuit that produces an output dc voltage equal to a four times of peak input ac voltage (V_m) is called a Quadrupler
- Four Diode and four capacitors are needed for Voltage Quadrupler Circuit



Gateway Classes

Working

(ii) During positive Half cycle

- Diode D_1 and D_3 are forward biased (ON)
- Diode D_2 and D_4 are reversed biased (OFF)
- So, capacitor C_1 is charged up to voltage V_m

$$V_{c_1} = V_m$$

(ii) During negative Half cycle

- Diode D_1 and D_3 are reversed biased (OFF)
- Diode D_2 and D_4 are forward biased (ON)
- So, capacitor C_2 is charges up to voltage $2V_m$

$$V_{c_2} = 2V_m$$

(iii) During second positive Half cycle

- Diode D_1 and D_3 are forward biased (ON)
- Diode D_2 and D_4 are reversed biased (OFF)
- So the capacitor C_3 is charges up to voltage $2V_m$

$$V_{c_3} = 2V_m$$

(iv) During second negative Half cycle

- Diode D_1 and D_3 are reversed biased (OFF)
- Diode D_2 and D_4 are forward biased (ON)
- So the capacitor C_4 is charges up to voltage $2V_m$

$$V_{c_4} = 2V_m$$

Gateway Classes

UNIT - 1 : DPP - 10

Q.1 With help of a neat diagram, explain the working of a voltage doubler circuit. AKTU 2017

Q.2 Describe with the help of circuit diagram working of voltage Tripler circuit. AKTU 2016

Q.3 Define Voltage Multiplier. Draw the circuit and explain the working of voltage Tripler and Voltage Quadrupler circuit AKTU 2022

Gateway Classes

AKTU

B.Tech : I-Year

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 11

Today's Target

- Zener Diode
- Zener Diode as a Shunt Regulator
- DPP
- PYQs

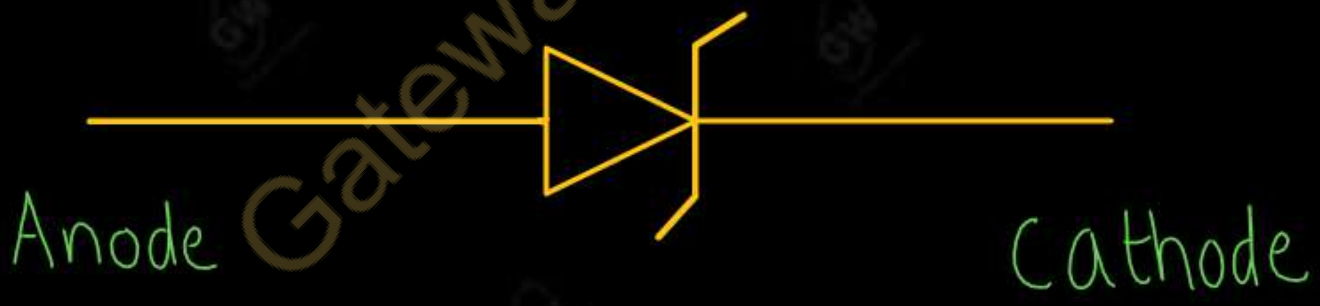
By Gulshan sir

Gateway Classes

Zener Diode

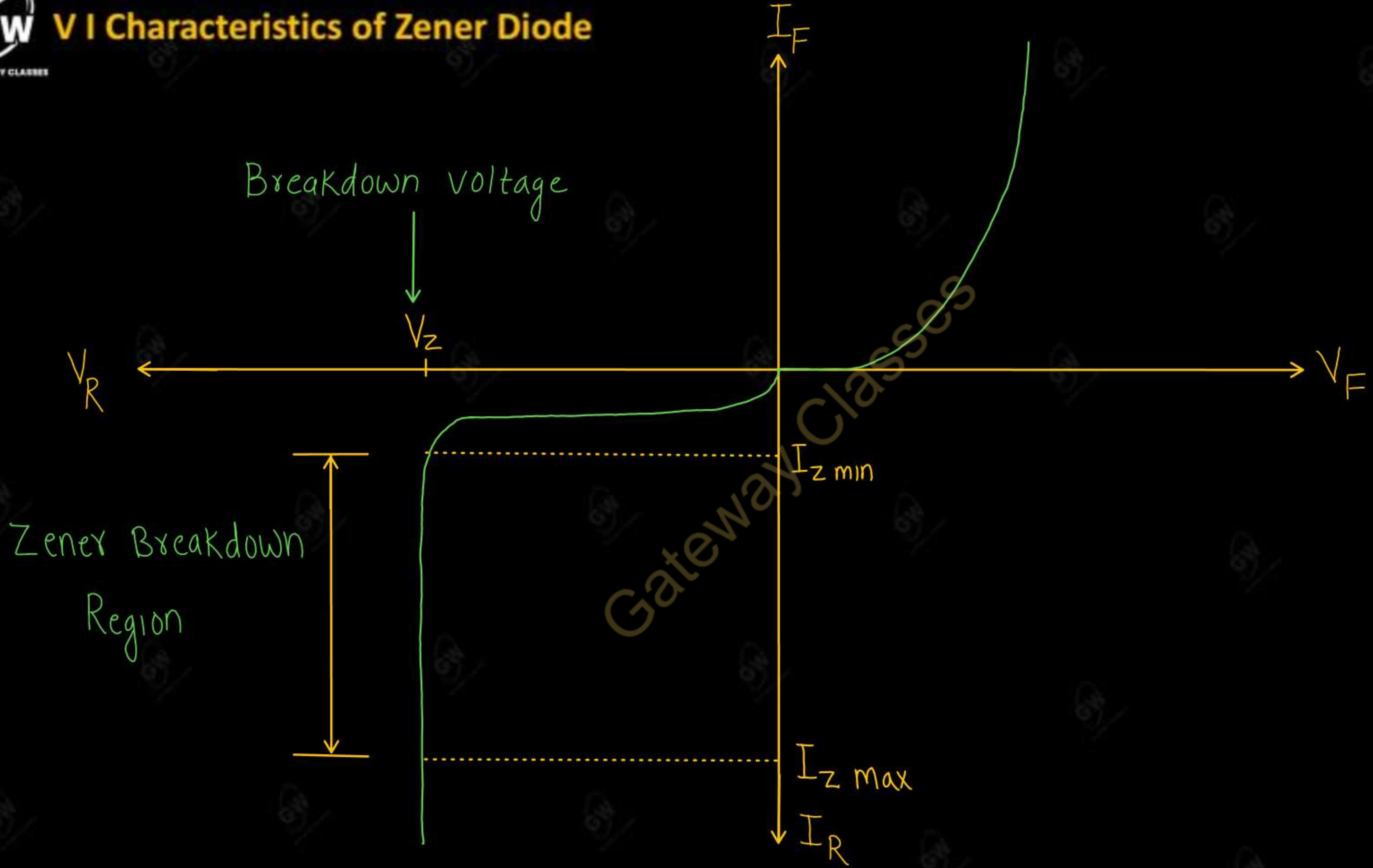
- A junction diode specially designed to operate only in the reverse breakdown region continuously is called a Zener diode
- Zener diode is a heavily doped P-N junction semiconductor diode.
- Zener diode in reversed biased condition (Zener breakdown region) is used as a voltage regulator

Circuit Symbol



Gateway Classes

V I Characteristics of Zener Diode



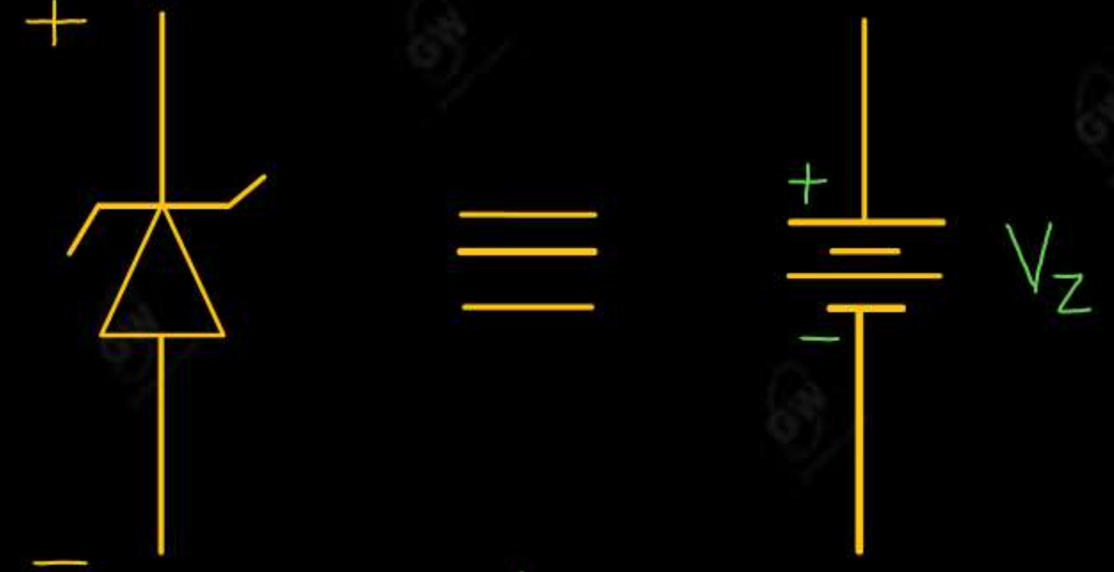
Gateway Classes

Equivalent circuit of Zener diode

(1) Reverse Biasing

(i) If $V < V_z$

(ii) If $V > V_z$



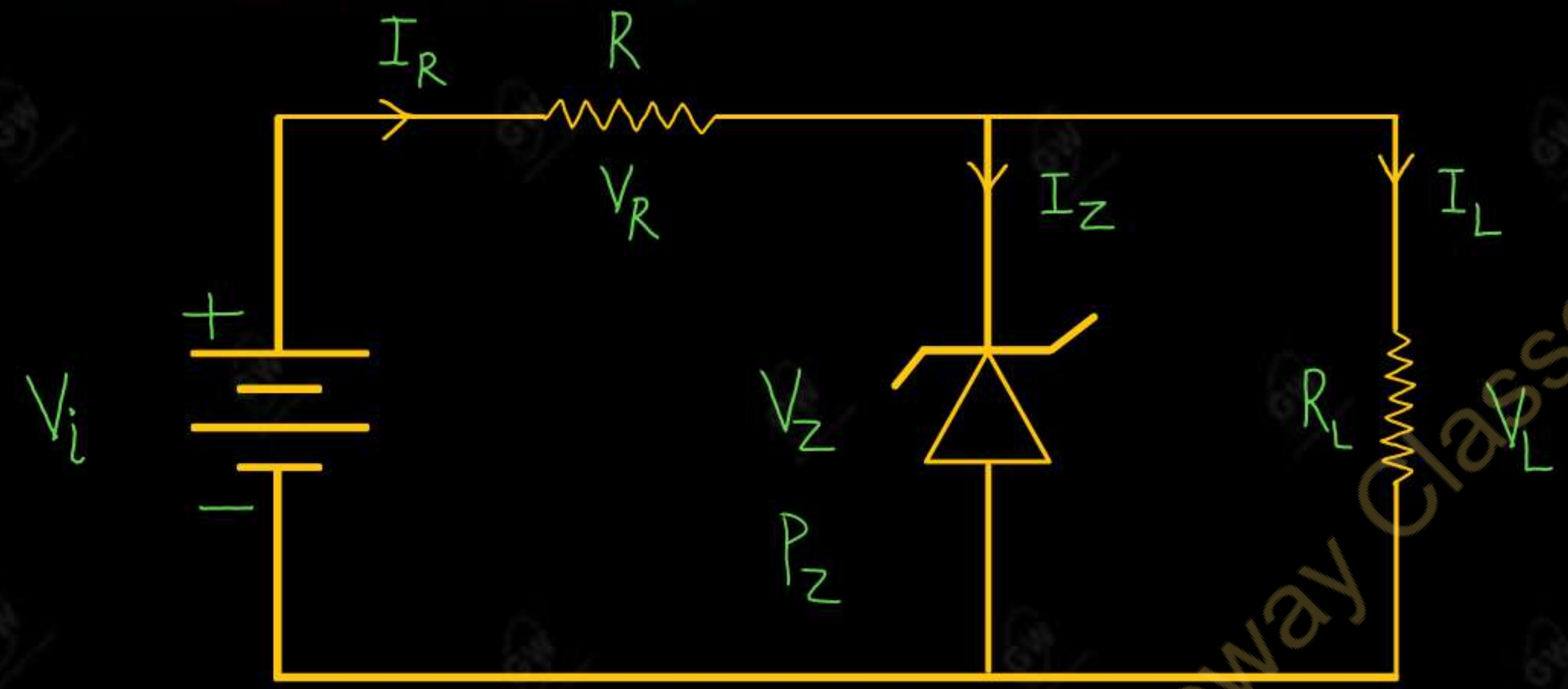
Voltage Regulator

(2) Forward Biasing

Gateway Classes

Zener Diode as a Shunt Voltage regulator

(1) Fixed V_i and Fixed R_L



Determine the state of Zener diode by removing it from the circuit and find voltage across load Resistor.

$$\boxed{V = V_L = \frac{R_L}{R_L + R} V_i}$$

(i) If $V > V_z$, diode is ON

(a) $V_L = V_z$

(b) $I_L = \frac{V_L}{R_L}$

(c) $V_R = V_i - V_Z$

(d) $I_R = \frac{V_R}{R}$

(e) $I_R = I_z + I_L$

(f) $P_Z = V_Z \; I_z$

Voltage Regulator

(i) If $V < V_z$, diode is OFF

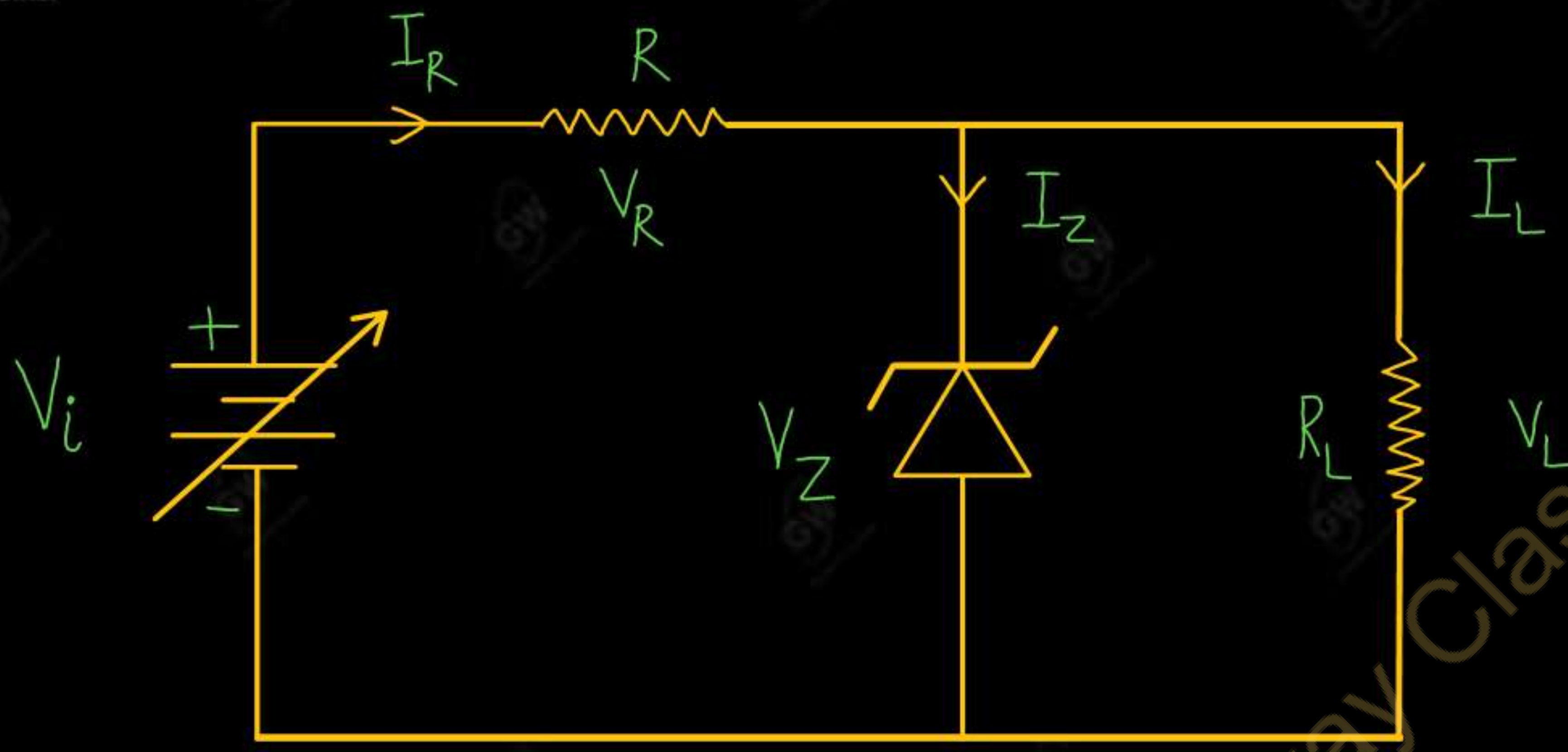
(a) $V_L \neq V_z$

(b) $I_R = I_L$

(c) $I_z = 0$

Not Voltage Regulator

(2) Variable V_i and Fixed R_L



Gateway Classes

(i) When V_i increases

- I_R increases
- I_L is constant and I_Z increases
- Voltage drop (V_R) across R increases
- So, output voltage remain constant

(ii) When V_i decreases

- I_R decreases
- I_L is constant and I_Z decreases
- Voltage drop across R decreases
- So, output voltage remain constant

Range of V_i that will maintain the Zener diode in "ON" state

(i) Minimum Voltage

$$V_L = \frac{R_L}{R + R_L} V_i$$

Put $V_L = V_Z$

$$V_Z = \frac{R_L}{R+R_L} V_{i\,min}$$

$$\boxed{V_{i\,min} = \frac{(R+R_L)\,V_Z}{R_L}}$$

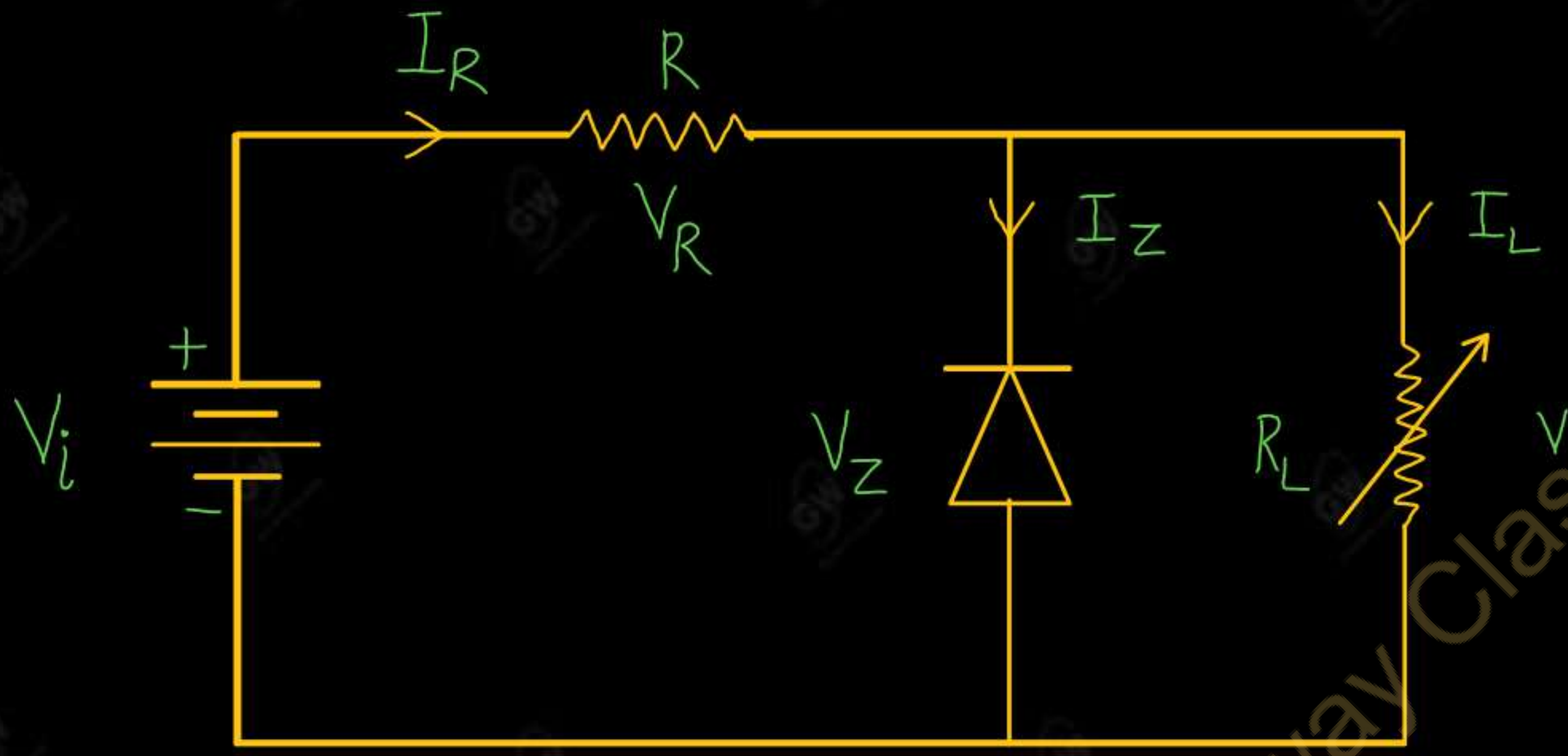
(ii) Maximum Voltage

$$I_{R\,max} = I_{ZM} + I_L$$

$$\boxed{V_{i\,max} = I_{R\,max} R + V_Z}$$

Gateway Classes

(3) Fixed V_i and Variable R_L



(i) When R_L increases

- I_L decreases
- I_R is constant and I_Z increases
- Voltage drop (V_R) across R is constant
- so, output voltage remain constant

(ii) When R_L decreases

- I_L increases
- I_R constant and I_Z decreases
- Voltage drop (V_R) across R is constant
- So, output voltage remain constant

Gateway Classes

Range of R_L that will maintain the Zener diode in "ON" state

(i) Minimum Load Resistance

$$V_L = \frac{R_L}{R_L + R} V_i$$

Put $V_L = V_Z$

$$V_Z = \frac{R_{L\,min}}{R_{L\,min} + R} V_i$$

$$\boxed{R_{L\,min} = \frac{V_Z R}{V_i - V_Z}}$$

$$\boxed{I_{L\,max} = \frac{V_Z}{R_{L\,min}}}$$

(ii) Maximum Load Resistance

$$\boxed{I_{L\,min} = I_R - I_{ZM}}$$

$$\boxed{R_{L\,max} = \frac{V_Z}{I_{L\,min}}}$$

Gateway Classes

Q.1 For the zener diode network shown in figure

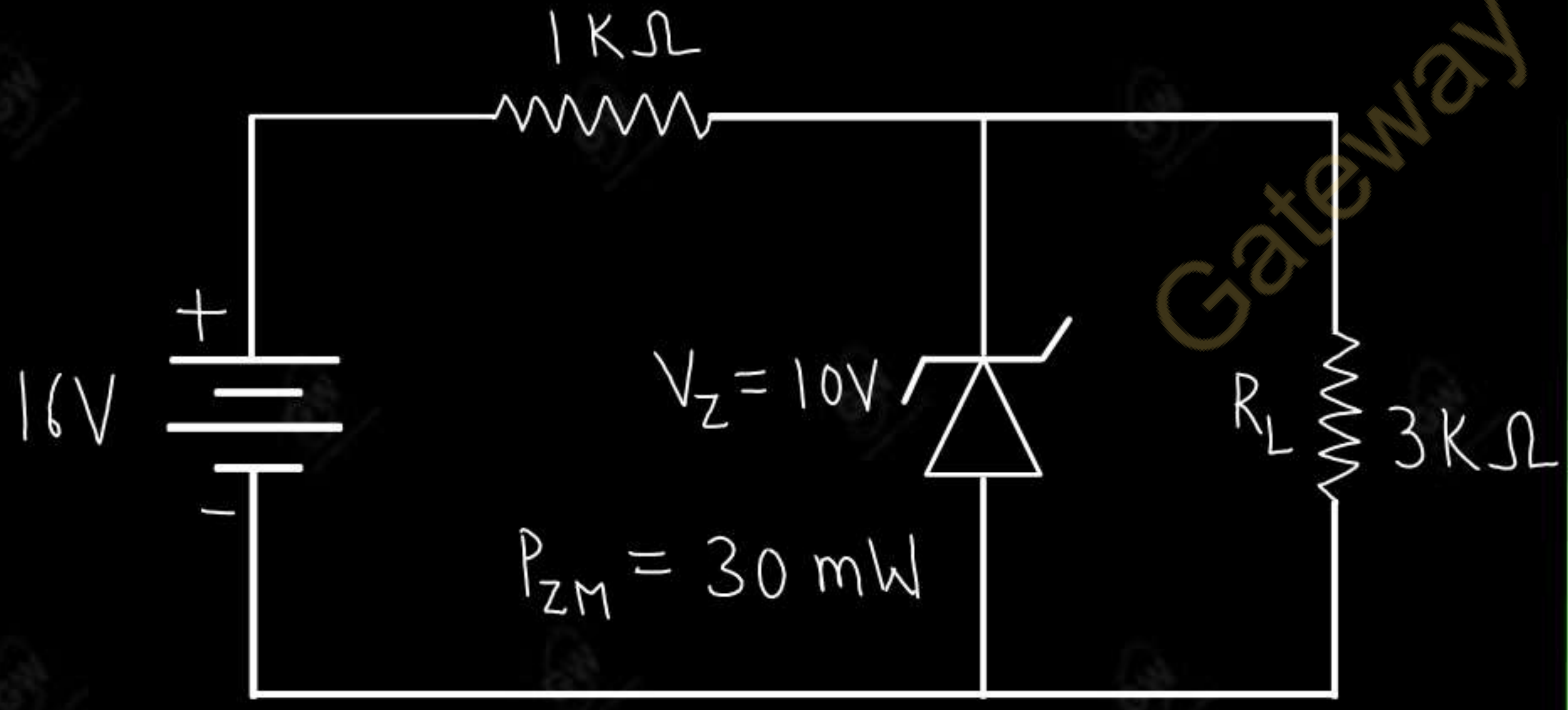
Determine

(i) Load voltage (V_L)

(ii) Voltage across series resistor (V_R)

(iii) Current through diode (I_Z)

(iv) Power dissipated by zener diode (P_Z)



Given

$V_i = 16\,V$

$R = 1\,K\Omega$

$V_Z = 10\,V$

$R_L = 3\,\Omega$

$$V = \frac{R_L}{R + R_L} V_i$$

$$V = \frac{3}{1+3} \times 16$$

$V = 12\,V$

Since, $V > V_Z$

So, Diode is ON

(i) $V_L = V_Z$

$$\boxed{V_L = 10\,V}$$

(ii) $V_R = V_i - V_Z$

$= 16 - 10$

$$\boxed{V_R = 6\,V}$$

(iii)

$$I_R = \frac{V_R}{R}$$

$$I_R = \frac{6}{1}$$

Gateway Classes

$I_R = 6\,mA$

$I_L = \frac{V_L}{R_L}$

$I_L = \frac{10}{3}$

$I_L = 3.33\,mA$

$I_Z = I_R - I_L$

$I_Z = 6 - 3.33$

$\boxed{I_Z = 2.67\,mA}$

(IV) $P_Z = V_Z I_Z$

$= 10 \times 2.67$

$\boxed{P_Z = 26.7\,mW}$

Gateway Classes

Q.2 For the zener diode network shown in figure

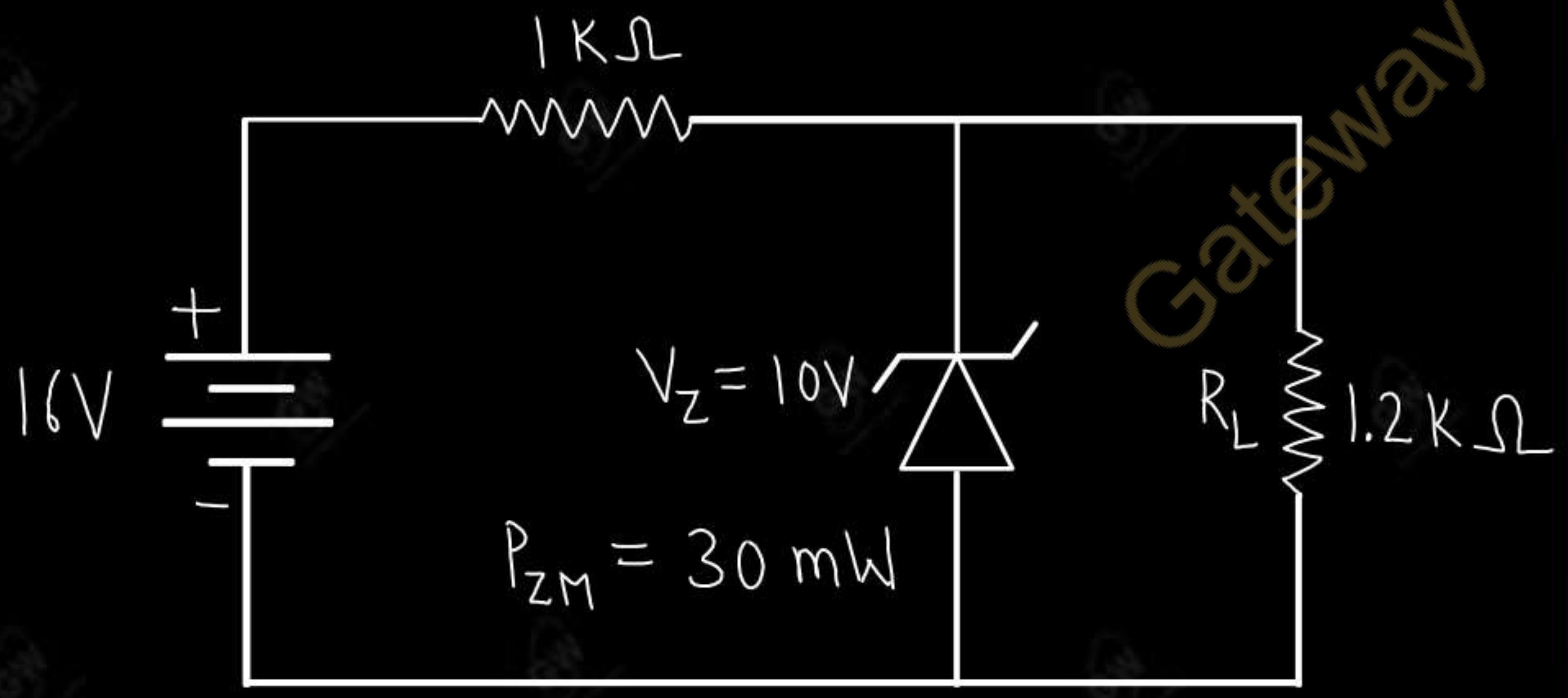
Determine

(i) Load voltage

(ii) Voltage across series resistor

(iii) Current through diode

(iv) Power dissipated by zener diode



Given

$V_i = 16\,V$

$R = 1\,K\Omega$

$V_Z = 10\,V$

$R_L = 1.2\,K\Omega$

$$V = \frac{R_L}{R_L + R_L} V_i$$

$$= \frac{1.2}{1.2+1} 16$$

$$= \frac{1.2}{2.2} \times 16$$

$V = 8.73\,V$

Since $V < V_Z$

So, Diode is OFF

(i) $V_L = V$

$$\boxed{V_L = 8.73}$$

(ii) $V_R = V_i - V_L$

$= 16 - 8.73$

$$\boxed{V_R = 7.27\,V}$$

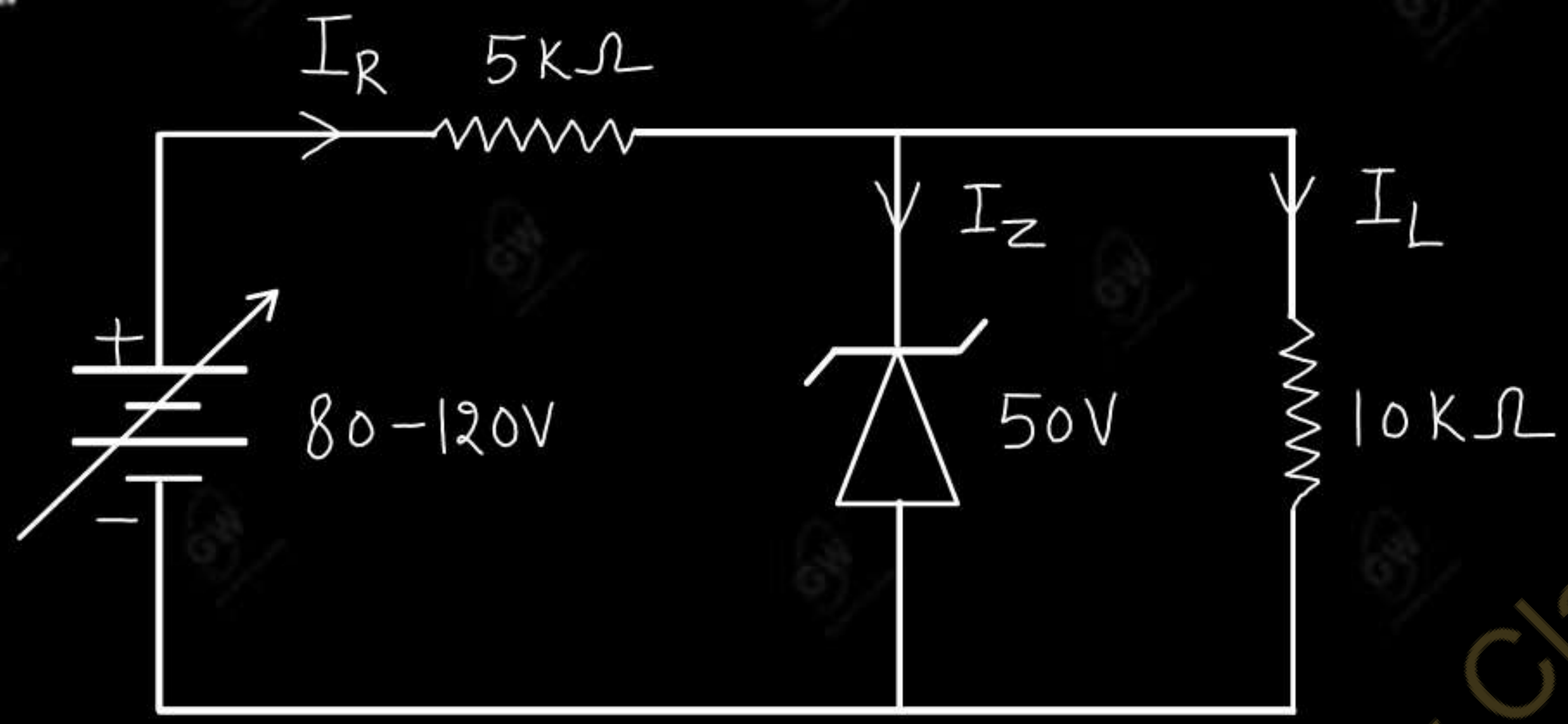
(iii) $\boxed{I = 0}$

(iv) $\boxed{P_Z = 0}$

Gateway Classes

Q 3 For the circuit shown in figure, find the maximum and minimum values of zener diode current.

AKTU-2021, 2025



Given

$R = 5\,k\Omega$

$R_L = 10\,k\Omega$

$V_Z = 50V$

$V_L = 50V$

$I_L = \frac{V_L}{R_L}$

$I_L = \frac{50}{10}$

$I_L = 5\,mA$

When $V_i = 80V$

$V_R = V_i - V_Z$

$= 80 - 50$

$= 30V$

$I_{R\,min} = \frac{V_R}{R}$

$= \frac{30}{5}$

$I_{R\,min} = 6\,mA$

$I_{Z\,min} = I_{R\,min} - I_L$

$= 6 - 5$

$\boxed{I_{Z\,min} = 1\,mA}$

Gateway Classes

When $V_i = 120V$

$$V_R = V_i - V_Z$$

$$= 120 - 50$$

$$= 70\,V$$

$$I_{R\,max} = \frac{V_R}{R}$$

$$= \frac{70}{5}$$

$$= 14\,mA$$

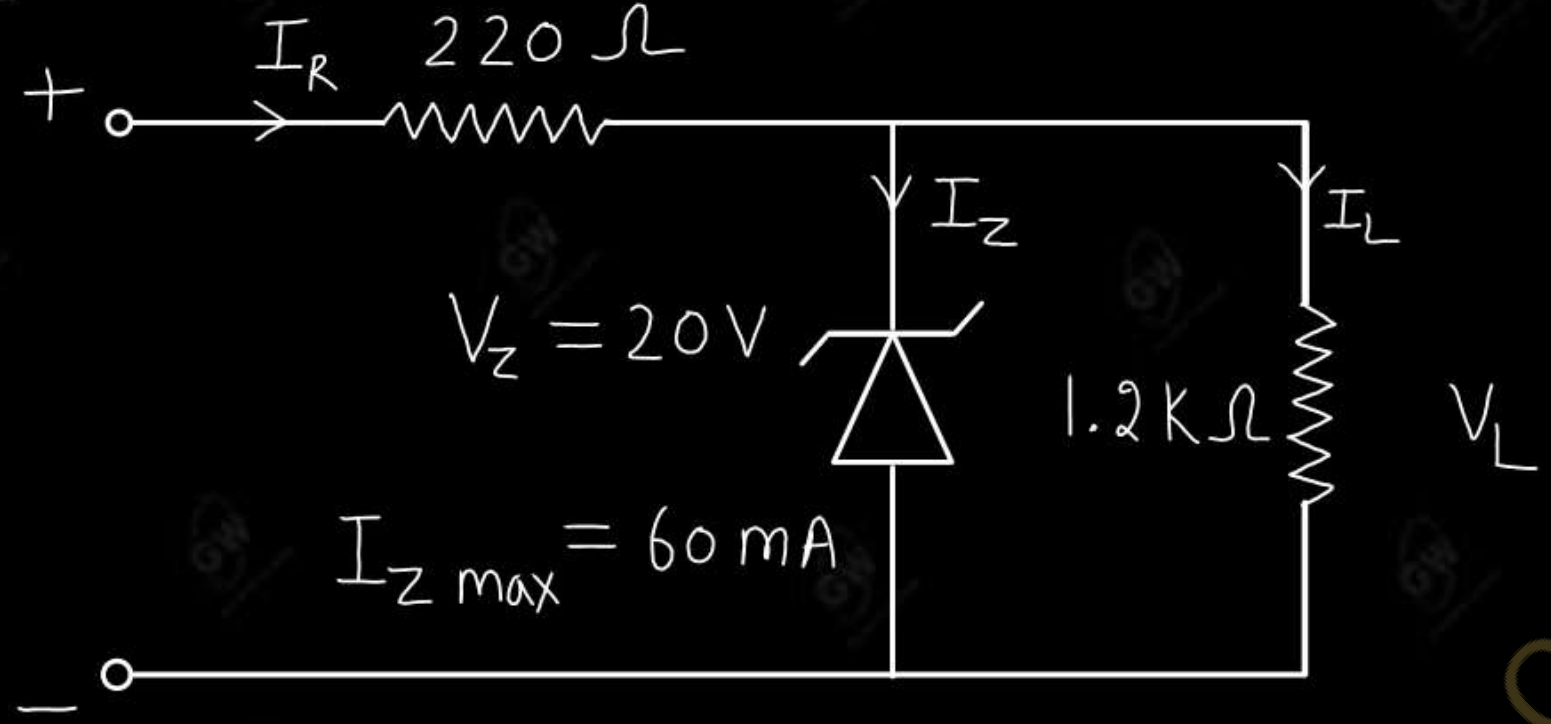
$$I_{Z\,max} = I_{R\,max} - I_L$$

$$= 14 - 5$$

$$\boxed{I_{Z\,max} = 9\,mA}$$

Gateway Classes

Q 4 Determine the range of V_i that will maintain the zener diode in "ON" state. **AKTU-2017-18**



Given

$R = 220\,\Omega$

$R_L = 1200\,\Omega$

$I_{Z\,max} = 60\,mA$

$V_Z = 20V$

$V_L = 20V$

Minimum Voltage

$$V_{i\,min} = \frac{(R_L + R)V_Z}{R_L}$$

$$= \frac{1200 + 220}{1200} \times 20$$

$$= \frac{1420 \times 20}{1200}$$

$$\boxed{V_{i\,min} = 23.67V}$$

Gateway Classes

Maximum Voltage

$$I_L = \frac{V_L}{R_L}$$

$$= \frac{20}{1.2 \times 10^3}$$

$$I_L = 16.67\ mA$$

$$I_{R\,max} = I_{Z\,max} + I_L$$

$$= 60 + 16.67$$

$$I_{R\,max} = 76.67\ mA$$

$$V_{i\,max} = I_{R\,max} R + V_Z$$

$$= 76.67 \times 10^{-3} \times 220 + 20$$

$$\boxed{V_{i\,max} = 36.87\ V}$$

Range of V_i

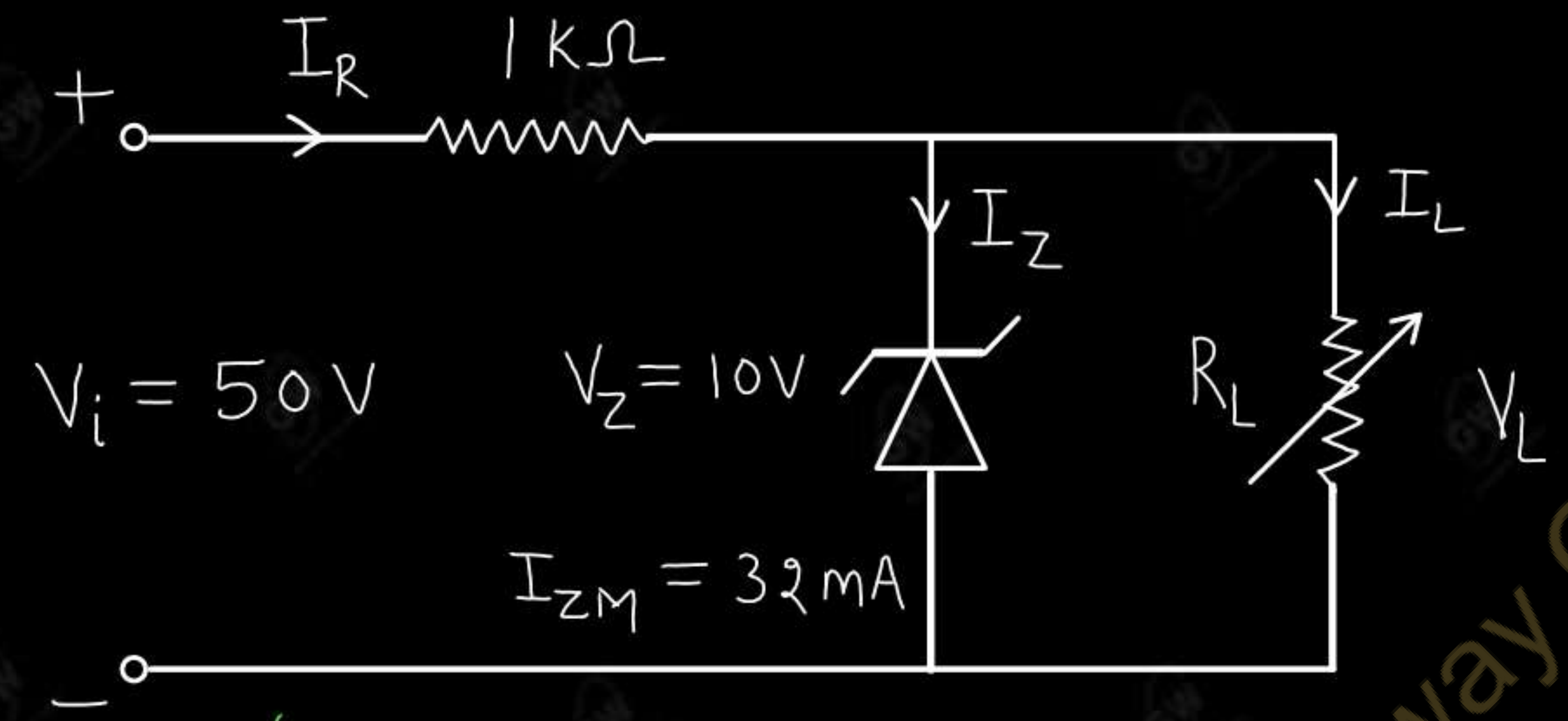
23.67 V to 36.87 V

Gateway Classes

Q 5 (i) Find the range of R_L and I_L that will maintain a constant output 10V.

(ii) Also determine the maximum wattage rating of the zener diode for given circuit.

AKTU-2017-18



Given

$R = 1\,K\Omega$

$V_i = 50V$

$V_Z = 10V$

$V_L = 10V$

$I_{ZM} = 32\,mA$

$$V_R = V_i - V_Z$$
$$= 50 - 10$$
$$= 40\,V$$

$$I_R = \frac{V_R}{R}$$
$$= \frac{40}{1}$$
$$= 40\,mA$$

$$I_{L\,min} = I_R - I_{ZM}$$
$$= 40 - 32$$
$$\boxed{I_{L\,min} = 8\,mA}$$

$$R_{L\,max} = \frac{V_L}{I_{L\,min}}$$
$$= \frac{10}{8}$$
$$\boxed{R_{L\,max} = 1.25\,K\Omega}$$

Gateway Classes

Minimum value of R_L

$$R_{L\,min} = \frac{V_Z R}{V_i - V_Z}$$

$$= \frac{10 \times 1}{50 - 10}$$

$$= \frac{10}{40}$$

$$\boxed{R_{L\,min} = 0.25\,K\Omega}$$

$$I_{L\,max} = \frac{V_L}{R_{L\,min}}$$

$$= \frac{10}{0.25}$$

$$\boxed{I_{L\,max} = 40\,mA}$$

Hence

Range of R_L

$$\boxed{0.25\,K\Omega \text{ to } 1.25\,K\Omega}$$

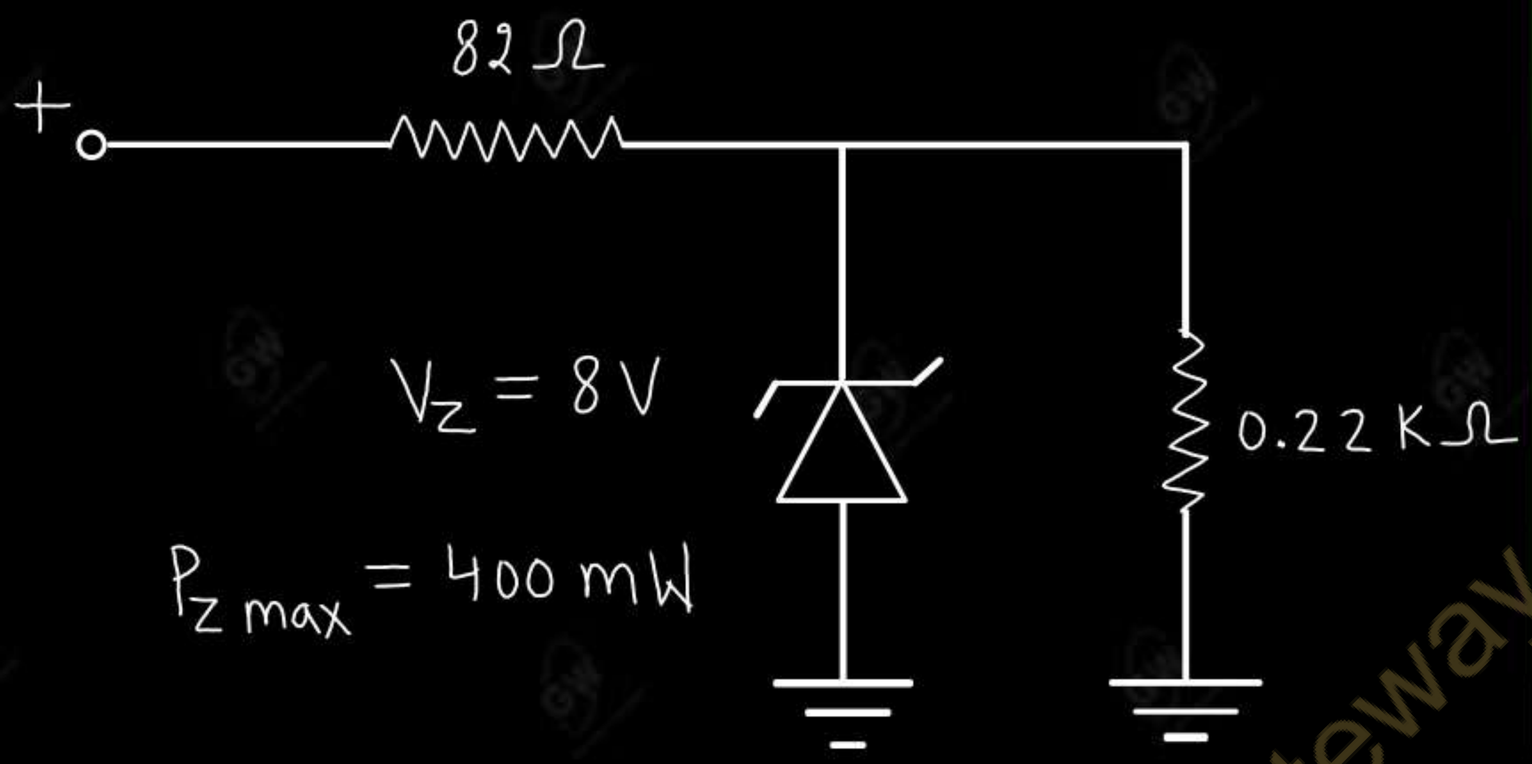
Range of I_L

$$\boxed{8\,mA \text{ to } 40\,mA}$$

Gateway Classes

Q 6 Determine the V-I characteristics of zener diode. Determine the network of figure given below, determine the range of V_i that will maintain V_L at 8V and not exceeded the maximum power rating of the zener diode.

AKTU-2022



$V_Z = 8\,V$

$P_{Z\,max} = 400\,mW$

Given

$R = 82\,\Omega = 0.082\,K\Omega$

$R_L = 0.22\,K\Omega$

$V_Z = 8V$

$V_L = 8V$

$I_L = \frac{V_L}{R_L}$

$= \frac{8}{0.22}$

$= 36.36\,mA$

Minimum votage

$V_{i\,min} = \frac{(R_L + R)\,V_Z}{R_L}$

$= \frac{(0.22 + 0.082) \times 8}{0.22}$

$\boxed{V_{i\,min} = 10.98\,V}$

$P_{Z\,max} = V_Z\, I_{Z\,max}$

$400 = 8 \times I_{Z\,max}$

$I_{Z\,max} = 50\,mA$

$I_{R\,max} = I_{Z\,max} + I_L$

$= 50 + 36.36$

$$I_{R\,max} = 86.36\,mA$$

Maximum voltage

$$V_{i\,max} = I_{R\,max} R + V_Z$$

$$= 86.36 \times 10^{-3} \times 82 + 8$$

$$\boxed{V_{i\,max} = 15.08\,V}$$

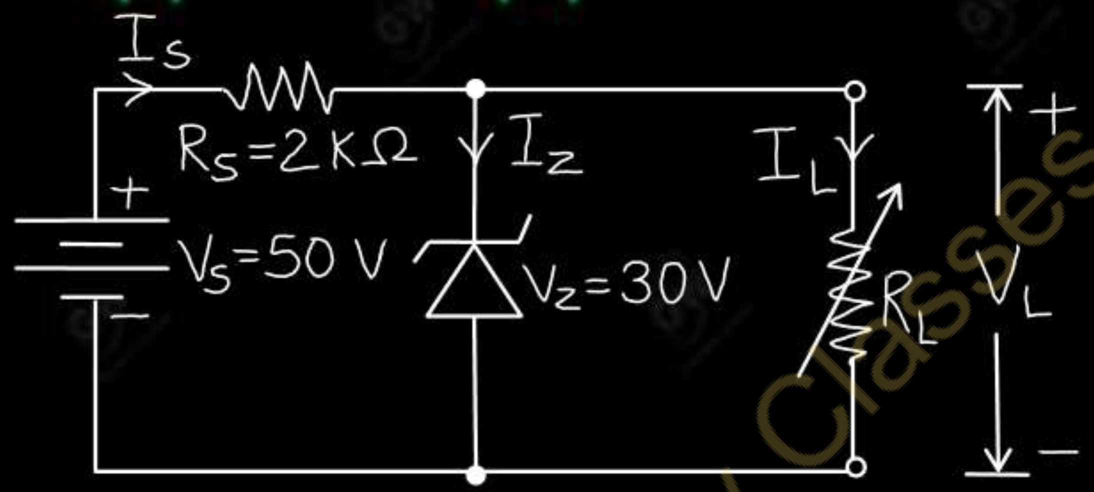
Range of V_i

$$\boxed{10.98 \text{ to } 15.08\,V}$$

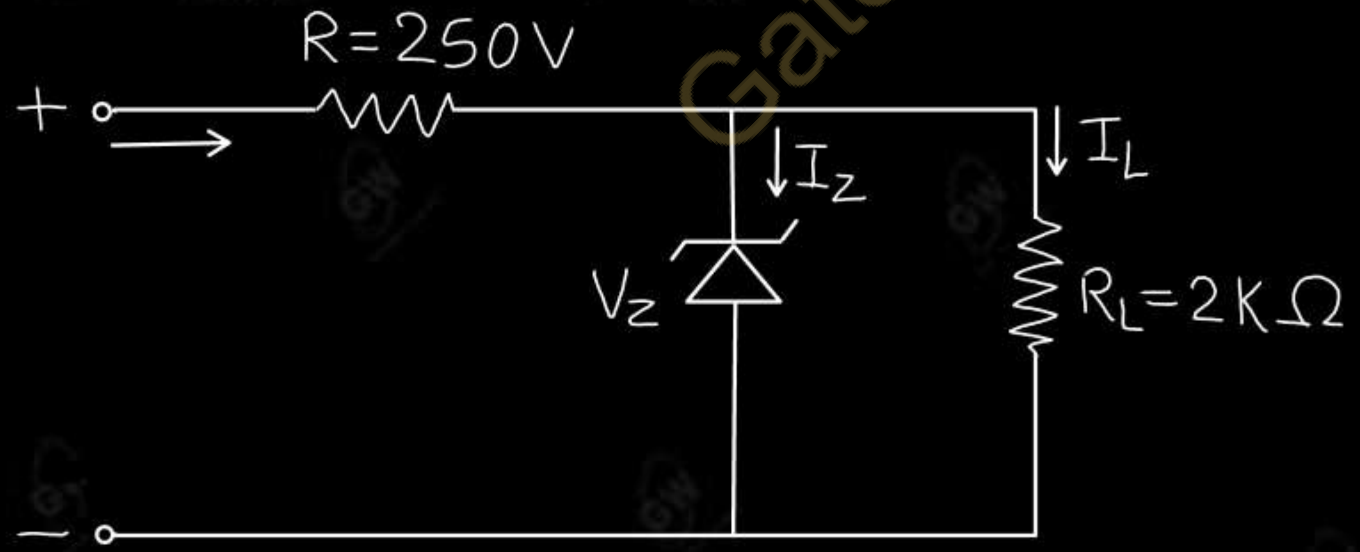
Gateway Classes

UNIT - 1 : DPP - 11

Q.1 Figure shows the circuit of zener diode shunt regulator. Determine the zener current if the load resistance R_L is (i) 30 KΩ (ii) 10 KΩ (iii) 3 KΩ



Q.1 Determine the range of input voltage (V_s) for the zener diode to remain in ON state shown in following figure. $V_z = 20V$, $I_{z\,max} = 50\ mA$, $R_z = 0\,\Omega$ **AKTU-2023**



Gateway Classes

AKTU

B.Tech : I-Year

GATEWAY CLASSES

Electronics Engg.

UNIT -1 : (i) Semiconductor Diode (ii) Diode Application

(iii) Special Purpose two terminal Devices

Lecture - 12

Gateway Classes

Today's Target

- Special Purpose two terminal Devices
- DPP
- PYQs

By Gulshan sir

UNIT-1

Semiconductor Diode

Depletion layer, V-I characteristics, ideal and practical Diodes, Diode Equivalent Circuits, Zener Diodes breakdown mechanism (Zener and avalanche)

Diode Application

Diode Configuration, Half and Full Wave rectification, Clippers, Clampers, Zener diode as shunt regulator, Voltage-Multiplier Circuits

Special Purpose two terminal Devices

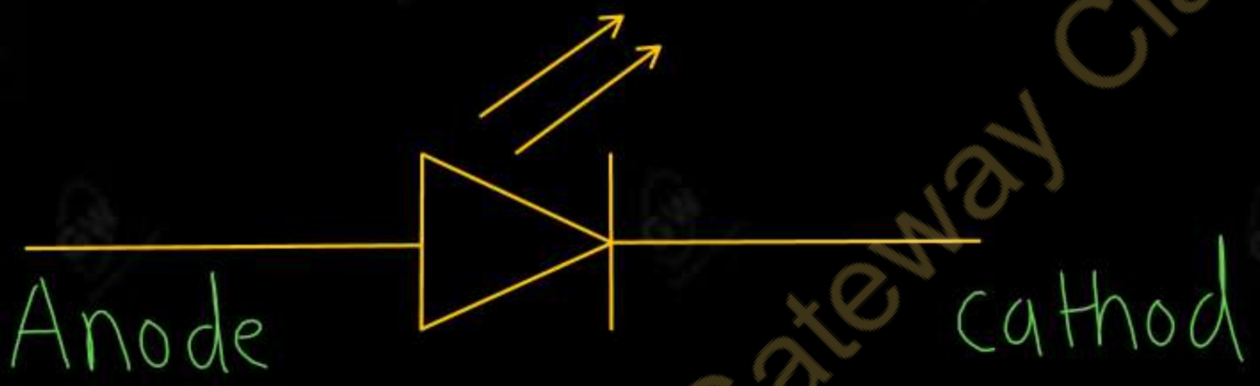
Light-Emitting Diodes, Photo Diodes, Varactor Diodes, Tunnel Diodes.

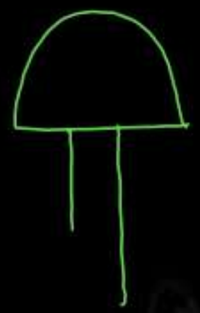
Gateway Classes

Light Emitting Diode (LED)

- Light emitting diode is a special diode that emit light under forward biasing
- LEDs are made of Gallium Arsenide (Ga As), Gallium Arsenide Phosphide (Ga As P) and Gallium phosphide (Ga P)
- Silicon and Germanium are not used since they are essentially heat producing materials and are very poor in producing light.

Symbol of LED



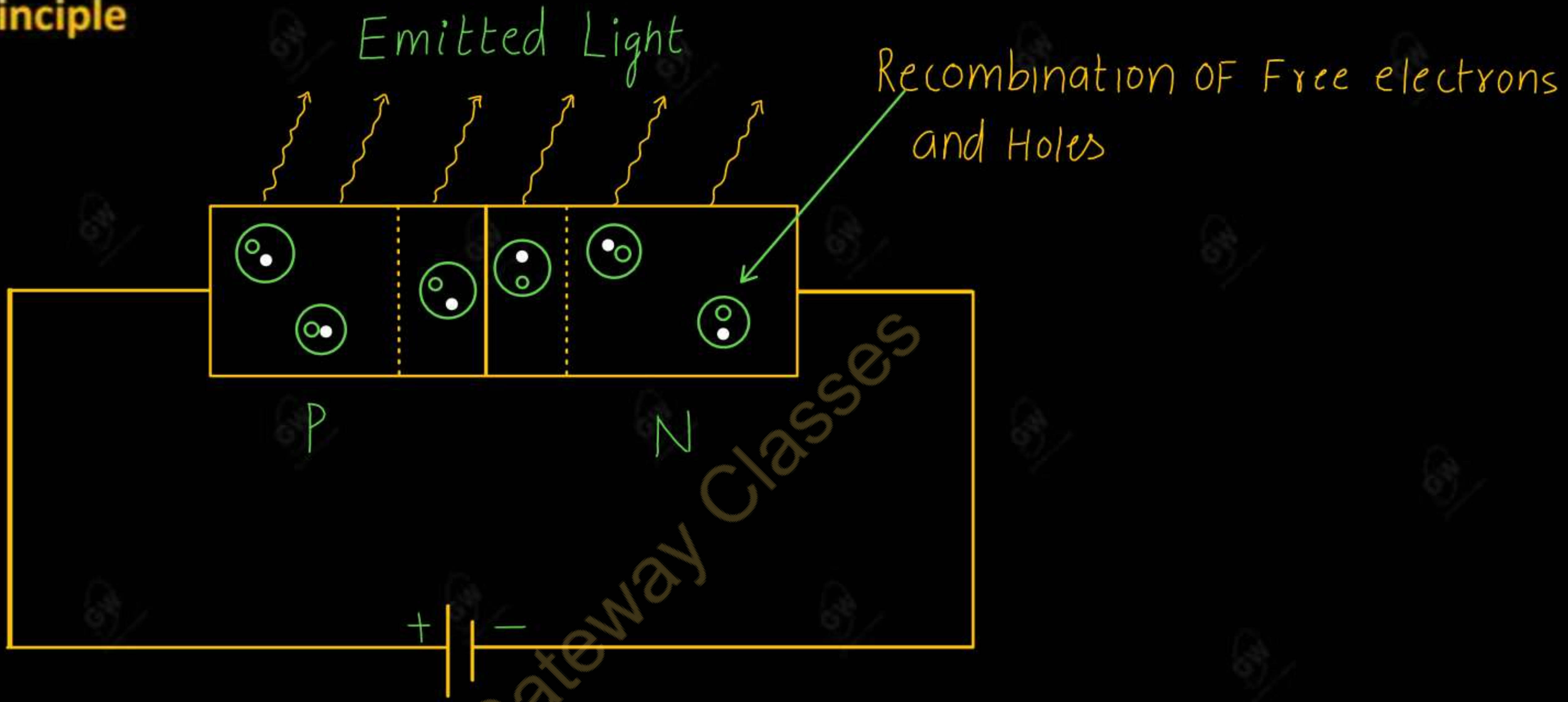


Gateway Classes

LED Materials and its color

S. No	Material used	Color of emitted light
1	GaAs	Infrared (IR)
2	GaAsP	Red or Yellow
3	GaP	Red or green

Working Principle



- When LED is forward biased then holes in P-region and electron in N-region start to cross the junction and recombine with each other.
- These free electrons reside in the conduction band and hence at a higher energy level than the holes in the valence band

Gateway Classes

When the recombination take place, these electrons return back to the valence band which is at a lower energy level than the conduction band

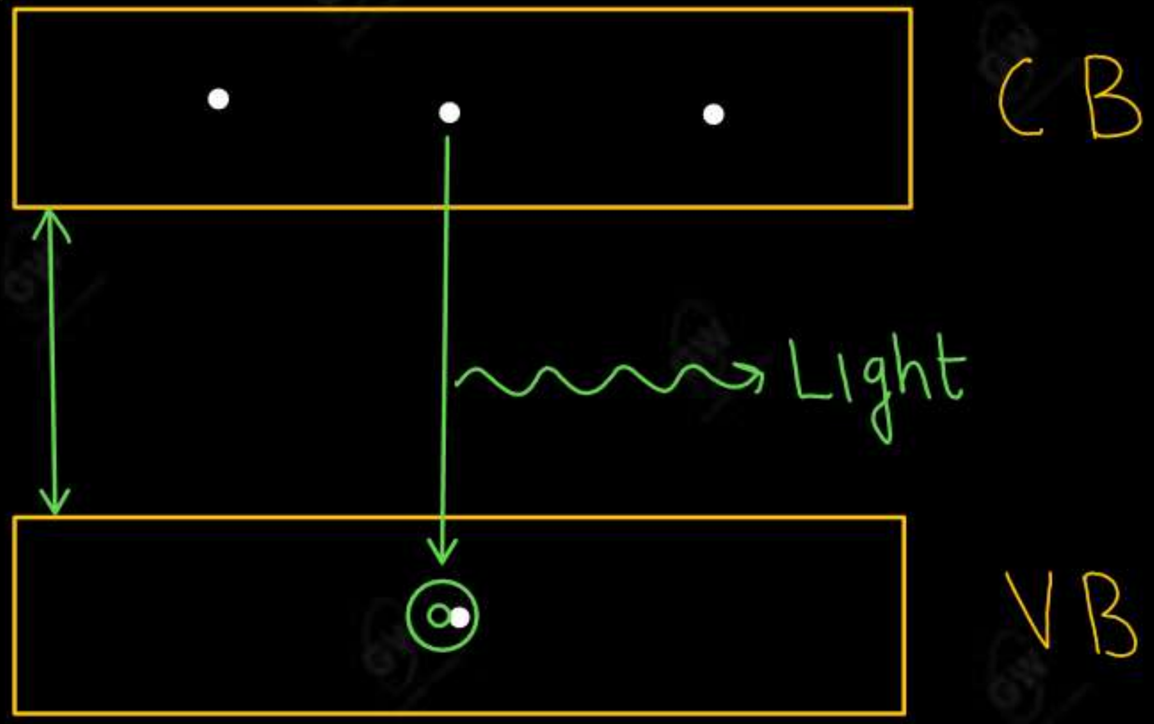
- While returning back, the recombining electrons give away the excess energy in the form of light. This process is known as electroluminescence. This is the principle of operational LED
- Simple Diode (diode of Si or Ge) produce heat in recombination process. But LED produce light in recombination process

Applications

(1) Used in digital clocks

(2) Used in calculators

(3) Used in mobile, TV display



(4) Light source in optical fiber communication

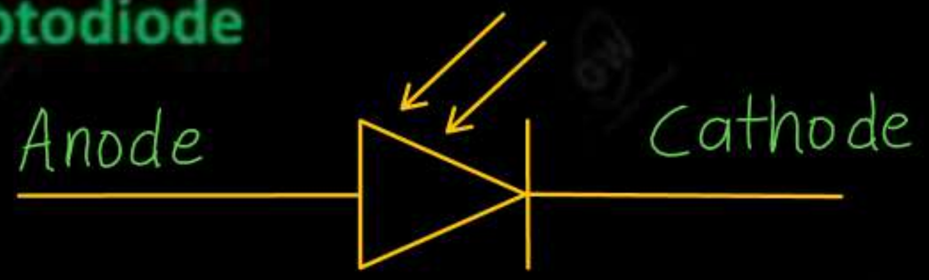
(5) Used in Seven Segment Display

Gateway Classes

Photo Diode

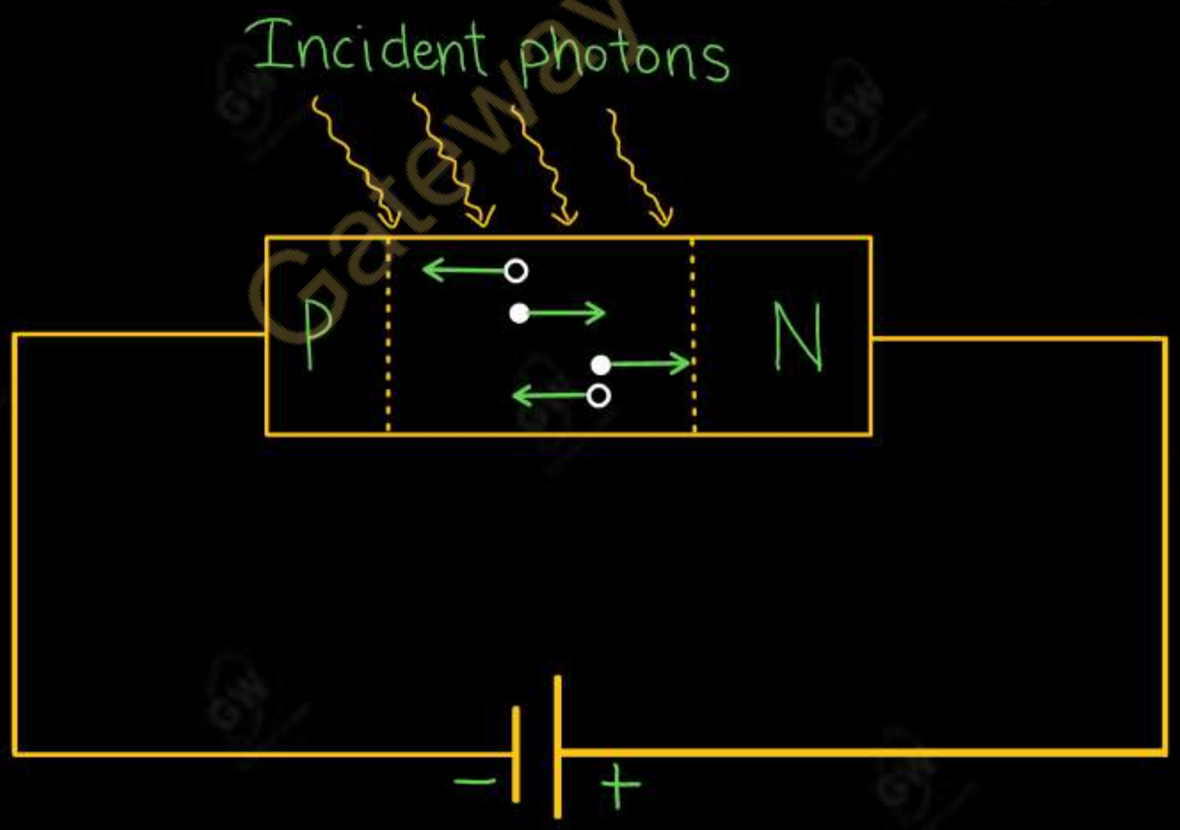
- A photodiode is a semiconductor P-N junction device that converts light into electric current.
- Photodiodes are sensitive to light, making them suitable for detecting optical signal.

Symbol of photodiode



Working

- When a light or photon is used to illuminate PN junction then photon hits the immobile ions present in the depletion layer.



Gateway Classes

- If the energy of photon is greater than 1.1eV then covalent bond will break. So, electron hole pair are generated.
- Due to electric field, electron hole pairs move away from the junction. Hence, holes move towards anode and electron move towards cathode to produce photocurrent. This entire process is known as photoelectric effect.
- Types of photodiode : (1) P-N photodiode (2) P-I-N photodiode (3) Avalanche photodiode

Applications

(1) solar cell panels

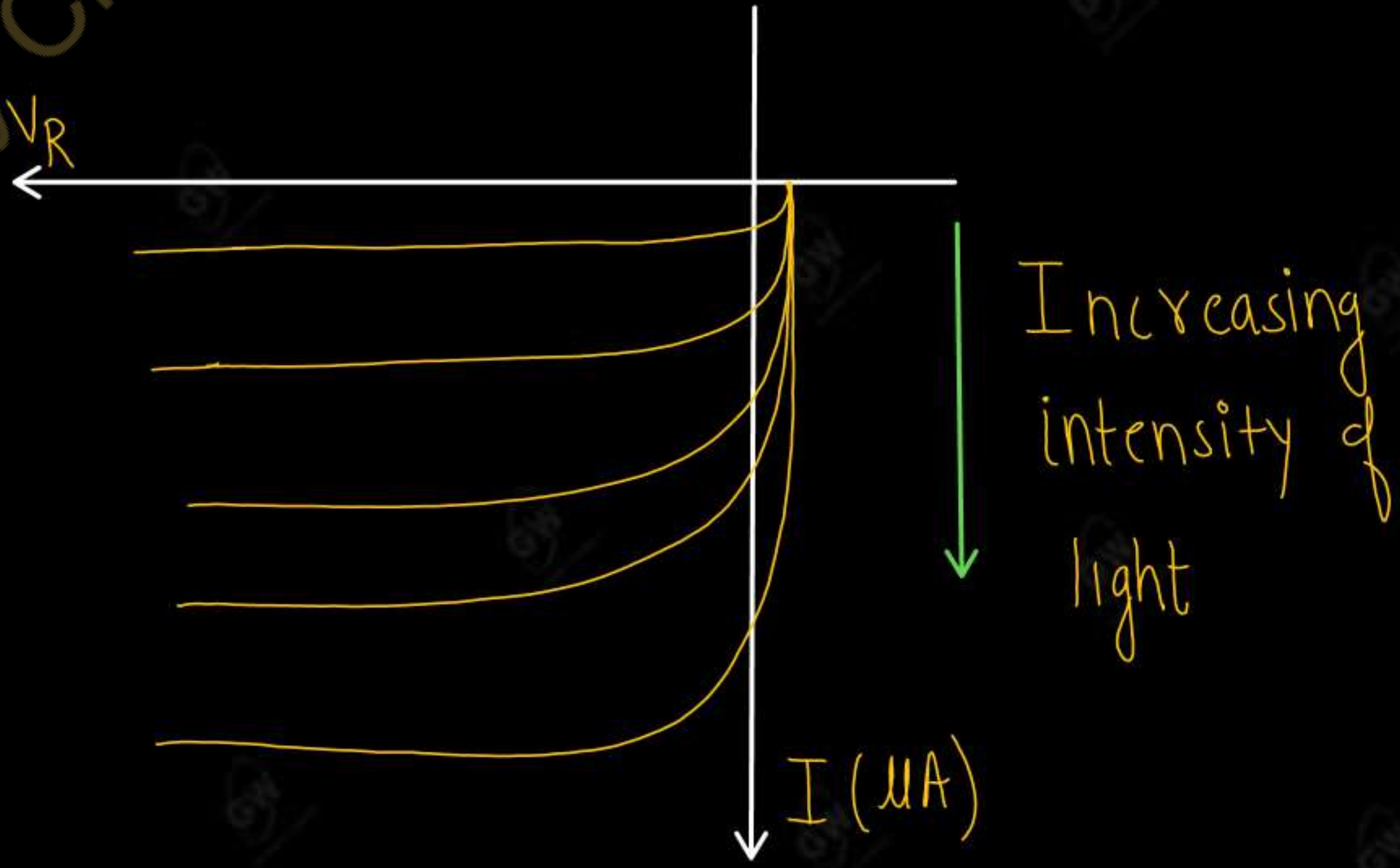
(2) optical communication system

(3) Medical devices

(4) camera light

(5) Light sensor

(6) optical switches

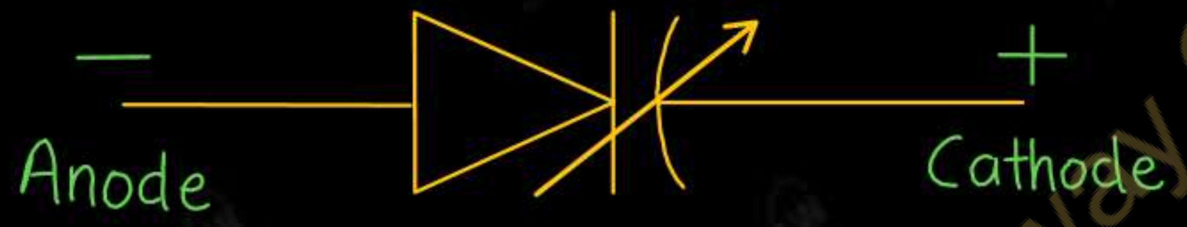


Gateway Classes

Varactor Diode

- A varactor diode is a reversed biased PN junction diode that exhibits variable capacitance
- It is used as a voltage-variable capacitor.
- It is also known as a Varicap, Voltcap or Tuning diode.

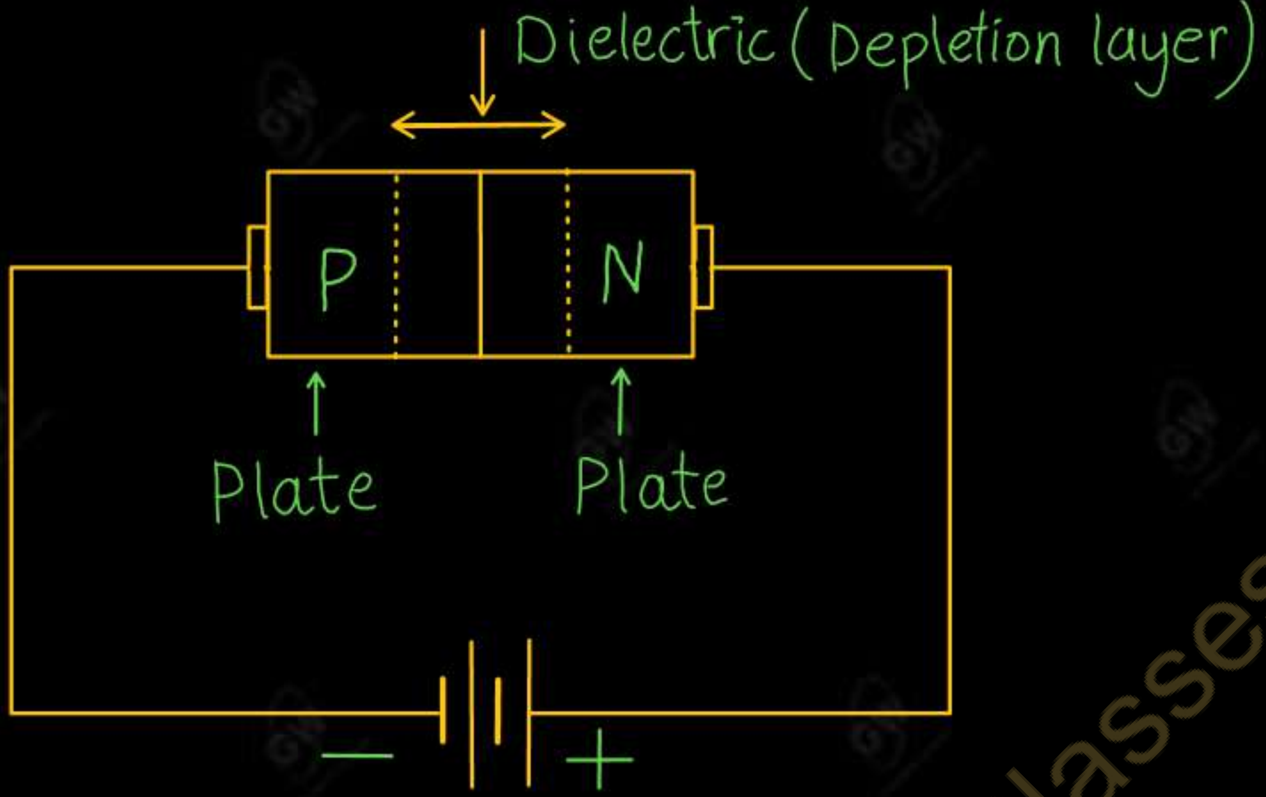
Symbol of varactor diode



Working

- Varactor diode is used in reverse bias condition
- Depletion layer created by the reverse bias acts as a capacitor dielectric whereas the P and N regions act as the capacitor plates

Gateway Classes



➢ So, there exist a capacitance, space charge capacitance or depletion region capacitance.

It is given by

$$\boxed{C_T = \epsilon \frac{A}{W}}$$

Where

ϵ = Dielectric constant

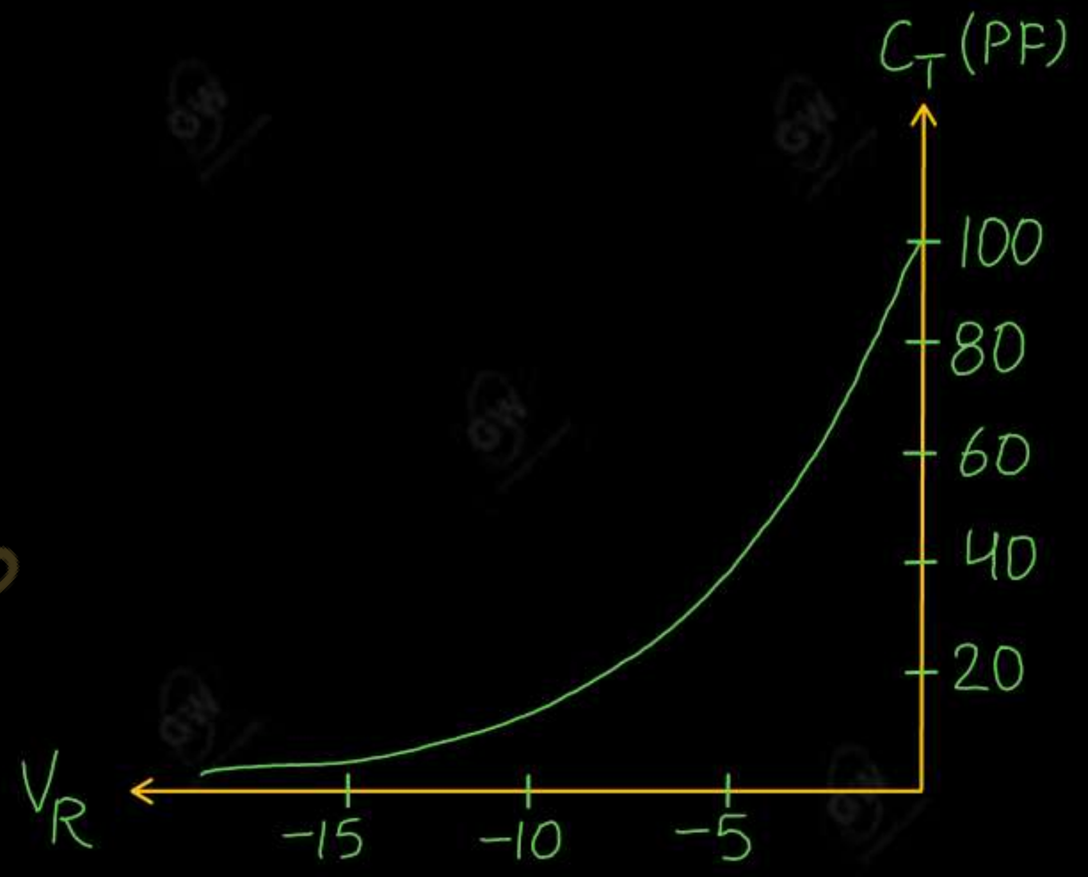
A = capacitor plate Area

W = Depletion layer width

Gateway Classes

Characteristics

- When the reverse bias voltage increases, the depletion layer width increases and capacitance decreases.
- When the reverse bias voltage decreases, the depletion layer width decreases and capacitance increases.
- Therefore, capacitance can be controlled by applied voltage.



Capacitance versus Reverse voltage

Application

(1) FM Modulator

(2) Tuning circuit

(3) T V receiver

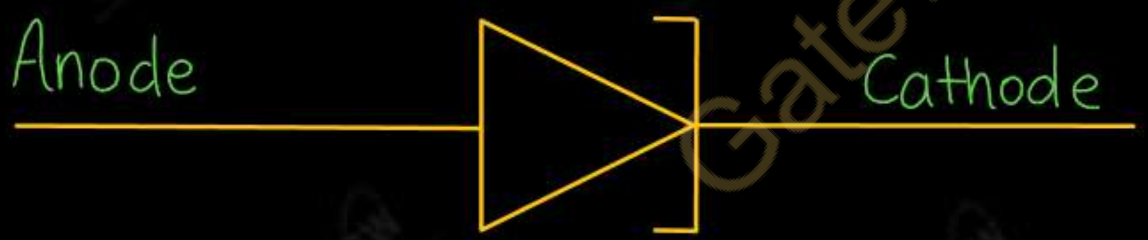
(4) Radio receiver

Gateway Classes

Tunnel Diode

- Tunnel diode is a P-N junction diode that exhibit negative resistance between two values of forward voltage
- Negative resistance means the Tunnel diode has a region where the current decreases as the voltage increases
- The operation of a tunnel diode is depend upon the tunneling effect and hence the name tunnel diode

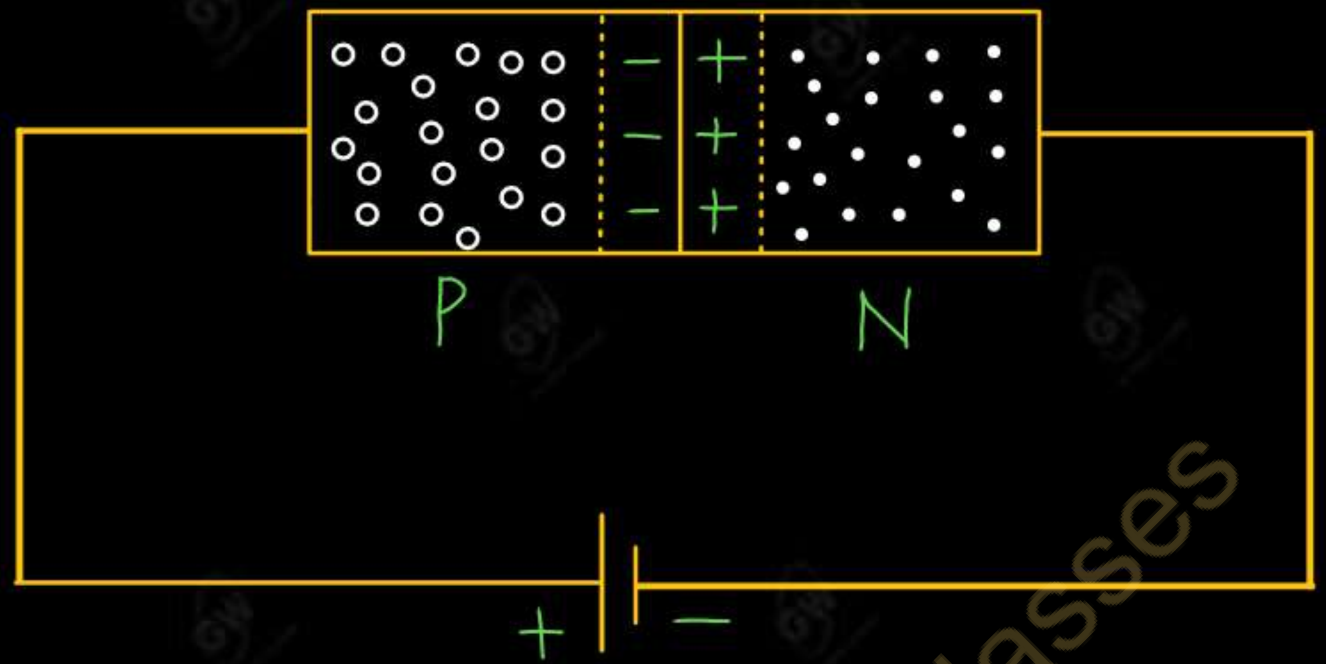
Symbol of tunnel diode



The semiconductor materials used for constructing the Tunnel diode are Germanium or Gallium Arsenide

Gateway Classes

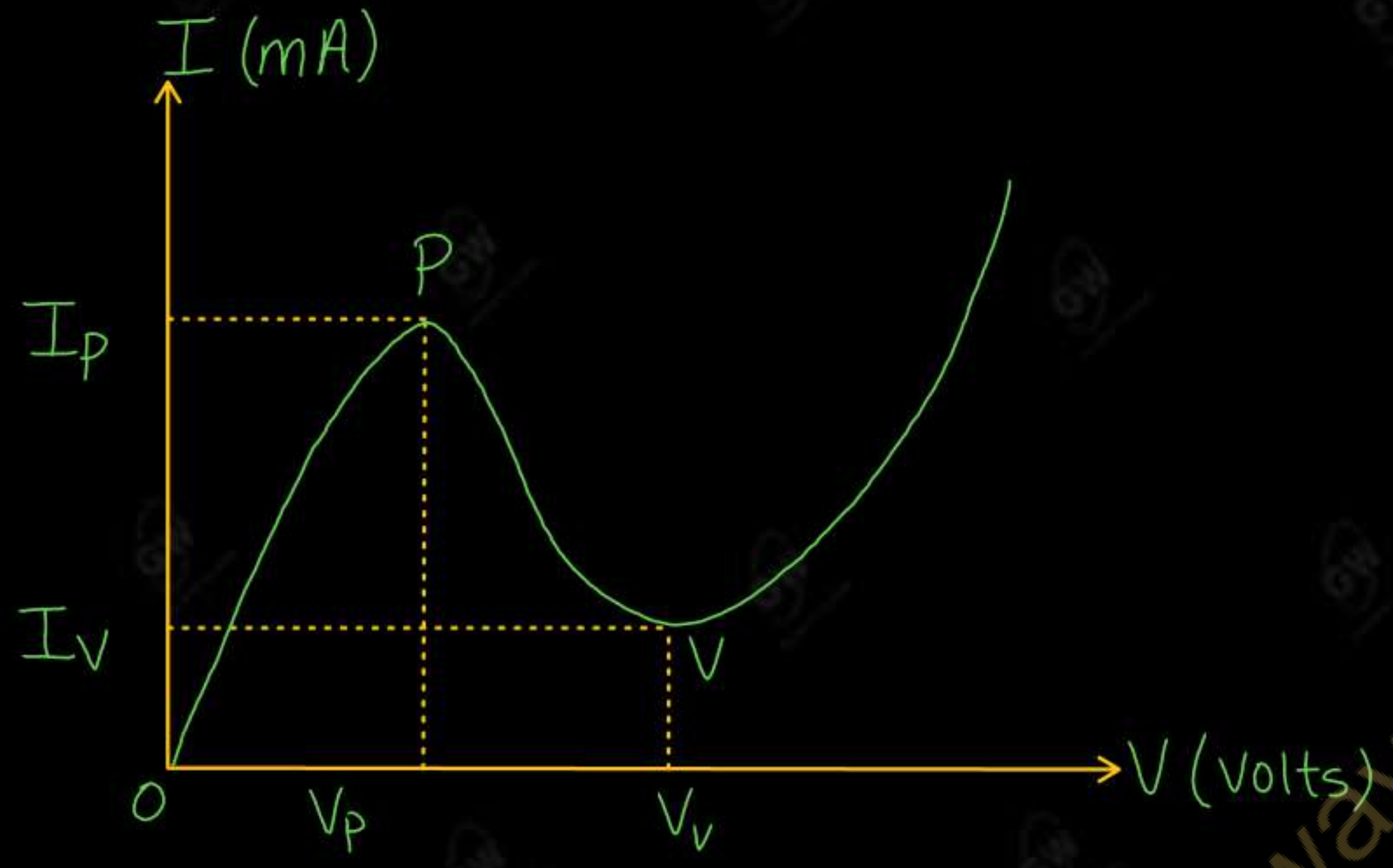
Working



- Tunnel diode is a heavily doped P-N junction diode. The doping of Tunnel diode is 1000 times greater than a simple diode. This heavy doping causes a large number of majority charge carriers. So that depletion layer become narrow and electron tunnel are formed through potential barrier
- If doping is very high then according to quantum mechanics an electron with energy less than barrier energy can penetrate the barrier or cross the barrier. This effect is called tunneling effect.
- Electrons tunnel through the potential barrier, allowing current to flow

Gateway Classes

V-I characteristic of Tunnel Diode



$V_P \rightarrow$ Peak point voltage

$V_V \rightarrow$ Valley point voltage

$I_P \rightarrow$ Peak point current

$I_V \rightarrow$ Valley point current

Point O to P : Current increases till point P at very low voltage due to tunneling effect

Point P to V : Current decreases till point V. At point V current is minimum and diode shows negative resistance

After Point V : Tunnel diode works as normal diode

Gateway Classes

Advantages of Tunnel Diode

- It consume low power
- Long life
- High speed device
- It produces Low noise

Disadvantages of Tunnel Diode

- No isolation between the input and output
- Tunnel diode can not be fabricated in large numbers

Applications of Tunnel Diode

- Tunnel diode are used as an ultra-high speed switch
- Tunnel diode are used in FM receivers
- Tunnel diode are used as logic memory storage device
- Tunnel diode are used in relaxation oscillation circuit

Gateway Classes

UNIT - 1 : DPP - 12

Q.1 What is the difference between the P-N junction diode and Light-Emitting Diodes AKTU 2024

Q.2 What do you mean by varactor diode? Write its one application AKTU 2023

Q.3 What is negative resistance of tunnel diode? AKTU 2025

Q.4 Describe structure and working Principle of Light-Emitting Diodes (LEDs).Discuss their applications in electronics AKTU 2024

Q.5 Explain working and characteristics of the Tunnel diode with the help of a neat diagram AKTU 2018

Q.6 Explain working of the following with the help of suitable diagram

(i) LED (ii) Photo Diode AKTU 2022

Gateway Classes

AKTU B.Tech I-Year, II-Year, III-Year

Gateway Classes

GATEWAY CLASSES

Courses in Application for the Session:2025-26

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4	Fundamental of electronics engineering	I/II
5	Engineering Mathematics – III	III / IV
6	Engineering Mathematics – IV	III / IV
7	Discrete Mathematics OR Discrete Structures and Theory of Logic (DSTL)	III

Gateway Classes